

# JetHE: Prepared Composition on Ramified Homomorphic Ciphertexts

**Abstract.** We study exact composition of encrypted truncated power series using ramified plaintext packing. Multiplication performs a packed convolution, but composition also needs coefficient movements that multiplication alone cannot express. JetHE interleaves a classical small-characteristic recursion so that each level uses one ciphertext product and one downward Hasse map. We give a norm-one integral lift of this map and prove a sharp, attained  $r + 1$  functional-payload minimum for dyadic order- $r$  transport in a universal single-layer model.

Preparing part of the recursion at the input holders trades their work and traffic against encrypted depth and public keys. Under a specified discrete dual Ring-LWE source, a Gaussian family has adaptive IND-CPA security and  $\tilde{O}(L^{3/2})$  complete bit work. Over the same fixed field, exact one-layer bilinear compilers with affine preprocessing, binary-linear recovery and full-domain private input encodings require  $\Omega(L^2)$  work. Shared-key matrix evaluation gives a further theoretical improvement when sufficiently many fresh batches are available together.

A separate ternary/CBD20 implementation computes sixteen length-256 compositions with 35 inputs, 19 products and one terminal ciphertext. Its complete warm median is 1.730 seconds; a same-backend one-layer control takes 7.710 seconds under its own parameter profile. The saving in total computation and traffic comes with larger public material and memory. The analysis and implementation quantify how coefficient transport and input preparation determine the value of encrypted depth.

**Keywords:** Homomorphic encryption · Ring-LWE · Power series composition · Hasse derivatives · Algebraic complexity

## 1 Introduction

JetHE is a leveled compiler for exact series composition. It computes  $f(g(z)) \bmod z^L$  over a binary extension field, with  $g(0) = 0$ . Separate input holders may prepare the two operands; an external evaluator computes on their encryptions and returns encrypted coefficients to a trusted recipient. The central implementation question is how to combine polynomial multiplication with the coefficient movements that composition requires.

Ramified packing gives a useful starting point: one ciphertext product computes an entire truncated convolution in every lane. Our organizing idea is to interleave the branches of Bernstein’s small-characteristic recursion [5] so that one such product updates a whole recursion level. This makes a downward Hasse derivative the remaining coordinate operation. We realize that operation by functional key switching and account for its keys and noise. The input holders can

also prepare several levels in advance, exchanging more preparation and encryption for less encrypted depth. The paper follows this tradeoff from the algebraic identity to security, complete work and a measured realization.

These results share a compiler but use different parameter profiles. The reading map below separates their roles; Table 3 later states their precise guarantees.

| Layer                       | Role in the argument                                                                                                                                                   |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shared algebra and compiler | Packing, interleaving and owner preparation specify the computation (Sections 3–5).                                                                                    |
| Reference transport         | Relative-basis switching gives a baseline schedule for the sharp Hasse identity and resource comparison (Sections 5 and 8).                                            |
| Gaussian family             | A distinct sampling law and growing parameter family support the security and complete-work theorems (Sections 7 and 8).                                               |
| Measured terminal profile   | The ternary/CBD20 instance retains the final product ciphertext for recovery; its reduction, correctness and timings are specified separately (Sections 6, 7.1 and 9). |

*Interleaving and optimal functional transport.* We interleave every branch at recursion level  $j$  into one carrier  $C_j$ . The update is

$$C_j = C_{j+1} + (g^{2^j} + z^{2^j})D_z^{[2^j]}C_{j+1} \pmod{z^L}.$$

Thus one product updates all branches. The downward Hasse derivative is the essential linear operation: nilpotence prevents implementing it by division by  $z^{2^j}$ . We construct an integral lift of coefficient norm one and realize it by relative-coordinate functional key switching. For dyadic order  $r$ , a universal single-layer realization requires at least  $r+1$  functional payloads when the source-secret identity is charged. An explicit identity attains the bound, including odd-modulus signs and negacyclic wrap. This is an optimality theorem within the stated functional-transport model.

*A complete cryptographic compiler.* Owner preparation collapses  $k$  initial recursion levels into one ciphertext-product layer. Put  $d = \log_2 L$  and  $\tau = d - k$ . The compiler uses  $2^k - 1 + \tau$  products,  $2^{k+1} - 1 + \tau$  encrypted inputs, and depth  $\tau + 1$ . Its acyclic independent-key graph supports a reduction for the full public transcript. For a growing Gaussian family, we explicitly reduce to uniform-secret discrete dual Ring-LWE under the declared parameter generator, including bounded sampling and reduction time. The same algebraic construction admits a separate ternary/CBD20 implementation. Its fresh-helper phase independence reduces a sufficient tail-noise amplification from  $\Theta(N^2)$  to  $O(NT^2)$  at threshold  $T$ . The two source laws and their parameter guarantees are specified separately.

**Table 1.** Arithmetic and cryptographic antecedents, and the results built on them.

| Established ingredient                                                | Result developed here                                                                         |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Small-characteristic composition [5]; fast classical composition [28] | One interleaved carrier per level and an encrypted preparation–depth–work tradeoff.           |
| Relative-basis functional switching [15, Sec. 4.6]                    | A norm-one Hasse lift, sharp payload count, and support-sensitive noise accounting.           |
| Leveled HE [11]; polynomial plaintext quotients [22]                  | An acyclic compiler with an explicit source reduction and all-party resource bounds.          |
| Fast matrix arithmetic [3]                                            | Exact shared-key batch transport, including integer-arithmetic costs and the batch threshold. |

*A separation between compiler interfaces.* For the fixed field  $\mathbb{F}_{2^{16}}$ , the Gaussian family attains  $\tilde{O}(L^{3/2})$  complete bit work, including setup and every party’s computation. An exact one-layer bilinear compiler with affine preprocessing and binary-linear recovery needs  $\Omega(L^2)$  work under the full-domain private-encoding contract. The lower bound allows arbitrary independent owner preparation and mixing across jobs. Its key ingredient is a growing coefficient-function space over a fixed finite field; a secure-encoding argument converts width to complete work. The result quantifies the benefit of encrypted depth for this interface.

Shared switching keys also expose an exact matrix computation across fresh batches. Known rational matrix algorithms further improve the complete-work exponent when sufficiently many batches are available together. Section 8 gives the exponents, availability threshold and source conditions, and extends the one-layer lower bound to joint preparation. The arithmetic algorithms are established; the bound charges their use within the encrypted workflow.

*An implemented instance and its costs.* Our primary implementation computes sixteen length-256 compositions using 102 hint rows, 35 inputs, and 19 products at depth five. It returns one three-component ciphertext and achieves a 1.730-second complete warm median. OpenFHE 1.5.1 BGV adapters for the same exact outputs have medians of 15.795 and 12.885 seconds under their specified configurations. A custom one-prime control using the same arithmetic backend takes 7.710 seconds. JetHE reduces preparation traffic and total computation while using more public material and memory; the one-prime control has faster evaluator time. The evaluation states the tuning, security, and measurement differences needed to interpret these observations.

*Position among existing techniques.* Table 1 separates the ingredients from their use here. The central connection is that interleaving specifies a downward map, the sharp transport theorem prices its functional payloads, and the prepared schedule balances those costs against owner work and depth. A compatible conventional HE base can inherit this compiler; our separation concerns the specified one-layer interface, not every conventional HE program. Section 8 compares the reference and optimized transports within the same cryptographic model.

The supplement supplies complete derivations, source transformations and their quantitative limits, alternative compiler ledgers, implementation equivalence results, and all reported measurements.

## 2 Functionality, notation, and threat model

Write  $\Phi_m$  for the  $m$ th cyclotomic polynomial. For a field  $K$ , let  $\mathcal{J}_L(K) = K[z]/(z^L)$ . A jet records the  $L$  coefficients of a polynomial after discarding terms of degree at least  $L$ . Addition is coefficientwise and multiplication is truncated convolution. Throughout the construction,  $L = 2^d$ ,  $K = \mathbb{F}_{2^{16}}$ , and  $h = 16$  independent jets occupy one plaintext carrier. The Hasse derivative is

$$D_z^{[r]} \left( \sum_i a_i z^i \right) = \sum_{i \geq r} \binom{i}{r} a_i z^{i-r},$$

where the binomial coefficients are reduced in  $K$ . We apply it to the unique degree-below- $L$  representative.

**Definition 1 (Prepared composition).** *For each lane  $i$ , an outer owner holds  $f_i \in K[z]_{<L}$  and an inner owner holds  $g_i \in zK[z]_{<L}$ . Each owner noninteractively prepares from its own input and public parameters, and encrypts with fresh coins under the prescribed public keys. The evaluator returns encrypted data from which the recipient recovers every coefficient of  $f_i(g_i(z)) \bmod z^L$  for all  $i$ .*

The condition  $g_i(0) = 0$  makes substitution well-defined on the truncated-series quotient: discarded terms remain of degree at least  $L$  after substitution. The primary construction returns one completed composition ciphertext. Comparisons also consider terminal statistics followed by specified plaintext reconstruction; that reconstruction is part of the measured workload. Separate operand ownership describes the permitted preparation, and does not imply independently generated owner keys or a multi-key security guarantee.

*Trust and leakage.* A key authority generates the independent vertex keys, supplies owner public keys and evaluator switching material, and provides the recipient's terminal decryption key. The adversary is a passive evaluator with all public parameters, public and evaluation keys, prepared input ciphertexts, and computed ciphertexts. It has no secret key or decryption oracle. Lengths, occupied lanes, preparation shapes, key policy, and traffic are public. Input confidentiality is proved against this adversary. The recipient is trusted with respect to input privacy; circuit privacy, recipient privacy, malicious-input consistency, and verifiable evaluation are outside the guarantee. In particular, returning only a composition ciphertext does not prove that its secret-key holder learns only that composition.

*Outsourced symbolic computation.* One deployment consistent with this model has two data services holding coefficient arrays for an authorized analysis unit. They prepare independently; an external server performs the encrypted products and coordinate transport, and the unit decodes the completed composition. A completed output avoids making that unit assemble and combine a larger collection of intermediate statistics. The analysis charges the services' preparation and the unit's recovery, so the benefit of this work placement is assessed by total cost. This workflow motivates the interface; the experiments measure composition itself, rather than an application-specific deployment.

*Correctness quantifiers.* The algebraic compiler is exact on the full stated input domain. Deterministic ciphertext bounds apply to all coins within the specified supports when admitted. The smaller measured parameter schedule instead uses a probabilistic bound for at most 1024 input batches fixed independently of setup and encryption randomness. These batches can reuse keys but cannot be chosen as functions of public keys or earlier execution results. Adaptive IND-CPA security does not enlarge this fixed-input correctness statement.

*Cost conventions.* Multiplicative depth counts dependent ciphertext–ciphertext product layers. A depth-one layer may contain many products, executed serially on a single processor. Warm time is the complete second batch under an existing setup, with fresh encryption. First-use time is setup plus the first batch. MiB means  $2^{20}$  bytes. Ring dimension, plaintext capacity, logical ciphertext count, and component count are reported separately.

### 3 Ramified packing and its codec

#### 3.1 A fully charged binary tensor codec

For the construction below take  $L = 2^d \geq 2$  and

$$E = \mathbb{Z}[b]/(\Phi_{257}(b)), \quad R_L = E[t]/(t^L + 1), \quad N = 256L.$$

Coprime conductors identify  $R_L$  with  $\mathbb{Z}[\zeta_{514L}]$ . Use the integral basis  $t^i b^v$  for  $0 \leq i < L$ ,  $1 \leq v \leq 256$ . Since  $\text{ord}_{257}(2) = 16$ , ordinary CRT on the  $b$  axis maps  $E/2E$  to  $K^{16}$ , where  $K = \mathbb{F}_{65536}$ . Let  $C$  be its  $256 \times 256$  binary matrix in fixed field coordinates, and let  $\chi(C)$  count XORs in the chosen linear circuit. Put  $z = t + 1 \pmod{2}$ .

**Theorem 1 (Tensor Taylor codec).** *The decoder  $F \mapsto (F(1 + z, \beta_i))_{i=1}^{16}$ , for roots  $\beta_i$  from the sixteen distinct doubling orbits, is an algebra isomorphism  $R_L/2R_L \rightarrow \mathcal{J}_L(K)^{16}$ . Decoding and encoding respectively use*

$$L\chi(C) + \frac{Nd}{2}, \quad L\chi(C^{-1}) + \frac{Nd}{2}$$

*bit XORs, plus linear movement. Dense matrices give  $O(256N + N \log L)$  bit operations and  $O(N + 256^2)$  bits of coefficient workspace and matrix storage, excluding optional tables and runtime overhead.*

*Proof.* CRT handles the  $b$  axis, and  $t^L + 1 = (t + 1)^L$  in characteristic two. If  $c_j$  are the  $t$  coefficients, the coefficient of  $z^r$  after translation is  $\sum_{j:r \subseteq j} c_j$ , where containment is on binary index bits. For every bit  $a$ , add  $c_{r+2^a}$  into  $c_r$  for indices with  $r_a = 0$ . This subset butterfly has  $Ld/2$  additions of 256-bit vectors. Translation by one is its own inverse in characteristic two. The CRT and Taylor maps act on different tensor axes and commute. Their counts add; one length- $L$  buffer and the two fixed matrices suffice. The underlying subset transform is classical [8].

The implementation fixes the CRT matrices and charges their construction, validation, and inversion during setup.

*One packed example.* At  $L = 256$ , one plaintext contains sixteen independent jobs, each with 256 arbitrary  $K$  coefficients: 4096 field symbols, or 65536 bits. A fresh two-polynomial ciphertext carries all of them. Multiplying two such ciphertexts computes sixteen full truncated convolutions. The binary tensor-basis representative does not restrict those field coefficients to 0 or 1. A conventional slotwise layout can instead hold the 4096 individual field symbols, or encode a larger terminal evaluation representation; merely filling those slots does not make slotwise multiplication the same convolution. Its actual compiler, physical ciphertext count and decoding costs must be priced.

### 3.2 Automorphisms and native circuit semantics

A cyclotomic automorphism can combine a lane permutation, field isomorphism and invertible triangular parameter substitution. A coefficient-preserving lane rotation must undo those extra actions; it is not assumed free. The explicit action is in Appendix A.1.

**Corollary 1 (Topology preservation).** *An arithmetic circuit over  $\mathcal{J}_L(K)^h$  using addition and multiplication has the same addition/multiplication topology under an algebraic codec into the plaintext ring. Its encrypted realization has the same number of nonlinear layers, provided all required alignments and key transitions are provisioned and the correctness budget below is admitted.*

*Proof.* Apply the algebra isomorphism at every wire and induct on the circuit DAG. Key management preserves each wire’s plaintext but has separately charged cost.

This is a preservation theorem for a given circuit, not optimality against public specialization, prepared inputs, or another arithmetic compiler.

## 4 An explicit acyclic leveled realization

We instantiate the binary tensor family of Theorem 1 using conventional exact-HE ingredients [11]. We first specify the finite-support laws used by the implementation, together with their deterministic correctness bounds. The Gaussian

family in Section 7 uses the same algebraic interface with different sampling laws and corresponding bounds. The basis, source law and acyclic key policy are explicit parameters of each realization.

#### 4.1 Setup, keys, encryption and evaluation

*Setup and key generation.* For odd  $q$ , write  $R_q = R_L/qR_L$ . Polynomial coefficient norms and sampling distributions refer to the integral tensor basis of Section 3. Choose the public codec, odd modulus schedule and a directed acyclic key graph. At each vertex independently sample a secret with coefficients uniform in  $\{-1, 0, 1\}$ . Sample each error coefficient from  $\text{CBD20} = \text{Binomial}(40, 1/2) - 20$ , independently unless the same published row is being reused. Its support bound is  $\beta = 20$ . Small polynomials are sampled once and then reduced into every RNS channel; independent small errors per channel would be a different scheme. Publish a public key  $(b, a) = (-as + 2e_{\text{pk}}, a)$ , with uniform  $a \in R_q$ , only at vertices used for owner encryption.

For an edge to a fresh destination  $s'$ , an evaluation bank for a predecessor payload  $v$  consists of

$$(b_j, a_j) = (-a_j s' + 2e_j + \gamma^j v, a_j), \quad 0 \leq j < g_q.$$

Masks and errors are fresh across rows. The gadget represents each centered source coefficient exactly as  $x = \sum_j d_j \gamma^j$ , with  $|d_j| \leq D_g$ . Here  $g_q$  is the number of gadget digits at modulus  $q$ . Admission checks that these digits suffice for every centered coefficient. For polynomial digits write  $\text{Ext}(x, \mathbf{H}) = \sum_j d_j (b_j, a_j)$ .

*Encode, encrypt and decrypt.* Encode all lanes by the inverse tensor codec and take the unique binary coefficient representative  $\mu \in R_L$ . With fresh ternary  $u$  and errors  $e_0, e_1$ , encrypt under  $(b, a)$  as

$$\text{Enc}(\mu) = (bu + 2e_0 + \mu, au + 2e_1) \in R_q^2.$$

To decrypt, center the coefficients of  $c_0 + c_1 s \pmod{q}$ , take coefficient parity, then apply the tensor decoder. A ciphertext's metadata records its key vertex and modulus; a different label is not a key switch.

*Eval.* Add compatible-key ciphertexts componentwise. A raw product of  $(c_0, c_1)$  and  $(d_0, d_1)$  under  $s$  is  $(c_0 d_0, c_0 d_1 + c_1 d_0, c_1 d_1)$ , with phase evaluated at  $(1, s, s^2)$ . Rekey it to a fresh  $s'$  using *both* payload banks  $s, s^2$ . An identity alignment uses the payload  $s$  alone. For an automorphism  $\sigma$ , first apply  $\sigma$  to both components, then use the bank for  $\sigma(s)$  under the fresh destination. Every needed branch alignment is an actual provisioned edge. Same-key relinearization and backward key edges are not part of this acyclic interface. The Hasse transport needed by composition is specified in Section 5.

## 4.2 Uniform correctness and modulus admission

**Lemma 1 (Tensor product norm and fresh error).** *In the fixed integral basis, for  $\kappa = 511L$ ,*

$$\|xy\|_\infty \leq \kappa\|x\|_\infty\|y\|_\infty.$$

*Every fresh ciphertext has a phase  $\mu + 2e \pmod{q}$  with  $\|e\|_\infty \leq B_{\text{fresh}} = (2\kappa + 1)\beta$ .*

*Proof.* In the odd axis first convolve modulo  $b^{257} - 1$ . Eliminating the constant coefficient by  $1 = -\sum_{v=1}^{256} b^v$  makes each output the difference of a sum of at most 255 products and a sum of at most 256 products. The negacyclic  $t$  convolution contributes at most  $L$  such terms. The encryption phase error is  $e_{\text{pk}}u + e_0 + e_1s$ ; the norm bound and ternary supports give the stated expression. No probabilistic independence is needed for this deterministic correctness argument.

At modulus  $q$  put

$$\Delta_q = \kappa g_q D_g \beta, \quad P(B, C) = \kappa(B + C + 2BC) + \left\lfloor \frac{\kappa + 1}{2} \right\rfloor.$$

The following bounds include carries from restoring binary representatives:

| Operation                          | Output error bound                                       |
|------------------------------------|----------------------------------------------------------|
| Addition                           | $B + C + 1$                                              |
| Raw product                        | $P(B, C)$                                                |
| One-bank identity alignment        | $B + \Delta_q$                                           |
| Two-bank product rekeying          | $P(B, C) + 2\Delta_q$                                    |
| Automorphism and one-bank rekeying | $B + 1 + \Delta_q$                                       |
| Public multiplication by $w$       | $AB + \lfloor (A + 1)/2 \rfloor, A = \kappa\ w\ _\infty$ |

For the automorphism row,  $b \mapsto b^v$  permutes the odd basis and  $t \mapsto t^u$ ,  $u$  odd, is a signed permutation; its norm is one. For another basis or public linear map use its actual operator norm instead.

*Derivation.* Expanding  $(\mu + 2e)(\nu + 2f)$  gives error  $\mu f + \nu e + 2ef$  before restoring the binary representative of  $\mu\nu$ . Its clean product has norm at most  $\kappa$ , giving the displayed carry. Each bank adds  $\sum_j d_j e_j$ , bounded by  $\Delta_q$ . An automorphism is norm one here, but a negative binary coefficient incurs a carry of size one. Public multiplication has norm at most  $A$ , giving its analogous carry. Correlations induced by reused hints do not invalidate these deterministic inequalities.

*Parity-preserving prime drop.* For  $Q = p_*q$  with odd  $p_*$ , write  $v = x \pmod{p_*}$  in  $[0, p_*)$  and set  $r = v$  if  $v$  is even, and  $r = v - p_*$  otherwise. Replace each ciphertext coefficient by  $(x - r)/p_* \pmod{q}$ . Here  $r$  is even and  $|r| < p_*$ ; the result is independent of the representative modulo  $Q$ . This is modulus switching, not bootstrapping [11]. For a two-component phase  $\mu + 2e$  the new error is

$$e' = \frac{2e - r_0 - r_1s - (p_* - 1)\mu}{2p_*}, \quad \|e'\|_\infty \leq \left\lfloor \frac{2B - 1 + (\kappa + 2)p_*}{2p_*} \right\rfloor.$$

The numerator is divisible by  $p_*$  by the phase identity and is even; the support and product bounds prove the inequality. Apply drops only to two-component states, after rekeying, and admit the result at the new modulus. In RNS arithmetic, invert the dropped channel to choose  $r$ , lift and transform it in each retained channel, and multiply by  $p_*^{-1}$  there. These transforms and lifts are charged; there is no free full-modulus change.

**Theorem 2 (Uniform admitted leveled correctness).** *Fix a finite public DAG schedule over this interface, exact gadgets and the stated support bounds. Propagate  $B_{\text{fresh}}$  through the displayed recurrences, including all alignments, Hasse transports as below, and drops. If every complete state at modulus  $q_v$  satisfies  $q_v > 2 + 4B_v$ , honest evaluation decrypts to the scheduled full-jet arithmetic for every admitted input and every realization within the specified supports. The induction uses only the declared leveled interfaces and admission bounds.*

*Proof.* Each operation preserves its asserted affine phase and obeys the indicated bound, so induction on the public DAG gives the invariant at every state. Since  $\|\mu\|_\infty \leq 1$ , the inequality makes  $\mu + 2e$  the centered representative of its phase. Coefficient parity therefore recovers  $\mu$ . The codec is an algebra isomorphism, giving the full-jet result.

Admission is public and precedes execution. An inadmissible manifest or sampler exhaustion produces an explicit abort, not a secret-dependent retry or an excluded experimental outcome. The theorem assumes correct arithmetic; OS randomness, machine faults and side channels are not consequences of it.

*Parameterized scope.* The template applies to any dyadic  $L$  and finite supported circuit. For fixed radix and circuit, its no-drop bound is a polynomial in  $g_q$ : addition/switching are linear and multiplication composes polynomials. Taking a sufficient exact gadget with  $g_q = O(1 + \log q)$  therefore permits an odd  $q > 2 + 4B_{\text{max}}$  for every fixed circuit, since  $q$  eventually exceeds that polynomial in  $\log q$ . Direct exact arithmetic needs no NTT assumption. This is a correctness-admissible family, not a demonstrated security-parameter versus depth tradeoff. The finite prime list and four-limb executable do not instantiate all members. Hardness of the chosen ring/law/modulus family remains the explicit assumption of Section 7.

## 5 A prepared compiler for private full-jet composition

The construction places all recursive subproblems in one carrier as interleaved module coordinates, sharing the same inner polynomial. These coordinates are not independent shorter algebra factors. Classical small-characteristic composition [5] supplies the recursion; we realize each level on encrypted packed data with norm-controlled coefficient transport and explicit key/preparation costs.

### 5.1 One carrier, one product and one Hasse map per level

Let  $L = 2^d$  and  $f, g \in K[z]_{<L}$  with  $g(0) = 0$ . For  $r = 2^j$ , define  $G_j(X) = \sum_{i < L/r} g_i^{2^j} X^i$ . The binary composition recursion starts with  $R_{d,t} = f_t$  and sets

$$R_{j,t}(X) = R_{j+1,t}(X^2) + G_j(X)R_{j+1,t+r}(X^2) \pmod{X^{L/r}}, \quad 0 \leq t < r.$$

Splitting the outer coefficients into even and odd subsequences proves this recursion by induction;  $G_j(z^r) = g(z)^r \pmod{z^L}$  and the inner series has zero constant term. Interleave its entire level in  $C_j(z) = \sum_{t < 2^j} z^t R_{j,t}(z^{2^j})$ . Then  $C_d = f$  and  $C_0 = f(g) \pmod{z^L}$ .

**Theorem 3 (Interleaved composition update).** *For  $r = 2^j < L$  and  $U_j = g^{2^j} + z^{2^j}$ ,*

$$C_j = C_{j+1} + U_j D_z^{[r]} C_{j+1} \pmod{z^L}.$$

*Consequently owner-prepared  $U_j$  permit  $d$  native products,  $d$  Hasse maps and  $d$  additions on one packed carrier.*

*Proof.* The parity of  $\binom{a}{r}$  is the  $j$ th bit of  $a$ . Hence  $O = D_z^{[r]} C_{j+1}$  selects its bit- $j$ -one terms and shifts them down by  $r$ , while  $C_{\text{even}} = C_{j+1} + z^r O$  selects the bit-zero terms. Multiplication by the shared  $g^{2^j} = G_j(z^r)$  preserves the residue classes modulo  $r$ . The preceding recursion says  $C_j = C_{\text{even}} + g^{2^j} O$ , which is the stated identity. Apply it separately in all packed lanes.

*A length-eight example.* The unprepared recursion follows

$$C_3 = f \xrightarrow{r=4} C_2 \xrightarrow{r=2} C_1 \xrightarrow{r=1} C_0 = f(g) \pmod{z^8}.$$

For the middle step, write the *current* state as  $C_2 = \sum_{i=0}^7 a_i z^i$ . The order-two Hasse map selects indices 2, 3, 6, 7 and shifts them down by two:

$$\begin{aligned} O &= D_z^{[2]} C_2 = a_2 + a_3 z + a_6 z^4 + a_7 z^5, \\ C_1 &= C_2 + (g^2 + z^2)O = (a_0 + a_1 z + a_4 z^4 + a_5 z^5) + g^2 O \pmod{z^8}. \end{aligned}$$

Thus the update replaces the selected terms'  $z^2$  factor by  $g^2$ . The coefficients  $a_i$  already incorporate the order-four update and need not equal the coefficients of  $f$ . The downward shift is essential: a mask alone cannot implement it, and  $z^2$  is not invertible. Cancellation here is in the binary plaintext algebra; integer noise is bounded separately.

*Why additional ring depth does not replace transport.* For raw packed inputs, addition, multiplication, constants and ring automorphisms preserve a mixed-ideal invariant at every circuit depth. Let  $J = (z)$  in the product of jet algebras. Changing  $f$  within  $J$  and  $g$  within  $J^{L-1}$  makes each output's four-input mixed difference lie in  $J J^{L-1} = 0$ . Fixed additive reconstruction preserves zero, but composition on  $f \in \{0, z\}$  and  $g \in \{0, z^{L-1}\}$  has mixed difference  $z^{L-1} \neq 0$ . Thus raw-input ring circuits require an additional capability, regardless of depth (Proposition 16). Owner preparation and functional coefficient transport provide that capability here; compatible conventional HE can supply the same operations.

## 5.2 A norm-one lift and its exact key-switch interface

Put  $S_r = E[t^{2r}] \subset R_L$ . The relative basis is  $1, t, \dots, t^{2r-1}$ , and define the integral map

$$H_r(t^i b^v) = \begin{cases} t^{i-r} b^v, & \text{bit } j \text{ of } i \text{ is one,} \\ 0, & \text{otherwise.} \end{cases}$$

It has norm one, preserves binary representatives and reduces to  $D_z^{[r]}$ : Hasse differentiation commutes with translation  $z = t + 1$  modulo two. It is  $S_r$ -linear because shifting by a multiple of  $2r$  preserves the selected bit; negacyclic wrap gives the same sign on both sides.

**Lemma 2 (Full-dimensional Hasse transport).** *For a phase  $c_0 + c_1 s = \mu + 2e \pmod{q}$ , publish  $2r$  banks with payloads  $H_r(t^i s)$  under a fresh independent full-ring destination  $h_r$ . Write  $c_1 = \sum_{i < 2r} t^i a_i$  over  $S_r$ . Then*

$$(H_r(c_0), 0) + \sum_{i < 2r} \text{Ext}(a_i, H_i)$$

*has plaintext  $D_z^{[r]} \mu$  and error at most  $B + \Delta_q$ . It uses  $2rg_q$  rows, each consisting of two full-ring elements.*

*Proof.*  $S_r$ -linearity gives the clean phase  $H_r(c_0 + c_1 s)$ . Every digit of  $a_i$  is supported on multiples of  $2r$ , so its product with a full-ring row error has norm at most  $511(L/(2r))D_g\beta$ . Summing  $2rg_q$  terms gives  $\Delta_q$ . The original error is mapped with norm one, and  $H_r(\mu)$  needs no representative carry.

Relative-basis functional switching [15, Sec. 4.6] supplies the mechanism. The lift and support-sensitive bound are derived above; masks, errors and destination secrets remain full-dimensional. Subring digits do not reduce the hardness dimension.

## 5.3 A sharp bound for Hasse transport

Use  $R_L = E[t]/(t^L + 1)$  and the integral basis of Section 3, and fix a dyadic order  $r < L$ . Let  $q > 1$  be odd and put  $A = R_L/qR_L$ . A norm-one integral lift  $F : R_L \rightarrow R_L$  is  $\mathbb{Z}$ -linear, reduces to the binary Hasse map  $H_r$  modulo two, and has induced coefficient infinity norm at most one. It induces a map on  $A$ .

We count fixed functional payloads in identities

$$F(cs) = u(c) + \sum_{\ell=1}^m d_\ell(c) \phi_\ell(s) \quad \text{for every } c, s \in A. \quad (1)$$

The coefficients depend on the public multiplier  $c$ , but not on the secret  $s$ . Payload functions  $\phi_\ell : A \rightarrow A$  may be nonlinear. The sending-secret identity is charged if it is one of these payloads. The free-identity variant additionally permits an uncharged term  $v(c)s$ . This universal functional model specifies one realization layer, rather than all possible ciphertext algorithms or a restricted sampled secret set.

**Theorem 4 (Minimum over norm-one Hasse lifts).** *Every norm-one integral lift  $F$  requires  $m \geq r + 1$  in (1). The minimum over choices of lifts is attained by the standard  $H_r$ , with payloads*

$$s, H_r(s), H_r(ts), \dots, H_r(t^{r-1}s).$$

*In the free-identity variant, the minimum over choices of lifts is  $r$ . These statements hold for every odd  $q$ , without requiring  $E$ -linearity of the competing lifts.*

*Proof.* The matrix of  $F$  has integral entries and absolute row sums at most one. Reduction modulo two specifies one odd entry in each nonzero Hasse row. That entry is therefore 1 or  $-1$ , and all other entries vanish. In particular,  $F(t^i b) = 0$  for  $i < r$ , whereas  $F(t^i b) = \pm t^{i-r} b$  for  $r \leq i < 2r$ . Both  $t$  and  $b$  are units in  $A$ .

Subtracting the identity at  $s = 0$  removes  $u(c)$  and replaces  $\phi_\ell$  by  $\phi_\ell - \phi_\ell(0)$ . Evaluate the  $r + 1$  functions

$$F(t^r s), \quad F(t^i s) \quad (0 \leq i < r)$$

at  $s = b$  and  $s = t^{r-j}b$  for  $0 \leq j < r$ . The first column is  $(\pm b, 0, \dots, 0)^t$ . In the remaining  $r$  by  $r$  block, row  $i$  and column  $j$  have entry zero for  $i < j$  and  $\pm t^{i-j}b$  for  $i \geq j$ . There is no wrap in that block because  $1 \leq i + r - j \leq 2r - 1 < L$ . It is triangular with unit diagonal. Consequently the full matrix has unit determinant.

Equation (1) factors this evaluation matrix through  $m$  payload columns. Reduce its entries modulo any maximal ideal of  $A$ . Its rank over the resulting field is  $r + 1$ , whereas the factorization has rank at most  $m$ . This proves the lower bound. Only evaluated entries are reduced: the payload functions and the lift need not descend to functions on that residue field.

For attainment, write

$$c = \sum_{j < r} t^j (u_j + t^r v_j), \quad u_j, v_j \in E[t^{2r}].$$

The standard lift is  $E[t^{2r}]$ -linear and satisfies the integral identity  $H_r(t^r y) = y - t^r H_r(y)$ . Therefore

$$H_r(cs) = H_r(c)s + \sum_{j < r} (u_j - t^r v_j) H_r(t^j s). \quad (2)$$

This identity includes negacyclic wraps and uses no division. It supplies the  $r + 1$  counted payloads at every modulus.

For the free-identity lower bound, replace the first function in the evaluation matrix by  $s$ . Its first column is now  $(b, 0, \dots, 0)^t$ , and the same block gives rank  $r + 1$ . There are at most  $m + 1$  columns when the free identity is included, so  $m \geq r$ . Equation (2) attains this bound.

*From payloads to noisy transport.* Signed partial permutations of shared gadget digits implement the attaining coefficients with error increment at most  $3\Delta_q/2$  and  $(r + 1)g_q$  rows. The original transport uses  $2rg_q$  rows and increment  $\Delta_q$ .

The support proof is in Appendix A.2.1. The selected profile uses the attaining transport above order one and the original at order one.

For one reference tail stage, let  $c$  encrypt  $C_{j+1}$  under  $s$  and let  $u_j$  freshly encrypt  $U_j$  under the independent Hasse destination  $h_r$ . The evaluator follows this path, with plaintexts shown below their ciphertexts:

$$\underbrace{c}_{C_{j+1}} \xrightarrow{\text{Hasse switch } s \rightarrow h_r} \underbrace{h}_{D_z^{[r]} C_{j+1}} \xrightarrow{\times u_j} \underbrace{p}_{U_j D_z^{[r]} C_{j+1}} .$$

Rekey the raw product  $p$  through the  $h_r, h_r^2$  banks to  $s'$ , align  $c$  through its  $s \rightarrow s'$  bank, and add to obtain  $C_j$ . The successor secrets  $h_r$  and  $s'$  are independent of  $s$  and each other; the edges are  $s \rightarrow h_r$ ,  $h_r \rightarrow s'$ , and  $s \rightarrow s'$ . The stage has  $(2r + 3)g_q$  rows and sufficient output bound

$$T_q(B) = B + 3\Delta_q + P(B + \Delta_q, B_{\text{fresh}}) + 1. \quad (3)$$

This charges alignment and both product banks. It does not assume same-key relinearization or free relabeling of the bypass branch.

#### 5.4 Collapse an owner-prepared prefix into one nonlinear layer

Choose  $1 \leq k \leq d$ , put  $\tau = d - k$ ,  $M = 2^k - 1$ , and  $J = \{\tau, \dots, d - 1\}$ . For each subset  $S \subseteq J$ , the outer owner prepares  $f_S = D^{[\delta_S]} f$ ,  $\delta_S = \sum_{j \in S} 2^j$ . For nonempty  $S$ , the inner owner prepares  $W_S = \prod_{j \in S} U_j$ . All these prefix inputs are freshly encrypted under one initial public key. The inner owner also supplies the  $\tau$  remaining tail multipliers under their Hasse destination public keys.

**Theorem 5 (Owner-product prepared compiler).** *Sum the  $M$  raw products  $W_S f_S$  together with  $f_\emptyset$ , then rekey that three-component sum once. Follow it by the  $\tau$  tail stages of (3). Subject to public correctness admission, the result contains every coefficient of every  $f_i(g_i) \bmod z^L$ . The construction uses*

$$\begin{array}{ll} M + \tau & \text{private products,} \\ \tau + 1 & \text{online nonlinear layers,} \\ 2\tau + 2 & \text{independently sampled keys,} \\ 2^{k+1} - 1 + \tau & \text{prepared input ciphertexts.} \end{array}$$

Writing  $q_0$  for the prefix modulus and  $q_r$  for each tail modulus, the number of published hint rows is

$$2g_{q_0} + \sum_{r=2^{\tau-1}, \dots, 1} (2r + 3)g_{q_r}.$$

The prefix's sufficient error bound is

$$B_P = B_{\text{fresh}} + M\kappa(2B_{\text{fresh}} + 2B_{\text{fresh}}^2) + \left\lceil \frac{M\kappa}{2} \right\rceil + 2\Delta_{q_0}. \quad (4)$$

*Proof.* For  $i < j$ ,  $U_j$  has only exponents divisible by  $2^j$ , so  $D^{[2^i]}(U_j X) = U_j D^{[2^i]}X$ . Hasse derivatives with disjoint index bits compose without a binomial factor. Expanding the  $k$  high-bit updates therefore gives exactly  $f_\emptyset + \sum_{S \neq \emptyset} W_S f_S$ . Its prepared products are computed by one owner, not by an uncharged encrypted multiplication subroutine. The remaining updates give composition by Theorem 3.

Each raw product contributes error at most  $\kappa(2B_{\text{fresh}} + 2B_{\text{fresh}}^2)$  before representative reduction. The clean aggregate has coefficients in  $[-M\kappa, M\kappa + 1]$ , so its *single* final carry is at most  $\lceil M\kappa/2 \rceil$ . The unmultiplied term contributes  $B_{\text{fresh}}$  and the two aggregate rekeying banks add  $2\Delta_{q_0}$ . This proves (4). There is one prefix destination, followed by two fresh keys per tail. Summing their rows and operations proves the inventory. Correctness follows from Theorem 2.

## 6 The selected terminal construction

Our primary instance has  $L = 256$ ,  $k = 4$ , and tail orders 8, 4, 2, 1. It uses the paid-identity Hasse transport at orders eight, four, and two, and the original two-bank transport at order one. At the last stage, the Hasse output and bypass share their identity bank. With  $C_{\text{even}}(x) = x - tH_1(x)$  and  $S(x) = C_{\text{even}}(x) - tH_1(x)$ , the identity

$$H_1(xs) = S(x)H_1(s) + H_1(x)s$$

implements this sharing. Add the bypass directly to the raw product under the final helper key  $h_1$ . The recipient evaluates  $c_0 + c_1h_1 + c_2h_1^2$  before parity and tensor decoding. Keeping a raw product is standard HE practice [12]; here it removes the last result key and three final switching banks. The output is terminal and the construction uses no bootstrap.

**Table 2.** The single selected JetHE profile. A stage’s width applies to all its gadget banks. Prime values are given in Appendix B.2.

| Stage           | Prefix | Tail 8 | Tail 4 | Tail 2 | Tail 1 |
|-----------------|--------|--------|--------|--------|--------|
| RNS primes      | 3      | 3      | 3      | 2      | 2      |
| Radix width $w$ | 45     | 45     | 60     | 40     | 60     |
| Gadget digits   | 4      | 4      | 3      | 3      | 2      |
| Bank families   | 2      | 12     | 8      | 6      | 2      |
| Hint rows       | 8      | 48     | 24     | 18     | 4      |

There are nine independent secrets in topological order

$$(s_0, s_1, h_8, s_2, h_4, s_3, h_2, s_4, h_1),$$

and five owner public keys, at  $s_0, h_8, h_4, h_2, h_1$ . Thirty-one prefix ciphertexts are under  $s_0$ , and each tail receives one freshly encrypted multiplier under its helper key. Only one prime is dropped, after the tail-four result and before the tail-two Hasse transport.

## 6.1 Why the smaller noise bound applies

The next helper key disappears from the Hasse ciphertext’s phase, even though its individual components depend on that key. This distinction is the reason a fresh multiplier can be analyzed more tightly than a product of arbitrary dependent ciphertexts. At order  $r = 2^j$ , write  $\mu_H$  for the Hasse output’s binary representative and  $e_H$  for its phase error. Its next multiplier is  $U_j$ .

The complete phase argument is Lemma 7; its proof uses the acyclic key order above.

For clarity, the resulting local bound can be stated explicitly. If  $\|e_H\|_\infty \leq B$ , put  $a = 1 + 2B$  and

$$C = \left\lceil T\sqrt{40Na^2} \right\rceil, \quad \eta = \left\lceil T\sqrt{(16/3)NC^2 + 40Na^2} \right\rceil.$$

The tensor multiplication matrix has squared row norms at most  $4\|A\|_2^2$ . Conditioning on the independent phase  $A$  bounds  $Ae_{\text{pk},r}$  and  $Ae_1$  by  $C$ . Conditional on those two polynomials, the remaining ternary variables  $u, h_r$  and error  $e_0$  remain independent. Their variance proxies  $2/3, 2/3, 10$  give the bound  $\eta$  for  $(Ae_{\text{pk},r})u + (Ae_1)h_r + Ae_0$ . The product error is therefore bounded by  $P_T(B) = \kappa B + \lceil \kappa/2 \rceil + \eta$ , including the binary-representative carry. Each of the three local bad events has probability at most  $2N \exp(-T^2/2)$ ; cap failures inherited from an earlier stage are charged to that stage. Appendix B.2 gives the row-norm calculation, concentration details, and exact recurrence.

**Theorem 6 (Fixed-input correctness of the selected profile).** *With independent uniform ternary secrets and ephemerals, independent  $\text{CBD}_{20}$  primitive errors, and ideal sampling coins, the profile of Table 2 correctly computes every coefficient of sixteen length-256 compositions for any collection of at most 1024 fixed input batches, except with probability below  $2^{-129}$ . It uses 102 hint rows, 35 input ciphertexts, 19 products, and one three-component output, at multiplicative depth five.*

*Proof (Proof outline).* Each row of multiplication by  $A$  in the actual tensor basis has squared Euclidean norm at most  $4\|A\|_2^2$ . The primitive laws give a fresh-error cap  $F = 34347$  at threshold  $T = 15$ . Before each tail product, its Hasse output has a chosen integer phase independent of the next helper secret and its owner public-key error. Conditioning on this phase, then on two primitive-error products, permits a second concentration step. The resulting product bound satisfies  $P_T(B) \leq 42NT^2(B + 1)$ . Shared Hasse/bypass row errors are bounded separately and their caps added. Appendix B.2 gives the complete argument and all 22 admitted state bounds. The final cap satisfies

$$B_{\text{out}} = 72824530783593675310113562548789870 < q_2/4.$$

Each of the 59 polynomial bad events has probability at most  $2N \exp(-225/2)$ . The stated batch union is below  $2^{-129}$ .

This uses the fresh-operand data flow to improve a sufficient modulus bound. For comparison, the deterministic tail recurrence has amplification  $\Theta(N^2)$  for these fixed primitive supports; the fixed-input concentration bound has amplification  $O(NT^2)$ . The selected widths then reduce gadget storage while respecting every intermediate admission inequality. Their benefit is supported by the complete measured workflow, rather than by assuming that minimum row count implies minimum runtime.

The raw public arrays contain 284 MiB of hints and 13 MiB of owner public keys. Inputs occupy 103 MiB, and the three-component output occupies 3 MiB. Setup also charges the recipient’s cached key square, codec tables, fixed multipliers, and serialization. These array counts exclude framing and other process memory, which are measured separately.

## 7 Security from a discrete dual Ring-LWE source

Table 3 identifies the profiles used below and in the evaluation. A correctness failure bound, a source reduction and a numerical hardness estimate are different statements.

**Table 3.** Security and evidence profiles. All concern confidentiality from the passive evaluator; numerical security is not matched across the measured programs.

| Profile           | Laws and parameters                                                       | Guarantee and evidence                                                                                                                              |
|-------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Gaussian family   | Canonical discrete Gaussian; $N = 256L$ , growing $L$ , generated modulus | Adaptive IND-CPA under the specified dual Ring-LWE source; negligible admitted correctness error; asymptotic complete-work bounds, no timings.      |
| JetHE-102         | Tensor ternary secrets/ephemerals, CBD20 errors; $L = 256$ , $k = 4$      | Exact measured-source reduction; ideal-coin failure $< 2^{-129}$ for at most 1024 fixed batches; measured runtime, no full-source 128-bit estimate. |
| One-prime control | Same primitive laws and algebra; separate key schedule and modulus        | Its own source and correctness analysis; same-backend one-layer measurements.                                                                       |
| OpenFHE BGV       | Library ternary secrets and Gaussian errors; two specified parameter sets | Requested <code>HEStd_128_classic</code> ; exact-output adapter and measured runtime, not a transferred JetHE theorem.                              |

*Route from the source to the public transcript.* We start from ordinary uniform-secret dual Ring-LWE rows. Scaling the codifferent and using a unit-mask pivot gives the small-secret Gaussian law used by the compiler. The acyclic key graph then permits replacing setup and hint rows in reverse topological order, keeping

reused aliases and projections joint. Once public keys are uniform, fresh encryption coins permit a final sequence of hybrids for adaptive prepared-message queries. We specify the source and its loss below; the measured ternary/CBD20 law is treated in Section 7.1.

Write  $D_{A,s}(x) \propto \exp(-\pi\|x\|^2/s^2)$  for the discrete Gaussian on  $A$ , in the canonical Euclidean norm. For  $\lambda \geq 256$  set

$$L = 2^{\lceil \log_2 \lambda \rceil}, \quad N = 256L, \quad D = 257L, \\ R = \mathbb{Z}[t, b]/(t^L + 1, \Phi_{257}(b)), \quad \mu_\lambda = D_{(1-b)R, \sqrt{\lambda D}}.$$

Every vertex secret, encryption ephemeral and primitive error is an independent  $\mu_\lambda$  draw. RNS projections use the same integer polynomial, not independent channel samples.

*Public parameters and source assumption.* Fix a public PPT generator  $\mathcal{P}$ , independent of subsequent secret, error and source randomness. Its output  $\theta$  specifies  $q_* = \prod_{i=1}^\ell p_i$ , where the distinct certified odd primes satisfy  $p_i \neq 257$  and  $p_i \geq 2N\ell$ ; vertex moduli divide  $q_*$ . It also specifies a static acyclic key manifest and fixed efficiently computable, polynomially bounded row/query caps  $\bar{m} \geq \max(2, \max_v n_v)$  and  $\bar{H} \geq |V_{\text{rows}}| + \sum_v Q_v$ , with  $\bar{H}$  a power of two. Here  $n_v$  counts incoming hint and published public-key rows, and  $Q_v$  counts owner-encryption queries. Dimensions, bit lengths, payloads and evaluation work are polynomially bounded; explicit generator failure has negligible probability and the same failure-only transcript in both games.

Secret-dependent *setup/key material* consists of declared affine rows with fresh independent masks/errors. Each payload is simulatable from predecessor secrets and permitted pre-existing data, without the destination secret or the rows being replaced. Aliases and projections are generated jointly. Encryption queries use fresh independent coins and published keys; both messages are validated before the challenge bit is consulted, have matching public metadata, and depend on secrets only through allowed public information. Private diagnostics and decryption oracles are excluded.

Our assumption concerns  $\bar{m} + \lambda$  ordinary unshifted source rows, jointly with  $\theta$ :

$$(a_i, a_i s_0 + e_i), \quad a_i \leftarrow R/q_* R, \quad s_0 \leftarrow R^\vee/q_* R^\vee, \quad e_i \leftarrow D_{R^\vee, \sqrt{\lambda/D}}.$$

The masks and secret are mutually independent uniform draws in their respective quotients, independent of the iid errors; second coordinates are reduced modulo  $q_* R^\vee$ . Let  $\epsilon_{\mathcal{P},n}^G(T)$  bound the absolute distinguishing gap for  $n$  source rows at time  $T$ . Their joint distribution is computationally indistinguishable from independent uniform pairs for every uniform PPT distinguisher. This is

a specified discrete dual Ring-LWE family, not an assumption that the whole JetHE transcript is secure.

**Corollary 2 (Conditionally secure admitted leveled family).** *For the preceding generator, law and public interface, Gaussian JetHE has polynomial-time algorithms and adaptive multi-query IND-CPA security under the stated source assumption. With absolute challenge-bit output gap as advantage, its reduction satisfies*

$$\text{Adv}_{\text{JetHE}}^{\text{gap}}(T) \leq 2\bar{H} \epsilon_{\mathcal{P}, \bar{m} + \lambda}^{\text{G}}(T + \text{poly}(\lambda)) + \text{negl}(\lambda).$$

*It is correct except negligibly when every complete state passes  $q_v > 2 + 4B_v$  using the recurrences of Theorem 2, with primitive coefficient cap  $\lambda$ , fresh bound  $F = (2\kappa\lambda + 1)\lambda$ , and bank bound  $\Delta_q = \kappa g_q D_g \lambda$ . A two-component prime drop instead uses  $\lfloor (2B - 1 + (\kappa\lambda + 2)p_*) / (2p_*) \rfloor$ .*

*Proof.* The codifferent is  $R^\vee = ((1 - b)/D)R$ . Reserve  $\lambda$  source rows for the first unit-mask pivot  $(a_0, b_0)$ , choosing it from masks alone; use a disjoint response block. The prime condition gives unit probability at least  $1/2$ , hence pivot failure at most  $2^{-\lambda}$ . For response rows put  $a'_i = -a_0^{-1}a_i$ . Then

$$A_i = 2a'_i, \quad B_i = 2D(b_i + a'_i b_0) = A_i(De_0) + 2De_i.$$

Multiplication by  $D$  maps the source errors exactly to  $\mu_\lambda$ ; the transformed secret is an independent draw from that law. In the null experiment the transformed rows remain independent uniform pairs. Projection to  $q_v \mid q_*$  and discarding rows supplies every needed small-secret game.

For a fixed challenge bit, process key vertices in reverse topological order, replacing all rows incoming to one destination by uniform pairs. Its outgoing secret-dependent rows have already been replaced, so the reduction can hide this destination secret while sampling all other secrets and all incoming payloads. It generates every alias from the same challenge rows. Encryption under the selected public key needs no destination secret. Once all public keys are uniform, replace each encryption by a two-row ephemeral-secret challenge: its two multipliers are the uniform key elements, and its message is a known additive shift. Fresh coins allow adaptive, correlated prepared messages. Both bit paths reach the same endpoint.

Uniformly select one of  $2\bar{H}$  padded hybrid positions and complement the adversary's output on the bit-one path. Signed telescoping proves the displayed single uniform reduction, including randomly generated parameters. Bounded ideal-bit Gaussian samplers have negligible discrepancy jointly over all honest and simulation calls; both losses are charged, not identified. The explicit loss and sampler construction are in Appendix C.3.6. Gaussian tails give primitive coefficient cap  $\lambda$  with failure  $\text{poly}(\lambda)2e^{-\pi\lambda/2}$ ; deterministic admission then proves correctness. Privacy is never conditioned on this event.

## 7.1 The measured source and its exact reduction

JetHE-102 samples independent uniform ternary coefficients for secrets and ephemerals and independent  $\text{CBD}_{20}$  errors in the tensor basis. These laws define

the source  $D_{q,n}$  consisting of  $n$  pairs  $(A_i, A_i S + 2E_i)$  with independent uniform masks and one shared secret. The public-view hybrid of Theorem 9 applies directly to this source. Including owner public keys, the nine incoming row counts are

$$(1, 8, 37, 12, 16, 9, 10, 9, 5).$$

Projection from  $q_3$  and row deletion supply every required game. For  $B$  batches the bound is

$$\text{Adv}^{\text{gap}}(T) \leq 2(9 + 35B)\epsilon_{q_3, 37}(T + T_{\text{sim}}) + 2\delta_B,$$

where  $\delta_B$  charges implemented sampling and abort deviations. The ideal two-batch factor is 158. This reduction describes the measured law independently of the Gaussian-family theorem.

Further source results identify both reductions and their quantitative limits. A ternary normal-form bridge uses one extra source row and charges an error-law distance below  $2^{-28}$  (Proposition 9). Restricted CRT distances, block embeddings, and full-filter support bounds are proved in Appendices C.1.4, D.2 and D.3. Their scopes do not supply a full-source 128-bit estimate. The generic attack-model calculations, including incomplete dual/hybrid computations, are reported with the implementation parameters in the supplement.

## 8 Computational and Space Comparison

We charge setup, key generation, owner preparation/encoding/encryption, evaluation, transmission and recipient recovery separately. Fewer ciphertext products need not mean lower complete time or memory. Appendix I.4 makes the recorded recipient-work and public-material/batch-traffic tradeoffs explicit without assigning an application-specific resource threshold.

### 8.1 A Complete Prepared Workflow

Put  $d = \log_2 L$ ,  $\tau = d - k$ ,  $M = 2^k - 1$ ,  $I = 2M + 1 + \tau$  and  $H = g_q(2^{\tau+1} + 3\tau)$ . These count prepared products, inputs and reference hint rows at a common admitted modulus  $q$ , with gadget length  $g_q$ . Write  $b_q = \lceil \log_2 q \rceil$  and let  $\mathcal{M}_q, \mathcal{D}_q$  price a ring product and polynomial gadget decomposition in bits. Sharing the state-component digits between Hasse transport and bypass alignment gives

$$C_{\text{eval}} \leq [3(M + \tau) + 2H]\mathcal{M}_q + (2 + 3\tau)\mathcal{D}_q + O((M + \tau + H)Nb_q).$$

Key generation uses at most  $H + 2\tau + 2$  ring products; encryption of all inputs uses  $2I$ , and recovery one. Owners use  $M - k$  truncated products per lane and  $O(hL(M + d))$  other field operations/moves. A tensor codec costs  $\mathcal{C} = O(N(256 + d))$  bit operations. Thus schoolbook arithmetic gives the complete finite-support reference bound

$$\Pi(q) + O((M + \tau + H)N^2b_q^2 + IC + h((M - k)L^2 + L(M + d))\mathcal{F}_K),$$

where  $\mathcal{F}_K$  prices field operations and  $\Pi(q)$  charges modulus selection. This includes ideal-bit sampling in expectation, not fixed-time exact sampling. No transform roots are needed. With compatible channel roots, the implemented radix-two/Rader backend instead prices a ring product by  $O(\ell_q N \log N)$  modular word operations; CRT, digits, channel lifting, allocations and prime drops remain charged.

The optimized implementation reorders the independent first-axis butterflies over contiguous tensor lanes. Proposition 33 proves identical canonical outputs with a 512 KiB temporary at full length. It retains this asymptotic ring-product bound; the separate complete-time measurement in Table 28 is an implementation result, not an additional complexity separation.

Two-polynomial public material occupies  $2N \sum_v (m_v + p_v) b_{q_v}$  bit-packed bits, with  $m_v$  hint rows and  $p_v \in \{0, 1\}$  indicating a public key. Inputs use  $2INb_q$  bits; plaintext variants use  $O((M + d)N)$  bits. A streamed evaluator needs  $O(Nb_q)$  additional state, not including keys, retained inputs or backend scratch. Actual 64-bit RNS arrays replace  $b_q$  by  $64\ell_q$ . Propositions 18 and 19 give the exact ledgers and counting proofs.

*Comparison with compatible conventional transport.* Fix the same ring, basis, primitive laws, acyclic keys, prepared inputs and admitted modulus  $q$ , and put  $n_\tau = 2^\tau$ . Relative-basis functional switching and the attaining Hasse identity give, respectively,

|                         | Reference              | Attaining                   |
|-------------------------|------------------------|-----------------------------|
| Hasse rows at order $r$ | $2rg_q$                | $(r + 1)g_q$                |
| Hasse error increment   | $\Delta_q$             | $3\Delta_q/2$               |
| Complete hint rows      | $g_q(2n_\tau + 3\tau)$ | $g_q(n_\tau + 1 + 4\tau)$ . |

The complete counts retain reference transport at order one and charge the prefix, both product-rekeying banks, and separate bypass alignment. Both schedules retain  $M + \tau$  products,  $I$  inputs and depth  $\tau + 1$ ; these are the two-component-output schedules, before the terminal savings of Section 6. The attaining identity reduces row storage but increases the sufficient Hasse error bound; admission and digit work must therefore be checked before inferring a runtime gain. On a modulus chain each row is charged at its own  $q, g_q$ . Appendix A.3 realizes both in a compatible exact-HE base extended with the same functional keys: identical primitive laws couple their complete transcripts to the corresponding JetHE schedules. This establishes portability of the compiler, rather than an independent timing result or a lower bound on deeper competitors.

**Corollary 3 (Balancing the charged reference schedule).** *Fix  $L = 2^d \geq 2$ , a common modulus  $q$  and gadget length  $g_q \geq 1$ . For  $1 \leq k \leq d$ , the reference evaluator's ring-product coefficient is*

$$R(k) = 3(2^k - 1 + d - k) + 2g_q(2^{d-k+1} + 3(d - k)).$$

*As an optimization of this count function,*

$$\min_{1 \leq k \leq d} R(k) = \Theta(\sqrt{g_q L} + g_q).$$

An exact minimizer is the first  $k < d$  for which  $3 \cdot 2^k - 2g_q L/2^k - (6g_q + 3) \geq 0$ , or  $d$  if none exists. Equality gives a tie between  $k$  and  $k + 1$ .

*Proof.* Put  $x = 2^k$ . Then  $R = 3x + 4g_q L/x + (6g_q + 3)(d - k) - 3$ . The arithmetic-geometric mean bound gives  $R \geq 4\sqrt{3g_q L} - 3$ , while the hint term gives  $R \geq 4g_q$ . For  $g_q \geq L$ , choose  $k = d$ , obtaining  $R = 3L + 4g_q - 3 = O(g_q)$ . For  $1 \leq g_q \leq L$ , choose  $k = \lceil \log_2 \sqrt{g_q L} \rceil$ . Then  $\sqrt{g_q L} \leq x < 2\sqrt{g_q L}$  and  $(6g_q + 3)(d - k) \leq (9g_q/2) \log_2(L/g_q) = O(\sqrt{g_q L})$ . This proves the upper bound. Finally,  $R(k + 1) - R(k) = 3 \cdot 2^k - 2g_q L/2^k - (6g_q + 3)$  strictly increases with  $k$ , which proves the discrete rule.

This elementary consequence optimizes one charged reference count, including hint consumption; it is not a faster classical composition algorithm or a lower bound on conventional HE. Using its optimizer cryptographically requires admission at that same  $q, g_q$ . If only a subset of prefixes is admitted, minimize over that subset instead. Changing the modulus changes gadget length and kernel prices. The 149-row implementation has a separate ledger, and the following theorem supplies the complete bit-work and storage statement for its own Gaussian reference family.

**Corollary 4 (Gaussian Security and Complete Resources Together).**

*In the family of Corollary 2, fix the gadget radix independently of  $\lambda$  and choose  $k = \min\{d, \lceil d/2 + \log_2(d + 1) \rceil\}$ . An explicitly generated no-drop prime has  $b_q = O(d^2)$ . Under the discrete dual Ring-LWE assumption for this generator, the complete prepared workflow is IND-CPA secure and correct except negligibly, with*

$$\Pi + O(N\sqrt{L}\log^5 L) \text{ bit work,} \quad O(N\sqrt{L}\log^3 L) \text{ retained bits.}$$

*Work includes Gaussian-table setup, key generation, owner preparation, encryption, evaluation and recovery. Retained bits include keys, inputs, plaintext variants, authority secrets and shared sampling tables. Transmission is separately charged;  $\Pi$  is bounded ordinary-prime generation.*

*Proof.* The coefficient cap  $\lambda$  gives  $F = O(\lambda^3)$  and affine-tail multiplier  $O(\lambda^4)$ . The prefix has constant term  $O(2^k \lambda^7)$ ; iterating the  $\tau$  fresh-operand tails therefore needs  $b_q = O(k + (\tau + 1) \log \lambda)$ , including the  $g_q = O(b_q)$  hint term. For a sufficiently large width  $b$  of that order, draw  $2^b + U$  for a uniform  $b$ -bit integer  $U$  and test primality deterministically. At most  $\lceil 8bd^2 \rceil$  trials fail with probability at most  $e^{-4d^2}$ , by the prime-density bound of [17, Theorem 6.9]; primality is polynomial-time [1]. This generator is the one used in the hardness assumption.

Fixed-axis Gaussian tables cost  $O(\sqrt{\lambda} d^7)$  setup bit operations and  $O(\sqrt{\lambda} d^3)$  stored bits. Rounded cumulative tables use  $O(d^3)$  work per scalar draw with negligible joint discrepancy. There are  $H + 3\tau + 3 + 3I$  polynomial draws; uniform-coordinate rejection also has a bounded negligible-failure implementation. Now  $2^k = O(\sqrt{L}d)$ ,  $2^\tau = O(1 + \sqrt{L}/d)$  and  $b_q = O(d^2)$ . Lemma 21 gives exact ring products in  $O(Nd^4)$  bits and owner jet products in  $O(Ld^2)$  bits, without

channel roots. Substituting in the full arithmetic, codec and sampling counts proves the work bound. Retained objects use  $O(N\sqrt{L}d^3)$  bits; fast arithmetic scratch  $O(Nd^4)$  is absorbed. Proposition 8 gives the sampler, generator and admission derivation, and Appendix E.5.2 charges the faster exact backend.

## 8.2 Why one product layer needs quadratic preparation

The proof has three steps. First, exhibit quadratically many independent coefficient functions of the inner operand. Second, force the one-layer interface below to expose enough prepared owner coordinates to span them. Third, use full-domain correctness and message privacy to show that encoding those coordinates still costs quadratic work. The private-encoding contract enters in this last step.

*Step 1: independent coefficient functions.* Put  $P_{i,j}(g) = [z^j]g^i$ . Even when all coefficients of  $g$  are Boolean, these functions have a  $K$ -span of dimension at least

$$C_L = 5L^2/64 + (d+1)L/16 = \Omega(L^2).$$

The field is fixed; the argument therefore avoids a growing-field hypothesis. Here is the coefficient witness behind the bound.

Let  $V_r = \text{span}_{\mathbb{F}_2}\{z, \dots, z^r\}$  and  $E_r(Y) = \prod_{a \in V_r} (Y+a)$ . The additive identity  $E_{r+1}(Y) = E_r(Y)^2 + E_r(z^{r+1})E_r(Y)$  permits a triangular coefficient analysis. On Boolean  $g = \sum_v X_v z^v$ , the first potentially nonzero coefficient of  $E_r(g)$  is at  $\nu_r = 2^{r+1} - 1$ , and

$$[z^{\nu_r+j}]E_r(g) = X_{r+1+j} + R_{r,j}(X_1, \dots, X_{r+j}).$$

Indeed the linear term has valuation  $2^{r+1} - r - 2$ ; every term of degree at least two uses only smaller-index variables at this output degree.

To build the independent family, set  $R = d - 3$  and  $b_* = L/8$ . For  $b_* \leq i < 2b_*$ , multiply the  $E_r$  with indices in the binary support of  $i$ . Its valuation is  $\nu(i) = 2i - \text{wt}(i)$ . For  $0 \leq j < L - \nu(i)$  the coefficient at degree  $\nu(i) + j$  has leading variable monomial

$$X_{R+1+j} \prod_{r < R: i_r=1} X_{r+1}.$$

Take the largest  $j$  in a linear relation. The displayed monomials are distinct for distinct  $i$ , forcing every coefficient to vanish; descend in  $j$ . All these functions lie in the span of the  $P_{i,j}$ , and their number is  $\sum_{i=b_*}^{2b_*-1} (L - 2i + \text{wt}(i)) = C_L$ . These rescaled Carlitz products are classical; their coefficient independence within the truncation budget is the ingredient used here (Lemma 19).

*Step 2: necessary prepared-coordinate width.* For a mixed bilinear gate with operands  $A(f) + B(g)$  and  $C(f) + D(g)$ , subtract its values on  $(f, 0), (0, g), (0, 0)$ . The surviving terms are bilinear in centered owner coordinates. Setting  $f = z^i$  shows that all  $P_{i,j}$  must lie in the span of those inner coordinates. Affine preprocessing cannot create more independent coordinate functions; hence their initial width is at least  $hC_L$ .

*Step 3: the length of a private encoding.* Correctness on all  $V$ -bit carrier messages permits at most  $2^{V-t+1}$  decodable strings of length at most  $V - t$ . Public length indistinguishability transfers this counting bound from uniform messages to the actual prepared distribution. Writing the encrypted stream therefore costs  $\Omega(V) = \Omega(L^2)$  under the full-domain encoding contract. The complete statements and proofs are Lemma 20 and Corollary 9.

**Main theorem 1 (Preparation, Depth and Work).** *Let  $L = 2^d \geq 256$ ,  $K = \mathbb{F}_{2^{16}}$ ,  $h = 16$  and  $N = 256L$ . Under the source and generator assumption of Corollary 4, the Gaussian compiler uses  $O(\sqrt{L} \log L)$  products and initial ciphertexts, depth  $d - k + 1$ , and  $\tilde{O}(L^{3/2})$  complete computational bit work. Here  $k = \min\{d, \lceil d/2 + \log_2(d + 1) \rceil\}$ .*

*For a polynomial-time compiler exact at the plaintext-functional level on the same full input domain, allow arbitrary independent owner preparation, affine preprocessing, one parallel layer of fixed binary-bilinear primitives, binary-linear recovery and unilateral bypasses. Public maps and shapes are fixed independently of inputs. This class requires  $\Omega(L)$  calls and inputs, when each operand and initial carrier holds at most  $N$  plaintext bits. If its initial encoded stream is correct and message-private over the full joint carrier message space, with negligible errors, producing that stream requires  $\Omega(L^2)$  bit work except negligibly and in expectation. Explicit materialization of the prepared plaintext bits is an alternative sufficient condition for the same work lower bound.*

*Proof.* The upper statement combines Corollary 4 with  $\Pi = \text{poly}(\log L)$  and the prepared inventory. For the lower statement, Corollary 8 requires  $hC_L = \Omega(L^2)$  initial inner-owner bits and total operand width, where  $C_L = 5L^2/64 + (d + 1)L/16$ . Dividing operand widths by  $2N$  and input width by  $N$  proves the call and input bounds. The secure-length lemma and Corollary 9 charge the encoded stream even with fused preparation/encryption or joint compression. Writing explicitly materialized plaintext bits gives the alternative.

This synthesizes a separation between specified compiler interfaces. The work ratio grows as  $\Omega(\sqrt{L}/\log^5 L)$ ; no finite crossover is asserted. The full-domain condition concerns recovery of every carrier tuple, not high entropy of the actual prepared inputs. Encodings correct only on a structured image, deeper competitors and nonlinear recovery remain outside the lower bound. Conventional HE can adopt the upper compiler. At  $L = 256$  the Gaussian choice is  $k = 8$ ; the measured  $k = 4$  profile has different laws and is not an instance of this asymptotic bound.

### 8.3 Joint evaluation over shared transport keys

For multiple fresh batches, each Hasse transport is a matrix product between one public switching matrix and a matrix of ciphertext digits. Relative-coordinate expansion makes this identity exact over the quotient ring: changing its evaluation order leaves the resulting ciphertexts identical. This exposes shared-key arithmetic to fast matrix multiplication.

**Main theorem 2 (Complete work for joint batches).** *Under the Gaussian source assumption for each chosen prefix and a polynomial reuse horizon, put  $x = 2^k$ ,  $n_\tau = L/x$ , and let  $B \geq n_\tau$  fresh batches be available together. For any fixed rational matrix exponent  $\theta > \omega_{\mathbb{Q}}$ ,  $\theta > 2$ , the compiler has complete bit work*

$$O(N[n_\tau + Bx + Bn_\tau^{\theta-2}] \log^7 L).$$

For  $\theta = 593/250$ , choose  $k$  nearest to  $93d/343$ . Then

$$W(B, L) = O(BL^{436/343} \log^7 L), \quad B \geq n_\tau, \quad n_\tau = \Theta(L^{250/343}).$$

The joint one-layer comparison class of Main theorem 1, with full-domain private encodings, requires  $\Omega(BL^2)$  work.

*Proof.* At relative width  $e$ , a  $(2e)$ -by- $(eg_q)$  public matrix multiplies an  $(eg_q)$ -by- $B$  digit matrix over a degree- $N/e$  quotient ring. Kronecker packing and a fixed rational bilinear kernel implement this in  $O(Bg_q N e^{\theta-2} \log^4 L)$  bit operations. Clear denominators and divide exactly in the integers, so no modular inverse is required. Summing the dyadic widths costs  $O(BN n_\tau^{\theta-2} \log^6 L)$ . Setup, sampling, owner preparation, encryption, codecs, traffic and recovery cost  $O(N[n_\tau + Bx] \log^7 L)$ . Balancing  $x n_\tau = L$  gives the stated exponents. The coefficient-function witness and secure-encoding argument extend over the Cartesian product of batch inputs, yielding the lower bound even with joint preparation. Detailed proofs are Propositions 22 and 24 and Corollary 12.

The rational kernel follows from the published exponent  $\omega < 2.371339$  [3]; our choice has strict slack. This is a theoretical batch improvement, with potentially large kernel constants and an explicit availability threshold. The evaluated implementation uses ordinary transforms. Appendix F.2 reports finite matrix experiments and the resulting preparation and memory costs. Conventional exact HE can implement the same compiler (Appendix A.3).

The comparison identifies the value of encrypted depth under a specified reconstruction interface. If the recipient instead receives and decrypts both original operands, classical composition gives a softly linear complete protocol; Appendix F.1 accounts for that different work placement. Such a relay is relevant when raw-input delivery and recipient-side composition are permitted. It does not perform the delegated computation; excluding it on recipient-privacy grounds would require a guarantee beyond the present evaluator-confidential model. It is not a main-text experimental baseline. The sharp  $2L-1$  slotwise multiplication bound, prepared contraction controls, and complete storage/crossover formulas are developed in Appendices E.2 and E.5.

## 9 Implementation and evaluation

We evaluate sixteen length-256 compositions over  $\mathbb{F}_{2^{16}}$ , checking all 4096 output symbols against a complete reference vector. The primary profile, *JetHE-102*, uses the construction of Section 6, exact 64-bit RNS arithmetic, radix-two/Rader

transforms, shared gadget decomposition, contiguous tensor butterflies, and pre-computed fixed multipliers. Lazy transform arithmetic follows established techniques [26]. The source-equivalent transform implementations and their separate measurements are detailed in Appendix H.2.2.

### 9.1 OpenFHE BGV adapters for exact composition

The OpenFHE programs use stock version 1.5.1 [34], built in Release mode with GCC 16.2.0, 64-bit backend 4, and OpenMP and native-CPU tuning disabled. A finite parameter/layout selection contains twenty candidates; two BFV configurations fail at the library interface because of their sizing. The reported BGV programs use the selected packing layouts and BSGS settings. JetHE additionally uses the tensor-layout, gadget-width and fixed-multiplier optimizations above; compiler optimization effort is therefore unequal.

*Summed BGV* uses dimension 16384, HYBRID switching, FLEXIBLEAUTOEXT, and baby width 16. *Terminal BGV* uses dimension 32768 and baby width 32, returning individual product statistics for recipient reconstruction. Both have ciphertext-product depth one and plaintext modulus 65537. Their exact extension-field adapter is important: this prime field is different from  $\mathbb{F}_{2^{16}}$ .

Represent a field element by a binary polynomial of degree below 16. The inner owner prepares  $[z^j]g^i$ , and the outer owner supplies  $f_i$ . Thirty-two evaluations in  $\mathbb{F}_{65537}$  encode the integer polynomial products. Inverse transformation and canonical lifting recover coefficients bounded by  $256 \cdot 16 = 4096 < 65537$ ; parity and reduction by the defining degree-16 polynomial recover the exact field answer. Appendix I.3 proves this adapter. Summed BGV uses 2168 encryptions, 2040 products, 128 relinearizations, 120 rotations and eight outputs. Terminal BGV uses 1021 encryptions, 510 products and 511 outputs, without rotations or relinearization. Its charged recipient reconstruction trades evaluator work for output traffic. These output interfaces provide evaluator confidentiality under their respective assumptions; no recipient-privacy claim is made.

Both BGV configurations request `HEStd_128_classic`, with uniform ternary secrets and error standard deviation approximately 3.19. Their ciphertext moduli have 92 and 87 bits, respectively, and auxiliary switching moduli have 120 bits. JetHE-102 has dimension 65536, a 180-bit creation modulus and a 120-bit terminal modulus. Its ternary/CBD20 source and reduction are specified in Section 7.1. The measurements use distinct source assumptions and are not numerically security matched.

### 9.2 Measurement protocol and complete costs

The measurement platform is an AMD Ryzen 9950X WSL host. Each process uses one thread pinned to logical CPU zero. Each setup serves a first batch and then a second, warm batch, both freshly encrypted. The BGV configurations and *JetHE-149*, a four-prime two-component-output reference profile, have three independent setups each. JetHE-102 and the one-prime control have four

**Table 4.** Complete local workflow, seconds. Each row uses all independent setups of its selected configuration. Warm shows median [minimum, maximum]; first use is the median of setup plus that setup’s first batch. Rows use the parameter and output interfaces specified in the text; compiler tuning and numerical security are not matched.

| Configuration        | Setups | Warm median [range]     | First use |
|----------------------|--------|-------------------------|-----------|
| JetHE-102            | 4      | 1.730 [1.710, 1.746]    | 3.804     |
| OpenFHE summed BGV   | 3      | 15.795 [15.652, 15.862] | 16.013    |
| OpenFHE terminal BGV | 3      | 12.885 [12.854, 12.903] | 12.681    |
| One-prime control    | 4      | 7.710 [7.696, 7.729]    | 7.826     |
| JetHE-149            | 3      | 4.734 [4.690, 4.818]    | 10.239    |

**Table 5.** Median serialized public, input, and output payloads and process peak memory, in MiB. Public material is reusable setup; input and output traffic recur per batch. Peak RSS includes implementation allocations.

| Configuration        | Public  | Input    | Output  | Peak RSS |
|----------------------|---------|----------|---------|----------|
| JetHE-102            | 297.002 | 103.001  | 3.000   | 499.344  |
| OpenFHE summed BGV   | 40.765  | 1628.028 | 4.008   | 78.840   |
| OpenFHE terminal BGV | 1.501   | 1532.455 | 767.105 | 43.453   |
| One-prime control    | 1.000   | 346.015  | 131.011 | 63.395   |
| JetHE-149            | 559.002 | 137.001  | 2.000   | 809.945  |

each. These are separate measurement series without exclusive-host or paired-observation guarantees across series. JetHE-149 identifies the native profile measured alongside the BGV adapters; it also quantifies the change in construction and resources.

Complete local time includes owner preparation, interpolation and codecs, sampling, encryption, evaluation, counting-sink serialization, decryption and recipient reconstruction. First use additionally charges context, codec tables, all key material, recipient key preparation and public-material serialization. Process startup, fixture/reference construction, validation and physical network transfer are excluded. The fixed synthetic workload includes edge cases; the small-sample medians and ranges characterize these configurations, not a workload distribution or scaling law.

JetHE-102 has a complete warm median of 1.730384 seconds. Summed and terminal BGV take 15.795136 and 12.884548 seconds, giving ratios of 9.128 and 7.446. These ratios compare the specified programs with unequal compiler tuning and distinct security parameters. JetHE’s 297 MiB public material and 499 MiB peak RSS exceed both BGV adapters; its 103 MiB input and 3 MiB output reduce their communication volumes. Network time is not inferred from serialized sizes.

**Table 6.** Cost decomposition for the same-backend comparison, in seconds. Each entry is the median over four setups. Other batch work is the per-run residual after the three timed role phases, including preparation, codecs and serialization. Separate phase medians need not sum to the complete median. Setup is paid once, and warm work for each fresh batch.

| Phase                 | JetHE-102 | One-prime control |
|-----------------------|-----------|-------------------|
| Setup                 | 2.029544  | 0.087457          |
| Warm owner encryption | 0.907677  | 3.690296          |
| Warm evaluator        | 0.597183  | 0.092010          |
| Warm recipient        | 0.028432  | 1.435881          |
| Other warm batch work | 0.194515  | 2.491134          |

### 9.3 Control with the same arithmetic backend

A one-prime bilinear control uses the same ramified algebra, codec, primitive source laws and optimized arithmetic library as JetHE-102. It has 346 inputs, 86 depth-one products and 88 terminal outputs, including two unilateral corrections. Its interpolation, owner work, linear recovery and exact correctness conditions are given in Appendices G.1 and G.3.7.

Complete warm medians are 1.730384 seconds for JetHE-102 and 7.710294 for the control, a ratio of 4.456. First-use medians are 3.803819 and 7.826098 seconds. Table 6 shows the work placement: fewer inputs reduce encryption and preparation, and the completed output reduces recipient work. The control evaluates faster and uses only 1 MiB public material and 63 MiB peak RSS. JetHE’s gain is in the complete workflow; its setup, evaluator time and memory are larger. This one-layer control explains the saving but does not cover deeper conventional compilers.

All sixteen batches in this comparison recover every coefficient. Independent untimed phase checks validate the admitted intermediate states. The complete observations appear in Appendix J.2. The supplement also reports the reference profiles, HELib adapters, alternative contraction constructions, and unsuccessful matrix/cache experiments. Those results delimit where arithmetic improvements translate into a complete-workflow gain.

## 10 Conclusion

JetHE gives an explicit preparation–depth–work tradeoff for encrypted series composition. Interleaving concentrates a classical recursion into one ramified carrier, and a norm-one Hasse lift attains the sharp functional-payload minimum. The Gaussian construction has a reduction from a specified discrete dual Ring-LWE source and subquadratic complete work. Fixed-field coefficient-function bounds separate that work from exact one-layer bilinear compilation under its full-domain encoding contract; joint evaluation strengthens the upper bound when sufficiently many fresh batches are available together.

The ternary/CBD20 realization demonstrates the corresponding practical compiler: sixteen length-256 compositions complete in a 1.730-second warm median, with reduced encrypted input and output traffic and larger public-material and memory costs than the configured OpenFHE adapters. These are program-level measurements under distinct source assumptions; the same-backend control locates the saving in preparation, encryption and recovery, while its evaluator is faster. The proven phase-independence and arithmetic-equivalence results explain how its smaller parameter schedule and implementation retain correctness. Its supported application is terminal outsourced symbolic computation with a passive evaluator and trusted receiving side. Recipient-private or independently keyed delivery requires additional protocols; multi-key FHE [31] provides a precedent for that extension. The present results establish a fully charged use of ramified packing, from owner preparation through encrypted coordinate transport to exact recovery. Compatible exact-HE bases can adopt the compiler and its transport improvement; the proven separation is between specified compiler interfaces, rather than between scheme names.

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## Supplementary proofs and measurements

The supplement is organized by topic. Appendices A–D give the algebra, correctness arguments and source-specific security analysis. Appendices E–G develop the comparison interfaces, complete-work bounds and conventional controls. The remaining appendices collect implementation refinements, measured comparisons and reproducibility details. The selected 102-row, uniform-width 134-row and 149-row profiles retain separate statements and parameter assumptions.

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## A Representation and Hasse transport

### A.1 General Ramification and Coordinate Details

The general decomposition explains the binary tensor specialization used in Section 3. Its factorization and coordinate choices are public setup, not encrypted operations.

Let  $m = p^k s$  with  $k \geq 1$  and  $\gcd(p, s) = 1$ , and put  $L = p^{k-1}(p-1)$ . For  $s > 1$ , let  $d = \text{ord}_s(p)$  and  $h = \varphi(s)/d$ ; for  $s = 1$  put  $d = h = 1$ .

**Lemma 3 (Cyclotomic Ramification).** *In  $\mathbb{F}_p[X]$ ,  $\Phi_{p^k s}(X) = \Phi_s(X)^L$ .*

*Proof.* Use the integer identity  $\Phi_{ps}(X) = \Phi_s(X^p)/\Phi_s(X)$ . Modulo  $p$ , Frobenius changes its numerator to  $\Phi_s(X)^p$ . For  $k > 1$ , substitute  $X^{p^{k-1}}$  in  $\Phi_{ps}$  and apply Frobenius again. The resulting exponent is  $p^{k-1}(p-1)$ .

**Theorem 7 (Local decomposition and confluent coordinates).** *There is a publicly computable  $\mathbb{F}_p$ -algebra isomorphism*

$$\mathbb{F}_p[X]/(\Phi_m) \simeq \prod_{i=1}^h \mathcal{J}_L(\mathbb{F}_{p^d}), \quad \varphi(m) = hdL.$$

*If  $\Phi_s \bmod p = \prod_i f_i$  and  $\alpha_i \in \mathbb{F}_{p^d}$  is a selected root of  $f_i$ , an explicit choice is*

$$F \mapsto (F(\alpha_i + z) \bmod z^L)_i = \left( \sum_{r < L} D_X^{[r]} F(\alpha_i) z^r \right)_i.$$

*Its inverse is the unique confluent interpolation class modulo  $\Phi_m$ .*

*Proof.* Since  $p \nmid s$ , the factors  $f_i$  are distinct, separable and of degree  $d$ . CRT first gives the factors  $A_i = \mathbb{F}_p[X]/(f_i^L)$ . Each is local, with maximal ideal generated by  $\pi_i = f_i(X)$  and residue field  $\mathbb{F}_{p^d}$ . The polynomial  $Y^{p^d} - Y$  has derivative  $-1$ ; successive lifting through the nilpotent ideal gives a unique root over each residue. These roots form a coefficient field. Every element of  $A_i$  has a unique expansion  $\sum_{r < L} \tilde{a}_r \pi_i^r$ : existence follows by successive reduction, and the  $dL$ -dimensional spaces have the same size.

In the displayed substitution,  $f_i(\alpha_i + z) = f'_i(\alpha_i)z + O(z^2)$  has valuation one. The coefficient field maps onto the constants, and subtracting the constant lift of  $X \bmod f_i$  from  $X$  gives an element mapping to  $z$ . Thus  $A_i \rightarrow \mathbb{F}_{p^d}[z]/(z^L)$  is surjective and hence an isomorphism. The Hasse expansion gives its coordinates. Finally,  $hdL = \varphi(s)\varphi(p^k) = \varphi(m)$ .

The isomorphism depends on public root/field choices. The alternative  $f_i$ -adic coordinate differs by the generally nonlinear triangular change  $\pi_i = f_i(\alpha_i + z)$ ; these maps must not be conflated in a cost claim. For example, a coefficient-field lift can be computed by raising any residue representative to  $p^{da}$  with  $p^{da} \geq L$ . Public factorization, root selection, lifting and interpolation are setup costs, not free HE operations.

**A.1.1 Finite Tensor-Codec Ledger** Matrix construction, root and field validation, and  $O(256^3)$  binary inversion are charged once per public representation. The checked matrices have  $\chi(C) = 31984$  and  $\chi(C^{-1}) = 32000$ ; their packed storage is 16384 bytes. At  $L = 256$ , decoding/encoding use 8450048/8454144 logical XORs. These are operation counts, not elapsed time. Under the identification  $X = bt$ , the original univariate Hasse coordinate at  $\beta_i$  differs from this tensor coordinate by scaling degree  $r$  by  $\beta_i^{-r}$ ; that public field arithmetic is additional.

**A.1.2 General Automorphism Action** For  $a \in (\mathbb{Z}/m\mathbb{Z})^*$ , the action  $X \mapsto X^a$  induces a permutation of the local factors, a field isomorphism, and the parameter substitution

$$z \mapsto (\alpha_i + z)^a - \alpha_i^a = a\alpha_i^{a-1}z + O(z^2).$$

The linear coefficient is nonzero, so this substitution is invertible modulo  $z^L$ . This follows by substituting in the confluent coordinates and relabeling the Frobenius orbit of  $\alpha_i^a$ . A pure coefficient-preserving lane rotation would additionally need to undo the field and triangular maps; we do not assume that correction is one free native operation.

## A.2 A universal single-layer comparison for Hasse transport

Use  $R_L = E[t]/(t^L + 1)$  and the integral basis of Section 3, and fix a dyadic order  $r < L$ . Let  $q > 1$  be odd and put  $A = R_L/qR_L$ . A norm-one integral lift

$F : R_L \rightarrow R_L$  is  $\mathbb{Z}$ -linear, reduces to the binary Hasse map  $H_r$  modulo two, and has induced coefficient infinity norm at most one. It induces a map on  $A$ .

We count fixed functional payloads in identities

$$F(cs) = u(c) + \sum_{\ell=1}^k d_\ell(c) \phi_\ell(s).$$

The coefficients depend on the public multiplier  $c$ , but not on the secret  $s$ . Payload functions  $\phi_\ell : A \rightarrow A$  may be nonlinear. The sending-secret identity is charged if it is one of these payloads. The free-identity variant additionally permits an uncharged term  $v(c)s$ . This universal functional model specifies one realization layer, rather than all possible ciphertext algorithms or a restricted sampled secret set.

The sharp minimum and its proof are Theorem 4. The following refinements address sampled secrets and implementation costs.

Equality is an optimization over lift choices; it does not assert that every signed lift has an  $r + 1$ -payload factorization. The following distributional extension uses the same unit minor. It does not rely on an exact identity at any particular sparse secret.

**Proposition 1 (Independent ternary secrets and exact identities).** *Fix the lift, coefficient functions and common payload functions in (1) independently of the secret. Draw every coefficient of  $s$  in the stated integral basis independently and uniformly from  $\{-1, 0, 1\}$ . Let  $\mathcal{E}_c$  be the event that the exact identity holds at multiplier  $c$ . If  $k < r + 1$ , then*

$$\Pr \left[ \bigcap_{c \in \{1, t, \dots, t^r\}} \mathcal{E}_c \right] \leq 3^{-(r+1-k)}. \quad (5)$$

*With a free identity and  $k < r$ , omit  $t^r$  from the test set. The joint-success bound is then  $3^{-(r-k)}$ . Consequently, the maximum per-test failure probability is at least*

$$\frac{1 - 3^{-(r+1-k)}}{r + 1} \quad (\text{paid}), \quad \frac{1 - 3^{-(r-k)}}{r} \quad (\text{free}).$$

*The payload functions may be nonlinear. The statements also hold after averaging over auxiliary randomness independent of  $s$ .*

*Proof.* Reduce evaluated ring elements by a map  $\pi : A \rightarrow K$  onto a residue field. Its characteristic is odd. In the paid case, set  $m = r + 1$  and let  $Y(s)$  be the vector with entries  $F(t^r s), F(s), \dots, F(t^{r-1} s)$ . Writing  $X$  for the vector of ternary coefficients gives  $\pi(Y(s)) = TX$  for a matrix  $T$  over  $K$ . The unit minor in the preceding proof uses the source-basis columns  $b, t^r b, t^{r-1} b, \dots, tb$ ; hence  $T$  has row rank  $m$ , even when  $F$  is not  $E$ -linear.

Let  $D$  be the  $m \times k$  matrix of the evaluated coefficients  $\pi(d_\ell(c))$ , and let  $a$  collect  $\pi(u(c))$ . Joint success implies  $TX - a \in \text{col}(D)$ , regardless of the payload

functions. Choose a rank- $(m - k)$  matrix  $W$  with  $WD = 0$ . Since  $T$  has full row rank,  $WT$  has rank  $m - k$ . Joint success therefore implies  $WTX = Wa$ . Select  $m - k$  columns giving an invertible minor and condition on all other source coefficients. At most one assignment in  $K^{m-k}$  to the selected coordinates solves this equation. Their three values are distinct in odd characteristic, and their independence gives probability at most  $3^{-(m-k)}$ . No payload function is assumed to descend to a function on  $K$ .

For free identity, augment the  $r$  queried functions by a first component  $s$ . The same minor gives rank  $r + 1$ . The coefficient matrix now has at most  $k + 1$  columns, including the free identity, and the affine term's first coordinate is zero. The same argument gives codimension at least  $r - k$  and probability at most  $3^{-(r-k)}$ . The per-test statements follow by a union bound, without independence between tests. Conditioning first on independent auxiliary randomness proves the randomized version.

With one fewer paid payload, some fixed test has exact-identity failure at least  $2/(3(r + 1))$ . This is an algebraic statement for independent ternary coefficients, not a lower bound on noisy decryption failure or on the distribution of incoming ciphertext multipliers. Independence matters: in  $\mathbb{Z}[t]/(t^2 + 1)$ ,  $s = X(1 + t)$  for one uniform ternary  $X$  satisfies  $H_1(s) = H_1(ts) = X$ , so one payload serves both paid tests with probability one. Arbitrary restricted secret supports, coefficients selected using secret-dependent public data, and different payload families for different tests remain outside the proposition.

Gadget expansion, digit supports and primitive row errors are absent from this algebraic objective. In particular, inserting the attainer into an independent-key transport requires charging the identity bank and deriving its own full error bound. The next proposition gives that bound in the current unscaled phase convention; the original transport remains available through Lemma 2.

### A.2.1 An admitted material tradeoff in the current phase convention

**Proposition 2 (Paid-identity transport with coherent digits).** *Use the source-component radix digits already computed for bypass alignment. For  $c_1 = \sum_{i < r} t^i(u_i + t^r v_i)$ , apply  $H_r$  and the maps  $c_1 \mapsto u_i - t^r v_i$  to each digit. Publish independent fresh-error banks under the Hasse destination  $k$  for  $s$  and  $H_r(t^i s)$ ,  $i < r$ . The resulting Hasse ciphertext has error at most  $B + 3\Delta_q/2$  and uses  $(r + 1)g_q$  rows. Its complete composition stage has bound*

$$T'_q(B) = B + 3\Delta_q + P(B + 3\Delta_q/2, B_{\text{fresh}}) + 1. \quad (6)$$

*The identity bank is distinct from the bypass bank, whose destination is the subsequent result secret.*

*Proof.* These coefficient maps are signed partial permutations. Applied to the already extracted integer digits, they preserve the magnitude bound  $D_g$  and reconstruct the intended multiplier exactly. This does not assume that a new canonical radix decomposition commutes with negation. The multiplier  $H_r(c_1)$

has support on  $L/2$  possible  $t$  positions. Each  $u_i - t^r v_i$  has  $L/r$  possible positions, totalling  $L$  across  $i$ . The support bound of Lemma 2 therefore charges  $\Delta_q/2$  for the identity rows and  $\Delta_q$  for the other rows. Equation (2) gives the clean phase  $H_r(\mu) + 2H_r(e)$ . There is no representative carry because  $H_r$  preserves binary representatives. This proves the Hasse bound. Bypass alignment still contributes  $B + \Delta_q$ , and the product's two rekeying banks contribute  $2\Delta_q$ . Adding these branches and their final carry gives (6).

At order one both transports use two functional banks; the original has the smaller sufficient error increment. Using the new transport only at orders eight, four and two in the selected  $L = 256$ ,  $k = 4$  profile gives 149 rows, compared with 203, on the same chain  $(4, 4, 4, 3, 2)$  and radix  $2^{48}$ . The raw hint arrays occupy 542 MiB instead of 754 MiB; the five owner public keys remain 17 MiB. The final sufficient bound increases from 16392681195061871150589051193478792 to 18825417735414023944722075316721744, and every complete intermediate still satisfies  $q > 2 + 4B$ . These are correctness and array inventories, not runtime or security estimates.

The existing compact digit spectra also implement the changed multipliers. At a full dyadic evaluation point  $x$ , the relative spectra  $U_i, V_i$  give  $U_i - x^r V_i$  and  $\sum_{i < r} x^i V_i$ , respectively. Thus the change needs no additional NTT or coefficient decomposition. With cached public powers and one streamed full multiplier, its construction adds one dimension- $N$  modular product and addition per changed digit/channel after accounting for the removed row operations:  $52N$  of each in this profile. This is an operation count; actual allocation and memory traffic require measurement. Fresh identity-row errors are essential to the stated source interface; adding old hint rows would change that interface.

### A.2.2 Why the serial compiler cannot reuse one common-pair bank

The sending and receiving keys are part of a functional-switching bank's interface. In the serial compiler the pairs are  $(s_1, k_8)$ ,  $(s_2, k_4)$ ,  $(s_3, k_2)$  and  $(s_4, k_1)$  at  $L = 256$ ,  $k = 4$ . These are different independently sampled secrets. A bank for several individual derivative orders at one common pair cannot be shared across these four edges merely because the plaintext maps have related spectra.

Regrouping the updates introduces another constraint. Let  $B$  consist of distinct dyadic orders below  $L$ . Put  $D = \sum_{r \in B} r < L$  and  $\delta = g + z$  in characteristic two. The descending updates expand as

$$\prod_{r \in B}^{\text{descending updates}} (I + \delta^r D^{[r]}) f = \sum_{d \subseteq D} \delta^d D^{[d]} f.$$

Here  $d \subseteq D$  means that every set bit of  $d$  occurs in  $D$ . For a later, smaller order  $r < h$ , every positive derivative of  $\delta^h$  below  $h$  vanishes, and Hasse operators with disjoint index bits compose to their summed order. Expanding successively proves the identity, including its mixed derivatives. For example, grouping orders two and one also requires  $D^{[3]} f$ .

**Proposition 3 (Restriction on explicit Taylor-closed preparation).** *If one universal layer explicitly prepares norm-one lifts of all derivatives in this expansion at one key pair, it needs at least  $D$  nonidentity payloads even with a free identity term. Consequently grouping the four tail stages into such blocks cannot reduce their sum of these lower bounds below  $8 + 4 + 2 + 1 = 15$ .*

*Proof.* Subtract the identity at  $s = 0$  to remove the constant term. The family includes order  $D$ . For any norm-one lift  $F_D$ , evaluate the  $D+1$  functions  $F_D(t^i s)$  at  $s = t^{D-j}b$ , with  $0 \leq i, j \leq D$ . Entries above the diagonal vanish: their monomial exponent is nonnegative and smaller than  $D$ , so it cannot contain all the set bits of  $D$ . Diagonal entries are  $\pm b$ . Thus this matrix is triangular with unit determinant, regardless of possible wraps below the diagonal. Factoring through  $k$  payloads and the free identity gives rank at most  $k+1$  after reduction modulo a maximal ideal. Hence  $k \geq D$ . For a partition into disjoint blocks, their values of  $D$  sum to fifteen.

This concerns materializing the Taylor-closed derivative family in the stated single-layer model. It does not constrain a fused composition algorithm that avoids those outputs, additional layers, larger-norm lifts, or algorithms on another representation. Identity banks, gadget rows, errors and products still require their own accounting. The classical Taylor identity and binary recurrence are not new composition algorithms.

### A.3 A Conventional Exact-HE Realization of the Compiler

The compiler’s resource bounds are available to a conventional exact-HE base extended by the specified functional keys. This statement requires more than support for the same plaintext quotient. Fix the integral basis, affine phase, actual small-vector laws, gadgets and acyclic key graph of Section 4. Supply full-ring rows  $(-ak + 2\epsilon + \gamma^j v, a)$  for every declared predecessor payload  $v$ , with fresh independent destination secrets, masks and errors. Ring arithmetic, two-bank product rekeying, separate bypass alignment and parity-preserving drops are then exactly the already specified operations.

Relative-basis functional switching [15, Section 4.6] supplies the reference transport of Lemma 2 with  $2rg_q$  rows and increment  $\Delta_q$ . The paid factorization in Proposition 2 uses  $(r+1)g_q$  rows and increment  $3\Delta_q/2$ , with its coherent source digits. These facts depend on the lift and support calculation, not the base scheme’s name. The Hasse identity bank and the bypass bank have different destinations; neither is omitted in this transfer.

For either transport separately, couple the augmented conventional base with JetHE using identical primitive randomness and public arithmetic. Induction over the same prepared schedule then gives identical ciphertexts, phase expressions and public transcripts. This couples two realizations of the same transport; it does not identify the reference transport with the paid factorization. At a common gadget length  $g$ , writing  $x = 2^k$ ,  $\tau = d - k$  and  $E = 2^\tau$ , the reference has  $g(2E + 3\tau)$  hint rows. The paid choice at orders above one, retaining the reference at order one, has  $g(E + 1 + 4\tau)$  rows. Both keep  $x - 1 + \tau$  private products,

$\tau + 1$  nonlinear layers,  $2\tau + 2$  independent keys and  $2x - 1 + \tau$  owner ciphertexts. For a chain, charge each row and operation at its actual modulus and gadget. The smaller row total does not imply a time ratio: the paid lift has the stated larger sufficient error increment, and any resulting change of modulus, gadget length or drop schedule must be charged separately.

The joint matrix arithmetic also transfers on these same rows. Thus the complete upper bound of Proposition 23 is available to this augmented conventional realization, under its own admitted Gaussian source, sufficient batch horizon and stated arithmetic costs. No independent timing measurement results from this coupling. A different sampler, integral-basis law, key graph or multiplication policy requires its own correctness and source analysis; the measured and Gaussian laws are not identified by a common plaintext algebra.

*Why functional transport is an additional capability.* Let  $A = \prod_{a < h} K[z]/(z^L)$  and let  $J = (z)$  denote its nilradical, lane by lane. The following elementary observation applies to an existing packed input, with all other inputs and auxiliaries held fixed.

**Proposition 4 (Hasse gate restriction).** *Every fixed finite plaintext circuit over  $A$  using ring addition, multiplication, public constants and ring automorphisms preserves congruence modulo  $J^r$ . Identity plaintext operations may be included. For  $1 \leq r < L$ , such a circuit cannot implement  $D^{[r]}$  on every input, regardless of its depth.*

*Proof.* Every automorphism preserves the nilradical and its powers. Addition preserves ideal congruence, and  $ab - a'b' = (a - a')b + a'(b - b')$  proves preservation by multiplication. Constants and identity gates preserve it as well, so induction proves the circuit assertion. Inputs 0 and  $z^r$  in one lane are congruent modulo  $J^r$ , but their Hasse images differ by 1 in that lane.

This is an interface obstruction, not a lower bound on arbitrary ciphertext algorithms or a new cryptanalytic result. Functional switching, conversion through another plaintext algebra, programmable refresh and auxiliaries already encoding a non-congruence-preserving function lie outside its gate set. This first-order invariant alone does not exclude composition: if  $f - f', g - g' \in (z^r)$  and  $g, g' \in (z)$ , then  $f(g) - f'(g') \in (z^r)$ . The first difference factors through  $g^r$  and the second through  $g - g'$ . The owner-preparation model is broader still.

*A phase convention is not a multiplication algorithm.* For odd  $q$ , the unit  $\delta_q = (q - 1)/2$  has inverse  $-2$ . Componentwise multiplication  $C_q(c) = \delta_q c$  changes the phase to  $\delta_q \mu - e$ . A transformed key row has phase  $\delta_q \gamma^j v - \epsilon$ . Using gadget  $\gamma_j' = \delta_q \gamma^j$  and decomposing  $\delta_q^{-1} c_1'$  therefore reproduces the old digits. The Hasse map commutes with the integer  $\delta_q$ , so both transports conjugate exactly under  $C_q$ . The corresponding raw product is

$$\text{Raw}'(c', d') = \delta_q^{-1} \text{Raw}(c', d'),$$

and a modulus edge becomes  $C_q \text{Drop}_{Q \rightarrow q} C_Q^{-1}$ . For odd  $q \mid Q$ ,  $\delta_Q \bmod q = \delta_q$ , respecting RNS aliases. Modular halving/doubling adds linear coefficient passes; it is not free.

This public change of representation does not identify the algorithm with ordinary BFV/GBFV rounded multiplication. Generalized BFV’s principal plaintext quotients are eligible, but its scaled-and-rounded arithmetic has its own error and key-policy obligations [22, Section 3.3]. Likewise, default coefficient Gaussians, smaller-ring masks or cyclic relinearization do not inherit the present source reduction. Under the exact coupling the old reduction and query caps remain; a public invertible conversion adds its time cost without changing advantage. No numerical security match, unmodified-library interoperability or scheme-exclusive speedup follows.

Finite computations verify phase identities, digit reconstruction, the change of convention and ideal-congruence preservation in small rings. They support implementation consistency for those cases; the general coupling follows from the argument above. The coupled realization is not an independently implemented BFV/BGV benchmark. It shows that the quantified compiler and lift improvements are portable to a base with the stated capabilities. Their benefit is the change in transport and complete resource costs under this interface, rather than an advantage derived from the base scheme’s name.

## B Terminal profiles and correctness

### B.1 Uniform-width terminal profile

The last update need not return to a new result key. Write  $x = E(x) + tH_1(x)$  and  $S(x) = E(x) - tH_1(x)$ . The integral identity

$$H_1(xs) = S(x)H_1(s) + H_1(x)s$$

lets the Hasse output and the bypass share the identity bank from  $s_4$  to  $h_1$ . Return their sum with the raw product under  $h_1$ , in three components. This removes the final bypass bank, two product-switching banks and  $s_5$ . The recipient computes  $c_0 + c_1h_1 + c_2h_1^2$  before the unchanged linear codec. Keeping a raw product in three components is a standard HE capability [12, Secs. 5.1–5.4]; the relevant specialization here is the shared Hasse/bypass key graph and its bound. Shared row errors are correlated and are charged accordingly. Terminal delivery does not supply encrypted continuation.

For the finite ternary/CBD<sub>20</sub> law, a more precise analysis also reduces the sufficient creation moduli. The essential fact is local: the Hasse output has a chosen small integer phase independent of the fresh helper secret and its as-yet unused owner public-key error. This fact follows from cancellation of the helper and row masks in the phase; it does not hold for the ciphertext components themselves.

**Theorem 8 (Fixed-input three-prime terminal compilation).** *Use the prescribed length-256, sixteen-lane,  $k = 4$  circuit with independent ternary secrets*

and ephemerals, independent  $\text{CBD}_{20}$  primitive errors, and ideal sampler coins. Replace the reference profile's chain  $(4, 4, 4, 3, 2)$  and radix  $2^{48}$  by  $(3, 3, 3, 2, 2)$  and radix  $2^{44}$ , using the terminal construction above. For any collection of at most 1024 fixed input batches independent of all setup and encryption coins, the probability of any decryption failure is below  $2^{-128}$ . The profile has nine independent secrets, five owner public keys, 134 hint rows, 35 input ciphertexts and one three-component output.

*Proof (Proof outline).* In the actual tensor basis, each row of multiplication by  $A$  has squared norm at most  $4\|A\|_2^2$ . The exact coefficient laws then give fresh encryption cap 34347 and support-sensitive switching caps. At each tail, condition on its Hasse phase, then on two products with independent primitive errors. The remaining helper secret, ephemeral and error stay independent, permitting a second concentration step. No independence between the bypass and Hasse errors is assumed. The integer recurrence in Appendix B.3 admits all 22 intermediate states; at most 59 polynomial events per batch give failure at most  $1024 \cdot 59 \cdot 2N \exp(-225/2) < 2^{-128}$ .

This is a probabilistic, fixed-input guarantee, unlike the reference's all-coins admission. It does not cover inputs chosen from public keys or earlier execution results. The conditional privacy reduction remains separate from correctness: the exact-law  $q_3$ , 46-row premise suffices for  $2(9 + 35K_{\text{batch}})\epsilon_{q_3,46} + 2\delta_{K_{\text{batch}}}$  at the complete simulation time. This is not 128-bit source qualification.

*Theoretical reason for the parameter saving.* For this tensor-ring family,  $\kappa = \Theta(N)$ . The deterministic tail bound has amplification  $\kappa(1 + 2F_{\text{det}}) = \Theta(N^2)$ , where  $F_{\text{det}} = 20(2\kappa + 1)$ . With concentration threshold  $T \geq 1$ , the new raw-product cap satisfies  $P_T(B) \leq 42NT^2(B + 1)$  (Lemma 8). For a prescribed collection of  $E$  polynomial events and target failure  $\epsilon$ ,  $T^2 \geq 2\ln(2NE/\epsilon)$  suffices. Thus the sufficient tail amplification is  $O(N \log(NE/\epsilon))$ . This improves a correctness bound and can reduce modulus and gadget costs. It is not a lower bound on noise, an asymptotic runtime separation, or an exclusive property of JetHE: the same analysis is available to another realization of this key graph.

## B.2 Correctness of the selected 102-row profile

This subsection proves Theorem 6 under ideal independent coins. It uses the existing interleaved composition, parity preserving prime drop, and paid Hasse identities. The input collection is fixed independently of setup and encryption randomness. Batches have no data dependence on earlier executions. Public keys may be reused for this fixed collection. A union bound does not require independent batches.

**B.2.1 The basis and primitive coefficient laws** Let  $R = \mathbb{Z}[t, b]/(t^{256} + 1, \Phi_{257}(b))$ , in the basis  $t^i b^j$ ,  $0 \leq i < 256$ ,  $1 \leq j \leq 256$ . Write  $N = 65536$ ,

$\kappa = 130816$  and  $T = 15$ . At a stage of width  $w$ , set  $D = 2^{w-1}$ . Put  $q_a = \prod_{i=0}^{a-1} p_i$ , where

$$\begin{aligned} p_0 &= 1152921504002872321, & p_1 &= 1152921503566671361, \\ p_2 &= 1152921503264686081. \end{aligned}$$

Only  $q_3$  and  $q_2$  occur. For the current stage set  $g_a = \lceil \log_{2^w} q_a \rceil$ ; the widths and digit counts follow Table 2. Thus  $D$  and  $g_a$  in the local inequalities below always refer to that stage. Signed radix digits have magnitude at most  $D$ , including the last quotient: extracting the first  $g_a - 1$  digits in  $[-D, D)$  from a centered coefficient with  $|x| < q_a/2$  leaves a final quotient in  $[-D, D]$  because  $q_a < 2^{wg_a}$ . Keeping the positive endpoint is necessary.

**Lemma 4 (Multiplication rows in the actual tensor basis).** *Every row of multiplication by  $A \in R$  has squared Euclidean norm at most  $4\|A\|_2^2$ . Consequently a coefficient bound  $a$  gives row bound  $4Na^2$ , and support size  $m$  with bound  $D$  gives  $4mD^2$ .*

*Proof.* Extend  $A_{i,j}$  by  $A_{i,0} = 0$ , taking the  $b$  index modulo 257. For an output coordinate  $(u, v)$  and input coordinate  $(i, j)$ , put  $h = (u - i) \bmod 256$  and let  $\sigma$  be the sign of the negacyclic  $t$  wrap. The row entry is

$$\sigma(A_{h,v-j} - A_{h,-j}).$$

Indeed  $b^{257} = 1$  and the coefficient of  $b^0$  is removed using  $1 = -\sum_{j=1}^{256} b^j$ . Each of the two index maps is injective on its nonzero entries. Apply  $(x-y)^2 \leq 2x^2 + 2y^2$  and sum over  $(i, j)$ . This is a deterministic ring identity, not a random-matrix model.

A centered variable has variance proxy  $v$  here when its moment generating function is at most  $\exp(vt^2/2)$  for every real  $t$ . The primitive error law has proxy 10, since  $\cosh(t/2)^{40} \leq \exp(5t^2)$ . Uniform ternary has proxy  $2/3$ :  $(1 + 2 \cosh t)/3 \leq \exp(t^2/3)$ . For the latter inequality, compare the even Taylor coefficients using  $2 \cdot 3^{k-1} k! \leq (2k)!$  for  $k \geq 1$ ; the inequality propagates by induction since  $3(k+1) \leq (2k+2)(2k+1)$ . Independent centered variables add their squared-coefficient-weighted proxies. If every coordinate of a polynomial has proxy at most  $v$ , then

$$\Pr[\|X\|_\infty > T\sqrt{v}] \leq 2N \exp(-T^2/2). \quad (7)$$

No independence among output coordinates is used.

**B.2.2 Fresh encryption and switching increments** The fresh encryption error is  $e_{\text{pk}}u + e_0 + e_1s$ . Condition on the independent ternary  $u, s$ . By Lemma 4, the independent primitive error sources give coordinate proxy at most  $80N + 10$ . Thus

$$F = \left\lceil T\sqrt{80N + 10} \right\rceil = 34347.$$

Public-key reuse correlates different encryptions, but each marginal bound holds and their failure events can be united.

Each switching bank is used only at its prescribed circuit position. The source ciphertext supplying its digits has no dependence on that bank's primitive row errors. Conditioning on this source state therefore leaves independent centered row errors. Digits have bound  $D$ . For  $b$  full banks, and for the paid Hasse map, respectively, define

$$\Delta_a(b) = \left\lceil T\sqrt{40ND^2g_ab} \right\rceil, \quad \Delta_{H,a} = \left\lceil T\sqrt{60ND^2g_a} \right\rceil. \quad (8)$$

The  $r$  nonidentity Hasse multipliers each have support  $N/r$ ; the identity multiplier has support  $N/2$ . These signed partial permutations preserve  $D$ , giving total support  $3N/2$  per digit. This proves the second expression. The terminal Hasse map and bypass share identity-row errors: apply separate bounds and add their caps. Their outputs are not treated as independent.

The source-independence statement refers to the execution's data flow, not to conditioning on the entire public transcript. A gadget digit is not declared independent of a row error merely because that error was sampled during setup. Here the digit-generating state uses only earlier vertices and their relevant banks, and not this unused bank.

**B.2.3 A chosen phase independent of the next helper** Lemma 7 supplies the phase independence used below.

For a deterministic cap  $\|e_H\|_\infty \leq B$ , put  $a = 1 + 2B$  and

$$C = \left\lceil T\sqrt{40Na^2} \right\rceil, \quad \eta = \left\lceil T\sqrt{\frac{16}{3}NC^2 + 40Na^2} \right\rceil. \quad (9)$$

Condition on  $A$ , restricted to  $\|A\|_\infty \leq a$ . Two applications of (7) bound  $Ae_{\text{pk},r}$  and  $Ae_1$  by  $C$ . Condition next on these polynomials. The remaining  $u, h_r, e_0$  are independent with proxies  $2/3, 2/3, 10$ . Their combined expression

$$(Ae_{\text{pk},r})u + (Ae_1)h_r + Ae_0$$

has coordinate proxy at most  $(16/3)NC^2 + 40Na^2$ , giving the third event. The raw product's remaining error consists of  $e_H\mu_U$  and the binary representative carry. Hence its sufficient cap is

$$P_T(B) = \kappa B + \lceil \kappa/2 \rceil + \eta. \quad (10)$$

These are local conditional bounds. Form the first two bad events as  $\|A\|_\infty \leq a$  intersected with the respective bound failure, and the third as that event with both preceding bounds satisfied and the last bound failing. Each has probability at most  $2N \exp(-T^2/2)$ . A failure of the cap on  $A$  is charged to an earlier event in the recurrence. Do not first condition on a global good event including later helper encryptions: that would destroy the independence needed here. The same precaution applies to other batches.

**Lemma 5 (Sufficient tail amplification).** *For  $N \geq 1$ ,  $T \geq 1$ ,  $B \geq 0$  and the tensor-ring family above with  $\kappa \leq 2N$  and  $\lceil \kappa/2 \rceil \leq N$ ,  $P_T(B) \leq 42NT^2(B+1)$ .*

*Proof.* The first ceiling gives  $C \leq 8T\sqrt{N}(1+2B)$ . Thus the square root inside the second ceiling is at most  $\sqrt{1144/3}NT(1+2B) < 20NT(1+2B)$ . Consequently  $\eta \leq 20NT^2(1+2B) + 1$ , and

$$P_T(B) \leq 42NT^2B + 21NT^2 + 1 \leq 42NT^2(B+1).$$

This compares sufficient bounds under different correctness quantifiers; it is not an attained-noise or general algorithmic lower bound.

**B.2.4 Exact recurrence and admission** The prefix uses width 45 and cap

$$B_{P,\text{raw}} = F + 15\kappa(2F + 2F^2) + \lceil 15\kappa/2 \rceil.$$

Switching its accumulated raw product adds  $\Delta_3(2)$ . This prefix bound is deterministic conditional on the fresh caps; it permits correlation through the common root public-key error. At tails 8, 4, 2, 1, use the following stage parameters:

$$(a, w) = (3, 45), (3, 60), (2, 40), (2, 60),$$

respectively. Before tail two, apply the parity-preserving drop

$$B \leftarrow \left\lfloor \frac{2B - 1 + (\kappa + 2)p_2}{2p_2} \right\rfloor.$$

At each tail set  $H = B + \Delta_{H,a}$  and  $A = B + \Delta_a(1)$ . The product cap is  $P_T(H) + \Delta_a(2)$  for nonterminal tails and  $P_T(H)$  for the terminal tail. The new state cap is this product cap plus  $A + 1$ . At order one the stated Hasse cap is conservative for the original two-bank map; adding the bypass cap also charges its shared row errors without assuming independence.

Every complete state in Table 7 satisfies  $q_a > 2 + 4B$ . Per batch, charge 35 fresh encryptions, one prefix switching event, four Hasse increments, four bypass increments, three nonterminal product-switching increments, and three product events at each of four tails. These are 59 polynomial bad events. Outside their union the recurrence proves correctness by induction. For at most 1024 fixed batches the total bound is

$$1024 \cdot 59 \cdot 2N \exp(-225/2) < 2^{-129}.$$

The marginal bounds and union do not require batch independence. Input independence from setup and encryption coins is used in the phase-independence lemma. A successful finite execution corroborates the implementation but cannot empirically measure a failure probability of this magnitude.

### B.3 Correctness of the uniform-width 134-row profile

This subsection proves Theorem 8 under ideal independent coins. It uses the existing interleaved composition, parity preserving prime drop, and paid Hasse identities. The input collection is fixed independently of setup and encryption randomness. Batches have no data dependence on earlier executions. Public keys may be reused for this fixed collection. A union bound does not require independent batches.

**B.3.1 The basis and primitive coefficient laws** Let  $R = \mathbb{Z}[t, b]/(t^{256} + 1, \Phi_{257}(b))$ , in the basis  $t^i b^j$ ,  $0 \leq i < 256$ ,  $1 \leq j \leq 256$ . Write  $N = 65536$ ,  $\kappa = 130816$ ,  $T = 15$ , and  $D = 2^{43}$ . Put  $q_a = \prod_{i=0}^{a-1} p_i$ , where

$$\begin{aligned} p_0 &= 1152921504002872321, & p_1 &= 1152921503566671361, \\ p_2 &= 1152921503264686081, & p_3 &= 1152921503096916481. \end{aligned}$$

Only  $q_3$  and  $q_2$  occur in this construction;  $q_4$  also names the conservative common source premise. Let  $g_a = \lceil \log_{2^{44}} q_a \rceil$ , so  $g_3 = 5$  and  $g_2 = 3$ .

**Lemma 6 (Multiplication rows in the actual tensor basis).** *Every row of multiplication by  $A \in R$  has squared Euclidean norm at most  $4\|A\|_2^2$ . Consequently a coefficient bound  $a$  gives row bound  $4Na^2$ , and support size  $m$  with bound  $D$  gives  $4mD^2$ .*

*Proof.* Extend  $A_{i,j}$  by  $A_{i,0} = 0$ , taking the  $b$  index modulo 257. For an output coordinate  $(u, v)$  and input coordinate  $(i, j)$ , put  $h = (u - i) \bmod 256$  and let  $\sigma$  be the sign of the negacyclic  $t$  wrap. The row entry is

$$\sigma(A_{h,v-j} - A_{h,-j}).$$

Indeed  $b^{257} = 1$  and the coefficient of  $b^0$  is removed using  $1 = -\sum_{j=1}^{256} b^j$ . Each of the two index maps is injective on its nonzero entries. Apply  $(x - y)^2 \leq 2x^2 + 2y^2$  and sum over  $(i, j)$ . This is a deterministic ring identity, not a random-matrix model.

A centered variable has variance proxy  $v$  here when its moment generating function is at most  $\exp(vt^2/2)$  for every real  $t$ . The primitive error law has proxy 10, since  $\cosh(t/2)^{40} \leq \exp(5t^2)$ . Uniform ternary has proxy  $2/3$ :  $(1 + 2 \cosh t)/3 \leq \exp(t^2/3)$ . For the latter inequality, compare the even Taylor coefficients using  $2 \cdot 3^{k-1} k! \leq (2k)!$  for  $k \geq 1$ ; the inequality propagates by induction since  $3(k+1) \leq (2k+2)(2k+1)$ . Independent centered variables add their squared-coefficient-weighted proxies. If every coordinate of a polynomial has proxy at most  $v$ , then

$$\Pr[\|X\|_\infty > T\sqrt{v}] \leq 2N \exp(-T^2/2). \quad (11)$$

No independence among output coordinates is used.

**B.3.2 Fresh encryption and switching increments** The fresh encryption error is  $e_{\text{pk}}u + e_0 + e_1s$ . Condition on the independent ternary  $u, s$ . By Lemma 6, the independent primitive error sources give coordinate proxy at most  $80N + 10$ . Thus

$$F = \left\lceil T\sqrt{80N + 10} \right\rceil = 34347.$$

Public-key reuse correlates different encryptions, but each marginal bound holds and their failure events can be united.

Each switching bank is used only at its prescribed circuit position. The source ciphertext supplying its digits has no dependence on that bank's primitive row errors. Conditioning on this source state therefore leaves independent centered row errors. Digits have bound  $D$ . For  $b$  full banks, and for the paid Hasse map, respectively, define

$$\Delta_a(b) = \left\lceil T\sqrt{40ND^2g_ab} \right\rceil, \quad \Delta_{H,a} = \left\lceil T\sqrt{60ND^2g_a} \right\rceil. \quad (12)$$

The  $r$  nonidentity Hasse multipliers each have support  $N/r$ ; the identity multiplier has support  $N/2$ . These signed partial permutations preserve  $D$ , giving total support  $3N/2$  per digit. This proves the second expression. The terminal Hasse map and bypass share identity-row errors: apply separate bounds and add their caps. Their outputs are not treated as independent.

The source-independence statement refers to the execution's data flow, not to conditioning on the entire public transcript. A gadget digit is not declared independent of a row error merely because that error was sampled during setup. Here the digit-generating state uses only earlier vertices and their relevant banks, and not this unused bank.

### B.3.3 A chosen phase independent of the next helper

**Lemma 7 (Phase independence).** *Immediately before the multiplication at order  $r = 2^j$ , the Hasse ciphertext has a formal integer phase lift  $A = \mu_H + 2e_H$  independent of the helper secret  $h_r$ , its owner public-key error  $e_{\text{pk},r}$ , and the owner encryption coins for  $U_j$ . Here  $A$  is congruent to the ciphertext phase modulo  $q_a$ ; it is not the unreduced integer expression  $c_0 + c_1h_r$ .*

*Proof.* The incoming state, including any prime-drop rounding, is computed from earlier vertex secrets, their banks and the earlier owner inputs. None of this data uses  $h_r$ , its public key, or the encryption of  $U_j$ . The fixed plaintext condition rules out additional dependence introduced by input selection. The Hasse digits therefore have the same exclusion.

The phase of each outgoing Hasse row equals its functional source payload plus twice its primitive row error, modulo  $q_a$ . Substitution in the paid Hasse identity cancels the masks and  $h_r$ . Choose the lift

$$A = H_r(\mu + 2e) + 2 \sum_{\ell,j} d_{\ell,j} e_{\ell,j},$$

where  $\mu + 2e$  is the chosen incoming lift. The integer identity holds modulo  $q_a$ , which is sufficient.  $H_r$  preserves binary representatives, so this is  $\mu_H + 2e_H$  without a new representative carry. Every variable on this expression's right-hand side is independent of the excluded helper and owner sources. Canonical representatives of the two ciphertext components may depend on  $h_r$ ; no assertion is made about their unreduced integer phase. On the admitted event, the chosen lift is the small lift needed for decryption correctness.

For a deterministic cap  $\|e_H\|_\infty \leq B$ , put  $a = 1 + 2B$  and

$$C = \left\lceil T\sqrt{40Na^2} \right\rceil, \quad \eta = \left\lceil T\sqrt{\frac{16}{3}NC^2 + 40Na^2} \right\rceil. \quad (13)$$

Condition on  $A$ , restricted to  $\|A\|_\infty \leq a$ . Two applications of (11) bound  $Ae_{\text{pk},r}$  and  $Ae_1$  by  $C$ . Condition next on these polynomials. The remaining  $u, h_r, e_0$  are independent with proxies  $2/3, 2/3, 10$ . Their combined expression

$$(Ae_{\text{pk},r})u + (Ae_1)h_r + Ae_0$$

has coordinate proxy at most  $(16/3)NC^2 + 40Na^2$ , giving the third event. The raw product's remaining error consists of  $e_H\mu_U$  and the binary representative carry. Hence its sufficient cap is

$$P_T(B) = \kappa B + \lceil \kappa/2 \rceil + \eta. \quad (14)$$

These are local conditional bounds. Form the first two bad events as  $\|A\|_\infty \leq a$  intersected with the respective bound failure, and the third as that event with both preceding bounds satisfied and the last bound failing. Each has probability at most  $2N \exp(-T^2/2)$ . A failure of the cap on  $A$  is charged to an earlier event in the recurrence. Do not first condition on a global good event including later helper encryptions: that would destroy the independence needed here. The same precaution applies to other batches.

**Lemma 8 (Sufficient tail amplification).** *For  $N \geq 1$ ,  $T \geq 1$ ,  $B \geq 0$  and the tensor-ring family above with  $\kappa \leq 2N$  and  $\lceil \kappa/2 \rceil \leq N$ ,  $P_T(B) \leq 42NT^2(B+1)$ .*

*Proof.* The first ceiling gives  $C \leq 8T\sqrt{N}(1+2B)$ . Thus the square root inside the second ceiling is at most  $\sqrt{1144/3}NT(1+2B) < 20NT(1+2B)$ . Consequently  $\eta \leq 20NT^2(1+2B) + 1$ , and

$$P_T(B) \leq 42NT^2B + 21NT^2 + 1 \leq 42NT^2(B+1).$$

This compares sufficient bounds under different correctness quantifiers; it is not an attained-noise or general algorithmic lower bound.

**B.3.4 Exact recurrence, admission and batch union** The raw prefix cap is

$$B_{P,\text{raw}} = F + 15\kappa(2F + 2F^2) + \lceil 15\kappa/2 \rceil.$$

Use the deterministic prefix product inequality to handle the shared root public-key error without an independence claim. Relinearization adds  $\Delta_3(2)$ . At orders 8, 4, 2, 1 use creation moduli  $q_3, q_3, q_2, q_2$ . Before order two, drop  $p_2$  with the existing deterministic rule

$$B \leftarrow \left\lfloor \frac{2B - 1 + (\kappa + 2)p_2}{2p_2} \right\rfloor.$$

Then set

$$H = B + \Delta_{H,a}, \quad A_{\text{bypass}} = B + \Delta_a(1),$$

take raw cap  $P_T(H)$ , and add  $\Delta_a(2)$  except at order one. The final addition has cap  $A_{\text{bypass}} + B_{\text{product}} + 1$ . Only one prime drop is used in this variant.

All roots and divisions in Table 8 use exact integer arithmetic, including the rational  $1/3$ . Every state satisfies  $q_a > 2 + 4B$ , with minimum whole-bit margin one. In particular the final cap is

$$B_{\text{out}} = 121957389274810025244769488583297324 < q_2/4.$$

This is a correctness admission, not a source-security estimate.

Per batch, charge 35 fresh encryptions, one prefix relinearization, four Hasse increments, four bypass increments, three tail relinearizations, and three product events at each of four tails. These are 59 polynomial events. Outside their union, induction through the stated recurrence and the deterministic prime-drop rule proves every admitted phase bound and the final output. For  $K_{\text{batch}} \leq 1024$ , the union bound is

$$K_{\text{batch}} 59 2N \exp(-225/2), \quad \log_2(1024 59 2N \exp(-225/2)) < -129.42.$$

This completes the fixed-input proof. It does not establish correctness for adaptive key-dependent plaintexts, arbitrary evaluator programs, or continuation from arbitrary returned ciphertexts. Implementation checks cannot empirically validate a failure probability of this size.

**B.3.5 Independent-key privacy and material accounting** The vertex order and incoming row counts, including owner public keys, are

$$(s_0, s_1, h_8, s_2, h_4, s_3, h_2, s_4, h_1), \quad (1, 10, 46, 15, 26, 15, 10, 9, 7).$$

Every payload depends only on an earlier source vertex. The reverse-topological reduction of Theorem 9 therefore applies. Every creation modulus divides  $q_3$ , and the maximum incoming count is 46. Coordinate projection and row deletion from the exact-law  $q_3$ , 46-row game supply every setup and two-row encryption game. Thus the separately published profile satisfies

$$\text{Adv}^{\text{gap}} \leq 2(9 + 35K_{\text{batch}})\epsilon_{q_3,46} + 2\delta_{K_{\text{batch}}}, \quad (15)$$

at adversary time plus complete simulation overhead. This is a conditional privacy reduction, not numerical hardness. The four-prime  $q_4$  premise still suffices

by projection; the actual maximum modulus is 180 bits. For two batches the factor is 158, so an ideal-law scheme gap  $1/4$  entails a source gap at least  $1/632$ . This implication asserts no attack cost. The reduction neither identifies OS randomness with ideal coins nor licenses joint publication of different profiles' banks under shared keys. The separate Gaussian theorem has a different source law; its adaptive security statement does not enlarge this finite-law correctness theorem.

*Honest source inventory for the 134-row profile.* For two batches, setup uses 391 uniform prime-coordinate vectors, nine ternary secrets and 139 primitive errors. The 70 encryptions add 70 ternary ephemerals and 140 errors. The stopped-test proof of Lemma 10, stopping before a ninth refill pass per vector, therefore uses

$$B_2^{(3)} = N[8(391 + 79) + 279] = 264699904$$

64-bit words. With  $\rho_{\max} < 2^{-29}$ , its ideal stop probability is at most  $N(391\rho_{\max}^8 + 79 \cdot 2^{-512}) < 2^{-207}$ . Hence the logical word-source deviation is at most the gap of the corresponding bounded adaptive source test plus this stop probability. Complete processing time and other implementation deviations must still be charged. This is a bound for a stopped test, not a cap on the unchanged OS implementation or the local hybrid-simulation catalogue. The older 81-row reference's concrete time envelope is not being assigned to the new source interface here.

There are 30 bank families and 134 rows: two prefix families, and  $r + 1$  Hasse families at each tail plus three transition families for each nonterminal tail. Two full polynomials per row at its creation modulus give 378 MiB of raw hints. Five owner public keys add 13 MiB. The 31 prefix inputs use three channels each; the four tail inputs use 3, 3, 2, 2, giving 103 MiB of raw inputs. The single three-component output uses two channels, giving 3 MiB. These are explicit array costs; serialization framing, public tables, recipient work and transport are additional charged quantities.

#### B.4 Explicit Prepared-Modulus Synthesis

**Proposition 5 (Explicit No-Drop Modulus Synthesis).** *Use one modulus and gadget length throughout the prepared compiler. Put  $F = B_{\text{fresh}}$ ,  $c = \lfloor (\kappa + 1)/2 \rfloor$ ,  $\Delta_0 = \kappa D_g \beta$ , and*

$$a = 1 + \kappa(1 + 2F), \quad H = F + M\kappa(2F + 2F^2) + \lceil M\kappa/2 \rceil, \quad \Sigma_\tau = \sum_{j=0}^{\tau-1} a^j.$$

*The empty sum is zero. The final sufficient error bound is exactly*

$$\begin{aligned} B_\tau &= \mathcal{U} + \mathcal{V}g, & \mathcal{U} &= a^\tau H + (\kappa F + c + 1)\Sigma_\tau, \\ & & \mathcal{V} &= \Delta_0(2a^\tau + (a + 2)\Sigma_\tau). \end{aligned}$$

For radix  $\gamma = 2^w$ ,  $w \geq 1$ , take  $D_g = \gamma/2$ . A sufficient explicit choice is

$$b = 2 \lceil \log_2(8(\mathcal{U} + \mathcal{V} + 1)) \rceil, \quad 2^b < q < 2^{b+1} \text{ prime}, \quad g = \left\lceil \frac{b+1}{w} \right\rceil + 1.$$

Such a prime exists, and every intermediate state satisfies correctness admission. For fixed radix and error support,  $\log_2 q = O(k + (\tau + 1) \log(1 + \kappa))$ . This is a correctness upper bound, not NTT-prime selection or a security-qualified parameter family.

*Proof.* The other operand of every tail product is fresh. Consequently the existing recurrence is affine in the current bound:

$$T_q(B) = aB + (\kappa F + c + 1) + (a + 2)\Delta_0 g, \quad B_P = H + 2\Delta_0 g.$$

Iterating gives the formula. These positive bounds increase, and dominate the fresh, transported, raw and aligned intermediate bounds. An extra balanced-radix digit represents every centered coefficient at the chosen modulus, while  $g \leq b + 2$ . Put  $S = \mathcal{U} + \mathcal{V}$  and  $u = b/2$ . Since  $u \geq 4$ ,  $2^u > 2u + 2$  and  $2^u \geq 8(S + 1)$ , whence

$$2^b > 8(S + 1)(b + 2) > 2 + 4S(b + 2) \geq 2 + 4B_\tau.$$

Bertrand's postulate supplies a prime in the displayed interval. For fixed radix and support,  $F = O(\kappa)$ ,  $a = O(\kappa^2)$ ,  $H = O(2^k \kappa^3)$  and  $\Delta_0 = O(\kappa)$ ; the geometric sum then proves the bit-length bound. Direct exact ring arithmetic suffices for this existence argument; no transform roots or numerical hardness are inferred.

**Corollary 5 (No-Drop Modulus Bound).** *For fixed radix and primitive supports, the prepared compiler has a correctness-admitted common prime modulus with  $\log_2 q = O(k + (\tau + 1) \log(1 + \kappa))$ . This is a correctness upper bound, not NTT-prime selection or a security-qualified family.*

*Proof.* Every tail multiplies the current state by a fresh operand, so its bound is affine:  $T_q(B) = aB + c_0 + c_1 g$ , with  $a = O(\kappa^2)$  and polynomial  $c_0, c_1$  for fixed supports. Iteration gives  $B_\tau = \mathcal{U} + \mathcal{V}g$  with  $\log(\mathcal{U} + \mathcal{V}) = O(k + (\tau + 1) \log(1 + \kappa))$ . Choose  $b = 2 \lceil \log_2(8(\mathcal{U} + \mathcal{V} + 1)) \rceil$  and a prime  $2^b < q < 2^{b+1}$ . Since  $g \leq b + 2$ , every intermediate bound obeys  $q > 2 + 4B$ . Proposition 5 gives the exact constants and domination inequalities.

At one common modulus  $q$  with common gadget length  $g$ , the hints occupy  $(4 \cdot 2^\tau + 6\tau)gN$  ring coefficients, or that quantity times  $\lceil \log_2 q \rceil$  bits in bit-packed serialization. For a modulus chain, sum each row's storage at its actual creation modulus. This is an attained preparation-depth-key tradeoff, not a lower bound on conventional HE. The owners must compute, encode, encrypt and upload every prepared variant. Retaining all inputs costs  $O((2^{k+1} - 1 + \tau)N \log q)$  bits. Streaming the prefix needs three accumulators, in addition to current inputs, rows, digits and arithmetic scratch.

Source-component digits can be shared between Hasse transport and alignment: coefficientwise gadget extraction commutes with grouping relative coordinates. Then  $(2 + 3\tau)N$  coefficients undergo internal gadget decomposition. Final export adds  $2N$  coefficient reconstructions, not additional gadget digits. The unshared  $(2 + 4\tau)N$  decomposition count is a valid unshared upper bound, not the attained shared-digit count.

Common-pair sharing of several individual Hasse orders does not give the same saving on these serial edges: each stage changes the sending and receiving secrets. Regrouping stages also introduces mixed derivatives. Appendix A.2.2 makes this obstruction precise for explicit norm-one Taylor-closed preparation, while leaving fused conventional composition algorithms eligible.

### B.5 Admission and terminal interfaces for the dense core

This subsection distinguishes theory-first admission, encrypted functional checks and the required matched product and circuit measurements. None supplies concrete security qualification. Use the binary field and independent-owner inputs for the product and circuit workloads of Appendix E.2, with the lengths and occupancies specified in Appendix I.1. Native always allocates sixteen jet lanes, including when only one job is occupied.

*Do not require an unnecessary terminal key switch.* A raw terminal product is already decryptable from  $c_0 + c_1s + c_2s^2$  by its recipient. Horner evaluation takes two ring products, versus one for a canonical two-component ciphertext; the extra component is also charged in output traffic. No  $s^2$  evaluation key is published for that final gate. This option is available to both competitors. It changes terminal delivery, not the requested coefficients or the allowed owner preparation. It does not supply a reusable canonical ciphertext. This is a standard terminal-output optimization, not a new relinearization technique.

Using the deterministic recurrences of Section 4, the following profiles pass with the specified 60-bit primes and gadget width 48:

| Workload       | $L$ | Raw terminal prime counts | Rekeyed terminal counts |
|----------------|-----|---------------------------|-------------------------|
| Product        | 16  | 1                         | 2                       |
| Product        | 256 | 2                         | 2                       |
| Affine product | 256 | 2                         | 2                       |
| Tree, by level | 256 | (3, 2, 1)                 | (4, 3, 2)               |

These are the first admitted choices in a finite ranking by initial limb count, hint bytes, sum of limb counts and then lexicographic chain. Only nonincreasing prefixes of the four specified primes and the fixed balanced circuit are compared: four one-level choices or twenty three-level choices per terminal policy. This is not global optimality. Every product/rekey and intermediate two-component prime drop is admitted.

For example, the raw product bounds are 1749150067436408 at  $L = 16$  and 7163676697149945728 at  $L = 256$ . Admission requires  $q > 2 + 4B$ . At length

16, removing the final two banks permits the one-prime carrier. Its raw public key, two inputs and terminal output occupy respectively 64, 128 and 96 KiB, excluding metadata and codec work. At length 256, product requires no hints with raw delivery; its terminal output has 3 MiB rather than the two-component alternative’s 2 MiB.

For the deep circuit, rekey the four first-level products under one fresh secret and drop from three to two primes; rekey the two second-level products under another fresh secret and drop to one prime. Return the last product raw. The banks are shared across nodes at their level, with fresh independent rows at setup. This has three independent secrets, fourteen hint rows and 36 MiB of raw RNS hints, instead of four secrets, twenty-four rows and 76 MiB for the rekeyed-terminal control. Both execute six prime drops; the raw route executes six rekeys instead of seven. It uses 12 fresh input ciphertexts, 36 MiB of input arrays, a 3 MiB public key and a 1.5 MiB raw output. These are array volumes, not peak process memory.

The single-level raw routes publish one owner public key and no hints. For two batches their ideal privacy factors are 10 for product and 14 for shallow arithmetic circuit. The raw deep graph has incoming row counts (1, 8, 6), giving factor 54 and a largest creation game of eight rows at the three-prime top modulus. These instantiate Theorem 9; they do not establish the hardness of those games. In particular, a different ring dimension does not inherit the measured composition profile’s assessment by relabeling.

The finite parameter search enumerates the specified chains for all six primary cells and both terminal policies, retaining rejected candidates. Native phase checks validate all six cells under the selected profiles, independently of conventional parameter selection.

*Conventional finite admission.* For terminal CRT use  $K = 32$  points at length 16,  $K = 512$  at length 256 for product/shallow, and  $K = 2048$  for the depth-three circuit. With  $S$  field slots, there are  $B = \lceil JK/S \rceil$  groups. One raw product per group suffices for product; shallow adds one ciphertext per group. Deep evaluates seven products and four additions per group. No encrypted interpolation or rotation is required. Each owner prepares its own evaluations; the recipient performs the inverse transform and retains the requested coefficients.

The four conventional conductors and first admitted prime counts are below. The product/shallow column applies to both batched product lengths and Affine product; each product single-job counterpart also passes at its batched choice. In a deep policy  $xy$ ,  $x$  and  $y$  indicate relinearization after product levels one and two. A dash means the predeclared finite family is exhausted, not that no implementation or secure parameter choice exists.

| $m$   | $\varphi(m)$ | $S$  | product/shallow | $nn$ | $ny$ | $yn$ | $yy$ |
|-------|--------------|------|-----------------|------|------|------|------|
| 4369  | 4096         | 256  |                 | 1    | –    | –    | –    |
| 13107 | 8192         | 512  |                 | 1    | 3    | 2    | –    |
| 21845 | 16384        | 1024 |                 | 2    | 3    | 2    | 3    |
| 65535 | 32768        | 2048 |                 | 2    | 3    | 3    | 3    |

The underlying requested-bit sequence is 20, 60, 120, 180, 240, with actual primes recorded separately; a request is not the actual modulus bit length. Within each carrier/policy, all lower eligible requests are tested or rejected before selecting the first passing one. This is not a runtime ranking or global parameter optimum. A library affine heuristic screens the actually published modulus:  $Q$  for zero-switch-key routes,  $QP$  when switching rows are published. Both the library’s default estimate and the effective screen are retained; neither is 128-bit qualification.

Per group, policies  $nn, ny, yn, yy$  return respectively 9, 3, 5, 3 components, use 0, 2, 4, 6 relinearizations, and require 0, 3, 1, 1 shared switching matrices. The  $ny$  matrices cover  $s^2, s^3, s^4$ ; a nine-component raw terminal decryption requires no published keys for those secret powers. These conventional same-key matrices do not inherit the native independent-secret theorem. HELib’s legal small-prime terminal sets are retained; only special primes are removed. Differing addends are explicitly brought to the accumulator’s set. All resulting costs and terminal sizes are charged.

The dense BSGS anchor admits the  $m = 13107$  two-prime profile. Across 16 jobs, it uses 2176 input encryptions, 2048 raw products, 2040 additions, 120 relinearizations and 120 rotations, with 16 shared matrices and eight three-component outputs. Its triangular diagonals include every term of the dense fixture, not the composition-specific omitted diagonal.

Across the primary family and this anchor, 32 admitted configurations pass two encrypted batches each. This count includes eight single-job product counterparts. Of 31 retained rejections, 23 fail a public noise bound before decryption and eight complete matching-output checks but miss the terminal margin. No observed wrong-output rejection occurs. These are repeated tests of six fixed public fixtures, not independent plaintext-distribution samples. Functional admission does not replace the final timing protocol or measurements.

## C Public-view reductions and Gaussian constructions

### C.1 Exact-Law Public-View Reduction and Measured Inventory

Fix the integral tensor basis and independent integer-polynomial secret and error laws  $\chi_s, \chi_e$ , with owner ephemerals also drawn from  $\chi_s$ . The measured algorithms of Section 4 use ternary secrets and CBD20 errors; the Gaussian family below instead uses  $\chi_s = \chi_e = \mu_\lambda$ . Define  $D_{q,n}$  by the joint samples  $(A_j, A_j S + 2E_j)_{j=1}^n$ , with uniform independent masks,  $S \leftarrow \chi_s$ , and independent  $E_j \leftarrow \chi_e$ . Its comparison experiment consists of independent uniform pairs in  $R_q^2$ . Let  $\epsilon_{q,n}(T)$  bound the absolute output-probability gap of a time- $T$  distinguisher. This names an exact distributional Ring-LWE assumption, not every distribution called Ring-LWE. Since  $q$  is odd, scaling both coordinates by  $2^{-1}$  gives the equivalent normalized  $\chi_e$ -error game with the same secret law.

For a data challenge bit  $b_*$  define  $\text{Adv}^{\text{gap}} = |\Pr[Z = 1 \mid b_* = 0] - \Pr[Z = 1 \mid b_* = 1]|$ . The corresponding best-oriented success-minus-one-half is half this gap. Write  $\delta_{K_{\text{batch}}}$  for the whole implemented-view deviation from exact laws, for

each fixed bit, at the declared reuse horizon. Sampler/PRG losses and explicit aborts must be accounted for here; privacy is not conditioned on a good-noise event.

### C.1.1 The acyclic transcript reduction

**Theorem 9 (Public-view privacy with reuse).** *Consider a fixed acyclic key manifest with independent vertex secrets. All secret-dependent public objects must be declared affine rows with fresh masks/errors; each payload depends only on predecessor secrets and permitted pre-existing data, not on the destination secret or the row values being replaced. Public projections/aliases are generated from these same rows. At vertex  $v$ , let  $n_v$  count incoming hint rows and its public key if published, and let  $Q_v$  bound its owner-encryption queries under one common challenge bit. Queries are permitted only where a public key is published. Both challenge messages are validated before consulting the challenge bit. Pairs have matching public metadata, and their messages may depend on secret keys only through the permitted public information. Under these exact simulation conditions,*

$$\text{Adv}^{\text{gap}} \leq 2 \sum_v \epsilon_{q_v, n_v} + 2 \sum_v Q_v \epsilon_{q_v, 2} + 2\delta_{K_{\text{batch}}}.$$

*Every term is evaluated at adversary time plus its full simulation overhead. This is the exact-local-simulation bound: approximate local samplers require their additional endpoint losses, separately from the honest-view  $\delta$ . Public evaluation is postprocessing and introduces no extra privacy hybrid.*

*Proof.* Fix  $b_*$ . Replace the implemented law by the exact law at cost  $\delta_{K_{\text{batch}}}$ . Process vertices in reverse topological order. At a chosen destination its outgoing secret-dependent rows are already uniform, so the reduction need not know that destination secret. It samples all other secrets and knows every incoming payload. Changing  $a$  to  $-a$  identifies each unshifted row with  $D_{q_v, n_v}$ ; adding the known payload preserves uniformity in the comparison branch. Generate aliases and modulus projections jointly from those same rows. Encryption queries under the current public key can still be answered using fresh coins, without its secret.

Once all public keys are uniform, hybridize each owner encryption using its two-sample ephemeral-secret game. Its public multipliers are the two uniform key elements, its ephemeral has the same  $\chi_s$  law as  $S$ , and its message is an additive shift. Fresh coins permit the query-by-query hybrid despite correlations between prepared messages. Fixed per-vertex query caps allow an unused challenge index to be simulated without a distinction. The resulting common endpoint is independent of  $b_*$ . The per-bit cost is  $\delta_{K_{\text{batch}}} + \sum_v \epsilon_{q_v, n_v} + \sum_v Q_v \epsilon_{q_v, 2}$ ; comparing the two bits doubles it.

Direct replacement by uniform rows is important. Comparing a real payload to a zero payload *through* uniform rows can cost two decisional gaps; the common endpoint alone does not justify one. Evaluation/serialization, all other keys and encryptions, and public projections are part of reduction time. Fixed-key cycles,

undeclared correlated rows, private diagnostics and same-ephemeral prepared encryptions are outside this theorem.

**C.1.2 Exact inventory of the measured construction** Here  $\epsilon_{q,n}$  again refers specifically to the measured ternary/CBD20 laws, not the Gaussian source above. The table gives creation moduli and query caps, not ciphertext-rescaling operations. Result vertices publish no unnecessary owner public keys.

The revised paid-identity profile retains these ten secrets, five owner public keys, moduli and query caps. Its incoming row counts, in the same vertex order, are 1, 10, 46, 15, 26, 15, 13, 12, 7, 9: 149 hints plus five public-key rows, with maximum 46. The changed Hasse payloads depend only on the predecessor secret, and every new identity row has its own independent error. The same reverse-topological proof therefore applies. The 81-row source assumption below is conservative for either profile considered separately; it does not authorize publishing their union under common keys.

Ordinary coordinate reduction of a  $\mathcal{D}_{q_4,n}$  challenge modulo  $q_a \mid q_4$  gives exactly  $\mathcal{D}_{q_a,n}$ : uniform masks project to uniform masks, and the same integer secret/error laws reduce to the prescribed ones. This challenge projection is not parity-preserving ciphertext rescaling and does not generate independent extra samples. Discarding unused rows reduces every game in the table to  $\mathcal{D}_{q_4,81}$ . Thus for  $K_{\text{batch}}$  fresh input batches with the table’s per-vertex caps,

$$\text{Adv}^{\text{gap}} \leq 2(10 + 35K_{\text{batch}})\epsilon_{q_4,81} + 2\delta_{K_{\text{batch}}}. \quad (16)$$

For  $K_{\text{batch}} = 2$ , the reduction factor is 160, not 90. With ideal coins, a scheme gap of  $1/4$  therefore implies an exact-game gap of at least  $1/640$ , at the charged reduction time. This is an implication, not evidence that such distinguishing requires  $2^{128}$  work. Additional batches increase the query loss; they do not extend any ciphertext’s admitted depth.

*Public interface and optional concrete assessment.* The declared view includes the full manifest, all keys/hints with their joint RNS projections, challenge ciphertexts and public evaluation/serialization. It excludes secret keys, decryption/phase oracles, and private-input or recovered-plaintext digests. The two-batch harness does not enforce a deployment key lifetime. Its synthetic fixtures and private validation digests are not authorized leakage for arbitrary inputs. Larger reuse horizons retain the general query-dependent bound.

Appendix C.2 supplies a bounded ideal-coin reference and a source-interface comparison for the measured finite tuple; its simulation costs are not measured HE costs. The independent-coin model does not by itself justify an OS source, a CSPRNG or the harness’s exported interface. These are separate implementation premises.

A concrete bit-security estimate is a separate question. In particular, the measured tensor-basis CBD20 errors are not canonical spherical Gaussians [36]. The exact source law and the complete public view need their own assessment;

no numerical hardness conclusion follows from the conditional reduction or successful ciphertext execution.

Source and deployment premises are separate: a cryptanalytic assessment may be conditional on ideal coins or a stated CSPRNG assumption, but does not thereby bound  $\delta_2$ , enforce the reuse horizon or isolate private diagnostics in the measured harness. If a material finding motivates changing the profile, its noise, depth and total resources must be repriced before retaining the measured efficiency claim. The present execution comparison is not security-matched.

**C.1.3 Exact source scope of the 149-row profile** The measured profile uses  $R = \mathbb{Z}[t, b]/(t^{256} + 1, \Phi_{257}(b))$ , of rank  $N = 65536$  and conductor 131584, in the basis  $\{t^i b^j : 0 \leq i < 256, 1 \leq j \leq 256\}$ . Its vertex secrets and owner ephemerals have independent uniform  $\{-1, 0, 1\}$  coefficients. Primitive errors have independent  $\text{CBD}_{20} = \text{Bin}(40, 1/2) - 20$  coefficients, of variance 10 and support  $[-20, 20]$ . Every error is one integer polynomial projected jointly into its RNS channels. The ideal-source statement does not identify the measured OS random generator with independent uniform words.

The exact source game has rows  $(A, AS + 2E)$  modulo  $q$ . Because  $q$  is odd, multiplying *both* coordinates by  $2^{-1}$  gives the equivalent game  $(A', A'S + E)$  with the same secret and error laws and an independent uniform mask  $A'$ . This normalization removes the public error multiplier; it does not replace  $\text{CBD}_{20}$  by a Gaussian or change its variance to that of  $2E$ .

**Corollary 6 (A 46-row source for the 149-row profile).** *For the 149-hint-row profile alone, under the exact-law simulation hypotheses of Theorem 9,*

$$\text{Adv}_{\text{paid}}^{\text{gap}} \leq 2(10 + 35K_{\text{batch}})\epsilon_{q_4, 46} + 2\delta_{K_{\text{batch}}}.$$

*As in that theorem, the source advantage is evaluated at the adversary's time plus the complete simulation overhead. In particular, for two batches and ideal coins the multiplicative loss is 160, and a scheme gap of 1/4 entails a source-game gap of at least 1/640.*

*Proof.* Use the topological order  $(s_0, s_1, h_8, s_2, h_4, s_3, h_2, s_4, h_1, s_5)$ . The incoming hint-plus-public-key counts are  $(1, 10, 46, 15, 26, 15, 13, 12, 7, 9)$ . All linear, quadratic and Hasse payloads depend only on the source vertex, which precedes their destination; every primitive row uses its own fresh mask and error. Thus the reverse-topological reduction applies unchanged. The maximum incoming count is 46, every creation modulus divides  $q_4$ , and there are 35 owner encryptions per batch. Coordinate projection and row deletion reduce every local source game, including each two-row encryption game, to the exact  $q_4$ , 46-row game. Substituting these counts in the theorem proves the claim.

An 81-row source is also a sufficient common premise: deleting 35 challenge rows supplies the 46-row game, with the deletion cost charged. This observation concerns each profile separately. It does not authorize publishing the 203-row and 149-row banks under shared keys or treating their union as a 46-row transcript. It changes neither the physical key graph nor any measured parameter.

*Numerical assessment remains separate.* The complete Gaussian-family theorem uses a different specified source law. It gives no numerical hardness certificate for these tensor-basis ternary/CBD<sub>20</sub> rows. Likewise, expanding each ring row into  $N$  coefficient equations does not make its multiplication matrix a matrix of independent uniform entries. The nonfinite capped-optimizer outputs were limited by their search cap and are not evidence of infinite attack cost.

A separate cost calculation evaluates the one-published-owner-row nearest-plane route at the actual 240-, 180- and 120-bit moduli. Its lattice and capped target radius are exact; its GSA basis-shape and reduction-work predictions are heuristic. These are model-dependent attack-cost witnesses, not achieved attacks, optimized whole-transcript costs or hardness lower bounds. In particular, their large work exponents do not set a numerical security level or make the timing comparison security matched.

**C.1.4 Exact-law protection for restricted CRT observations** This supporting source analysis concerns the exact ideal CBD20 law above. It supplies no full-view hardness or numerical security level. Chen et al. [13, Sections 2.8 and 5] study residue-field statistical attacks and give the shifted-binomial cosine characteristic function. Their small-modulus Theorem 3 assumes  $q < m^2$  for a power-of-two conductor  $m$ ; the three primes exceed  $512^2$ . We instead use an explicit cyclotomic norm bound and count nonzero frequency columns. These are established mathematical tools, specialized here to the actual tensor law and joint prime channels.

Let  $L = 256$ ,  $J = 256$  and  $Q$  be a product of the three ciphertext primes. Every  $\ell \mid Q$  satisfies  $\ell \equiv 1 \pmod{512 \cdot 257}$ . A scalar CRT position is a pair  $(\alpha, \beta)$  in  $\mathbb{F}_\ell$ , where  $\alpha$  has order 512 and  $\beta$  has order 257. The full CRT map is a bijection. Choose at most  $k$  distinct positions at each prime, allowing different choices between primes, and let  $\rho$  be their joint projection. Its alphabet has size  $H \leq Q^k$ . The same integer error is projected into every prime channel.

**Proposition 6 (Restricted CRT error distance).** *For  $1 \leq k \leq 256$  and an error  $E$  with independent tensor-coordinate CBD<sub>20</sub> coefficients,*

$$\text{TV}(\rho(E), U) \leq \frac{Q^{k/2}}{2} \exp\left(-20\pi^2 \left\lfloor \frac{256}{k} \right\rfloor Q^{-k/128}\right). \quad (17)$$

*The same bound holds for  $2E$  and any independent additive shift. At  $Q < 2^{180}$  and  $k = 3$ , it is strictly below  $2^{-1028}$ .*

*Proof.* Fix a nontrivial character of the output alphabet, with coefficients  $c_{\ell,a}$  at selected pairs  $(\alpha_{\ell,a}, \beta_{\ell,a})$ . The centered frequency of tensor coefficient  $(i, j)$  is

$$\delta_{ij} = \left\langle \sum_{\ell,a} \frac{c_{\ell,a} \alpha_{\ell,a}^i \beta_{\ell,a}^j}{\ell} \right\rangle_{[-1/2, 1/2]}, \quad v_{ij} = Q\delta_{ij} \in \mathbb{Z}.$$

For a fixed column  $j$ , form the integer  $F_j = \sum_{i=0}^{L-1} v_{ij} \zeta^{-i}$  in  $\mathbb{Z}[\zeta]$ , with  $\zeta$  a primitive 512th root. The inverse powers are an integral basis. For any primitive 512th root  $\gamma$  modulo  $\ell$ , the length- $L$  geometric sum gives

$$F_j(\gamma) = L \frac{Q}{\ell} \sum_{a: \alpha_{\ell,a} = \gamma} c_{\ell,a} \beta_{\ell,a}^j \pmod{\ell}.$$

At most  $k$  of the  $L$  prime-ideal evaluations can be nonzero. The integer  $F_j$  belongs to at least  $L - k$  distinct degree-one prime ideals above each  $\ell$ . If  $F_j \neq 0$ , its nonzero integral norm is therefore divisible by  $Q^{L-k}$ . Hence  $|\text{Norm}(F_j/Q)| \geq Q^{-k}$ . Embedding orthogonality and the arithmetic-geometric mean inequality give

$$L \sum_i \delta_{ij}^2 = \sum_{\sigma} |\sigma(F_j/Q)|^2 \geq L |\text{Norm}(F_j/Q)|^{2/L} \geq L Q^{-2k/L}.$$

This step does not assume the character is nonzero at every prime.

Choose a prime with a nonzero character coefficient and a nonzero group of coordinates sharing one  $\alpha$ . Their column sequence has the form  $\sum_{a=1}^s c_a \beta_a^j$ , with  $s \leq k$  distinct nonzero  $\beta_a$ . It cannot have  $s$  consecutive zeros, by invertibility of the corresponding Vandermonde matrix. Therefore no  $k$  consecutive complete columns vanish, and at least  $\lfloor 256/k \rfloor$  columns have  $F_j \neq 0$ . It follows that  $\sum_{i,j} \delta_{ij}^2 \geq \lfloor 256/k \rfloor Q^{-k/128}$ .

For a CBD20 coefficient the characteristic function is  $\cos(\pi x)^{40}$ . The inequality  $|\cos(\pi x)| \leq \exp(-\pi^2 x^2/2)$  for  $|x| \leq 1/2$  follows by integrating  $\tan(\pi x) \geq \pi x$  and using symmetry. Independence of the original tensor coefficients thus bounds every nontrivial output character by the exponential in (17). Finite-group Plancherel and Cauchy-Schwarz give half the square root of  $H - 1$  times this bound. Uniform ring inputs project to product-uniform outputs, which proves the claim. Since 2 is a unit, multiplication by 2 permutes the character set. Fixed translations preserve distance; averaging proves the independent-shift case.

For the numerical claim, certify a dyadic lower bound  $u = a/2^{64}$  to  $Q^{-k/128}$  by

$$a^{128} Q^k \leq 2^{8192} < (a+1)^{128} Q^k.$$

Using  $\pi > 157/50$  and  $\ln 2 < 347/500$ , a strict distance exponent is

$$\left\lfloor \frac{98596}{347} \left\lfloor \frac{256}{k} \right\rfloor u - \frac{k \lceil \log_2 Q \rceil}{2} + 1 \right\rfloor.$$

Replacing  $Q$  by  $2^{180}$  in the energy term and taking  $k = 3$  gives 1028. The source record certifies the integer powers and rational constants, using Machin's identity and alternating arctangent bounds for  $\pi$ , and a degree-ten exponential lower bound for  $\ln 2$ .

*Restricted transcript and its boundary.* Suppose an observer receives all primitive masks and, from each body, at most three positions per prime chosen before

observing that body. The choice may depend on previous restricted bodies. Conditioning on all secrets, masks and allowed history leaves the current independent error unchanged and its remaining phase a fixed shift. Sequential coupling therefore gives distance at most  $m2^{-1028}$  from uniform bodies for  $m$  fresh primitive rows. It is below  $2^{-1022}$  for the 46-row source game and below  $2^{-1020}$  for the 134-hint-row setup, including its five public-key rows in this *restricted observation model*.

This does not include full bodies, extra probes, current-response-dependent choices, owner ciphertexts or arbitrary public evaluations. It cannot replace the full exact-law source assumption. Real source-interface deviations remain separate from the ideal-word statement. In particular, small CRT marginals need not determine the full error law: an exact rank-four conductor-12 example modulo 13 has all four scalar marginals within  $2^{-42}$  of uniform, but full-vector distance greater than  $1/8$ , as computed by exact integer convolution.

Finite arithmetic verification also covers all prime and root certificates, all 41 CBD20 masses, 648 small frequency fixtures with 1366 exact nonzero integer norms, and 4,718,592 spectral entries in 36 full-size fixtures. The fixtures include inactive channels and forced consecutive-column cancellation. These computations corroborate the identities on the tested cases; they do not enumerate all production characters or measure a cryptographic implementation or attack.

## C.2 An explicit strict-time reference reduction

This subsection bounds the reduction’s auxiliary work for the exact two-batch measured-law game of Appendix C.1. It neither changes the measured sampler nor assesses the hardness of the resulting Ring-LWE game. The reference simulator is intentionally elementary; its cost is not an implementation benchmark. Chosen-message queries supply binary coefficient representatives. Preparing those queries, including owner preprocessing or a codec invoked by the adversary, is adversary work. There is no private decryption oracle. Only the fixed, bounded query schema is admitted; malformed or over-cap requests do not invoke an unbounded parser or an extra evaluation service.

**C.2.1 A fixed catalogue covers every neighboring hybrid** The honest-run random-tape budget does not suffice: uniform replacement rows and ciphertexts need two uniform components. Instead, reserve the following independent catalogue before running the adversary. Query slots are indexed by key and occurrence number, including unused slots.

| Reserved role                  | Uniform prime | Ternary | CBD20 |
|--------------------------------|---------------|---------|-------|
| 208 setup rows and ten secrets | 1542          | 10      | 208   |
| 70 permitted encryption slots  | 548           | 70      | 140   |
| Total                          | 2090          | 80      | 348   |

Here a prime vector has  $N = 65536$  coordinates in one RNS channel. The setup’s limb-weighted row count is 771; the query slots have weight  $62 \cdot 4 + 2(4 + 4 +$

$3 + 2) = 274$ . Reserve two uniform polynomials per row/query, as well as the displayed honest-law coins, and discard whichever draws a hybrid does not use. The respective uniform-vector counts at the four primes are  $(556, 556, 520, 458)$ .

The base-game challenge supplies its own correlated coordinates. Their reserved local substitutes are discarded; in particular a dummy secret cannot replace the unknown challenge secret. Projections, sign changes, known payload shifts and public postprocessing use no extra randomness. In a setup hybrid, predecessor secrets are known; outgoing rows whose payloads depend on the unknown destination secret have already been uniformized, while incoming rows are supplied by the base challenge. In a data hybrid, the selected public key uses the challenge masks across all its queries, not newly sampled masks at each encryption. The two challenge values answer only the designated query slot and are discarded if that slot is unused.

**Lemma 9 (Strict sampling with transcript-independent abort).** *Generate this fixed catalogue from independent uniform 64-bit words. For a prime or ternary vector, use the declared rejection map with at most eight refill passes; CBD20 uses one word per coordinate. On any failure output a fixed distinguisher bit. Then the word budget and abort probability  $a$  satisfy*

$$\begin{aligned} B_{\text{red}} &= N[8(2090 + 80) + 348] = 1160511488, \\ a \leq \alpha &:= N[2090\rho_{\text{max}}^8 + 80 \cdot 2^{-512}] < 2^{-204}, \\ \rho_{\text{max}} &= \max_{p|q_4} \frac{2^{64} \bmod p}{2^{64}} < 2^{-29}. \end{aligned} \tag{18}$$

*For either challenge distribution, conditional on catalogue completion the retained coins have exactly their ideal joint law. Consequently,*

$$\text{gap}_{\text{capped}} = (1 - a) \text{gap}_{\text{ideal}}.$$

*The same  $a$  applies in both challenge branches.*

*Proof.* A word is accepted below  $\lfloor 2^{64}/m \rfloor m$ , giving a residue uniform modulo  $m$  and independent of the acceptance indicator. For ternary  $m = 3$ , rejection has probability  $2^{-64}$ . CBD20 uses two disjoint twenty-bit popcounts and has no rejection. Assign one fresh word per refill to each still-unfilled position. A position fails all eight attempts with probability  $\rho^8$ ; a union bound over the fixed catalogue gives the stated  $\alpha$ . Each refill requests at most  $N$  words, giving  $B_{\text{red}}$ .

Rejection indicators are independent of accepted values. Conditioning on completion of every reserved vector therefore preserves all the accepted laws and their independence, including values later selected adaptively. Early termination at the first failed vector is equivalent to the corresponding event in the full virtual catalogue. This event is independent of the base challenge and exact simulated transcript. On failure the output bit is fixed, so subtracting the two branch probabilities gives the gap identity. If the catalogue instead depended on the realized queries or challenge values, this factorization would need a new

proof; separate endpoint-TV couplings alone can cost  $2\alpha$ . Finally  $2090 \cdot 2^{-216} + 80 \cdot 2^{-496} < 2^{-204}$ .

The budget is 9284091904 bytes of streamed random input, not a stored tape. It belongs to an ideal-word *reference reduction*. It is neither the honest-run cap of 468975616 words nor a bound on OS entropy, allocation, network waiting, backend failures, side channels or the implemented RNG gap.

**C.2.2 A conservative finite time and space envelope** Use a random-access bit-operation model: Boolean operations, bit copies and reads/writes are charged, and bounded integer address arithmetic is expanded into bit operations. This is not a sequential-tape memory-access bound. Integer arithmetic is not unit cost at arbitrary precision. Run the adversary once at its charged cost  $T$ ; the bounds below concern additional reference data processing, not a universal Turing-machine simulation theorem. All reference loops have fixed public bounds.

Use 256-bit residue registers and 512-bit product registers, since  $q_4 < 2^{240}$ . Fixed shift/add multiplication and binary long division fit within  $2^{26}$  bit operations per modular scalar product, with additions below  $2^{20}$ . For example, charging  $64w$  operations for a width- $w$  ripple pass and  $16w$  for a selection pass bounds the multiplication and division core at width  $n$  by

$$4n^2 + 64n(2n + 1) + 2n[80(2n + 1)] < 2^{26} \quad (n = 256).$$

The slack includes register copies and bounded control/index work.

A schoolbook tensor product has  $N^2$  scalar products and fewer than  $2N^2$  accumulator updates: exactly  $N^2/256$  pairs have odd-axis exponent zero and use  $1 = -\sum_{v=1}^{256} b^v$ . Charging  $2^{60}$  bit operations per ring product is therefore sufficient. The complete conservative product inventory is

$$208 + 5 + 2 \cdot 70 + 2(2 \cdot 203 + 4 \cdot 19) = 1317.$$

These terms respectively cover affine setup rows, quadratic source payloads, encryptions, and both batches' bank/raw-product evaluations. Uniformized or challenge-supplied rows can only remove such products.

Keep internal coefficients and reproduce the specified RNS/spectral order at the interface. Direct tensor evaluation/interpolation suffices. The specified roots are public constants; generate the small axis power tables by bounded multiplication, not a root-search loop. At most four modular products and four additions per tensor term per prime cost  $4N^2(4 \cdot 2^{26} + 4 \cdot 2^{20}) < 2^{63}$ . Charge  $2^{63}$  bit operations per full polynomial conversion, including coefficientwise CRT reconstruction, permutations and normalization. No dense  $N$ -by- $N$  transform matrix is stored. The fixed request/response schemas have 486/53 component frames. Allowing both complete requests, both directions of conversion, and both directions for all 162 base-challenge polynomials gives  $2[2(486+53)+162] = 2480$  conversions.

| Reference work                         | Conservative bit-operation allowance |
|----------------------------------------|--------------------------------------|
| Ring arithmetic                        | $1317 \cdot 2^{60}$                  |
| Full polynomial conversions            | $2480 \cdot 2^{63}$                  |
| $2^{16}$ additional coefficient passes | $2^{58}$                             |
| Fewer than $2^{31}$ sampled words      | $2^{53}$                             |
| Buffered transport and hashing         | $2^{50}$                             |

The coefficient passes cover payload maps and scaling, centered gadgets, prime drops, challenge normalization/projection, validation and bookkeeping. They exceed the fixed manifest’s individual loop counts. The word map uses at most  $2^{22}$  bit operations per word. For transport, allow  $2^{38}$  processed bits across repeated bounded scans. SHA-256 uses 512-bit blocks and 64 rounds [33];  $2^{20}$  operations per compression block, including padding/schema overhead in the total allowance, is sufficient. Hashing public ciphertext frames does not authorize hashing private plaintexts into the public view.

The displayed sum is below  $2^{75}$ ; take  $c_\star = 2^{80}$  to cover remaining bounded setup and bookkeeping slack. Streaming products, transforms and hashes permits  $S_\star = 2^{40}$  auxiliary bits: even  $2^{14}$  full coefficient/spectral buffers at  $256N$  bits occupy only  $2^{38}$  bits, while the 2518 retained catalogue vectors cost  $2518N64 < 2^{34}$  bits. Allowing 1078 four-limb packet frames costs another  $1078(4N64)$  bits;  $2^{36}$  bits for small axis tables and scratch still leaves the total below  $2^{40}$ . Adversary storage is separate. These loose bounds establish existence of a finite reference simulator, not the measured implementation’s time or memory complexity.

**Corollary 7 (Explicit strict-time two-batch reduction).** *For the assessed interface and bit-cost convention, the exact-law reduction of Theorem 9 can be instantiated with*

$$\text{Adv}^{\text{gap}}(T) \leq \frac{160}{1-\alpha} \epsilon_{q_4,81}(T + 2^{80}) + 2\delta_2(T), \quad \alpha < 2^{-204}. \quad (19)$$

*Proof.* For each of the ten setup and seventy data hybrids, divide the capped gap bound by  $1-a \geq 1-\alpha$  using Lemma 9. Each capped distinguisher runs once within  $T + c_\star$ . Sum the eighty per-bit transitions and compare the two challenge bits. Implementation deviation contributes  $2\delta_2$  as before.

Thus if  $\delta_2(2^{128}) \leq d_2 < 1/8$ , it suffices to establish  $\epsilon_{q_4,81}(2^{128} + 2^{80}) < (1-\alpha)(1/4 - 2d_2)/160$ . Neither this computational hardness requirement nor a numerical RNG/view bound is proved here. The reference sampler explicitly bounds simulation time and truncation; it does not qualify 128-bit security.

**C.2.3 A finite source assumption for the unchanged uncapped sampler** The measured random-word sampler appends accepted words and refills only missing coordinates, with no rejection cap. A cap is not required in the implementation to state a finite source-security reduction. Restrict attention to the logical public view, the assessed query caps, the verified word maps and arithmetic correspondence. Model the source as a service returning exactly the

requested number of bytes. Timing, API failures and other implementation deviations are separate; successful completion is not assumed by conditioning on it.

**Lemma 10 (Stopped-test sandwich).** *Let  $\rho_2(T+c; B_2)$  bound the real-source versus independent-uniform-word gap of every adaptive binary test of charged work at most  $T + c$  requesting at most*

$$B_2 = N[8(771 + 80) + 348] = 468975616$$

*64-bit words. The randomness contribution to the gap for an unchanged uncapped execution satisfies*

$$\delta_2^{\text{words}}(T) \leq \rho_2(T + c; B_2) + a_2, \quad a_2 \leq N[771\rho_{\max}^8 + 80 \cdot 2^{-512}] < 2^{-206}.$$

*Here the compared event is that the test eventually returns one, with nontermination counted as zero; no runtime or availability guarantee follows. For the bounded logical reference processing above,  $c = 2^{80}$  suffices.*

*Proof.* Follow the uncapped word requests and adaptive control flow, but stop before requesting a ninth refill pass for any vector. Define  $D^-$  to return zero at that stop and  $D^+$  to return one; when no stop occurs, both return the original test's bit. Each has the stated finite word budget: 771 prime-mask vectors, 80 ternary vectors and 348 CBD20 vectors. No unused challenge slot needs to be executed; charging every slot is an upper bound. Under ideal words, rejection at any position for eight passes has probability  $\rho^8$ . Union bounds over vectors/positions yield  $a_2$ .

Pathwise, including any divergent continuation after the stop,  $D^- \leq D \leq D^+$ . Writing  $\mathbb{E}_R, \mathbb{E}_U$  for the real and uniform sources, respectively,

$$\mathbb{E}_R D - \mathbb{E}_U D \leq (\mathbb{E}_R D^+ - \mathbb{E}_U D^+) + (\mathbb{E}_U D^+ - \mathbb{E}_U D) \leq \rho_2 + a_2.$$

Use  $D^-$  for the opposite direction. Only one bounded test is used in either direction, so there is no additional factor two. In particular, the proof does not presume a bound on the real source's cap-hit probability. The stopped computations have the same bounded public processing as a subcase of the preceding reference inventory; no OS wall time is charged as a bounded bit operation.

If other source/API/arithmetic/view deviations have bound  $\eta_2(T)$ , the full interface may use  $\delta_2(T) \leq \rho_2(T + 2^{80}; B_2) + a_2 + \eta_2(T)$ . Neither  $\rho_2$  nor  $\eta_2$  is assigned a numerical security level here. The source premise must cover its prior history and allowed side information; fresh Python objects do not imply freshly seeded independent OS generators.

The archive records Python 3.14.4 on a WSL2 kernel. The tagged CPython source [37] retries partial reads/interruption where appropriate, uses blocking `getrandom` when available and retains a fallback path. These are API facts, not an entropy or 128-bit theorem. The public serializer hashes ciphertext packets,

which are deterministic public postprocessing. The experimental instrumentation also emits timing and a recovered-plaintext digest; these diagnostics are excluded from the logical privacy view. The source analysis and stopped-test lemma concern that view and do not establish isolation of diagnostic data in a deployed system.

### C.3 Gaussian JetHE: a conditional family and a finite example

This subsection first describes the finite variant G16, distinct from the measured ternary/CBD20 implementation. It changes every secret, ephemeral and primitive error law, while retaining the ring, plaintext codec, affine algorithms and acyclic composition graph. Its purpose is to remove a specific sampling-law mismatch without automatically increasing the composition modulus. We give a canonical discrete-Gaussian reference, a finite reference sampler, and a high-probability correctness admission. None establishes the reference problem's concrete hardness or transfers the recorded timings to G16. The specialization uses the canonical geometry, normal form and Gaussian-sampling framework of Lyubashevsky–Peikert–Regev [32]; it is not a new Ring-LWE hardness theorem or a generic bootstrap. The following growing-family sections give conditional IND-CPA security and complete resource bounds.

**C.3.1 An exact ideal law, rather than covariance matching** Until the growing-family subsection,  $p = 257$ ,  $L = 256$ ,  $N = 256L = 65536$ ,  $D = Lp = 65792$ , and

$$R = \mathbb{Z}[t, b]/(t^{256} + 1, \Phi_{257}(b)), \quad g = 1 - b.$$

The canonical embedding has its ordinary Euclidean norm, including both members of each complex-conjugate pair. The Gaussian on  $\Lambda$  is

$$D_{\Lambda, s}(x) \propto \exp(-\pi \|x\|^2 / s^2);$$

the parameter  $s$  is *not* its scalar standard deviation. Let  $A_{256} = \{y \in \mathbb{Z}^{257} : \sum_{v=0}^{256} y_v = 0\}$ .

**Lemma 11 (Root-lattice identification).** *The map*

$$F(y) = \sum_{v=0}^{256} y_v b^v = \sum_{v=1}^{256} (y_v - y_0) b^v$$

*is a bijection from  $A_{256}$  to  $g\mathbb{Z}[b]$ . Independent root-lattice blocks  $Y_i \leftarrow D_{A_{256}, s}$ ,  $0 \leq i < 256$ , give exactly*

$$E = \sum_i t^i F(Y_i) \sim \mu_s := D_{gR, s\sqrt{D}}. \quad (20)$$

*Moreover  $R^\vee = (g/D)R$ , so scalar division by  $D$  sends this law exactly to  $D_{R^\vee, s/\sqrt{D}}$ .*

*Proof.* In the physical basis  $b^v$ ,  $1 \leq v \leq 256$ , the image consists of vectors  $e$  with  $\sum e_v \equiv 0 \pmod{p}$ . Its inverse is  $y_0 = -\sum e_v/p$ ,  $y_v = e_v + y_0$ . Evaluation at  $b = 1$  modulo  $p$  identifies this kernel with  $(1 - b)\mathbb{Z}[b]$ . The canonical Gram matrix of the physical tensor basis is  $LI_L \otimes (pI - J)$ . Thus

$$\|\sigma(E)\|^2 = L \sum_i \left( p \sum_v e_{i,v}^2 - \left( \sum_v e_{i,v} \right)^2 \right) = D \sum_i \|Y_i\|^2.$$

This proves the probability-law identity, not just an identity of second moments. The codifferent formula is the specialization of [32, Corollary 2.18]. Division by the rational integer  $D$  is a similarity, so it preserves spherical shape.

G16 takes  $s = 16$  and uses  $\mu_{16}$  independently for all ten vertex secrets, every ephemeral, and every primitive error. The same integer polynomial is reduced into all RNS limbs. As  $g$  is a unit modulo two, this ideal does not reduce the plaintext space  $R/2R$ . Public masks remain uniform in  $R_q$ . In particular, this is neither iid physical-coordinate Gaussian sampling nor independent Gaussian sampling in each RNS channel.

**C.3.2 An exact discrete reference reduction** Put  $r = 16/\sqrt{D}$ . Define  $\epsilon_{q,82}^G(T)$  as the output gap against 82 samples

$$(a_i, a_i s_0 + e_i \bmod qR^\vee), \quad s_0 \leftarrow R^\vee/qR^\vee \text{ uniform}, \quad e_i \stackrel{\text{iid}}{\leftarrow} D_{R^\vee, r},$$

compared with independent uniform pairs. This is a fully specified *discrete*, finite-quotient Ring-LWE assumption. It does not contain a real-number sampling oracle. Coordinates may be supplied in the basis  $t^i(b^v - 1)/D$  of  $R^\vee$ .

**Proposition 7 (Scalar normal-form bridge).** *For the four specified primes, let*

$$u_q = \prod_{p_\ell | q} (1 - 1/p_\ell)^N > 1 - 2^{-41}.$$

*An 81-row distinguisher for  $(A_i, A_i S + 2E_i)$  with  $S, E_i \stackrel{\text{iid}}{\leftarrow} \mu_{16}$  gives a distinguisher for the discrete reference whose gap is exactly  $u_q$  times its gap, before local simulation approximations. One source row is consumed.*

*Proof.* If  $a_0$  is not a unit, output a fixed bit. Otherwise put  $a'_i = -a_0^{-1}a_i$  and  $b'_i = b_i + a'_i b_0$ . In the real branch,  $b'_i = a'_i e_0 + e_i$ , with exactly the claimed common-secret and error laws; the uniform branch stays uniform. This is the discrete normal-form argument of [32, Lemma 2.24]. Since  $gR$  and  $qR$  are coprime, multiplication by  $D$  is a bijection from  $R^\vee/qR^\vee$  to  $R/qR$ . Output  $A_i = 2a'_i$  and  $B_i = 2Db'_i$ . Then  $B_i = A_i(De_0) + 2De_i$ . Both coordinates, not only the response, are doubled. The unit event has the same probability in both branches, proving the gap identity. Complete splitting gives the displayed product and its union bound. This unit loss is multiplicative; it is not claimed to be a  $2^{-128}$ -small statistical deviation.

The reverse-topological proof of Theorem 9 therefore applies to G16: vertex secrets and ephemerals have the same law, and every incoming error has that law. Projection to a smaller prime prefix retains the same integer samples. The largest instance is still 81 rows, and the two-batch factor is still 160. This supplies a reference-law bridge, not a bound on  $\epsilon_{q4,82}^G$ .

**C.3.3 A finite sampler and its explicitly charged cost** In the basis  $v_j = e_j - e_0$  of  $A_{256}$ ,

$$\tilde{v}_j = e_j - j^{-1} \sum_{h=0}^{j-1} e_h, \quad \|\tilde{v}_j\|^2 = (j+1)/j, \quad \frac{\langle v_k, \tilde{v}_j \rangle}{\|\tilde{v}_j\|^2} = \frac{1}{j+1} \quad (k > j).$$

The sequential Gaussian sampler takes  $j = 256, \dots, 1$ , starts with  $T = 0$ , samples an integer  $z_j$  from a one-dimensional Gaussian of parameter  $s_j = 16\sqrt{j/(j+1)}$  centered at  $-T/(j+1)$ , and updates  $T \leftarrow T + z_j$ . It returns  $y_0 = -T$ ,  $y_j = z_j$ . Its additional arithmetic is linear, and it uses  $N$  scalar draws per ring polynomial. The center-dependent scalar normalizers mean that it is not *exactly*  $D_{A_{256},16}$ .

Indeed, scalar Poisson summation bounds the relative normalizer fluctuation by

$$\zeta = 2 \sum_{k \geq 1} \exp(-128\pi k^2).$$

The likelihood-ratio range of all  $N$  successive draws is at most  $((1+\zeta)/(1-\zeta))^N$ . For  $N\zeta \leq 1/8$  its total variation from the target is at most  $4N\zeta < 2^{-557}$ . This is a quantitative specialization of sequential Gaussian sampling, not an appeal to an unspecified negligible function.

**Lemma 12 (Bounded scalar implementation).** *There is an explicit ideal-bit reference sampler for the 428-polynomial two-batch catalogue whose Gaussian-law deviation is below  $2^{-342}$ . Together with eight-pass uniform RNS rejection, the whole local catalogue deviation is below  $\nu = 2^{-203}$  per endpoint. It has a finite streamed random-word budget and fits the reference work and space bounds given below.*

*Proof.* Write each scalar center as  $m + f$ ,  $m \in \mathbb{Z}$  and  $f = h/(j+1) \in [0,1)$ . Each of at most  $2^{14}$  attempts draws a uniform nine-bit code. Codes at least 321 reject; otherwise set  $w = \text{code} - 160$ . Draw a uniform 512-bit integer  $U$  and accept  $z = m + w$  when  $U < T_w$ , where

$$0 \leq T_w \leq 2^{512} \exp\left(-\pi \frac{(w-f)^2}{s_j^2}\right), \quad 2^{512} \exp\left(-\pi \frac{(w-f)^2}{s_j^2}\right) - T_w < 2.$$

The value  $2^{512}$  is stored in a 513-bit register or handled by an always-accept sentinel. Exhausting the attempts returns an explicit abort, not a silently resampled or clipped physical polynomial. The offset cutoff does not itself bound the final physical coefficients by 160; their correctness event is analyzed separately below.

Here is a finite way to obtain each threshold. Use 600 fractional bits, 640-bit value registers and 1280-bit products. Outward-rounded Machin sums  $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$  with 256 terms each give an interval of width below  $2^{13}2^{-600}$ . Once a denominator power exceeds  $2^{600}$ , keep a sentinel instead of growing it: every later positive term rounds to  $[0, 1]$  last-place units. The crossing product is below  $2^{617}$ , so these steps fit the stated registers. Compute  $x = \pi((j+1)w - h)^2/(256j(j+1)) < 640$  and  $t = x/1024 < 1$ . An outward-rounded alternating Taylor sum for  $e^{-t}$  through degree 256, with the next-term remainder, followed by ten outward-rounded squarings, encloses  $e^{-x}$ . Clamp the intervals to  $[0, 1]$ . The initial  $t$  interval has width below  $2^{11}2^{-600}$ ; term recursion and summation give width below  $2^{22}2^{-600}$ , and the squarings give width below  $2^{32}2^{-600} < 2^{-520}$ . Flooring the lower endpoint at 512 fractional bits proves the threshold inequalities. A bound of one last-place unit suffices for the Taylor remainder; no large factorial register is required.

The exact scalar normalizer exceeds 11 by Poisson summation, since  $s_j \geq \sqrt{128}$ . Its tail outside  $[-160, 160]$ , after normalization, is below  $8e^{-100\pi} < 2^{-449}$ . The truncated unnormalized mass exceeds 9, so downward threshold rounding still leaves mass above 8. Acceptance per attempt is therefore at least  $1/64$ . The probability of exhausting all attempts is at most  $e^{-256} < 2^{-369}$ . Normalized weight rounding costs less than  $2^{-500}$ . These three losses sum to below  $2^{-367}$  per scalar, uniformly over all earlier sampled centers. Union coupling over  $428N = 28049408$  scalars, plus the sequential-sampler discrepancy, gives the stated  $2^{-342}$  bound.

The uniform part reserves the same 2090 prime vectors as in Appendix C.2, with eight attempts per coordinate; its failure is below  $2^{-204}$ . Combining the two losses gives  $\nu$ . The Gaussian abort can depend on preceding centers. Consequently we use an additive endpoint coupling, not the transcript-independent abort factorization available for the original uniform/ternary rejection maps.

*Reference costs, not production timings.* At most nine 64-bit words per attempt suffice. The full local catalogue uses at most

$$N(8 \cdot 2090 + 9 \cdot 2^{14} \cdot 428) = 4137149267968 < 2^{42}$$

streamed words. The Gaussian sampler requires this distinct word budget. It is deliberately loose and is not an entropy or OS-RNG guarantee. There are fewer than  $2^{39}$  attempts. Allowing  $2^{40}$  charged bit operations per threshold is sufficient for the bounded integer interval algorithm: fewer than  $2^{13}$  arithmetic primitives, each bounded by  $2^{27}$  using schoolbook multiplication/division and register/control passes. Sampling work is below  $2^{79}$ . The non-sampling subtotal in Appendix C.2 is below  $2^{75}$ ; another  $2^{73}$  covers the normal-form products, unit test/inversion in the fixed split representation, scalar basis conversion and setup constants. Word processing costs at most  $2^{42}2^{22}$ . The sum stays below  $2^{80}$ .

Stream and discard rejected coins. Keeping the original 2518 output-vector slots,  $2^{14}$  polynomial buffers, 1078 four-limb packet frames and  $2^{36}$  scratch bits still fits  $2^{40}$  auxiliary bits. The random tape itself must not be retained. Adver-

sary work and memory remain separate. This is a constructive reference envelope, not a measured implementation.

Thus a bounded ideal-coin G16 realization of the declared two-batch interface admits the conservative conditional bound

$$\text{Adv}^{\text{gap}}(T) \leq \frac{160}{u_{q_4}} \epsilon_{q_4, 82}^G(T + 2^{80}) + 2^{-194} + 2\delta_{\text{source}}(T). \quad (21)$$

For each of the 80 per-bit hybrid transitions, the bounded local simulator differs from its ideal counterpart by at most  $2\nu$  in gap; this costs  $320\nu$ . The actual finite catalogue contributes at most another  $2\nu$ , and  $322\nu < 2^{-194}$ . The unit factor applies only to the exact source bridge. The source term includes deviations from the declared ideal-word interface and any additional implementation behavior; it is not qualified by the reference algorithm. Here  $\delta_{\text{source}}(T)$  bounds one whole view using the Gaussian stream of at most 4137149267968 words, with its source-test time charged at  $T + 2^{80}$ ; the ternary catalogue has a different source budget. In particular, the sufficient source gap near  $1/640$  at time  $2^{128} + 2^{80}$  remains an unproved numerical assumption.

**C.3.4 Probabilistic fresh error preserves the four-prime schedule** A support-only substitution would set  $F = (2\kappa B_s + 1)B_e$  and make the current schedule fail when  $B_s = B_e = 160$ . That failure is an artifact of this sufficient envelope, not an observed wrong decryption. The Gaussian law permits a stronger bound.

**Lemma 13 (Physical coefficient and fresh-product bounds).** *For an ideal  $\mu_{16}$  draw, every physical coefficient satisfies  $\Pr[|E_{i,v}| > B] \leq 2e^{-\pi B^2/512}$ . For an independent pair  $E, S \leftarrow \mu_{16}$ , conditional on  $\|\sigma(S)\| \leq 16\sqrt{DN}$ , each physical coefficient of  $ES$  is subgaussian with parameter  $2^{17}$ . In particular its absolute value exceeds  $2^{20}$  with probability at most  $2e^{-64\pi}$ .*

*Proof.* For any centered lattice Gaussian, completing the square in its moment generating function bounds every linear functional by the continuous Gaussian expression: the translated lattice mass is at most its centered mass, since every cosine in its Poisson series is at most one. Applying this to  $y_v - y_0$ , whose representing vector has norm  $\sqrt{2}$ , proves the first bound. This does not assert that the approximate sampler is itself exactly subgaussian; its discrepancy is separately coupled.

The Hermitian dual of the physical basis vector  $t^i b^v$  is  $d_{i,v} = t^i(b^v - 1)/D$ . Therefore  $\|\sigma(d_{i,v})\|_\infty \leq 2/D$ . With  $\tau = 16\sqrt{D}$ , the coefficient of  $ES$  is a linear functional of  $E$  with Gaussian parameter at most

$$\tau \|\overline{\sigma(S)}\| \sigma(d_{i,v}) \leq \tau(2/D) \|\sigma(S)\| \leq 2 \cdot 16^2 \sqrt{N} = 2^{17}.$$

Chernoff's inequality gives the assertion. The centered discrete-Gaussian norm bound [32, Lemma 2.8] makes the excluded norm event at most  $2^{-2N}$ .

For an owner encryption, condition on the norms of its fixed secret  $s$  and public-key error  $e_{\text{pk}}$ . The fresh Gaussian factors are  $u$  in  $e_{\text{pk}}u$  and  $e_1$  in  $e_1s$ . This avoids assuming any Gaussian posterior for a key after seeing its public history. The bound holds uniformly over previous messages and thus covers adaptive message choices and reuse of the same key. Five owner secrets and five public-key errors give ten fixed norm events. Over 70 encryption slots, union coupling gives

$$\begin{aligned} & \Pr[\text{a required ideal event fails}] \\ & \leq 428N 2e^{-50\pi} + 140N 2e^{-64\pi} + 10 2^{-2N} < 2^{-200}. \end{aligned}$$

The first event caps all primitive coefficients, including the ten destination secrets, at 160. On these events every fresh bracketed error in a phase  $\mu + 2e$  is at most

$$F_G = 2^{21} + 160 = 2097312.$$

The finite ideal-coin catalogue discrepancy still leaves total correctness failure below  $2^{-200}$ . Privacy is not conditioned on these good events.

All subsequent inequalities are deterministic. Use  $\Delta_q = \kappa g_q 2^{47} \cdot 160$  in the unchanged prefix/tail recurrences, and replace the drop bound by

$$B \mapsto \left\lfloor \frac{2B - 1 + (160\kappa + 2)p_*}{2p_*} \right\rfloor.$$

With  $F = F_G$ , the schedule  $(q_4, q_4, q_4, q_3, q_2)$  and gadgets  $(5, 5, 5, 4, 3)$  admit every complete state and both drops. The final bound is

$$B_{\text{final}} = 7864022803520907862453734488627905, \quad \log_2 \frac{q_2}{2 + 4B_{\text{final}}} > 5.4.$$

The strict margin is checked by the integer inequality  $q_2^5 > 2^{27}(2 + 4B_{\text{final}})^5$ , not by a rounded logarithm.

**C.3.5 Arithmetic inventory and implementation costs** The composition compiler retains 19 private products, 35 inputs per batch, 203 hint rows and the same RNS/gadget arithmetic counts. Its named raw arrays remain 754 MiB of hints, 137 MiB of retained inputs and 17 MiB of public keys. Gaussian sampling work, any implementation caches and total RSS are extra questions; equality of these array counts is not equality of end-to-end performance. Neither the composition speed ratio nor the product and circuit measurements is a measurement of G16.

**C.3.6 A growing family with conditional IND-CPA security** The finite G16 inequalities above are not an asymptotic security statement. We now give such a statement for a specified family. The construction uses the same codec, affine encryption, acyclic switching and evaluation algorithms; only the public parameters and sampling law are parameterized. The result does not require a numerical attack-cost estimate.

**Definition 2 (Gaussian JetHE family and public generator).** For an integer parameter  $\lambda \geq 256$ , set

$$L_\lambda = 2^{\lceil \log_2 \lambda \rceil}, \quad N_\lambda = 256L_\lambda, \quad D_\lambda = 257L_\lambda, \\ R_\lambda = \mathbb{Z}[t, b]/(t^{L_\lambda} + 1, \Phi_{257}(b)), \quad g = 1 - b.$$

Put  $s_\lambda = \sqrt{\lambda}$  and

$$\mu_\lambda = D_{gR_\lambda, s_\lambda \sqrt{D_\lambda}}, \quad r_\lambda = s_\lambda / \sqrt{D_\lambda}.$$

Each vertex secret, owner ephemeral and primitive error is an independent  $\mu_\lambda$  draw. Every RNS projection of a draw comes from the same integer polynomial. In particular,  $\lambda = 256$  recovers the G16 sampling law; it does not assign G16 a bit-security level.

Fix a public probabilistic polynomial-time generator  $\mathcal{P}$  whose output  $\theta$  specifies the following data, independently of all secret sampling. Its maximum modulus is  $q_* = \prod_{i=1}^\ell p_i$ , with distinct certified odd primes  $p_i \neq 257$  and  $p_i \geq 2N_\lambda \ell$ . All vertex moduli divide  $q_*$ . The output includes an acyclic manifest, the public input/encoding interfaces and per-vertex owner-query caps  $Q_v$ . Its full secret-dependent public view satisfies the simulation conditions of Theorem 9, with  $\mu_\lambda$  replacing both finite laws. Payloads and all other allowed metadata must be computable without the secret hidden at the selected hybrid. No secret-dependent diagnostic output is added to this interface.

Dimension, row counts, query caps, modulus/payload bit lengths, total public evaluation work and all algorithms are polynomially bounded in  $\lambda$ . Let fixed efficiently computable, polynomially bounded functions  $\bar{m} \geq 2$  and  $\bar{H}$  satisfy

$$\max_v n_v \leq \bar{m}(\lambda), \quad |V_{\text{rows}}| + \sum_v Q_v \leq \bar{H}(\lambda)$$

for every output, where  $V_{\text{rows}} = \{v : n_v > 0\}$ . Choose  $\bar{H}$  to be a power of two by padding. These are bounds fixed before the game, not realized adaptive query counts. All vertices use the same ring and integer sampling law. A mixed-degree or cyclic manifest is not covered. The generator may return an explicit failure; it yields the same failure-only transcript in both games and must have negligible failure probability for a usable family.

*Exact hardness assumption.* For the public output  $\theta \leftarrow \mathcal{P}(1^\lambda)$ , let  $\text{GRLWE}_{\mathcal{P}, \bar{m}+\lambda}$  be the decisional problem whose real input, jointly with  $\theta$ , consists of  $\bar{m}(\lambda) + \lambda$  pairs

$$(a_i, a_i s_0 + e_i \bmod q_* R_\lambda^\vee), \\ a_i \leftarrow R_\lambda / q_* R_\lambda \text{ uniform}, \quad s_0 \leftarrow R_\lambda^\vee / q_* R_\lambda^\vee \text{ uniform}, \\ e_i \stackrel{\text{iid}}{\leftarrow} D_{R_\lambda^\vee, r_\lambda}. \quad (22)$$

The null input has independent uniform pairs in these two finite quotients. We assume that every uniform probabilistic polynomial-time distinguisher has negligible absolute output gap for these joint experiments. The assumption is about

ordinary unshifted, uniform-secret Ring-LWE rows, not the entire JetHE transcript. It is tied to this public generator and canonical discrete error law. Neither arbitrary choices of  $q_*$  nor an iid coefficient Gaussian or the measured CBD20 law are interchangeable with it. No worst-case ideal-lattice reduction is asserted here.

**Lemma 14 (Bounded sampling).** *There is a uniform bounded polynomial-time, ideal-bit sampler for  $\mu_\lambda$  whose statistical error per call is negligible in  $\lambda$ . The statement holds jointly over polynomially many calls, including all honest and hybrid-simulation coins.*

*Proof.* The root-lattice identification of Lemma 11 holds with  $L = L_\lambda$  and  $D = D_\lambda$ . It reduces one draw to  $L_\lambda$  independent  $D_{A_{256}, \sqrt{\lambda}}$  blocks. Use the displayed root basis and reverse sequential sampler, now with squared scalar width  $s_j^2 = \lambda j/(j+1)$ . Scalar Poisson summation bounds its normalizer fluctuation, uniformly over the preceding centers, by

$$\zeta_\lambda = 2 \sum_{k \geq 1} e^{-\pi \lambda k^2 / 2} = O(e^{-\pi \lambda / 2}).$$

Consequently its per-polynomial discrepancy is at most  $4N_\lambda \zeta_\lambda$  for sufficiently large  $\lambda$ . This uses a bound vanishing in  $\lambda$ , not a claim of negligibility in a fixed block dimension. It is also a specialization of [32, Lemma 2.9].

A bounded finite-coin realization is enough. Write a scalar center as  $m + f$ , with  $f = h/(j+1) \in [0, 1)$ . Retain integer offsets  $-\lambda, \dots, \lambda$  and approximate their unnormalized Gaussian weights downwards to  $4\lambda$  fractional bits, with error at most two last-place units per weight. Rational interval evaluation of  $\pi$  and the exponential, using range reduction and convergent series, takes polynomial bit work: the exponent is  $O(\lambda)$  and all precisions are  $O(\lambda)$ . Their total mass is bounded below by a constant; omitted tail mass is  $e^{-\Omega(\lambda)}$ , uniformly in  $f$ . Draw proportionally to the integer weights by a uniform integer below their sum, capping its binary rejection at  $\lambda$  attempts. Its abort probability is at most  $2^{-\lambda}$ , and normalization of rounded weights costs  $\text{poly}(\lambda)2^{-4\lambda}$  in variation distance. The cumulative center has polynomial bit length: each block has only 256 steps and bounded offsets. This gives the claimed bounded algorithm. Uniform ring coordinates likewise use binary rejection capped at  $\lambda$  attempts. Polynomial union coupling covers the entire simulation catalogue. Aborts are explicit and charged, not conditioned away. This is an existence/reference algorithm, not a constant-time production implementation.

**Lemma 15 (Reserved pivot).** *From the source in (22), a polynomial-time transformation produces  $\bar{m}$  rows  $(A_i, A_i S + 2E_i)$  with  $S, E_i \stackrel{\text{iid}}{\leftarrow} \mu_\lambda$  in the real branch and independent uniform pairs in the null branch, except for an explicit abort of probability at most  $2^{-\lambda}$ . Conditional on each public parameter and non-abort, these laws are exact.*

*Proof.* Since each  $p_i$  is prime to the conductor  $2L_\lambda \cdot 257$ ,  $R_\lambda/p_i R_\lambda$  is a product of finite fields with at most  $N_\lambda$  factors. A uniform element is a nonunit with

probability at most  $N_\lambda/p_i$ . CRT and a union bound give

$$u(q_*) := \Pr[a \in (R_\lambda/q_* R_\lambda)^\times] \geq 1 - N_\lambda \sum_i p_i^{-1} \geq \frac{1}{2}.$$

Complete splitting is unnecessary. Reserve the first  $\lambda$  source rows solely for pivot selection; select the first unit mask  $a_0$  from them, or abort if none exists. Selection uses only those masks. The remaining  $\bar{m}$  response rows are disjoint from this pool; all unused pivot information is discarded. Thus the selected error  $e_0$  keeps its exact independent Gaussian law. For a response row define

$$a'_i = -a_0^{-1}a_i, \quad b'_i = b_i + a'_i b_0, \quad A_i = 2a'_i, \quad B_i = 2D_\lambda b'_i.$$

The codifferent is  $(g/D_\lambda)R_\lambda$  and multiplication by  $D_\lambda$  induces a bijection to  $R_\lambda/q_* R_\lambda$  because  $(g, q_*) = 1$ . Hence  $S = D_\lambda e_0$ ,  $E_i = D_\lambda e_i$  have exactly law  $\mu_\lambda$  and  $B_i = A_i S + 2E_i$ . The null branch stays uniform. These are the scalar and normal-form arguments of Proposition 7, now with reserved pivots. The common abort probability is  $(1 - u(q_*))^\lambda \leq 2^{-\lambda}$ . This uniform bound avoids pulling a parameter-dependent unit probability outside the absolute value of an averaged signed distinguishing gap. Unit testing/inversion is polynomial-time linear algebra over the known prime fields; no factorization of an unknown integer is needed.

**Theorem 10 (Conditional security of the Gaussian family).** *Fix a generator satisfying Definition 2 and the stated Ring-LWE assumption. The Gaussian JetHE family, using the bounded ideal-bit samplers above, is adaptive multi-query IND-CPA secure for its declared acyclic public view and per-vertex query caps. More precisely, for every uniform probabilistic time- $T$  adversary  $\mathcal{A}$  there is a uniform source distinguisher  $\mathcal{B}$  of time  $T + \text{poly}(\lambda)$  such that*

$$\text{Adv}_{\text{GJetHE}, \mathcal{A}}^{\text{gap}} \leq 2\bar{H} \text{Adv}_{\text{GRLWE}, \mathcal{B}}^{\text{gap}} + 2^{-\lambda} + 2\eta_{\text{hon}} + 4\bar{H}\eta_{\text{sim}}. \quad (23)$$

Here  $\eta_{\text{hon}}$  bounds a coupling of the complete honest sampling catalogue, and hence its public view, per fixed challenge bit;  $\eta_{\text{sim}}$  bounds the whole local simulator's deviation at each source endpoint, uniformly over the padded hybrid choice. Both are negligible by Lemma 14 and the polynomial catalogue bounds. No correctness event is conditioned on.

*Proof.* First use ideal samples. For each fixed bit apply the reverse-topological row replacements and owner-encryption replacements of Theorem 9. At a destination, all outgoing secret-dependent rows have already been replaced, and all predecessor secrets needed for its incoming shifts can be sampled locally. Its remaining encryptions need only the public key. This hides all incoming rows jointly, not one gadget digit at a time. The modulus-divisor map and discarding surplus samples reduce every row game to the same  $\bar{m}$ -row source law. All aliases are computed from those same rows.

For the owner step, the two already uniform public-key coordinates are the two source masks. Source responses provide one encryption with ephemeral  $S$ ,

after adding the selected message to its first coordinate. The simulator answers every other query under this same public key using independent local coins, or uniform ciphertexts at already replaced query positions. Predeclare positions by vertex and local query index; an unused position causes no distinction. Adaptive matching-metadata messages and subsequent public evaluation are therefore included in the simulation. This is not a secret-dependent-message or decryption-oracle claim.

Pad each per-bit path to exactly  $\bar{H}$  transitions and select uniformly one of their  $2\bar{H}$  slots. A bit-zero slot outputs the adversary's bit; a bit-one slot complements it. Let  $\Delta(\theta)$  be the signed difference of the two ideal honest-game output probabilities conditional on public parameters. Without pivot failure, the source gap of this single signed-hybrid reduction is  $\mathbb{E}_\theta[\Delta(\theta)]/(2\bar{H})$  by telescoping to the common uniform endpoint. There is no advice selecting an unknown best hybrid. Run the same reserved-pivot rule at  $q_*$  for every slot, before projection. Its parameter-dependent failure  $f(\theta) \leq 2^{-\lambda}$  gives signed source gap  $\mathbb{E}_\theta[(1 - f(\theta))\Delta(\theta)]/(2\bar{H})$ . Since  $|\Delta(\theta)| \leq 1$ , multiplying back loses at most  $2^{-\lambda}$ , even for randomly generated moduli.

Finally implement the local simulator with bounded samplers. Its two endpoint deviations change the source gap by at most  $2\eta_{\text{sim}}$ ; the two honest endpoints cost at most  $2\eta_{\text{hon}}$ . This proves (23). Polynomial overhead includes other keys, payloads, encryptions, evaluation and serialization. The Ring-LWE assumption and polynomial  $\bar{H}$  make the whole bound negligible.

*Correctness and nonvacuous admission.* Security above does not depend on good noise. For correct evaluation let  $P(\lambda)$  bound all honestly sampled polynomials. The same root-lattice moment-generating-function argument as Lemma 13 gives

$$\Pr[|E_{i,v}| > B] \leq 2e^{-\pi B^2/(2\lambda)}.$$

With  $B_\lambda = \lambda$ , all primitive coefficients are bounded by  $B_\lambda$  except with probability at most  $2P(\lambda)N_\lambda e^{-\pi\lambda/2} + \eta_{\text{hon}}$ , which is negligible. This event includes secrets and ephemerals, not just errors. Use  $\kappa = 511L_\lambda$  and the deterministic recurrences of Section 4, with fresh bound and per-bank cost

$$F_\lambda = (2\kappa B_\lambda + 1)B_\lambda, \quad \Delta_q = \kappa g_q D_g B_\lambda.$$

At a prime drop  $q_{\text{old}} = p_* q_{\text{new}}$  replace the secret-one bound by

$$B \mapsto \left\lfloor \frac{2B - 1 + (\kappa B_\lambda + 2)p_*}{2p_*} \right\rfloor.$$

The other recurrences are unchanged. Require the public generator to admit every complete state by  $q > 2 + 4B$  before execution. The deterministic proof of Theorem 2 then gives correctness with negligible failure for every permitted input and evaluated circuit.

These conditions admit actual polynomial-time families. For example take two encryptions under one source key and one product switched using two binary

power banks to an independent destination key, with one prime  $\lambda^{12} < q < 2\lambda^{12}$ . Such a prime can be selected in time polynomial in  $\lambda$  by trial search and primality testing. With  $g_q = \lceil \log_2 q \rceil$ , a complete output phase has bound

$$C = \kappa(1 + 2F_\lambda)^2 + 4g_q\kappa\lambda.$$

For  $\lambda \geq 256$ ,  $\kappa \leq 1022\lambda$ ,  $F_\lambda < 2045\lambda^3$  and  $g_q \leq 13\lambda$  give  $4+2C < 2^{35}\lambda^7 < \lambda^{12} < q$ ; thus the required parity/no-wrap admission holds. The prime also satisfies  $q \geq 2N_\lambda$ . This is a nonvacuity example, not a performance-optimized choice or a proof of its Ring-LWE hardness. More generally the theorem covers any efficiently generated, polynomial-size admitted manifest family; it does not itself provide a modulus/depth tradeoff for every requested circuit. The sharper finite G16 admission above is separate and need not use the coarser  $B_\lambda$  substitution.

*What has, and has not, been proved.* The conclusion is conditional IND-CPA security under the precisely stated uniform-secret discrete dual Ring-LWE family, together with efficient sampling and negligible-failure correctness for admitted families. It is not an unconditional hardness theorem, a concrete bit-security estimate, circular/KDM security, or an efficient reusable bootstrap. Ideal independent coins are part of the algorithmic model; replacing them with a CSPRNG requires its separate computational assumption and implementation accounting. The Gaussian theorem and finite bridge use different distributions from the measured CBD20 implementation.

**C.3.7 A jointly admitted and costed Gaussian composition family** To include sampling in the complete-work bound, we give an explicitly costed reference realization of the ideal law  $\mu_\lambda$ . Throughout this subsection  $d = \lceil \log_2 \lambda \rceil = \log_2 L$ . Bit costs use a random-access model charging the operand bits read, written and operated on, including table entries and addresses; arbitrary-precision operations are not unit-cost. These are asymptotic algorithms, not measured backends.

**Lemma 16 (Shared tables with negligible law error).** *There is a bounded ideal-bit reference sampler with public preprocessing  $O(\sqrt{\lambda}d^7)$  bit operations and  $O(\sqrt{\lambda}d^3)$  shared table bits. Each subsequent polynomial draw costs  $O(Nd^3)$  bit operations and  $O(Nd)$  output/scratch bits, excluding the shared tables. Put  $W = \lceil \sqrt{\lambda}d \rceil + 1$  and  $J = 2W + 1$ . Its per-polynomial statistical distance from  $\mu_\lambda$  is at most*

$$\rho_\lambda = N(16 \cdot 2^{-2\lambda} + 2^{-4d^2} + 5J \cdot 2^{-8d^2}). \quad (24)$$

*This remains negligible jointly over polynomially many draws. No table depends on a secret or plaintext.*

*Proof.* Use the reverse root-basis sampler from Lemma 14. Every scalar fractional center is  $h/(j+1)$ , with  $1 \leq j \leq 256$  and  $0 \leq h \leq j$ . Thus  $\sum_{j=1}^{256} (j+1) =$

33152 tables suffice, independently of  $\lambda$ . Put  $W = \lceil \sqrt{\lambda} d \rceil + 1$  and  $p = 8d^2$ . For offsets  $w = -W, \dots, W$ , compute nonnegative integer weights approximating

$$2^p \exp\left(-\pi \frac{((j+1)w - h)^2}{\lambda j(j+1)}\right)$$

downwards with deficit less than two. The exponent and precision are  $O(d^2)$ . Range reduction to an argument in  $[0, 1]$ , a length- $O(p)$  outward-rounded exponential series, and  $O(\log p)$  squarings suffice. With  $O(p)$ -bit registers and school-book arithmetic the work per weight is  $O(p^3)$ . Extra  $O(\log p)$  guard bits absorb accumulated rounding; the interval construction of Lemma 12 provides such enclosures. There are  $O(\sqrt{\lambda} d)$  entries per table.

For integer weights  $v_1, \dots, v_K$ , store  $T_i = \lfloor 2^p (\sum_{j \leq i} v_j) / (\sum_j v_j) \rfloor$ , so  $T_K = 2^p$ . For a uniform  $p$ -bit integer  $U$ , select the first  $T_i > U$  by binary search. This produces a draw in *bounded*  $O(pd) = O(d^3)$  bit work, without rejection; its additional variation error is at most  $K2^{-p}$ . Storing each threshold requires  $O(p)$  bits. Computing the normalized thresholds takes  $O(p^2)$  per entry, within the preprocessing bound. Zero-width intervals are simply skipped. The integer part of the center is added after the table lookup; over 256 steps it stays  $O(W)$ , so coefficient bit lengths are  $O(d)$ .

We make the law error explicit. Every scalar squared width lies in  $[\lambda/2, \lambda]$ . Poisson summation gives normalizer fluctuation

$$\zeta \leq \frac{2e^{-\pi\lambda/2}}{1 - e^{-3\pi\lambda/2}} \leq 4 \cdot 2^{-2\lambda}.$$

For  $\lambda \geq 256$ ,  $N \leq 512\lambda$  implies  $N\zeta < 1/8$ . The sequential likelihood bound used above is therefore at most  $4N\zeta \leq 16N2^{-2\lambda}$ . For any fractional center, the omitted unnormalized mass is at most  $2(1 + \lambda/(2\pi W))e^{-\pi W^2/\lambda}$ , by a monotone tail integral. The full scalar mass is at least  $\sigma/2 > \sqrt{\lambda}/3$ . Thus the omitted probability is bounded by

$$(6/\sqrt{\lambda} + \sqrt{\lambda}/W)e^{-3d^2} \leq \frac{1}{2}e^{-3d^2} \leq 2^{-4d^2}.$$

Here  $d \geq 8$ ,  $\sqrt{\lambda} \geq 16$  and  $4 \log 2 < 3$ . A nearest retained offset has weight greater than  $1/2$ . Removing less than  $2J2^{-p}$  total mass and renormalizing therefore costs at most  $4J2^{-p}$  in variation; the rounded integer total is positive since  $J \leq 2^{d+2}$  and  $p = 8d^2$ . Cumulative rounding adds at most  $J2^{-p}$ . These bounds are uniform in all earlier scalar draws, so sequential coupling proves (24). A catalogue of  $G$  polynomial draws costs at most  $G\rho_\lambda$ ; simulation catalogues must use their own bounds, including adaptive query caps. The sampler changes neither the target canonical law nor the source assumption. Secret-dependent table accesses are not claimed to be side-channel resistant or a constant-time production algorithm.

**Proposition 8 (Gaussian prepared-workflow family).** *Fix the radix and a public polynomial-time computable choice  $1 \leq k = k(\lambda) \leq d$ , and put  $\tau = d - k$ ,  $M = 2^k - 1$ ,  $I = 2M + 1 + \tau$ . Apply Corollary 5 with*

$$F = (2\kappa\lambda + 1)\lambda, \quad \Delta_0 = \kappa D_g \lambda, \quad \kappa = 511L.$$

Its ordinary-prime choice has

$$b_q = O(k + (\tau + 1) \log \lambda), \quad H = g(2^{\tau+1} + 3\tau)$$

hint rows. There is a bounded polynomial-time public prime generator for this choice. Under the Ring-LWE assumption for that generator, the resulting family for prepared composition is IND-CPA secure and correct except with negligible probability. One full batch, including sampling tables, key generation, independent-owner preparation and recovery, has bit work

$$\begin{aligned} & \Pi + O((M + \tau + H)N^2b_q^2 + IC + h((M - k)L^2 + L(M + d))\mathcal{F}_K \\ & + \sqrt{\lambda}d^7 + (H + I + \tau + 1)Nd^3 + (H + \tau + 1)Nb_qd^2). \end{aligned}$$

Here  $h = 16$ ,  $K = \mathbb{F}_{2^{16}}$ , and  $\mathcal{C}, \mathcal{F}_K$  have the meaning of Proposition 19. Transmission and its schedule are separate charges;  $\Pi$  bounds prime generation as specified below.

*Proof.* The deterministic admission proof uses coefficient bounds, not the old ternary/CBD law. The Gaussian catalogue has coefficient cap  $\lambda$  except with negligible probability. Its substituted no-drop quantities satisfy

$$F = \Theta(\lambda^3), \quad a = \Theta(\lambda^4), \quad H_0 = \Theta(2^k \lambda^7), \quad \Delta_0 = \Theta(\lambda^2),$$

where  $H_0$  denotes the corollary's prefix error term, not a row count. Substitution in  $\mathcal{U}, \mathcal{V}$  proves the bit-length bound. All fresh, raw, transported and aligned complete states are admitted.

Let  $b$  be the synthesized interval exponent, so  $2^b < q < 2^{b+1}$ . It is at least 12 and its interval lies above  $2N$  in this family. The prime-count bounds of [17, Theorem 6.9] imply

$$\pi(2^{b+1}) - \pi(2^b) \geq 2^b/(2b) \quad (b \geq 12).$$

Draw at most  $\lceil 8bd^2 \rceil$  independent candidates  $2^b + U$ , where  $U$  is a uniform  $b$ -bit integer, and test primality deterministically. The composite left endpoint is rejected. Exhaustion has probability at most  $e^{-4d^2}$  and is an explicit public failure. If  $A(b)$  bounds one test, the bit cost is

$$\Pi = O(bd^2(A(b) + b)), \quad A(b) = \text{poly}(b)$$

by deterministic polynomial-time primality testing [1]. The distribution of the resulting public parameters is part of the Ring-LWE assumption; primality is not a hardness guarantee. Searching all integers in this interval is not the claimed PPT generator. No complete splitting, special transform roots or NTT prime density is used.

The algebraic operation counts in Proposition 19 are unchanged. Exactly  $H + 3\tau + 3 + 3I$  Gaussian polynomial draws cover  $2\tau + 2$  vertex secrets,  $I$  ephemerals and  $H + \tau + 1 + 2I$  errors. There are  $H + \tau + 1$  uniform-mask polynomials. Binary rejection capped at  $4d^2$  attempts per uniform coordinate

costs  $O(b_q d^2)$  bit operations and has negligible total failure. Lemma 16 prices the Gaussian calls. Their whole-catalogue coupling is included in the security and correctness errors, not conditioned away. The manifest and all counts are polynomial in  $\lambda$ , so Theorem 10 applies.

*Explicit one-batch failure budget.* Write  $G = H + 3\tau + 3 + 3I$  and  $U = H + \tau + 1$  for the Gaussian and uniform polynomial counts. Each uniform-coordinate rejection probability is below  $1/2$ , giving exhaustion at most  $2^{-4d^2}$ ; the prime-exhaustion bound is also at most  $2^{-4d^2}$ . For the ideal Gaussian catalogue the physical coefficient bound gives at most  $2GN e^{-\pi\lambda/2}$  probability of exceeding the primitive cap  $\lambda$ . Combining admission, this tail and the table coupling, the probability of public generation failure, source exhaustion or incorrect output in one declared full batch is at most

$$(UN + 1)2^{-4d^2} + G\rho_\lambda + 2GN2^{-2\lambda}. \quad (25)$$

This is a sampling/correctness bound; it excludes the source distinguishing advantage and does not condition the privacy game on successful admission.

The alternative  $d^2$  mask and  $\lceil 2bd^2 \rceil$  prime limits instead give  $(UN + 1)2^{-d^2}$  as the first term. At  $\lambda = 256$ ,  $k = 8$  and radix  $2^{48}$ , the same admission yields  $b_q = 195$ ,  $H = 12$ ,  $I = 511$ ,  $G = 1548$  and  $U = 13$ . The smaller-cutoff sufficient total budget is approximately  $2^{-44.300}$ , whereas (25) is strictly below  $2^{-229}$ . These are calculated upper bounds, not observed failure rates or security bits. On a shared random tape, the two samplers agree whenever the smaller cutoffs suffice; the larger cutoffs extend only the corresponding abort paths, with the larger worst-case constants charged above. The reference stores  $33152(2W + 1)(8d^2 + 1)$  threshold bits, or 550600848 raw bytes at this point. Full tables and a Gaussian HE execution have not been implemented here. The measured ternary/CBD20 profile uses a different source and is unchanged.

For  $k = \min\{d, \lceil d/2 + \log_2(d + 1) \rceil\}$ ,  $M + \tau + H = O(\sqrt{L} \log L)$  and  $b_q = O(\log^2 L)$ . The complete arithmetic and sampling bounds consequently give

$$C_{\text{batch}} = \Pi + O(N^2 \sqrt{L} \log^5 L), \quad S_{\text{retained}} = O(N \sqrt{L} \log^3 L).$$

The latter includes all prepared inputs, public/evaluation keys, retained plaintext variants and sampler tables. Retaining the authority's Gaussian vertex secrets costs  $O(N(2\tau + 2) \log \lambda)$  bits, rather than ternary storage, and is also absorbed. Streaming arithmetic adds  $O(Nb_q)$  state beyond these retained objects; this is not a measured RSS bound. The public table constants are large and are not a practical efficiency claim. This particular choice need not minimize a finite instance; all  $d$  values of  $k$  can instead be priced with the explicit ledger. Neither result is an optimum over other compilers or modulus chains.

## D Source-law boundaries and security assessment

### D.1 A quantitative ternary normal-form bridge

This supporting reduction simplifies the mixed-law source requirement; it does not establish its hardness. Use the ring  $R_q$ , modulus  $q = q_3$  and physical tensor

basis of Appendix D.2, with  $N = 65536$ . Write  $U$  for a scalar uniform trit and  $B_k$  for the difference of two independent  $\text{Binomial}(k, 1/2)$  variables. Let  $\epsilon_{T,m}(T)$  denote the maximum gap at time  $T$  for the  $m$ -row source with secret law  $U^N$  and error law  $B_{20}^N$ . Define  $\epsilon_{H,m+1}(T)$  for  $m+1$  rows on the same ring and modulus, with a uniform secret in  $R_q$  and error law  $U^N$ . Both null experiments have independent uniform bodies and masks. All small integer vectors are shared across RNS reductions. The homogeneous premise has narrower errors; it is not identified with a canonical Gaussian or power-of-two reference instance.

**Lemma 17 (A ternary summand in CBD noise).** *For every integer  $k \geq 1$  there is an integer distribution  $V_k$ , supported on  $[-(k-1), k-1]$ , independent of  $U$ , such that*

$$B_k = (1 - 4^{-k}) \text{Law}(U + V_k) + 4^{-k} \delta_0.$$

*The coefficients of the nonnegative integer polynomial*

$$Q_k(x) = \frac{(1+x)^{2k} - x^k}{1+x+x^2}$$

*sum to  $D_k = (4^k - 1)/3$  and, divided by  $D_k$  with exponent offset  $k-1$ , give the distribution  $V_k$ .*

*Proof.* One difference of two fair bits has law  $(3/4)U + (1/4)\delta_0$ . In the sum of  $k$  such independent mixtures, all selectors choose zero with probability  $4^{-k}$ . Otherwise reserve the first selected trit. It remains uniform and independent of the selectors and remaining trits; their sum defines  $V_k$ . For the table, the Laurent identity

$$x^{-(k-1)} Q_k(x) = \sum_{j=1}^k \binom{k}{j} (x^{-1} + 1 + x)^{j-1}$$

proves integrality and nonnegativity. Multiplying by  $x^{-1} + 1 + x$  verifies the displayed quotient and mixture.

Put  $\varepsilon = 2^{-40}$ ,  $P = B_{20}$ ,  $Q = \text{Law}(U + V_{20})$  and  $p_0 = \binom{40}{20}/2^{40}$ . Then

$$Q(z) = \frac{P(z) - \varepsilon \mathbf{1}_{z=0}}{1 - \varepsilon}, \quad v := \chi^2(Q\|P) = \frac{\varepsilon^2(1 - p_0)}{p_0(1 - \varepsilon)^2}.$$

For  $M = mN$  independent error coefficients, the likelihood-ratio second moment factors. Cauchy-Schwarz and a geometric-series bound give, when  $Mv < 1$ ,

$$\text{TV}(Q^M, P^M) \leq \frac{1}{2} \sqrt{(1+v)^M - 1} \leq \Delta_m := \frac{1}{2} \sqrt{\frac{Mv}{1 - Mv}}. \quad (26)$$

Indeed  $\binom{M}{j} \leq M^j$ . Reduction modulo  $q > 40$  is injective on each scalar support. Adding the identically distributed secret and masks and then applying the source map cannot increase distance. This argument uses the actual structured multiplication matrices.

There is also a bounded exact sampler conditional on nonabort. For each padding coefficient, draw a uniform 40-bit integer, rejecting only  $2^{40} - 1$ , with at most four attempts. On acceptance reduce it modulo  $D_{20} = 366503875925$ ; every residue has three accepted preimages. A lookup in the 39-entry cumulative table of  $Q_{20}$  then samples  $V_{20}$  exactly. Abort the whole reduction on any failed coefficient. The global success probability is

$$w_m = (1 - 2^{-160})^{mN}, \quad 1 - w_m \leq mN 2^{-160} < 2^{-138} \quad (m \in \{37, 46\}).$$

Success is independent of the accepted values and of the input branch, so conditional global success preserves the independent padding law. At most  $160mN$  random bits and six table comparisons per accepted coefficient are charged. This is a reference reduction sampler, not a change to honest JetHE sampling.

**Proposition 9 (Full-view ternary bridge).** *Let  $u = \Pr[a \in R_q^\times]$  for a uniform mask. For  $m = 37$  or  $46$ ,*

$$\epsilon_{T,m}(T) \leq \frac{\epsilon_{H,m+1}(T + C_m)}{uw_m} + \Delta_m, \quad u > 1 - 2^{-42}, \quad \Delta_m < 2^{-28}. \quad (27)$$

*The simulation cost  $C_m$  charges one full-ring unit test/inversion,  $2m$  ring multiplications,  $m$  ring additions,  $mN$  coefficient additions and the preceding bounded padding sampler.*

*Proof.* Reserve one source row  $(a_0, c_0)$  as pivot. If its mask is a nonunit, output a fixed bit. Otherwise, for the disjoint response rows set

$$A_i = -a_0^{-1}a_i, \quad C_i = c_i + A_i c_0 + V_i,$$

where the  $V_i$  have independent  $V_{20}$  coefficients. On padding abort output the same fixed bit. Discard the pivot before applying the target distinguisher to the  $m$  transformed pairs.

In the real branch  $c_i = a_i s_0 + e_i$ , and cancellation yields  $C_i = A_i e_0 + (e_i + V_i)$ . Pivot selection depends only on its mask. Thus  $e_0$  has exactly law  $U^N$ , the response masks are independent uniform elements, and the response errors independently have law  $Q^N$ . All these objects have the required independence. In the null branch, condition on the pivot, masks and padding. Each response body is still independent uniform; the displayed addition is a translation. The resulting null view is exactly the target null view, jointly with its masks.

The three primes are coprime to the conductor. Each reduction is a product of at most  $N$  finite fields, so  $1 - u \leq N \sum_{p|q} p^{-1} < 2^{-42}$ ; complete splitting is unnecessary. Conditional nonabort, only the real view differs from the target, by at most  $\Delta_m$ . The two abort events have the same probabilities in both branches and produce the same bit. Consequently the actual signed gap is  $uw_m$  times the conditional signed gap. The triangle inequality proves (27) with the stated costs.

Exact rational checks give  $\Delta_{37} < 1871/10^{12}$  and  $\Delta_{46} < 2086/10^{12}$ . The 102-row ideal two-batch scheme reduction has factor 158; a scheme gap  $1/4$  therefore

gives a mixed-source gap  $1/632$ . The new reference distinguisher retains gap strictly above  $1/633$ , since

$$(1 - 2^{-42})(1 - 2^{-138})(1/632 - 2^{-28}) > 1/633.$$

It needs 38 or 47 full-ring source rows and the charged additional time. Neither this comparison nor the small statistical bound is a numerical security certificate. The reference error variance is  $2/3$ , whereas CBD20 has variance 10, and the reference retains the original ring. Its hardness still needs assessment. The additive  $\Delta_m$  bound cannot establish the separately stronger  $\epsilon_{T,m} \leq 2^{-137}$  target.

Pivot cancellation is the standard normal-form mechanism of [32]; the mixture and explicit losses specialize it to the specified finite laws. We claim no new general normal-form theorem. Exact consistency checks compare 32 scalar families by two integer expansions, 5460 small sampler draws, 78 production table endpoints, and 810000 cancellation identities in a ring with zero divisors. Enumeration also covers 455625 real and 140625 null toy assignments and compares their complete public laws. These checks support the argument and its quantitative constants; they are not independent proof review, HE execution or source qualification.

## D.2 Quantitative limits of two source decompositions

We make two elementary source calculations explicit to delimit the remaining concrete-security problem. Neither is a numerical hardness certificate or a new attack. Write

$$S_q = (\mathbb{Z}/q\mathbb{Z})[t]/(t^{256} + 1), \quad R_q = S_q[b]/(\Phi_{257}(b)), \quad q = q_3.$$

The actual basis of  $R_q$  over  $S_q$  is  $b^1, \dots, b^{256}$ . Each coefficient of the secret is an independent uniform trit and each error coefficient is independent CBD20; one integer vector is reduced across all RNS primes. The normalized source is  $(A_i, C_i = A_i s + e_i)$  versus independent uniform pairs, with  $m$  rows. For the uniform-width and stage-width schemes it suffices to take  $m = 46$  and  $m = 37$ , respectively, with their stated scheme reductions.

### D.2.1 An exact transfer with an insufficient smaller premise

**Proposition 10 (Exact block embedding).** *An  $m$ -row  $R_q$  source distinguisher with gap  $\epsilon$  and time  $T$  gives a  $256m$ -row  $S_q$  source distinguisher with the same gap and time  $T + C_{\text{sim}}$ . The smaller source has dimension 256, one trit secret and CBD20 errors in its negacyclic basis.*

*Proof.* Put  $k = 256$  and fix  $1 \leq \ell \leq k$ . Given smaller-source pairs  $(a_{ij}, c_{ij})$  for  $1 \leq i \leq m$ ,  $1 \leq j \leq k$ , sample independent trit blocks  $\tau_j$  for  $j \neq \ell$  and set

$$s_{\text{rest}} = \sum_{j \neq \ell} \tau_j b^j, \quad A_i = b^{-\ell} \sum_j a_{ij} b^j, \quad C_i = \sum_j c_{ij} b^j + A_i s_{\text{rest}}.$$

The element  $b$  is a unit since  $b^{257} = 1$ . Assembly followed by multiplication by  $b^{-\ell}$  is therefore a bijection, making the  $A_i$  independent uniform  $R_q$  elements. In the real game  $c_{ij} = a_{ij}\sigma + e_{ij}$  for one unknown trit block  $\sigma$ , so

$$C_i = A_i(b^\ell \sigma + s_{\text{rest}}) + \sum_j e_{ij} b^j.$$

Both secret and error have exactly the target coefficient laws in its actual basis. In the null game, assembling the independent uniform  $c_{ij}$  gives independent uniform  $R_q$  bodies, and the translation by  $A_i s_{\text{rest}}$  preserves that law conditioned on all masks and sampled blocks. Apply the target distinguisher. The simulation samples 255 blocks, assembles  $mk$  pairs and performs  $m$  public ring products and unit multiplications. A direct implementation uses  $O(mk^2)$  operations over  $S_q$  plus assembly. No irreducibility modulo  $q$  or change of error law is required.

The direction is  $\epsilon_R(T; m) \leq \epsilon_S(T + C_{\text{sim}}; 256m)$ . It does not extract small-ring samples from an arbitrary JetHE instance. In particular, an attack on the smaller sufficient premise does not become an attack on JetHE through this reduction.

For  $m = 46$ , the smaller premise has 11776 samples. Retaining the constant coefficient of each smaller-source body gives exactly 11776 independent scalar LWE equations in dimension 256: the mask row is  $(a_0, -a_{255}, \dots, -a_1)$ , a signed permutation of uniform coefficients. This projection justifies an independent-equation model for that smaller premise. Nine evaluated parameter and cost-model combinations give finite predicted costs throughout; classical MATZOV uSVP gives  $\log_2 \text{rop} = 40.9091$  with block size 40. These are model outputs, not recovered secrets or work lower bounds. Core-only ADPS estimates omit substantial overhead at this block size and are not complete attack costs. This premise does not supply the intended 128-bit assurance. The  $m = 37$  embedding is also exact, but its 9472-sample premise was not separately modeled here.

At dimension 65536, eighteen parameter choices use surrogates with independent equations for one or 46 ring rows and the same trit/CBD20 laws. The bounded optimizer finds no finite point; its maximum searched block size is 1754. Repeating the primal estimation with a block-size cap of 10000 gives finite predictions. All nine points, at one, 37 and 46 row budgets and three cost models, are finite with block size 6070 and lattice dimension 129031. Classical MATZOV gives  $\log_2 \text{rop} = 1751.6483$ ; ADPS classical and quantum core-only extrapolations give 1772.44 and 1608.55. These are independent-equation/GSA model outputs, not security bits, executed attacks or lower bounds on work. The finite search ceiling explains the nonfinite optimizer outcome; the larger-cap predictions apply only to the named primal model. The block embedding and these numerical estimates are supporting diagnostics. They do not replace the exact source assumption by a standard Module-LWE assumption: the multiplication matrices of uniform  $R_q$  masks remain structured, not uniform module matrices.

**D.2.2 One restricted route to fixed subalgebras** Restricted-form masks play a role in subring reductions such as Ring-BKW [39]. The actual error law

and residue structure must also be retained [13]. The next calculation treats only linear combinations of the original masks with coefficients in  $S_q$  and fixed subalgebras of the  $b$  extension.

**Proposition 11 (Fixed-subalgebra mask restriction).** *For  $m$  independent uniform masks  $A_i \in R_q$ , with  $m = 46$  or  $37$ , except with probability less than  $2^{-4883}$  or  $2^{-5414}$ , respectively, the following holds simultaneously for all eight nontrivial subgroups  $H$  of the cyclic group  $b \mapsto b^a$ ,  $a \in (\mathbb{Z}/257\mathbb{Z})^*$ : if  $\sum_i \lambda_i A_i \in R_q^H$  with  $\lambda_i \in S_q$ , then every  $\lambda_i$  is zero. The coefficients may depend on all masks.*

*Proof.* The group permutes the basis  $b^1, \dots, b^{256}$  regularly. Its subgroups have orders  $2, 4, \dots, 256$ , so the invariant coefficient vectors are constant on orbits and have  $S_q$  rank  $d_H = 256/|H|$ . Differences from one representative in each orbit give a surjective map  $P_H : R_q \rightarrow S_q^{c_H}$  with kernel  $R_q^H$  and  $c_H = 256 - d_H \geq 128$ . Consequently the matrix with columns  $P_H(A_i)$  has independent uniform  $S_q$  entries.

At the three actual primes,  $t^{256} + 1$  has 256 distinct linear factors. Thus  $S_q$  is a product of 768 fields. In any component  $\mathbb{F}_p$ , a uniform  $c_H$ -by- $m$  matrix fails full column rank with probability at most

$$\sum_{j=0}^{m-1} p^{j-c_H} < 2p^{-(c_H-m+1)}.$$

Every  $p > 2^{59}$ . Union over  $8 \cdot 768 < 2^{13}$  pairs gives  $2^{14-59 \cdot 83} = 2^{-4883}$  for  $m = 46$  and  $2^{14-59 \cdot 92} = 2^{-5414}$  for  $m = 37$ , with strict inequalities. If all component matrices are injective, their product-ring map is injective. Hence  $P_H(\sum_i \lambda_i A_i) = 0$  forces  $\lambda = 0$  simultaneously for every  $H$ .

These exponents concern a mask-rank event, identical in the real and null source games. They are not security bits. The fixed subalgebras may split modulo  $q$ ; no field property was assumed. Arbitrary  $R_q$  coefficients, mask-dependent cosets, multiple automorphic copies, additional rows and more general transformations are outside the proposition. For example,  $A_2 A_1 - A_1 A_2 = 0$  is a full-ring cancellation, but applying it to bodies leaves  $A_2 e_1 - A_1 e_2$ , whose error geometry still needs an argument. Derived rows are not fresh independent samples. The proposition does not exclude all Ring-BKW, trace, lattice or hybrid attacks and does not assess the other attack classes.

The exact block-transfer checks cover 14534 unit images and as many null translations, together with independent secret and scalar-law identities. Exhaustive enumeration covers 1444 small mask-restriction matrices, and exact arithmetic verifies the production union inequalities. These finite checks support reproducibility of the stated calculations; they are not independent cryptographic review.

### D.3 Boundaries of the measured source assessment

The timed source and the homogeneous reference of Proposition 9 remain computational assumptions on the same ring. This subsection records failed direct transfer conditions and assesses an explicit nonunit-filter mechanism. The results concern the ideal source laws; they neither qualify the implementation's random source nor establish a numerical security level. Use  $S = \mathbb{Z}[t]/(t^{256} + 1)$ ,  $R = S[b]/\Phi_{257}(b)$ ,  $N = 65536$  and the three-prime modulus  $q = q_3$ . Source bodies are  $C_i = A_i s + E_i$ , with independent uniform masks and independent tensor-coordinate CBD20 errors. Null bodies are independent uniform  $R_q$  elements. All original masks are public.

**D.3.1 Limits of direct short-distribution transfer** The bounded-secret reductions of [10, Theorems 3.1–3.2] retain Gaussian errors and require rank growth. Even deleting their positive asymptotic slack and taking source rank one leaves  $3^d \geq p$  for search and  $3^d \geq p^2$  for decision at a prime  $p$ . At each of the three primes their least integer ranks are 38 and 76; the actual rank over  $R_q$  is one. The formal product-modulus floors 114 and 228 do not meet the cited prime-modulus hypothesis. The coefficient dimension is not the module rank: multiplication over  $S_q$  has the structured matrix law described in Appendix D.2.

There is an additional splitting condition in the cited decision theorem. Writing  $\nu = 512 \cdot 257$ , it requires  $\text{ord}_\nu(p) = \nu/\mu$  and  $p > (\eta s_1(\mu))^{\varphi(\mu)}$ , where  $s_1$  is the largest singular value of the cyclotomic Vandermonde matrix. Complete splitting forces  $\mu = \nu$ . That matrix has  $N^2$  entries of modulus one, so  $s_1(\nu) \geq \sqrt{N}$ . Already at  $\eta = 1$  the threshold exceeds or equals  $N^{N/2} = 2^{524288}$ , whereas the three primes are below  $2^{60}$ . The analogous degree-256 threshold is  $2^{1024}$ . This excludes a direct instantiation, not all modulus conversions or other reductions.

**Proposition 12 (Rank-one endpoint of the cited small-error bound).** *In the sufficient bound of [10, Theorem 4.1 and Lemma 4.3], target module rank one gives a bound greater than one whenever  $nm \geq 16$ . Thus this bound does not assess either the 38-row or 47-row homogeneous reference here, even granting its other distribution and modulus hypotheses.*

*Proof.* The cited parameters satisfy  $m > d \geq k \geq 1$  and  $\ell = m - d + k$ . At  $d = 1$  one has  $k = 1$  and  $\ell = m$ : the intermediate map  $[I_\ell \mid Y]$  is the identity and its Gaussian-column hybrid has no steps. With  $h = nm$ , the displayed error term is at least

$$\varepsilon_3 \geq \frac{(\eta\sqrt{2\pi e})^h}{|X|\sqrt{\pi h}}, \quad |X| \leq (2\eta + 1)^h, \quad \eta \geq 1.$$

Using  $\eta/(2\eta + 1) \geq 1/3$ ,  $\pi > 3$ ,  $e > 2$  and  $\pi < 4$  gives  $\varepsilon_3^2 > (4/3)^h/(4h)$ . The latter is greater than one at  $h = 16$  by integer comparison, and its consecutive ratio  $4h/(3(h + 1))$  exceeds one for  $h > 3$ . The remaining nonnegative losses cannot repair this sufficient bound.

The small-error result cited above is a search statement; a decisional source obligation also needs a valid search-to-decision step. None of these observations is an attack, an impossibility theorem for other transfers, or a change to the measured error distribution.

**D.3.2 Explicit short nonunits and full filtered bodies** Table 11 supplies short nonunits at the actual primes, indexed as in Appendix B.3. For the hexadecimal masks  $u_j, v_j$  in the table define

$$f_j(t) = \sum_{i=0}^{63} (\text{bit}_i(u_j) - \text{bit}_i(v_j)) t^i.$$

Their supports are disjoint within each row. Direct modular evaluation gives exactly one zero among the 768 subring CRT positions: the displayed  $\alpha_j$  at  $p_j$ . All other listed positions are nonzero. The polynomials have degrees 35, 55, 61 and weights 22, 29, 25. They are nonzero in  $S$  and have coefficients in  $\{-1, 0, 1\}$ . Their existence is unsurprising: the  $2^{60}$  binary polynomials of degree below 60 cannot map injectively into any current  $\mathbb{F}_{p_j}$ . The explicit certificates avoid relying on this existence argument for the subsequent identities.

Let  $g_j \in S_q$  be the idempotent for this one CRT position. Explicitly, reduce

$$\frac{(t^{256} + 1)/(t - \alpha_j)}{256\alpha_j^{255}} \quad \text{over } \mathbb{F}_{p_j},$$

then lift its coefficients by the scalar CRT idempotent that is one at  $p_j$  and zero at the other primes. Thus  $g_j^2 = g_j \neq 0$  and  $f_j g_j = 0$ . In  $R_q$ , the image  $g_j R_q$  has size  $p_j^{256}$ . Multiplying a row by  $f_j$  makes this component zero in both the real and null experiments; the zero alone gives no distinguishing test.

**Proposition 13 (Full-filter recovery and error support).** *The joint view  $(f_0 C, f_1 C)$  determines every  $C \in R_q$ . It preserves the full source’s statistical distance from the corresponding null image. Each individual map  $E \mapsto f_j E$  is injective on the actual CBD20 error support, preserving its full-vector entropy, collision probability and maximal point mass.*

*Proof.* The element  $f_0 + g_0$  is a unit: it is one at the zero of  $f_0$  and equals  $f_0$  elsewhere. Put

$$U = (f_0 + g_0)^{-1}, \quad h_0 = U(1 - g_0), \quad h_1 = f_1(\alpha_0)^{-1} g_0,$$

where the last scalar inverse is at  $p_0$ ; its lift is immaterial outside the support of  $g_0$ . Componentwise,

$$h_0 f_0 + h_1 f_1 = 1, \quad C = h_0(f_0 C) + h_1(f_1 C). \quad (28)$$

Both directions are deterministic and exact, proving the distance claim also jointly with all original masks and across all source rows. Recovery costs two

full-ring multiplications and an addition per body, besides public preprocessing; no equality of attack runtimes is asserted.

For the last claim choose a different prime  $p_k$ . The filter is a unit modulo  $p_k$ , so  $f_j E$  determines  $E \bmod p_k$ . Centering recovers every original coefficient in  $[-20, 20]$  because  $p_k > 40$ . Hence the map is injective on that support.

For weight  $w_j$ , each individual coefficient of  $f_j E$  has law  $\text{CBD}_{20w_j}$  and variance  $10w_j$ . The full-vector result shows why these wider marginals cannot be modeled as independent new errors. The filtered rows also are deterministic descendants, not fresh source samples. This leaves open whether exploiting their structure can improve a computational attack.

**D.3.3 The isolated quotient and its observation limit** The complementary view  $g_j C$  is equivalent to substituting  $t = \alpha_j \bmod p_j$  while retaining every  $b$ -coordinate. More generally select one  $t$ -root at each prime of a set  $P$ . This gives a complete degree-256 quotient per prime, rather than a scalar  $(t, b)$  CRT evaluation.

**Proposition 14 (Whole-quotient error bound).** *Let  $M \in \{37, 46\}$ . An observer receives all original masks and at most one distinct quotient query per body, returning all 256  $b$ -coordinates at one chosen  $t$ -root per selected prime. It receives no other body information. The complete view has the strict TV bounds in Table 12 relative to the projected null source. The bounds hold jointly with the secret if desired.*

*Proof.* For one original  $b$ -block, a nonzero character has an active prime subset  $K \subseteq P$  and denominator  $Q = \prod_{p \in K} p$ . Its centered integer frequency vector  $v$  defines  $F(\zeta) = \sum_{i=0}^{255} v_i \zeta^{-i}$  in the power-of-two cyclotomic field. At every active prime it vanishes at 255 roots. The norm argument of Proposition 6 gives

$$Q^{255} \mid \text{Norm}(F), \quad \sum_i (v_i/Q)^2 \geq Q^{-1/128}.$$

The exact CBD20 characteristic function and the cosine bound yield  $|\phi_K| \leq \exp(-20\pi^2 Q^{-1/128})$ . For  $k = 1, 2, 3$ , respectively, the rational numbers  $a_k = 7225/10000, 522/1000, 377/1000$  satisfy  $a_k^{128} 2^{60k} < 1$ . With the certified constants  $\pi > 157/50$  and  $\ln 2 < 347/500$ , this implies  $|\phi_K| < 2^{-b_k}$  for  $(b_1, b_2, b_3) = (205, 148, 107)$ . Plancherel therefore bounds one block's divergence by

$$X_P = \sum_{\emptyset \neq K \subseteq P} \prod_{p \in K} (p-1) 2^{-2b_{|K|}}.$$

This groups active characters without assuming independent RNS errors. Surjectivity makes the actual null quotient product-uniform.

Original  $b$ -blocks and rows are independent. For a schedule fixed from masks before observing bodies, tensorization and  $\binom{L}{j} \leq L^j$  give

$$\text{TV} < \frac{1}{2} \sqrt{\frac{256 M X_P}{1 - 256 M X_P}}.$$

If prime sets vary, use the largest  $X_P$  for the permitted size. Conditioning on masks and the secret turns each mask-secret term into a fixed translation. For adaptive choices, an unqueried body’s error is still independent of the permitted past. The uniformly valid per-query bound is  $\frac{1}{2}\sqrt{256X_P/(1-256X_P)}$ . Sequential coupling charges at most  $M$  such bounds. Exact rational inequalities at the three stated primes give the table; its denominators are positive. Repeated identical queries can be cached, but a second distinct query to an inspected body is outside the experiment.

*Conditional completion fails.* The marginal bound does not simulate the missing quotient after disclosing a full filtered body. Given  $(A, s, f_0C)$ , subtract  $f_0As$  and recover  $E$  through an untouched prime as above. Then  $g_0C$  is determined. Preserving  $(A, s, f_0C)$  from the real experiment but replacing  $g_0C$  by an independent uniform element of  $g_0R_q$  changes the enlarged joint view by exactly

$$1 - p_0^{-256}. \quad (29)$$

Conditional on the preserved entries, this compares a point mass with a uniform alphabet of size  $p_0^{256}$ . This is a revealed-secret diagnostic of the completion step, not a public-view attack. The weaker multi-prime upper bounds likewise are not observed distances or attacks. Full-body computational assessment remains necessary.

*Validation scope.* Exact checks verify the nonunit masks and all subring roots, the full- $q$  identities, error-support inversion, 11360 small frequency norms against independent determinants, fourteen production norm fixtures and the rational bounds. Twelve complete small-ring cases give the corresponding completion distance 16/17. These support the written arguments; they are not independent proof review, a new encrypted run or a numerical security certificate.

#### D.4 Heuristic security assessment

The lattice-estimator revision specified in [29] was evaluated on an independent-equation surrogate with  $37N$  scalar equations. Its relation to JetHE’s structured source is heuristic. The completed uSVP predictions use block size 6070 and dimension 129031; the BDD prediction uses block size 6071 and dimension 129276. They extrapolate named cost models far beyond measured lattice reduction experiments.

The additional classical cases used one thread, 4 GiB virtual memory, 60 CPU seconds, and 75 wall seconds per case. The block-size search ceiling was increased from 1754 to 10000. Both hybrid calculations were incomplete; the ordinary dual calculation encountered numerical overflow. These outcomes provide no missing attack-resistance evidence. Dual cost predictions also require explicit success hypotheses [16].

The structured ring, the exact source distribution, and the acyclic public transcript may admit behavior absent from the surrogate. No complete classical or quantum assessment is claimed. The ideal two-batch reduction maps a scheme gap of  $1/4$  to a source gap of at least  $1/632$ ; subtracting  $\log_2(158)$  from a constant-success work estimate alone does not account for that gap, simulation cost, or preprocessing. OpenFHE’s requested security setting and this conditional JetHE assessment are reported separately.

## E Interfaces, lower bounds and complete work

### E.1 Terminal recovery and the location of nonlinear work

The broad task protects inputs from the evaluator and permits terminal recovery. It does not protect them from the recipient. It therefore admits an encrypted relay: owners encrypt disjoint blocks containing  $f$  and  $g$ ; the evaluator adds them; the recipient decrypts the aggregate and computes  $f(g)$ . This is a correct alternative under unrestricted recipient work, not a confidentiality attack. A single aggregate ciphertext does not exclude it.

We separately define a restricted algorithmic interface for compiled outputs. Its purpose is to distinguish reconstruction of completed output statistics from recomputation on the two private operands. This is not an additional security guarantee, a claim that every natural codec belongs to the interface, or a latency lower bound. All reconstruction work and all returned data remain charged.

**Definition 3 (Linear recovery of admitted statistics).** *Let  $V = \mathcal{J}_L(K)^h$ , viewed as a vector space over  $\mathbb{F}_2$ . Before inputs are chosen, the public program fixes its output shapes and recovery maps. After ordinary correct HE decryption, recovery may use the following statistics and add their contributions in  $V$ :*

1. *binary-vector statistics, decoded by fixed  $\mathbb{F}_2$ -linear maps;*
2. *vectors  $y \in \mathbb{F}_p^n$  for odd prime  $p$ , transformed by a fixed public  $\mathbb{F}_p$ -linear map  $A$ . Every coordinate of  $Ay$  has a unique integer representative in a fixed public interval of width  $B$  with  $2B < p$ . This interval promise holds for every honestly prepared admitted input. Recovery takes these representatives modulo two and applies a fixed  $\mathbb{F}_2$ -linear map to  $V$ ;*
3. *unilateral bypasses, whose contributions depend on only one owner’s input. Each is decoded independently and then added; no nonlinear combination of the two owners’ data is performed by the recipient.*

*Different coordinates may have different intervals, and different blocks may use different primes. All maps, intervals and provenance declarations are input-independent. No restriction on output dimension or linear-map cost is implicit in this definition. Nonlinear lookup, branching on private statistics, or another private-operand composition circuit at the recipient is outside this interface.*

The lifted case is not described as an  $\mathbb{F}_2$ -linear map on arbitrary  $\mathbb{F}_p$  residues. Reduction to parity after a modular wrap need not be additive. The common-interval condition is material.

**Proposition 15 (An affine evaluator cannot implement composition).**

Let  $L \geq 2$  and let  $K$  be a field of characteristic two. Owners may noninteractively prepare arbitrary functions of their respective inputs. Fix input-independent public parameters and a valid key setup. Suppose the evaluator's clean plaintext operations are affine over the carrier fields, and terminal recovery has Definition 3. Such a program cannot correctly return  $f(g) \bmod z^L$  for every  $f \in K[z]_{<L}$  and  $g \in zK[z]_{<L}$ .

*Proof.* An affine evaluator, including any public linear maps in recovery, produces each statistic in the form  $P(f) + Q(g) + c$  over its carrier field. For any  $f_0, f_1, g_0, g_1$ , its alternating four-input difference is zero in that field. Binary-linear recovery preserves this property.

For an odd-prime coordinate, write  $t_{ab}$  for the integer lift at  $(f_a, g_b)$ . All four lifts belong to the same prescribed interval of width  $B$ , so

$$d = t_{11} - t_{10} - t_{01} + t_{00} \equiv 0 \pmod{p}, \quad |d| \leq 2B < p.$$

Consequently  $d = 0$  as an integer. Reducing modulo two and applying a fixed linear map again gives zero mixed difference. Unilateral bypasses have zero mixed difference as well. Their sum therefore satisfies

$$F(f_1, g_1) + F(f_1, g_0) + F(f_0, g_1) + F(f_0, g_0) = 0.$$

Set  $f_0 = g_0 = 0$  and  $f_1 = g_1 = z$ . Three compositions are zero and  $f_1(g_1) = z \neq 0 \pmod{z^L}$ , a contradiction. In multiple lanes, put this witness in one lane and zero in the others.

The proof concerns clean plaintext semantics, not an equality between noisy ciphertext coordinates. Keys and public maps are fixed independently of inputs. Encryption continues to use fresh independent coins in each execution. A common input-independent random tape may be used to couple the four hypothetical executions; it is not a recommendation to reuse coins. Standard correct affine HE operations have the stated clean semantics regardless of those coins. Nonlinear programmable bootstrapping, lookup or plaintext-semantic multiplication is not an affine operation, even when an implementation does not call it a ciphertext product.

*Necessity of the interval condition.* Over  $\mathbb{F}_7$ , an affine program can produce  $4a + 4b$  for  $a, b \in \{0, 1\}$ . Its canonical integer representatives are 0, 4, 4, 1. Their parities are 0, 0, 0, 1, computing  $ab$  without a server multiplication. These four values require width 4, and  $2 \cdot 4 \not\leq 7$ . Thus arbitrary scalar lifting or parity is not silently covered by the proposition.

*Conformance and limits.* The final-only native evaluator adds every bypass before returning one logical ciphertext: two components in the original profile and three in the terminal profile. Its CRT/Taylor plaintext codec is binary-linear. The prepared shrinking control's six terminal ciphertexts use public field-linear

return and bypass maps. Conventional HE with the same ramified plaintext algebra is also eligible if its implementation and correctness admission provide these statistics.

The summed prime-field adapter uses a public inverse 32-point transform, integer coefficients in  $[0, 4096]$  and  $p = 65537$ , followed by parity and binary-field polynomial reduction. Since  $8192 < 65537$ , it satisfies the lifted case. Returning the individual terminal products also satisfies this recovery interface: their evaluations are first added over  $\mathbb{F}_{65537}$ , and the constant coefficient is an outer-owner-only bypass. This policy returns individual summands; its output volume and recipient work are included in the comparison. The encrypted-input relay fails this interface because its recipient runs the nonlinear composition on both recovered operands. It remains eligible for the unrestricted task.

This proposition shows only that some non-affine joint computation is necessary under this interface. It neither lower-bounds its number of products nor proves an asymptotic, latency, security or optimality advantage for JetHE. It admits the existing prepared shrinking and same-algebra constructions. Small exhaustive checks and the modular-wrap counterexample are recorded separately; they do not replace this proof.

**E.1.1 Information determined by the prescribed output** The prescribed output constrains applications. Let  $A = K[z]/(z^L)$ , where  $L \geq 2$ . For fixed nonzero  $g \in zA$  of valuation  $e$ , substitution  $S_g : f \mapsto f(g)$  has rank  $m = \lceil L/e \rceil$  and kernel  $z^m A$ . Indeed,  $1, g, \dots, g^{m-1}$  have distinct leading degrees  $0, e, \dots, (m-1)e$  with nonzero leading coefficients, whereas  $g^m = 0$ . They are linearly independent by their least nonzero degrees. Consequently, knowing  $g$  and  $H = f(g)$  determines exactly the contributing prefix  $f_0, \dots, f_{m-1}$ ; the remaining coefficients are arbitrary. For  $g = 0$ , only  $f_0$  is determined.

If  $g_1 \neq 0$ , then  $m = L$  and the triangular substitution matrix has diagonal  $1, g_1, \dots, g_1^{L-1}$ . Thus known  $g$  and  $H$  determine all of  $f$ . Conversely, suppose  $f$  is known and  $f_1 \neq 0$ . In degree  $i$ , the coefficient  $g_i$  occurs in  $f(g)$  as  $f_1 g_i$ ; any occurrence in  $g^j$  with  $j \geq 2$  has degree at least  $i + j - 1 > i$ . All remaining terms depend on earlier coefficients of  $g$ , so these coefficients are uniquely recovered in increasing order. This argument holds in every characteristic and concerns prescribed plaintext output, not a failure of evaluator confidentiality.

Output alone does not generally determine the pair. Over  $|K| = Q$ , there are  $(Q-1)Q^{L-2}$  choices of  $g$  with  $g_1 \neq 0$ . Each gives exactly one  $f$  for every fixed  $H \in A$ , by invertibility of  $S_g$ . Hence there are exactly this many compatible pairs under that restriction. A recipient who is also an input owner must therefore be distinguished from a third party without either operand. Privacy from the other owner cannot be assumed merely because the server evaluates on ciphertexts.

For the unrestricted task, ordinary public-key encryption of both coefficient vectors followed by forwarding and recipient composition is an eligible upper construction. For  $h$  jobs it encrypts and decrypts  $2hL \log_2 Q$  bits when the known zero coefficients are retained; packing, forwarding and setup are charged separately. Classical composition adds  $O(hM_K(L) \log L)$  field operations, where

$M_K(L)$  is polynomial multiplication cost [28]. This is an arithmetic upper bound, not a new measured relay time or a security-matched comparison. The recorded Horner relay remains a separate execution. Neither the rank argument nor this upper bound proves that useful applications are absent; they require an application to justify its actual recipient constraints.

## E.2 Conventional compiler comparisons

The auxiliary product workload returns  $ab \bmod z^L$ . The shallow arithmetic circuit returns  $ab + c \bmod z^L$ ; the depth-three circuit returns  $\prod_{i=0}^3 (a_i b_i + c_i) \bmod z^L$ , using a balanced product tree. All operands are private full jets. A separate public-polynomial control has public coefficients and is identified explicitly. These workloads test native convolution and terminal reconstruction in addition to the primary private-composition task.

### E.2.1 A Tight Restricted Multiplication Comparison

**Definition 4 (Depth-one bilinear SIMD model).** *Put  $A = K[z]/(z^L)$  and require the product map on  $A^B \times A^B$ , with both batches independently variable in all coefficients. A conventional carrier has  $B$  field slots. Arbitrary public  $K$ -linear preprocessing may mix jobs and coefficients, but separately on each operand. The only nonlinear operations are one parallel layer of slotwise products of transformed left and right vectors, followed by arbitrary public  $K$ -linear postprocessing.*

**Theorem 11 (Capacity-normalized nonlinear work).** *The model requires at least  $2L - 1$  slotwise multiplication calls. If  $|K| \geq 2L - 1$ , this is attainable. At equal field-symbol capacity  $B = hL$ , native JetHE processes the same  $B$  jobs with  $L$  ciphertext multiplications, giving a nonlinear-call ratio  $2 - 1/L$ . With coefficient-form external plaintext layouts, both use  $2L$  input and  $L$  output ciphertexts for these  $B$  jobs; evaluator linear expansion/contraction is additionally charged.*

*Proof.* The algebra  $A^B$  has dimension  $BL$  and  $B$  maximal two-sided ideals. The classical Alder–Strassen bound [2,9] gives bilinear rank at least  $B(2L - 1)$ . Each slotwise call supplies at most  $B$  rank-one products, regardless of cross-job linear mixing, so at least  $2L - 1$  calls are necessary. Evaluate each operand polynomial at  $2L - 1$  distinct field points, multiply there, interpolate the degree-at-most- $2L - 2$  product and retain its first  $L$  coefficients. This attains the bound. Native packs  $h$  complete jets per carrier, hence uses  $B/h = L$  products. Both external coefficient layouts store  $BL$  field symbols per operand in carriers of capacity  $hL$ , proving the stated input/output counts.

The ratio prices nonlinear calls, not time. For complete multiplication prices  $M_J, M_B$  and all remaining charges  $A_J, A_B$  in one common unit, the modeled ratio is  $((2L - 1)M_B + A_B)/(LM_J + A_J)$ . Proposition 20 gives the exact break-even condition, including setup, owner work and communication in those charges.

The theorem excludes deeper, non-bilinear, unary and publicly specialized algorithms; it does not prove a composition speedup. Its equal-capacity carrier is abstract: a reduced single-cyclotomic plaintext with slot field exactly  $\mathbb{F}_{2^{16}}$  has at most 2048 slots, since its odd conductor divides  $2^{16} - 1$ . Products of carriers, larger fields and ramified conventional HE remain eligible. This finite ceiling gives no security-preserving asymptotic separation.

*Joint evaluation of fresh batches.* Put  $\beta = 250/343$ . With a retuned Gaussian prefix and  $B \geq E = \Theta(L^\beta)$  batches sharing one bank, Appendix F.3 gives complete work  $O(BL^{2-\beta} \log^7 L)$ , with  $2 - \beta \approx 1.271137$ , under that profile's source premise. The binary one-layer model with full-domain private input encodings needs  $\Omega(BL^2)$  work even with joint owner preparation (Corollary 12). This is a scoped depth/work tradeoff. Deeper conventional compilers can use the same matrix arithmetic; the relay remains eligible more broadly. At the same retuned source, it has  $O(BL \log^4 L)$  complete work, whereas this compiler writes  $\Omega(BL^{436/343})$  input bits under the specified format (Appendix F.4). The finite cached-matrix measurements did not meet the predeclared preparation-inclusive improvement threshold (Section F.2.4).

**E.2.2 Whole-Program Conventional Alternatives** For addition/product circuits, terminal CRT need not truncate after each gate: evaluation of the untruncated circuit followed by final projection recovers the same jet. At output degree  $D$ , use  $P > D$  field points. The implemented binary additive-subspace CRT costs  $O(P \log^2 P)$  field operations per transform. With  $J$  jobs and  $S$  field slots, put  $B = \lceil JP/S \rceil$ . Product uses  $B$  products and  $2B$  inputs; affine product uses  $B$  products and  $3B$  inputs; balanced tree uses  $7B$  products and  $12B$  inputs, all with  $B$  terminal outputs. No encrypted rotations/interpolation are required. This is a terminal interface, not arbitrary encrypted continuation.

Private composition instead admits encrypted matrix contraction. Let  $A_{ji} = [z^j]g^i$ . With  $S_B = hr$  slots and  $B = \lceil L/r \rceil$  blocks, triangular zeros give

$$P_B = \sum_{b=0}^{B-1} \min((b+1)r - 1, L - 1)$$

products,  $P_B + L$  inputs and  $B$  outputs at depth one. The  $L - 1$  encrypted  $f_i$  broadcasts are reused across blocks; one further input places  $f_0$  only at row zero. Independently admitted common-modulus raw products share two switching banks and one final relinearization per block. For  $r \mid L$ ,  $P_B = L^2/(2r) + L/2 - L/r$ ; thus  $L = 256, r = 32$  gives 1144 products, 1400 inputs and eight outputs. This strengthened analytic control is unmeasured. This is an upper witness, not a product or live-memory lower bound. A stronger mixed-input contraction pairs two required terms per product with unilateral owner corrections. At  $h = 16$ ,  $L = 256$  and 2048  $K$ -slots, it uses 86 products, 348 inputs and 90 terminal ciphertexts, with routing charged to the recipient. This attains a one-layer field-linear call minimum. For binary-linear reconstruction and  $|K| \geq L^2$ , the broader

mixed-operand lower bound is  $\lceil hL^2/(8S) \rceil$  calls at  $S$  slots. Appendix E.4 proves both scopes and gives the component ledger. Appendix G.3 admits and executes the paired route; its diagnostic time is not a matched ranking. Theorem 13 extends the argument to arbitrary bilinear carriers, including the native algebra: a single mixed layer needs at least 32 binary-bilinear calls at native capacity, versus the compiler’s 19 products at depth five. The same-algebra conventional control in Section G.3.4 executes 86 products with two primes and no evaluation keys, at substantial traffic cost. Proposition 29 places both under the same ordinary source premise with their distinct reduction losses. These are depth/count tradeoffs; deeper circuits and odd-prime lifted recovery remain outside the lower bounds, and total-time ranking is open.

The HELib adapter uses diagonal BSGS contraction [24]; at  $L = 256, b = 32$  each bundle uses 287 inputs, 255 products, eight group relinearizations and seven giant rotations. This split is the fastest complete-workflow median in the declared HELib 16/32/64 subset, not a global optimum. The OpenFHE parameter selection gives  $b = 16$  within its finite tuning set (Section 9.1); these are different implementations and measurement series. A separate prepared shrinking control uses 29 inputs, 26 field products at depth eight and six terminal ciphertexts. Appendix G.2 gives its full graph, admitted 25.94 GB public-row upper bound and conditional kernel costs. A padded-CRT layout with 31 inputs and 28 products improves this attained row bound to 20.59 GB (Appendix G.2.2); neither witness is a timing result or a conventional lower bound. Non-dense composition [5], conventional HE over the same ramified algebra [22], and terminal exact-word compilation [21] are also eligible. Appendix E.5 gives noise, setup, codec, traffic and streaming ledgers for these alternatives; Appendix G.4 states exact-word admission conditions. None supplies an all-HE lower bound or a measured Gaussian comparison.

**Table 7.** All selected-profile state bounds, recomputed with exact integer square roots and rational arithmetic. Margin is  $\lfloor \log_2(q_a/(2 + 4B)) \rfloor$ .

| State              | Primes | Error cap $B$                        | Margin |
|--------------------|--------|--------------------------------------|--------|
| Prefix raw         | 3      | 4629908247922347                     | 125    |
| Prefix switched    | 3      | 1213072497477100529                  | 117    |
| Hasse 8            | 3      | 2259614478764612093                  | 117    |
| Bypass 8           | 3      | 2067570446995681958                  | 117    |
| Product 8 raw      | 3      | 973616732712397903801259907          | 88     |
| Product 8 switched | 3      | 973616733920840493030438089          | 88     |
| Tail 8             | 3      | 973616735988410940026120048          | 88     |
| Hasse 4            | 3      | 973646434673483836309103339          | 88     |
| Bypass 4           | 3      | 973640984863231009135150665          | 88     |
| Product 4 raw      | 3      | 419522210294096346900965468301996273 | 59     |
| Product 4 switched | 3      | 419522210294130639988608297480899008 | 59     |
| Tail 4             | 3      | 419522211267771624851839306616049674 | 59     |
| Prime drop         | 2      | 363877514713622027                   | 59     |
| Hasse 2            | 2      | 392200387898680891                   | 59     |
| Bypass 2           | 2      | 387003043831272164                   | 59     |
| Product 2 raw      | 2      | 168990269722124150325428385          | 30     |
| Product 2 switched | 2      | 168990269754828587240663122          | 30     |
| Tail 2             | 2      | 168990270141831631071935287          | 30     |
| Hasse 1            | 2      | 169014519016651700180965904          | 30     |
| Bypass 1           | 2      | 169010069265213561927257481          | 30     |
| Product 1 raw      | 2      | 7282453061458360604490000621532388   | 2      |
| Tail 1             | 2      | 72824530783593675310113562548789870  | 2      |

**Table 8.** Exact sufficient error caps for the three-prime profile. Margin is  $\lfloor \log_2(q_a/(2 + 4B)) \rfloor$ . All strict inequalities hold. All entries use exact integer arithmetic.

| State        | Channels | Error cap $B$                        | Margin |
|--------------|----------|--------------------------------------|--------|
| prefix_raw   | 3        | 4629908247922347                     | 125    |
| prefix       | 3        | 680169852353496747                   | 118    |
| hasse8       | 3        | 1265204605220043931                  | 117    |
| bypass8      | 3        | 1157848727792929690                  | 117    |
| product8_raw | 3        | 545148026587476860840630666          | 89     |
| product8     | 3        | 545148027263016804946205066          | 89     |
| tail8        | 3        | 545148028420865532739134757          | 89     |
| hasse4       | 3        | 545148029005900285605681941          | 89     |
| bypass4      | 3        | 545148028898544408178567700          | 89     |
| product4_raw | 3        | 234891946317989088001397086734982783 | 60     |
| product4     | 3        | 234891946317989088676937030840557183 | 60     |
| tail4        | 3        | 234891946863137117575481439019124884 | 60     |
| drop2        | 2        | 203736287507933113                   | 60     |
| hasse2       | 2        | 656902258468874931                   | 58     |
| bypass2      | 2        | 573744753390335290                   | 59     |
| product2_raw | 2        | 283044314245822433073705616          | 30     |
| product2     | 2        | 283044314769093423717461398          | 30     |
| tail2        | 2        | 283044315342838177107796689          | 30     |
| hasse1       | 2        | 283044315796004148068738507          | 30     |
| bypass1      | 2        | 283044315712846642990198866          | 30     |
| product1_raw | 2        | 121957388991765709531922845593098457 | 1      |
| tail1        | 2        | 121957389274810025244769488583297324 | 1      |

**Table 9.** Reference profile: ten independent secrets, 203 hint rows, five public-key rows and 35 encryptions per batch.  $q_4$  has 240 bits.

| Vertex                                                                                                                                                                               | Creation modulus | Joint rows $n_v$ | Queries per batch |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|-------------------|
| Initial source                                                                                                                                                                       | $q_4$            | 1                | 31                |
| Prefix result                                                                                                                                                                        | $q_4$            | 10               | 0                 |
| Hasse $r = 8$ / its result                                                                                                                                                           | $q_4 / q_4$      | 81 / 15          | 1 / 0             |
| Hasse $r = 4$ / its result                                                                                                                                                           | $q_4 / q_4$      | 41 / 15          | 1 / 0             |
| Hasse $r = 2$ / its result                                                                                                                                                           | $q_3 / q_3$      | 17 / 12          | 1 / 0             |
| Hasse $r = 1$ / its result                                                                                                                                                           | $q_2 / q_2$      | 7 / 9            | 1 / 0             |
| $q_a = \prod_{i=1}^a p_i$ , with primes in this order:<br>$p_1 = 1152921504002872321$ , $p_2 = 1152921503566671361$ ,<br>$p_3 = 1152921503264686081$ , $p_4 = 1152921503096916481$ . |                  |                  |                   |

**Table 10.** Certified strict bounds  $2^{-b}$  for the joint error projection. Entries are distance exponents  $b$ , not security bits. The four-position bound at the full modulus is too weak for a cryptographic small-gap claim.

| Modulus width | $k = 1$ | $k = 2$ | $k = 3$ | $k = 4$ |
|---------------|---------|---------|---------|---------|
| 60 bits       | 52531   | 18930   | 9023    | 4838    |
| 120 bits      | 37920   | 9796    | 3258    | 1112    |
| 180 bits      | 27354   | 4998    | 1028    | 9       |

**Table 11.** Public nonunit certificates;  $u_j, v_j$  are hexadecimal.

| $j$ | $\alpha_j$         | $u_j$           | $v_j$            |
|-----|--------------------|-----------------|------------------|
| 0   | 693094493487536309 | 893a81218       | 4600209a3        |
| 1   | 54777715398356879  | 8004010020d400  | 66fb590ab1       |
| 2   | 995026217748702709 | 8b011c0c020004c | 2208a0020ac01000 |

**Table 12.** Strict complete-inventory quotient TV bounds, for either row count.

| Prime channels | Schedule fixed before bodies | Adaptive schedule |
|----------------|------------------------------|-------------------|
| 1              | $2^{-169}$                   | $2^{-166}$        |
| 2              | $2^{-82}$                    | $2^{-79}$         |
| 3              | $2^{-11}$                    | $2^{-8}$          |

**Table 13.** Mainstream estimates for the 37-row independent-equation surrogate. Numeric entries are returned  $\log_2$  work predictions, not JetHE security bits. Missing outcomes provide no security evidence.

| Algorithm     | Cost model            | Estimate or recorded outcome |
|---------------|-----------------------|------------------------------|
| Primal uSVP   | MATZOV classical      | 1751.65                      |
| Primal uSVP   | ADPS16 classical core | 1772.44                      |
| Primal uSVP   | ADPS16 quantum core   | 1608.55                      |
| Primal BDD    | MATZOV classical      | 2392.07                      |
| Primal hybrid | MATZOV classical      | No result within limits      |
| Ordinary dual | MATZOV classical      | Numerical overflow           |
| Dual hybrid   | MATZOV classical      | No result within limits      |

### E.3 Composition from Raw Packed Inputs

The first-order invariant in Proposition 4 excludes Hasse extraction but is also preserved by composition. A mixed-ideal argument gives a stronger restriction on the raw-input interface. Its ideal-product mechanism is classical: the ring-ideal commutator and four-entry term condition are discussed in Freese–McKenzie [19, Chapter 1]. We give the direct argument, including the automorphisms and recovery maps needed here, without claiming a new general algebra theorem.

**Proposition 16 (Raw composition restriction).** *Let  $A = \prod_{a < h} K[z]/(z^L)$ , where  $K$  is a field and  $L \geq 2$ . An evaluator receives the raw packed values  $f \in A$  and  $g \in (z)$ . Its fixed finite plaintext circuit uses ring addition, negation, multiplication, constants and ring automorphisms; plaintext-identity operations may be included. Other auxiliary values are fixed independently of  $f, g$ . Terminal recovery is any fixed additive map of the circuit’s output vector, with an affine offset and separately added unilateral functions permitted. Such a program cannot return lane-wise  $f(g) \bmod z^L$  for every input, regardless of circuit depth or output count. For  $L \geq 3$  the conclusion also holds when every inner lane has linear coefficient one.*

*Proof.* Let  $I, H$  be ideals preserved by every permitted automorphism. Vary two input groups within  $I$  and  $H$ , respectively, and write  $u_{ab}$  for a wire’s four values. Ordinary ideal-congruence preservation gives its first differences in the corresponding ideals. We prove by induction that

$$M(u) = u_{11} - u_{10} - u_{01} + u_{00} \in IH.$$

Input wires and constants have zero mixed difference. Addition and negation preserve the property. An automorphism preserves  $IH$  and is additive. For a product, work modulo  $IH$  and write  $u_{11} = u + a + b$  and  $v_{11} = v + c + d$ , where  $u = u_{00}$ ,  $v = v_{00}$ ,  $a = u_{10} - u_{00}$ ,  $c = v_{10} - v_{00} \in I$  and  $b = u_{01} - u_{00}$ ,  $d = v_{01} - v_{00} \in H$ . The product’s mixed difference is

$$(u+a+b)(v+c+d) - (u+a)(v+c) - (u+b)(v+d) + uv = ad + bc = 0 \pmod{IH}.$$

Thus the invariant holds at every wire and every depth.

Let  $J$  be the nilradical of  $A$ , and take  $I = J$ ,  $H = J^{L-1}$ . Automorphisms preserve these ideals and  $IH = J^L = 0$ . In one lane set  $f_0 = g_0 = 0$ ,  $f_1 = z$  and  $g_1 = z^{L-1}$ , with other lanes zero. Every circuit output has zero mixed difference, as does any additive recovery. Affine offsets and unilateral contributions cancel as well. The four compositions instead have mixed difference  $z^{L-1} \neq 0$ .

For the last assertion take  $g_0 = z$  in every lane and add  $z^{L-1}$  to  $g_1$  in the selected lane. When  $L \geq 3$ , both linear coefficients remain one. Their difference is still in  $H$ , and the target’s mixed difference is again  $z^{L-1}$ . The same contradiction follows.

For  $L = 2$ , the promise  $g(0) = 0$  and  $g_1 = 1$  fixes  $g = z$  in every lane; composition is then the identity on  $f$ . The last assertion therefore requires  $L \geq 3$ .

The annihilating ideals are essential to the recovery claim. The simpler witness  $f = g = z$  gives a target difference outside  $J^2$ , but, for  $L \geq 3$ , a fixed coefficient shift can map the ring product  $z^2$  to  $z$ . It does not by itself exclude additive recovery from that product. The present witness forces an exactly zero difference in every returned statistic. In characteristic two this includes arbitrary fixed binary-linear CRT/Taylor recovery, without a restriction on its cost or dimension. It does not silently include nonadditive parity extraction from odd-prime representatives or an intermediate change of plaintext algebra.

*Preparation and other eligible algorithms.* The proof also permits unilateral prepared vectors when the outer preparation's change on  $0, z$  lies coordinatewise in  $J$  and the inner preparation's corresponding change lies in  $J^{L-1}$ . Arbitrary owner preparation need not satisfy this condition. For example, supply each  $f_i \in K^h$  as a constant-in- $z$  ring value, together with raw  $g$ . Horner evaluation computes the complete composition using  $L - 1$  private products from these  $L + 1$  supplied ring values. The preparation of  $f_1$  maps  $0, z$  to  $0, 1$ , violating the  $J$  condition. These are naive circuit counts, not an optimized or admitted HE implementation.

Intermediate coefficient extraction, the paid functional transport, other representations and nonlinear recipient computation likewise remain eligible. In particular the unrestricted relay in Appendix E.1 is unaffected. The current prepared JetHE compiler lies outside the excluded gate interface, and a conventional base supplying the same functional keys retains Appendix A.3's transfer and costs. This proposition does not extend the arbitrary-preparation one-layer comparison to every depth, establish a time lower bound for all HE, or provide a security result. It shows why more ring-multiplication depth alone does not supply the missing capability under the stated semantics.

Finite computations check the four corners, small-ring multiplication closure, multiple lanes, field automorphisms and the coefficient-prepared escape. These are finite algebra checks supporting the written argument, not independent proof verification or encrypted performance measurements.

#### E.4 One-layer prepared contraction: bounds and a paired control

This subsection strengthens the conventional prepared-contraction comparison. It permits arbitrary independent owner preparation and mixed-owner operands. The lower bounds concern a specified plaintext-functional layer, with linear maps free; the attained upper witness separately charges its terminal components. None is a lower bound for all conventional HE or total time.

**Definition 5 (One layer of prepared field products).** *Let  $K = \mathbb{F}_q$ ,  $q = 2^s$ , and  $L = 2^d \geq 2$ . There are  $h$  independently variable composition jobs. Each owner may prepare arbitrary  $K$ -valued functions of all its own inputs. The evaluator performs  $m$  scalar products*

$$Z_a = (A_a(f) + B_a(g))(C_a(f) + D_a(g)), \quad 1 \leq a \leq m,$$

where every displayed function is unilateral. All other joint processing is a fixed  $K$ -linear map in the field-linear model, or a fixed  $\mathbb{F}_2$ -linear map in the binary-linear model. Arbitrary decoded unilateral bypasses may be added. In the owner-separated submodel, each product has one outer-only and one inner-only operand. Parameters, maps and shapes are fixed independently of inputs. A carrier with  $S$   $K$ -slots supplies at most  $S$  scalar products per ciphertext product.

Public additive preprocessing preserves this form, including maps linear over  $\mathbb{F}_2$  applied to each owner's prepared values. Constants can be absorbed in the unilateral functions. This model allows unlimited terminal output and linear recipient work. It excludes odd-prime lifting/parity, larger-field products, native nilpotent slots, further joint product layers and nonlinear recipient computation. In particular, the deeper prepared shrinking and native compilers are outside the lower bounds.

**Lemma 18 (Coefficient-function space).** *Write  $g(z) = \sum_{v=1}^{L-1} X_v z^v$  and  $P_{i,j}(X) = [z^j]g(z)^i$ , for  $1 \leq i, j < L$ . If  $q \geq L$ , the nonzero  $P_{i,j}$  are independent as  $K$ -valued functions, and their number is*

$$D_L = (L^2 - 1)/3.$$

*If  $q \geq L^2$ , their Frobenius-closed  $K$ -span has dimension  $sL^2/4$ . Across  $h$  jobs both dimensions multiply by  $h$ .*

*Proof.* Put  $i = 2^a u$  with  $u$  odd. Then  $P_{i,j} = 0$  unless  $2^a \mid j$  and  $j \geq i$ . For  $j = 2^a k \geq i$  it is  $P_{u,k}^{2^a}$ . When  $k > u$ ,  $P_{u,k}$  contains  $X_1^{u-1} X_{k-u+1}$  with coefficient  $u = 1 \bmod 2$ ; when  $k = u$ , it contains  $X_1^u$  with coefficient one. Thus these are exactly the nonzero positions. They have distinct pairs of ordinary and weighted degrees  $(i, j)$ , where  $X_v$  has weight  $v$ , and hence disjoint monomial supports. Their individual-variable degrees are below  $L \leq q$ . The univariate root bound, iterated over variables, makes evaluation on  $K^{L-1}$  injective for these reduced polynomials. Their number is

$$\sum_{a=0}^{d-1} \sum_{\substack{1 \leq u < 2^{d-a} \\ u \text{ odd}}} (2^{d-a} - u) = \sum_{a=0}^{d-1} 4^{d-a-1} = (L^2 - 1)/3.$$

For the second claim take all  $s$  Frobenius powers of  $P_{u,j}$  with  $u$  odd and  $u \leq j < L$ . There are  $sL^2/4$  such functions. Every even-indexed coefficient is a Frobenius power of one already in this family, so they span the desired closure. We prove independence by unique monomials. Represent polynomial functions canonically with exponents in  $\{0, \dots, q-1\}$ ; a positive exponent  $e$  reduces to  $1 + ((e-1) \bmod (q-1))$ .

For  $j > u$ , choose the  $t$ -th Frobenius image of  $X_1^{u-1} X_v$ ,  $v = j - u + 1 \geq 2$ . Suppose it also occurs in the  $t'$ -th Frobenius image of  $P_{u',j'}$ , where  $u'$  is odd and  $u', j' < L$ . Undoing the latter Frobenius gives exponent  $2^r \bmod (q-1)$  on  $X_v$ , where  $r = t - t' \bmod s$ . Its degree is below  $L$ , so  $2^r < L$ . The  $X_1$  exponent is

then  $(u-1)2^r$  without wrap, since it is below  $L^2 \leq q$ . The candidate degree is  $u2^r$ ; oddness forces  $r = 0$ . Thus  $u' = u$ ,  $j' = j$  and  $t' = t$ . The case  $u = 1$  simply has no  $X_1$  factor.

For  $j = u$  the marker is the Frobenius image of  $X_1^u$ . A competing odd degree  $u' < L$  would satisfy  $u' = u2^r \bmod (q-1)$ . Now  $s \geq 2d$ . If  $0 < r < d$ , there is no wrap and this degree is even. If  $d \leq r < s$ , write  $u2^r - u' = k(q-1)$ . Both  $u'$  and  $k$  are below  $L \leq 2^r$ ; reducing modulo  $2^r$  yields  $k = u'$ . Consequently  $u = u'2^{s-r}$ , also contradicting oddness. Only  $r = 0$  is possible. The marker then belongs only to  $P_{u,u}$ . Each selected monomial therefore occurs with coefficient one in exactly its own row, even after polynomial-function reduction. This proves independence. Across jobs the nonconstant monomials use disjoint variables.

**Theorem 12 (One-layer prepared-composition requirements).** *In Definition 5, exact composition for all inputs requires, in the field-linear model with  $q \geq L$ ,*

$$m \geq \lceil hD_L/2 \rceil \quad (\text{mixed}), \quad m \geq hD_L \quad (\text{owner-separated}).$$

*The owner-separated bound is attained. In the binary-linear model with  $q \geq L^2$ , the corresponding lower bounds are*

$$m \geq \lceil hL^2/8 \rceil \quad (\text{mixed}), \quad m \geq hL^2/4 \quad (\text{owner-separated}).$$

*Divide the right-hand sides by  $S$  and round up to obtain ciphertext-product lower bounds for  $S$  field slots. The binary-linear bounds are not asserted to be exact optima.*

*Proof.* Take the mixed difference between  $f = e_i$  and  $f = 0$ , and between  $g$  and  $g = 0$ , placing  $e_i$  in one job. For each output coordinate, the target is  $P_{i,j}(g)$  and all unilateral bypasses disappear. A product contributes

$$(A_a(e_i) - A_a(0))(D_a(g) - D_a(0)) + (C_a(e_i) - C_a(0))(B_a(g) - B_a(0)).$$

Thus field-linear reconstruction places all  $hD_L$  target functions in the span of at most  $2m$  centered inner functions, or  $m$  in the owner-separated case. Lemma 18 proves the bounds. The separated attainer computes each nonzero  $f_i P_{i,j}(g)$  independently and routes it to its output coefficient. No linearity of owner preparation was assumed.

Every  $\mathbb{F}_2$ -linear map  $K \rightarrow K$  is  $\sum_{t=0}^{s-1} c_t x^{2^t}$ . Indeed, these maps are injectively represented by their coefficients by the root bound, and there are  $2^{s^2}$  of both. With binary-linear reconstruction, the target functions consequently lie in the span  $W$  of all Frobenius powers of the centered inner functions. This space is Frobenius-stable and has dimension at most  $2sm$ , or  $sm$  in the separated case. The entire target Frobenius closure lies in  $W$ ; its dimension is  $hsL^2/4$  by the lemma. Cancelling  $s$  proves the two remaining bounds. Free linear maps and arbitrary mixing of jobs do not increase the number of scalar products furnished by one carrier.

**Proposition 17 (Paired conventional upper witness).** *For dyadic  $L \geq 4$ , a mixed, field-linear prepared contraction uses*

$$P_L = (2L^2 + 3L - 8)/12$$

*scalar products per job. At  $h = 16$ ,  $L = 256$ ,  $K = \mathbb{F}_{65536}$  and  $S = 2048$ , it uses 86 ciphertext-product calls, attaining the mixed field-linear lower bound of Theorem 12.*

*Proof.* For each output degree  $j \geq 1$ , list its nonzero indices  $i$  in order and pair adjacent indices  $i, k$ . Put  $x = P_{i,j}(g)$ ,  $y = P_{k,j}(g)$ . The identity

$$(f_i + y)(f_k + x) + f_i f_k + xy = f_i x + f_k y$$

computes both desired terms with one joint product. The correction  $f_i f_k$  is outer-only and  $xy$  is inner-only; each owner aggregates its corrections per output coefficient. A final unpaired term uses one ordinary separated product. The outer bypass also includes  $f_0$ .

The number of nonzero terms in output  $j$  is  $t_j = j - (\text{oddpart}(j) - 1)/2$ , by summing the eligible odd indices at each power of two dividing  $j$ . For odd  $j$ ,  $t_j$  is odd exactly when  $j \equiv 1 \pmod{4}$ ; for even  $j$ , exactly when its odd part is  $3 \pmod{4}$ . There are  $L/4$  of the first kind and  $\sum_{a=1}^{d-2} L/2^{a+2} = L/4 - 1$  of the second. Thus exactly  $L/2 - 1$  counts are odd, while their sum is  $D_L$ . Hence  $P_L = \sum_j \lceil t_j/2 \rceil = (D_L + L/2 - 1)/2$  as stated. Pack the public product list consecutively into field slots and route results through a fixed linear map. At the displayed point,  $D_L = 21845$ ,  $P_L = 10986$ , and

$$\left\lceil \frac{16D_L}{2 \cdot 2048} \right\rceil = \left\lceil \frac{16P_L}{2048} \right\rceil = 86.$$

Padding the last carrier with zeros is permitted. For  $L = 2$  the direct count is one product per job.

*Charged terminal realization.* Put  $C = \lceil hP_L/S \rceil$  and  $B = \lceil hL/S \rceil$ . Each owner prepares  $2C$  product-operand carriers and  $B$  aggregated bypass carriers. Encrypt all  $4C + 2B$  inputs freshly under one common key. The evaluator makes  $2C$  operand additions and  $C$  products. A direct terminal route returns all  $C$  raw three-component products and  $2B$  two-component bypasses; the recipient decrypts and performs fixed field-linear routing/addition. This avoids presuming free encrypted scatter, reduction, relinearization or output compression. For ring rank  $n$  and common modulus width  $b$ , raw public-key, input and output sizes are respectively

$$2nb, \quad 2nb(4C + 2B), \quad nb(3C + 4B) \quad \text{bits.}$$

A common-key unscaled terminal implementation needs no evaluation key for these degree-two outputs. Its noise admission and source assessment are separate

obligations; neither is supplied by the functional proof. Codecs, tables, preparation, fresh encryption, transport framing and all recipient work remain charged. Returning all raw products is allowed by the restricted recovery interface and has a substantial traffic cost.

At the stated point  $C = 86$ ,  $B = 2$ : there are 348 input ciphertexts, 86 three-component product outputs and four two-component bypass outputs. The owner-separated field-linear minimum is 171 calls. With binary-linear reconstruction the proved lower bounds are instead 128 for separated and 64 for mixed operands; 86 is not proved optimal in that broader model.

*Scope and evidence.* This is a depth/interface tradeoff, not a time or all-HE separation. The odd-prime OpenFHE adapter has different scalar multiplication and lifted recovery semantics, so these  $K$ -product bounds do not apply to it. Nor does a fixed field satisfy  $q \geq L^2$  as  $L$  grows without bound. Deeper conventional schemes over JetHE’s own ramified algebra and the shrinking control remain eligible. The elementary polarization upper witness strengthens a conventional comparator rather than adding an exclusive JetHE capability.

Finite consistency checks compare small formal expansions and exact finite-field function ranks, and verify 262144 production Frobenius markers against 4194304 inverse-image candidates. Plaintext paired-contraction examples agree with direct composition. Small-field rank failures show why the field-size hypothesis matters, while omitting the unilateral corrections gives incorrect outputs. These finite algebraic checks do not replace the general proof. Appendix G.3 supplies a separate admission and encrypted direct-receiver trial, with diagnostic observations and its complete cost boundaries.

**E.4.1 General Bilinear Carriers and Initial Inputs** The mixed-difference argument also covers multiplication in a ramified algebra. The following extension prices operand dimensions, without restricting a primitive’s output dimension or its internal tensor rank.

**Theorem 13 (One-layer bilinear carrier bound).** *Keep the full prepared-composition input domain above. Permit arbitrary owner-only preparation, public affine preprocessing, one parallel layer of fixed bilinear primitives, fixed linear recovery, and arbitrary unilateral bypasses. No primitive output feeds another primitive and no other evaluator-side nonlinear operation is free.*

*In the  $K$ -linear model let primitive  $a$  have operand dimensions  $r_a, t_a$ . If  $|K| \geq L$ , then*

$$\sum_a (r_a + t_a) \geq H := h(L^2 - 1)/3.$$

*In the binary-linear model let its operand widths be  $a_a, b_a$  bits. If  $K = \mathbb{F}_{2^s}$  and  $|K| \geq L^2$ , then*

$$\sum_a (a_a + b_a) \geq T := hsL^2/4.$$

*With owner-separated operands, only the inner-owner operand widths are needed on the left. Moreover, if the initial inner-owner inputs contain  $v_K$  field coordinates or  $v_2$  bits and all subsequent preprocessing before the joint layer is affine, then respectively  $v_K \geq H$  or  $v_2 \geq T$ . Initial outer-owner inputs contain at least  $h(L-1)$  field coordinates or  $hs(L-1)$  bits. Extra input-dependent public data must be included in these input counts.*

*Proof.* Write one primitive's operands as  $A(f) + B(g)$  and  $C(f) + D(g)$ . The mixed difference at  $f = e_i, 0$  and  $g, 0$  is

$$\beta(A(e_i) - A(0), D(g) - D(0)) + \beta(B(g) - B(0), C(e_i) - C(0)).$$

Every recovered coordinate lies in the span of the centered coordinate functions of  $B, D$ . The target difference is  $[z^j]g^i$ ; all bypasses cancel. Lemma 18 gives dimension  $H$ , while each primitive supplies at most  $r_a + t_a$  generators. In the binary model, each fixed binary-linear map into a  $K$ -valued output is specified by the images of binary basis vectors. Thus the target functions lie in the  $K$ -span of at most  $a_a + b_a$  Boolean generators. This span is Frobenius-stable: squaring fixes a Boolean generator and squares its scalar coefficient. The target closure dimension  $T$  proved above gives the second bound. Owner separation removes the unused inner-owner operand in each case.

Affine preprocessing makes all centered inner-operand coordinates lie in the span of the initial centered inner-owner coordinates. The same arguments therefore bound  $v_K, v_2$ , even if an input is copied into many operands. For the outer owner, subtract the output at  $g = 0$  from that at  $g = z$  in each job. Its nonconstant coefficients are  $f_j$ ,  $1 \leq j < L$ . They are independent coordinate functions; their Frobenius closure has dimension  $hs(L-1)$  because the reduced monomials  $f_j^{2^u}$ ,  $0 \leq u < s$ , are distinct. Applying the symmetric input-span argument proves the outer bounds. Cross-job mixing changes neither argument.

At  $h = 16, L = 256, s = 16$ , the native algebra has 4096  $K$  coordinates, or 65536 bits, per carrier. Here  $H = 349520$  and  $T = 4194304$ . A mixed layer therefore needs at least 43  $K$ -bilinear calls or 32 arbitrary binary-bilinear calls; separated operands require 86 or 64. Initial input ciphertexts need at least 87 or 65 respectively, including one outer-owner input. The current compiler uses 16 outer plus 19 inner inputs and 19 products at depth five. This exhibits a depth tradeoff against the stated one-layer class, which now includes conventional multiplication in the very same algebra. It is neither a time lower bound nor an all-depth optimum: smaller moduli, sources, maps and terminal traffic still require a complete ledger. Nonlinear recipient recovery and odd-prime lifted arithmetic are outside the binary model.

Finite checks cover general mixed bilinear maps with cross-job preparation, binary Frobenius stability, truncated-polynomial products and inherited small-field rank boundaries. These finite checks do not replace the proof. The native-algebra upper control below uses 86 products with smaller moduli and no evaluation keys.

**E.4.2 A Growing Count Tradeoff over the Fixed Field** The preceding large-field dimensions do not apply without bound to the Gaussian family's fixed  $K = \mathbb{F}_{2^{16}}$ . The binary-linear hypothesis  $|K| \geq L^2$  ends at  $L = 256$ , and the field-linear hypothesis at  $L = 65536$ . A weaker dimension bound holds for every characteristic-two field and suffices for an asymptotic comparison on that same family.

**Lemma 19 (Boolean coefficient-function witness).** *Let  $L = 2^d \geq 8$  and let  $K$  have characteristic two. Restrict the coefficients of  $g$  to  $\mathbb{F}_2$ . The  $K$ -span of  $P_{i,j} = [z^j]g(z)^i$ ,  $1 \leq i, j < L$ , has dimension at least*

$$C_L = \frac{5L^2}{64} + \frac{(d+1)L}{16} \geq \frac{L^2}{16}.$$

For  $h$  independent jobs the lower bound is  $hC_L$ .

*Proof.* Let  $V_r = \text{span}_{\mathbb{F}_2}\{z, \dots, z^r\}$ , with  $V_0 = \{0\}$ , and put  $E_r(Y) = \prod_{a \in V_r} (Y + a)$ . Products over finite additive subgroups are classical additive polynomials [14, Theorem A.1]. Here the identity also follows directly from  $E_0(Y) = Y$  and

$$E_{r+1}(Y) = E_r(Y)^2 + E_r(z^{r+1})E_r(Y).$$

Write  $g = a + u$  with  $a \in V_r$  and  $u = \sum_{v \geq r+1} X_v z^v$ . On Boolean inputs,  $E_r(g) = E_r(u)$ . The coefficient  $A_r$  of  $u$  in  $u \prod_{a \in V_r \setminus \{0\}} (u + a)$  has leading coefficient one and valuation

$$\gamma_r = \sum_{\ell=1}^r \ell 2^{r-\ell} = 2^{r+1} - r - 2.$$

Put  $\nu_r = \gamma_r + r + 1 = 2^{r+1} - 1$ . The coefficient of  $u^b$ ,  $b \geq 2$ , has valuation at least  $\gamma_r - (b-1)r$ : each removed nonzero factor has valuation at most  $r$ . A monomial of  $u^b$  containing  $X_v$  has degree at least  $v + (b-1)(r+1)$ . Thus its contribution to degree  $\nu_r + j$  only uses  $v \leq r + j + 2 - b < r + 1 + j$ . The linear term  $A_r u$  gives  $X_{r+1+j}$  with coefficient one. Consequently

$$[z^j]z^{-\nu_r} E_r(g) = X_{r+1+j} + R_{r,j}(X_1, \dots, X_{r+j}),$$

and the constant coefficient is exactly  $X_{r+1}$ . These are polynomial functions on Boolean inputs; reducing  $X_v^2 = X_v$  preserves variable indices. Removing the vanishing prefix is only notation for public coefficient identities, not an encrypted division.

Set  $R = d - 3$  and  $B = 2^R = L/8$ . For  $B \leq i < 2B$ , let  $S(i)$  be its set of one-bit positions and  $w(i) = |S(i)|$ . Define

$$H_i(Y) = \prod_{r \in S(i)} E_r(Y), \quad \nu(i) = 2i - w(i).$$

These are rescaled classical Carlitz digit products: in the convention  $\psi_0(Y) = Y$ ,  $\psi_r(Y) = \prod_{\deg a < r} (Y - a)$  over  $\mathbb{F}_2[z]$ , and  $G_i = \prod_{r \in S(i)} \psi_r$ , one has  $E_r(Y) =$

$z^{2^r} \psi_r(Y/z)$  and  $H_i(Y) = z^i G_i(Y/z)$  [41, Section 1]. We use their coefficients below a fixed truncation budget; the polynomial family itself is classical. Since  $H_i$  has degree  $i < L$ , zero constant coefficient and coefficients in  $\mathbb{F}_2[z]$ , every coefficient of  $H_i(g)$  below degree  $L$  lies in the span of the original  $P_{a,b}$ . For  $0 \leq j < L - \nu(i)$ , put  $Q_{i,j} = [z^{\nu(i)+j}]H_i(g)$ . All selected  $i$  have highest bit  $R$ . Only the normalized degree- $j$  coefficient of the  $E_R$  factor can introduce the highest variable  $X_{R+1+j}$ ; all other factors must supply their constant coefficients. Hence

$$Q_{i,j} = X_{R+1+j} \prod_{r \in S(i), r < R} X_{r+1} + \text{a polynomial in } X_1, \dots, X_{R+j}.$$

The index is at most  $L - 1$ , since  $\nu(i) \geq 2B - 1$  and  $R + 2 \leq 2B$ . In any linear relation, take the largest  $j$  present. Distinct  $i$  give distinct multilinear monomials multiplying  $X_{R+1+j}$ , so all their coefficients vanish. Descending in  $j$  proves independence over  $K$ . Their number is

$$\sum_{i=B}^{2B-1} (L - 2i + w(i)) = BL - 3B^2 + 2B + \frac{RB}{2} = C_L.$$

Here  $\sum_i w(i) = B + RB/2$ , including  $R = 0$  with the latter term zero. All functions vanish at  $g = 0$ , so disjoint job variables multiply the dimension by  $h$ .

*Relation to classical polynomial-function counts.* Canonical representations and null ideals for polynomial functions over finite principal ideal rings are classical [6,38]. Bhargava's exposition [7, Section 7, Lemmas 12 and 14; Section 8] gives the Newton coefficient criterion and canonical representatives, and includes finite principal ideal rings in its framework. The scalar span in Lemma 19 is a different invariant. For  $R = \mathbb{F}_2[z]/(z^L)$ , let  $\mathcal{M}$  be the  $R$ -module of polynomial functions  $zR \rightarrow R$  vanishing at zero. The map  $H \mapsto [z^{L-1}]H$  has image exactly the scalar coefficient span:  $P_{i,j}$  is the image of  $z^{L-1-j}g^i$ . Its kernel need not be zero; at  $L = 8$  the two dimensions are 17 and 11, respectively. The Newton count for  $\mathcal{M}$  therefore cannot be substituted for the scalar dimension without analyzing this projection. The proof above supplies the needed coefficient independence for its selected classical digit products; neither that family nor the general valuation machinery is claimed as new. Exact computations at twelve finite lengths compare the module count with the scalar coefficient rank. They illustrate this distinction without supplying an exact general formula for the scalar rank.

**Corollary 8 (Fixed-field one-layer requirements).** *In Theorem 13's model, for any  $K = \mathbb{F}_{2^s}$  and dyadic  $L \geq 8$ , both the sum of binary operand widths and the initial inner-owner input width are at least  $hC_L$  bits. For carriers of at most  $N = hsL$  plaintext bits per operand this gives*

$$\#\text{calls} \geq \left\lceil \frac{C_L}{2sL} \right\rceil, \quad \#\text{inputs} \geq \left\lceil \frac{C_L}{sL} \right\rceil + 1.$$

*The  $K$ -linear requirements are  $hC_L$  field coordinates instead; owner separation charges only inner-operand widths in either model.*

*Proof.* Use the same mixed differences as in Theorem 13, restricting only the inner inputs to Boolean coefficients. Lemma 19 supplies  $hC_L$  independent  $K$ -valued targets. Each centered binary operand coordinate gives at most one generator for their  $K$ -span; affine preprocessing cannot increase the span of initial coordinates. There is no additional factor  $s$  from Frobenius closure here. The outer-input argument still uses  $g = z, 0$  and unrestricted  $f$ , requiring  $hs(L - 1)$  bits and therefore at least one outer carrier. The same argument counts field coordinates in the  $K$ -linear model.

In Corollary 4's already admitted family,  $h = s = 16$ ,  $N = 256L$ , and  $k = \min\{d, \lceil d/2 + \log_2(d + 1) \rceil\}$ . Its  $2^k - 1 + d - k$  products and  $2^{k+1} - 1 + d - k$  inputs are  $O(\sqrt{L} \log L)$ , at depth  $d - k + 1$ ; the one-layer requirements above are  $\Omega(L)$ . The ratio of the lower bound to these attained counts therefore grows as  $\Omega(\sqrt{L}/\log L)$  over the same fixed field, under the existing Gaussian source and generator assumption. This removes the large-field obstruction to a growing count/depth tradeoff. A complete-bit-work consequence follows under either of the encoding conditions stated below. Deeper conventional compilers over the same algebra remain eligible.

The constants are asymptotic evidence, not a practical parameter claim. At  $L = 256$  this weaker bound gives only one call and three inputs; the large-field bound supplies 32 and 65. The Gaussian balanced choice has  $k = 8$  there, unlike the measured  $k = 4$  finite-law profile. For example, at  $L = 2^{28}$  the fixed-field lower bounds are 655361 calls and 1310722 inputs, while the Gaussian schedule has 524296 products, 1048584 inputs and depth ten. This is a calculated family point; no encryption at that size was run.

**Lemma 20 (Secure encoding length).** *Let an input-independent setup supply public encoding information and secret decoding keys. An encoder  $E$  accepts every  $V$ -bit message, with a fixed public shape, and outputs a possibly variable-length string. All input-dependent information needed to decode belongs in that string. Let  $\epsilon$  be the decoding failure probability on uniform  $U \in \mathbb{F}_2^V$ . For an efficient selected-message sampler  $M$  of the same shape, let  $\Delta$  bound the absolute event-probability gap of the public length test between  $E(M)$  and  $E(U)$ . For  $0 \leq t \leq V$ ,*

$$\Pr[|E(M)| \leq V - t] \leq \epsilon + 2^{1-t} + \Delta.$$

*Full-domain message indistinguishability makes  $\Delta$  negligible.*

*Proof.* Fix the public setup and decoding keys. There are  $2^{V-t+1} - 1$  strings of length at most  $V - t$ . Writing  $D$  for the decoder, the probability of short and correct encoding of uniform  $U$  is

$$2^{-V} \sum_{|c| \leq V-t} \sum_m \Pr[E(m) = c] \Pr[D(c) = m] \leq 2^{-V} \sum_{|c| \leq V-t} 1 < 2^{1-t}.$$

This also permits randomized decoding. Average over setup, add its uniform-message correctness error, and apply the length test to transfer the bound to  $M$ . No entropy premise about  $M$  is used. This is the elementary counting and IND-CPA argument, not a new encryption primitive.

**Corollary 9 (Prepared-input work).** *Assume the binary model of Corollary 8, with fixed  $h = s = 16$  and a polynomial-time prepared-input family. Suppose the inner owner’s initial encoded stream is correct and message-private over its full carrier message space, with negligible errors in Lemma 20. Producing this stream requires  $\Omega(L^2)$  bit work except negligibly, and in expectation, even with fused preparation/encryption or joint compression. Alternatively, explicit materialization of the prepared plaintext bits already gives this lower bound deterministically. Under its existing source and generator assumption, the deeper Gaussian workflow uses  $\tilde{O}(L^{3/2})$  complete computational bit work.*

*Proof.* The full encoded message width is  $V \geq hC_L$ ; padding can only increase it. Apply Lemma 20 with  $t = \lceil V/2 \rceil$ . Except negligibly the stream has more than  $\lfloor V/2 \rfloor$  bits, whose writes cost  $\Omega(L^2)$ . In the alternative case, writing the  $hC_L$  prepared plaintext bits suffices. The Gaussian upper bound is  $\Pi + O(N\sqrt{L}\log^5 L)$  with  $N = 256L$  and  $\Pi = \text{poly}(\log L)$ , hence  $o(L^2)$ .

The ratio grows as  $\Omega(\sqrt{L}/\log^5 L)$ , with no finite-time crossover asserted. The encoding condition concerns recovery of every carrier message, not only the prepared image. It holds for ordinary independently encrypted carriers and for efficient ciphertext compression that preserves full-domain recovery and privacy. One may combine all initial decoding keys for the counting argument, without giving those keys to a real participant. A separate plaintext buffer is unnecessary. All transmitted input-dependent descriptors belong in the charged stream.

This does not rule out encodings correct only on a structured image. For example, sparse-vector compression can preserve privacy while guaranteeing recovery only on a declared family [18, Definition 4 and Remark 2]; underlying IND-CPA alone does not give the compressed format full-domain correctness. Finite examples exhibit correct-but-short encodings without privacy, and private one-time projections with correctness only on a promised subspace. They illustrate why both privacy and full-domain correctness are needed in the encoding lemma; they are not additional homomorphic constructions. Deeper protocols and nonlinear recipient recovery remain outside the one-layer comparison. Conventional HE can adopt the deeper compiler and classical fast arithmetic; the separation belongs to the specified compiler interfaces, not exclusively to an encryption primitive.

Independent root-product and additive-recurrence computations agree on the tested cases. Exact symbolic computations give witness ranks 7, 25, 92 and 348 at  $L = 8, 16, 32, 64$ ; point evaluations through  $L = 256$  verify 3918 leading-variable derivatives. Computations of the full scalar rank and a separate candidate family also exhibit a rank pattern and a dependent family, respectively. No general exact-rank formula or independent family beyond the proved witness is inferred from those observations. These are finite algebraic checks, not homomorphic executions or cryptographic hardness estimates.

## E.5 Detailed Computational and Space Ledgers

Native multiplication implements a useful algebraic operation, but this alone does not order the costs of complete encrypted programs. We first charge a leveled implementation, then distinguish a restricted multiplication bound from the attainable compilers for private composition.

**E.5.1 Whole-circuit accounting** At modulus  $q$ , let  $b_q = \lceil \log_2 q \rceil$ , let  $\ell_q$  be its RNS limb count, and price one ring product by  $M_q$ . Let  $D_q$  include centered CRT, exact gadget extraction and channel lifting of one polynomial. For coefficient-form inputs, an identity bank can be applied in

$$S_q \leq D_q + 2g_q M_q + O(g_q \ell_q N)$$

word operations, in a fixed-word RNS model. This is an upper bound; prepared spectra and shared transforms can improve it. Raw ciphertext multiplication costs  $3M_q + O(\ell_q N)$  using Karatsuba’s three polynomial products. Rekeying its two nonconstant components uses two banks and two decompositions. Addition costs  $O(\ell_q N)$ ; a public ring multiplier uses two ring products. An automorphism costs its actual coefficient map plus one bank. It is  $O(\ell_q N) + S_q$  only when that map is a signed permutation. Branch alignment costs  $S_q$  and does not change the plaintext by relabeling alone.

For the tensor family, compatible radix-two/Rader transforms give  $M_q = O(\ell_q N \log N)$  modular word operations; this assumes the required roots exist in every channel. The four primes in Table 9 satisfy this condition at  $L = 256$ . A spectral prime drop in a two-component state costs two inverse transforms in the dropped channel, two forward transforms in each retained channel, and the pointwise corrections. It does not reconstruct the full modulus. For ordinary gadget conversion, Garner reconstruction separately costs  $N \binom{\ell_q}{2}$  modular multiply-reduces and  $N(\ell_q - 1)$  growing-integer multiply-adds, followed by centering and digit extraction. These primitives, memory passes and allocations are not free consequences of a transform count.

For a publicly scheduled circuit, sum these costs at each operation’s actual modulus, including every alignment, public map and drop. Depth affects that sum through the admitted chain and key graph; a depth number alone does not determine modulus size. The uniform recurrences in Section 4 supply correctness admission, not a security-qualified asymptotic family by themselves. The Gaussian-family corollary below supplies an explicit conditional join.

**Proposition 18 (Explicit storage and workflow ledger).** *Let  $m_v$  be the number of hint rows created at vertex  $v$ , and  $p_v \in \{0, 1\}$  indicate publication of an owner public key. Uncompressed, two-polynomial, bit-packed public material occupies*

$$2N \sum_v (m_v + p_v) b_{q_v} \quad \text{bits},$$

*before metadata. These are bit-packed coefficient-residue sizes; an actual 64-bit-word RNS ciphertext instead uses  $2N \cdot 64\ell_q$  bits. A bit-packed ciphertext at*

$q$  occupies  $2Nb_q$  bits. At most  $w$  live two-component ciphertexts use at most  $2wN \max_q b_q$  bits, in addition to keys, digits, transforms, owner arrays and back-end scratch. Retaining all ternary stage secrets needs  $O(N|V|)$  bits at the authority, not at the evaluator. For  $K$  input batches, complete serial workflow cost is

$$C_{\text{total}}(K) = C_{\text{setup}} + C_{\text{keygen}} + \sum_{j=1}^K (C_{\text{prep},j} + C_{\text{codec},j} + C_{\text{enc},j} + C_{\text{eval},j} + C_{\text{transfer},j} + C_{\text{recover},j}),$$

with one-time key distribution also charged to setup/transfer. Overlap or parallelism requires a different, explicit scheduling model.

*Proof.* Count two ring elements for every published row/key and ciphertext, at its creation or specified modulus. Sum the executed operations and the owner and recipient phases. Sharing a key saves storage but not every invocation that consumes it; a deterministic spectral copy saves no independent key row.

**Proposition 19 (Complete common-modulus prepared workflow).** *For the compiler of Section 5, put  $s_L = \log_2 L$ ,  $M = 2^k - 1$ ,  $\tau = s_L - k$ ,  $I = 2M + 1 + \tau$  and  $H = g(2^{\tau+1} + 3\tau)$ . At a fixed admitted modulus, let  $\mathcal{M}_q, \mathcal{D}_q$  be bit costs of a coefficient-form ring product and one polynomial gadget decomposition. An unhoisted evaluator satisfies*

$$C_{\text{eval}} \leq [3(M + \tau) + 2H]\mathcal{M}_q + (2 + 3\tau)\mathcal{D}_q + O((M + \tau + H)Nb_q).$$

Key generation uses at most  $H + 2\tau + 2$  ring products, all  $I$  fresh encryptions use  $2I$ , and two-component recovery uses one. In addition to codecs, the owners use  $M - k$  truncated polynomial products per lane and  $O(hL(M + s_L))$  other field operations/moves, including construction of all  $U_j$ . For fixed radix, direct schoolbook arithmetic gives the expected one-batch bit-work upper bound

$$\Pi(q) + O((M + \tau + H)N^2b_q^2 + IC + h((M - k)L^2 + L(M + s_L))\mathcal{F}_K),$$

where  $\mathcal{C} = O(N(256 + \log L))$  bounds a tensor codec,  $\mathcal{F}_K$  bounds the bit cost of one field operation/move, and  $\Pi(q)$  charges modulus selection/validation. Fixed public codec setup is charged once; its cost is absorbed by this bound for the fixed 256-axis family. Transmission is additionally charged by its actual byte volume and schedule. Retained plaintext variants need  $O((M + s_L)N)$  bits; all encrypted inputs need  $2INb_q$  bits. A streaming coefficient evaluator needs  $O(Nb_q)$  extra state, separately from public/evaluation keys, owner caches and retained inputs.

*Proof.* There are  $M + \tau$  raw products and  $H$  consumed hint rows, each applying two ring products. Sharing the state-component digits between Hasse transport and bypass alignment leaves  $2 + 3\tau$  decompositions. Setup forms  $H + \tau + 1$  masked rows/public keys and  $\tau + 1$  source-secret squares; the norm-one Hasse payload maps and scaling cost  $O((H + \tau)Nb_q)$  bits. Each encryption has two ring products and decryption one.

Repeated characteristic-two squaring builds all  $g^{2^j}$  with  $h(L - s_L - 1)$  field squarings. Padding, movement and the additions defining  $U_j$  fit  $O(hLs_L)$ . Each  $f_S$  is a mask and downshift. Reuse singleton  $W_{\{j\}} = U_j$  and construct every other subset once from a smaller subset, using  $M - k$  products. Schoolbook truncated multiplication uses at most  $L(L+1)/2$  field products per lane. Charge  $I$  encodings and one decoding.

In coefficient form  $\mathcal{M}_q = O(N^2 b_q^2)$  and  $\mathcal{D}_q = O(N b_q)$  for fixed radix; no transform roots are needed. Sampling uses  $(H + \tau + 1)N$  uniform- $q$  coordinates,  $(2\tau + 2 + I)N$  ternary coordinates and  $(H + \tau + 1 + 2I)N$  CBD20 coordinates. Uniform- $q$  and ternary rejection have acceptance probabilities greater than  $1/2$  and equal to  $3/4$ , respectively, so ideal-bit sampling has the stated expected bound. This is not fixed-time exact sampling or a real-RNG theorem. Streaming rows/digits and the three prefix accumulators gives the workspace bound. The prime-existence corollary does not itself bound  $\Pi(q)$ .

**E.5.2 Fast Exact Arithmetic for the Reference Family** The schoolbook bound above can be improved without changing a ciphertext, source distribution or public parameter. The following applies standard integer multiplication; it is not a new multiplication algorithm.

**Lemma 21 (Fast exact products).** *Work in the ring  $R_q = (\mathbb{Z}/q)[t, b]/(t^L + 1, \Phi_{257}(b))$ . Put  $N = 256L$ ,  $b_q = \lceil \log_2 q \rceil$  and  $w = 2b_q + \lceil \log_2 N \rceil + 1$ . A product in the physical basis  $t^i b^v$ ,  $0 \leq i < L$ ,  $1 \leq v \leq 256$ , costs*

$$O(\mathbf{M}_{\text{int}}(Nw) + Nw^2)$$

*bit operations, where  $\mathbf{M}_{\text{int}}(n) = O(n \log(n + 2))$ . For the fixed field  $K = \mathbb{F}_{2^{16}}$ , truncated length- $L$  polynomial multiplication costs  $O(L \log^2(L + 2))$  bit operations.*

*Proof.* Use the exact integer-multiplication theorem of [27]. With coefficient representatives in  $[0, q)$ , pack position  $(i, v)$  into digit  $513i + v$  in base  $2^w$ . Every convolution coefficient is at most  $N(q - 1)^2 < 2^w$ . The extension degree is at most 512, so neither carries nor exponent aliasing cross a digit or a length-513 block. Multiply two  $O(Nw)$ -bit integers and unpack. Fold  $t^L = -1$  and  $b^{257} = 1$ ; eliminate each constant coefficient  $c_0$  by replacing  $c_v$  with  $c_v - c_0$ , using  $1 = -\sum_{v=1}^{256} b^v$ . Each folded coefficient sums at most four unpacked coefficients. These operations cost  $O(Nw)$  bits, and ordinary modular long division costs  $O(Nw^2)$ . Packing and unpacking use sequential bit arrays, not repeated shifts of a growing integer. This computes the same physical-basis product even for composite  $q$ .

For the field, fix an irreducible binary polynomial of degree 16. Pack each binary coefficient at jet degree  $i$  and field degree  $v < 16$  into position  $32i + v$ , in base  $2^{\lceil \log_2(16L+1) \rceil}$ . Convolution counts at most  $16L$  bit products, and field degree at most 30 prevents block aliasing. After the integer product, take each digit's parity, reduce by the fixed field polynomial and truncate to jet degree

below  $L$ . The packed integers have  $O(L \log(L+2))$  bits, giving the stated bound including serialization.

In Proposition 19, replace the ring-product price by this lemma and the owners'  $L^2 \mathcal{F}_K$  product price by  $O(L \log^2(L+2))$ . All operation counts and codec prices are unchanged. For the Gaussian schedule with  $b_q = O(d^2)$  and  $M + \tau + H = O(\sqrt{L}d)$ , a ring product costs  $O(Nd^4)$ . All ring products, including key generation, every input encryption and recovery, therefore cost  $O(N\sqrt{L}d^5)$ . Owner products cost  $O(N\sqrt{L}d^3)$ , other preparation and codecs  $O(N\sqrt{L}d^2)$ . Gaussian table setup costs  $O(\sqrt{\lambda}d^7)$ , draws  $O((H+I+\tau+1)Nd^3)$ , and bounded uniform sampling  $O((H+\tau+1)Nb_qd^2)$ , as proved in Proposition 8. All are absorbed by  $O(N\sqrt{L}d^5)$  for  $d \geq 8$ . Prime generation remains the separate  $\Pi = \text{poly}(d)$  charge. Thus the complete bound is  $\Pi + O(N\sqrt{L} \log^5 L)$ . The Gaussian appendix gives the schoolbook implementation's costs; fast arithmetic yields the smaller upper bound proved here.

Retained objects still need  $O(N\sqrt{L}d^3)$  bits. A fast product may use  $O(Nd^4)$  workspace, bounded conservatively by its time; this is absorbed in the total because  $d = O(\sqrt{L})$ . The  $O(Nb_q)$  streaming-scratch bound describes the schoolbook backend, not this one. Serial writes of public material and ciphertexts are absorbed in their charged bit volume; transfer latency retains its separate scheduling model. Exact arithmetic preserves every same-random-tape ciphertext, so the Gaussian source assumption, reduction, admission and failure budget do not change. No complete Gaussian HE execution or timing comparison is asserted.

Arithmetic consistency was checked on 24 physical-ring cases using independent power-basis multiplication and polynomial long reduction, including composite moduli and both degree folds. Forty truncated-field products were also compared with direct convolution. These finite checks establish agreement on the tested cases; the preceding proof establishes the general identities and complexity bound. They do not benchmark the fast algorithm.

With one 64-bit residue word, the selected native profile has 754 MiB of hint arrays, 17 MiB of public keys, 137 MiB of input arrays and a 2 MiB terminal ciphertext. These are four separate quantities, not their sum interpreted as peak RSS. A streamed prefix needs three polynomial accumulators; the tail can stream relative digits. Retaining prepared spectra, temporary arrays or diagnostic states has additional cost.

The executed-call arithmetic ledger counts 317577472 modular multiplications and 571539456 additions/subtractions for spectral-input, spectral-output evaluation. Optional coefficient export adds 3017728 and 6422528, respectively. It also counts  $14N$  internal CRT reconstructions, 2162688 growing-integer products, 3997696 digit coefficients and 14024704 channel lifts. These counts price executed calls using explicit codelets, not machine cycles or a security estimate. Owner sampling, setup and codec work remain outside this evaluator subtotal.

Terminal delivery need not retain the last rekey. The raw three-component phase is decryptable by the recipient, with an additional ring product and differential output traffic. Appendix B.5 admits this option for the dense core on both

sides. In the native depth-three control it reduces the sufficient prime chain from  $(4, 3, 2)$  to  $(3, 2, 1)$  and raw hint storage from 76 MiB to 36 MiB. Functional checks confirm the tested outputs; they do not transfer a concrete-security guarantee.

**E.5.3 A tight normalized bound for generic multiplication** The main text states and proves Definition 4 and Theorem 11. The following ledger records their accounting consequences.

| Same $B = hL$ independent products | Native jets                                | Reduced SIMD |
|------------------------------------|--------------------------------------------|--------------|
| Useful field symbols per carrier   | $hL$                                       | $hL$         |
| Ciphertext multiplication calls    | $L$                                        | $2L - 1$     |
| Nonlinear depth                    | 1                                          | 1            |
| External inputs / outputs          | $2L / L$                                   | $2L / L$     |
| Additional linear work             | Actual codec/maps Evaluation/interpolation |              |

**Proposition 20 (Finite Whole-Ledger Crossover).** *Fix the common batch of Theorem 11. Let  $M_J, M_B > 0$  price multiplication under each required key policy, and let  $A_J, A_B \geq 0$  charge all other work in the same cost unit. The modeled native total  $LM_J + A_J$  is smaller than the conventional total  $(2L - 1)M_B + A_B$  exactly when*

$$\frac{M_J}{M_B} < 2 - \frac{1}{L} + \frac{A_B - A_J}{LM_B}.$$

*In particular, if  $A_J = 0$ , the condition  $M_J/M_B < 2 - 1/L$  is sufficient; at  $L = 256$  this threshold is 511/256. Unequal setup, codecs, owner work, transfers and recovery belong in the  $A$  charges, not in omitted costs.*

*Proof.* Subtract  $A_J$  from the strict total-cost inequality and divide by  $LM_B > 0$ . For the sufficient condition use  $A_B \geq 0$ . This is a cost-model identity, not a measured latency prediction or a bound on another compiler.

The equal-capacity comparison is an abstract carrier model, not a claim that every field/slot combination is a realizable single-cyclotomic backend. For example, a reduced binary cyclotomic algebra whose slot field is exactly  $K = \mathbb{F}_{2^{16}}$  has at most 2048 slots and dimension 32768. Indeed, writing its conductor as  $2^k c$  with  $c$  odd, reducedness forces  $k \leq 1$  and the field condition gives  $\text{ord}_c(2) = 16$ , hence  $c \mid 65535$  and  $\varphi(c) \leq \varphi(65535) = 32768$ . Thus the abstract  $B = hL = 4096$  carrier at  $h = 16, L = 256$  is not available in that restricted physical family. Products of carriers, larger extension fields and conventional HE using the ramified algebra itself remain eligible with their actual costs. This finite slot ceiling supplies no security-preserving asymptotic separation: the conventional dimension and its secret/error laws must also scale, possibly changing the carrier model and field-symbol utilization. The measured conventional composition control uses eight 512-slot bundles. The theorem is neither a latency lower bound nor a bound for deeper, non-bilinear, unary or publicly specialized algorithms.

In particular, it does not prove the composition speedup reported below. The integer-substitution route below is also outside the model: its encoding is not  $K$ -linear, and one word product is not a rank-one  $K$ -slot product.

**E.5.4 Terminal CRT can compile the entire dense circuit** The terminal comparator is not required to truncate after each product. The projection  $\mathbb{F}[z] \rightarrow \mathbb{F}[z]/(z^L)$  is a ring homomorphism. Thus, for a circuit consisting of additions and products, evaluate the same circuit on untruncated input polynomials and project only at the end. If its untruncated output degree is at most  $D$ , evaluations at  $P > D$  distinct field points determine that output and hence every requested truncated coefficient. This applies to the complete product and circuit circuits, not to arbitrary intermediate coefficient maps or an encrypted-continuation interface.

For the binary field take  $P = 2^r \leq |\mathbb{F}|$  points in an additive subspace. The implemented recursive CRT divides by subspace polynomials of degrees  $2^{j-1}$  with at most  $j$  nonzero terms. At recursion level  $j$ , division and the two child combinations use at most  $(P/2)(j+2)$  field multiplications across its nodes. Interpolation has the same upper bound. Consequently each full transform uses at most

$$T(P) \leq \frac{P}{4}(r^2 + 5r)$$

field multiplications and  $O(P \log^2 P)$  field operations. For  $P = 32, 512, 2048$ , the respective multiplication bounds are 400, 16128, and 90112. These are attainable codec bounds, not optimal transform lower bounds. The recursion and offset tables use  $O(P + \log^2 P)$  field elements; the implemented fixed-field logarithm tables additionally use  $O(|\mathbb{F}|)$  public space and initialization. HELib encoding and ciphertext arithmetic are separate costs, not included in this field-operation count.

For  $c$  independently owned inputs,  $J$  jobs and  $S$  full-field slots per conventional carrier, let  $B = \lceil JP/S \rceil$ . Owners perform  $cJ$  forward transforms and the recipient  $J$  inverse transforms, for at most  $(c+1)JT(P)$  transform multiplications. The streamed conventional evaluator uses respectively

|                | inputs | raw products | adds | outputs |
|----------------|--------|--------------|------|---------|
| product        | $2B$   | $B$          | 0    | $B$     |
| affine product | $3B$   | $B$          | $B$  | $B$     |
| balanced tree  | $12B$  | $7B$         | $4B$ | $B$     |

with  $P$  chosen above degrees  $2(L-1)$ ,  $2(L-1)$  and  $8(L-1)$ , respectively. There are no encrypted rotations or interpolation in these routes. Weight each raw component-pair product, alignment and switch by its actual modulus; the four deep policies and their distinct key/terminal inventories are in Appendix B.5.

If the conventional ring dimension is  $n$ , its uncompressed input traffic is  $2cBnb_Q$  bits before metadata. An  $a$ -part output at modulus  $q$  uses  $anb_q$  bits, summed over the actual terminal carriers; fixed-word RNS replaces  $b_q$  by  $64\ell_q$ . The implemented owner cache holds  $cJP$  field elements. Streaming one group

bounds ciphertext registers independently of the total number of groups, but does not remove the keys, higher-component states, encoder workspaces or that cache. Actual raw decryption is priced as executed by its backend, not by assuming a canonical two-part decryptor. The native arm uses its own ring dimension, admitted chain and lane occupancy. This ledger explains why terminal expansion and unused native lanes can favor different workloads. It does not turn a ciphertext-call ratio into a runtime or security-matched superiority theorem.

**E.5.5 Prepared composition has different competing algorithms** The selected compiler in Section 5 has  $2^k - 1 + \tau$  raw products and depth  $\tau + 1$ , where  $\tau = \log_2 L - k$ . At a common modulus/gadget its hint volume is  $(4 \cdot 2^\tau + 6\tau)gN$  coefficients and its input count is  $2^{k+1} - 1 + \tau$ . Increasing preparation reduces online depth and hints but increases encryption and traffic. This is an attained trade-off, not an optimal compiler or a new classical composition algorithm. Corollary 5 makes one sufficient no-drop modulus dependence explicit. With fixed radix/support and  $b_q = O(k + (\tau + 1) \log(1 + \kappa))$ , its hint storage is  $O((2^\tau + \tau)Nb_q^2)$  bits and its prepared-input storage is  $O((2^k + \tau)Nb_q)$  bits. These are correctness-family upper bounds, not optimized chains or a security-preserving scaling claim.

**Corollary 10 (An attained joint-storage choice).** *In the fixed binary tensor family, as  $L = 2^{s_L} \rightarrow \infty$  with fixed radix/support, choose  $k = \min\{s_L, \lceil s_L/2 + \log_2(s_L + 1) \rceil\}$ . The no-drop construction uses  $O(\sqrt{L} \log L)$  prepared inputs and private products, and  $O(N\sqrt{L}(\log L)^3)$  bits for all prepared inputs and hint rows together. With the schoolbook backend of Proposition 19, its expected one-batch bit work is  $\Pi(q) + O(N^2\sqrt{L}(\log L)^5)$ , including owner preparation and key generation.*

*Proof.* We have  $2^k = O(\sqrt{L}(s_L + 1))$ ,  $2^\tau = O(1 + \sqrt{L}/(s_L + 1))$  and  $b_q = O(s_L^2)$ . Substitute into the preceding storage bounds; the remaining  $O(Ns_L^5)$  term is absorbed since  $s_L^2 = O(2^{s_L/2})$ . Also  $M + \tau + H = O(\sqrt{L}s_L)$ , since  $g = O(s_L^2)$  and  $s_L^3 = O(\sqrt{L}s_L)$ . Substitution in the complete-work bound gives the time claim; field arithmetic has fixed bit cost and  $N = 256L$  here.

This is an attainable upper bound, not an optimum or a separation from conventional HE. It selects  $k = 8$  at  $L = 256$ , not the measured  $k = 4$ ; neither this asymptotic choice nor its modulus is security-qualified.

Corollary 4 gives the joint Gaussian-family version of this bound.

A conventional alternative writes  $A_{ji} = [z^j]g(z)^i$ , so composition is  $Af$ . Consider  $h$  jobs,  $S_B = hr$  field slots per carrier,  $1 \leq r \leq L$ , and  $B = \lceil L/r \rceil$  row blocks. Block  $b$  ends at  $j_b = \min\{(b+1)r - 1, L - 1\}$ . Since  $g(0) = 0$ ,  $A_{ji} = 0$  for  $i > j$ . The  $g$ -owner therefore encrypts only columns  $1, \dots, j_b$  in that block; the  $f$ -owner's  $L - 1$  encrypted broadcasts of  $f_i$ ,  $i \geq 1$ , are reused across all blocks. One further  $f$ -owner input places  $f_0$  only in each job's row-zero position in block zero; other accumulators start at public zero. This supplies  $A_{j0} = \mathbf{1}_{j=0}$  without

an encrypted product. The complete depth-one contraction needs

$$P_B = \sum_{b=0}^{B-1} j_b \text{ products,} \quad P_B + L \text{ inputs,} \quad B \text{ outputs.}$$

There are no encrypted rotations or Hasse maps. All raw outputs may share two power-switch banks to one independent destination key, with one final re-linearization per block. Fresh inputs share one source key and an independently admitted common modulus. For an unscaled parity profile with fresh bound  $F_B$ , ring norm factor  $\kappa_B$  and per-bank error  $\Delta_B$ , the maximum complete-block error is at most

$$F_B + (L-1)\kappa_B(2F_B + 2F_B^2) + \lceil (L-1)\kappa_B/2 \rceil + 2\Delta_B.$$

The corresponding no-wrap test must pass. This uniform bound permits reuse; input and key errors shared across outputs are not independent. Both owners remain independent. When  $r \mid L$ ,  $P_B = L^2/(2r) + L/2 - L/r$ . This strengthened analytical control is not a measurement or a replacement for the per-bundle measurements.

At ring dimension  $n_B$  and common modulus width  $b_B$ , its uncompressed input traffic is  $2n_B b_B(P_B + L)$  bits. If  $n_B = 16S_B$ , this is  $\Theta(16hL^2b_B)$ ; otherwise retain the actual dimension/slot ratio. The Gaussian native bound above may improve traffic against this explicitly materialized matrix endpoint, not every conventional compiler. Powers and columns can be streamed with  $O(hL)$  live owner field elements and  $B$  raw ciphertext accumulators, plus keys and backend workspace. There is no quadratic live-memory lower bound. Nor is output traffic automatically smaller: when  $r \mid L$  and  $n_B = 16hr$ ,  $n_B B = 256L = N$  for  $h = 16$ , before pricing the different output moduli. Classical non-dense composition algorithms [5] and conventional HE over the same ramified plaintext algebra [22] remain eligible alternatives.

For the diagonal alternative used in the measured comparison, define  $(\rho_s v)_j = v_{j+s \bmod L}$  and  $(d_s)_j = A_{j,j+s \bmod L}$ . The diagonal  $d_1$  is identically zero. For  $b \mid L$ , baby-step/giant-step regrouping gives

$$A^f = \sum_{a=0,b,\dots,L-b} \rho_a \left( \sum_{\substack{0 \leq r < b \\ a+r \neq 1}} \rho_{-a}(d_{a+r}) \odot \rho_r(f) \right).$$

Each summand at output  $j$  is  $A_{j,j+a+r} f_{j+a+r}$ , proving the identity. The inner owner prepares the twisted diagonals; the outer owner prepares the babies. The regrouping is classical [24]; here the diagonals are encrypted, so their products are ciphertext-ciphertext, not public-matrix multiplications. For  $1 < b < L$ , the schedule uses  $L + b - 1$  encryptions,  $L - 1$  products,  $L/b$  group relinearizations,  $L/b - 1$  giant rotations and  $L - 2$  additions per bundle. Physical rotations must preserve the full field value, not just slot indices.

Writing  $E, M, K_s, R_s, A_3, A_2$  for fresh encryption, raw product, relinearization, complete rotation including its switch, and raw/canonical addition, the  $L = 256, b = 16$  per-bundle subtotal is

$$271E + 255M + 16K_s + 15R_s + 239A_3 + 15A_2.$$

The executed schoolbook preparation of all powers of  $g$  costs  $O(L^3)$  field operations and  $O(L^2)$  field elements per job; this is an upper bound, not a claim that faster classical preparation is impossible. Prepared babies add  $O(bL)$  moves. Streaming gives  $O(bnb_q)$  bits of ciphertext registers at conventional dimension  $n$ , in addition to keys, owner matrices and backend scratch. Thus neither fewer inputs nor fewer multiplications establishes a complete memory ordering.

Input boundaries matter even without timings. In one explicit unscaled, two-prime, dimension-4096 column control, spectral-input evaluation plus coefficient output uses 112205824 modular multiplications, versus native's 320595200. Charging a particular three-transform encryption implementation reverses that subtotal to 2282181632 versus 635050752, respectively. These are differently priced phases of two attained algorithms, not the measured HELib profiles or an optimum against the lower-input BSGS control. We therefore report complete phase-separated measurements next.

**E.5.6 A further terminal route through exact integer words** Integer arithmetic need not emulate a field reduction at every gate. Proposition 31 embeds binary polynomial lifts in widely spaced integer digits, evaluates the whole fixed circuit without carries or wrap, and recovers the field result after decryption. Combining this with terminal CRT gives sufficient word widths 256 bits for product/shallow arithmetic circuit and 4096 bits for balanced tree. These are integer bounds, conditional on an admitted exact-word backend such as the triangle family [21].

The same construction handles composition's prepared field matrix. At an assumed dimension-65536 backend with sixteen 4096-bit word slots, pack eleven output rows of each job per word. Public triangular zeros and shared encrypted  $f_i$  inputs give 3268 word products, 3245 additions, 3524 input ciphertexts and 24 aggregate outputs, without encrypted rotations. All owner preparation, encoding, keys, level alignment, transfers, backend arithmetic and recipient blockwise recovery remain charged in Appendix G.4. Its word-operation counts are neither ring-product counts nor measured costs. This is an eligible unmeasured alternative, not a new HE primitive or an optimal compiler. The wider terminal integer output can reveal more to the recipient; eligibility uses our explicit absence of recipient privacy.

## F Joint evaluation and Gaussian relay comparisons

### F.1 A same-source relay for unrestricted recovery

The unrestricted task in Definition 1 permits the recipient to recover both operands and compose them. This gives a stronger asymptotic comparator than the restricted one-layer model.

**Proposition 21 (Complete work of the Gaussian relay).** *Use the ring, codec, Gaussian law and exact marginal modulus distribution of Corollary 4. Under its same source assumption, an encrypted-input relay solves the unrestricted task with conditional IND-CPA privacy and negligible correctness failure. At  $d = \log_2 L$ ,  $N = 256L$  and  $h = 16$ , one full batch costs*

$$\Pi_{\text{gen}} + O(Nd^4 + \sqrt{\lambda}d^7 + hLd^3) = \tilde{O}(L)$$

*bit operations, including setup, owners and recipient. Here  $\Pi_{\text{gen}} = \Pi + \text{poly}(d)$  charges interval synthesis and the original bounded prime generator. No native evaluation bank is generated.*

*Proof.* Retain the original synthesized prime interval and its bounded candidate routine. Its exponent is computed from the formulas of Proposition 5 in  $\text{poly}(d)$  work, without preparing any ciphertext or hint row. Publish one ordinary key  $(B, A) = (-As + 2e_{\text{pk}}, A)$ . Each owner encodes its unprepared operand into one binary physical polynomial and uses the existing affine encryption. Forward the two ciphertexts. The recipient decrypts and decodes both, then computes the  $h$  clear compositions.

The public source experiment projects to this one-vertex, two-query manifest by retaining the same modulus and discarding unused metadata and surplus source rows. Retain two rows plus the  $\lambda$  reserved pivots. Both real and null distributions are exactly the required ones; a distinguisher for this projection is a distinguisher for the original source. Thus no hardness premise for a different modulus distribution is introduced. The honest relay samples the marginal directly, rather than generating unused native keys. The existing Gaussian theorem applies with  $\bar{m} = 2$  and  $\bar{H} = 4$ , padding the one row and two encryption transitions. Its advantage bound has source factor eight and the same honest/simulation coupling terms. This is an asymptotic reduction, not an equal concrete-security estimate or privacy against the recipient.

There are eight Gaussian polynomial draws:  $s, e_{\text{pk}}$  and the three fresh polynomials for each encryption. There is one uniform mask  $A$ . On the primitive coefficient cap  $\lambda$ , each fresh error is at most  $F = (2\kappa\lambda + 1)\lambda$ , and the original modulus already admits every fresh state. With the table discrepancy  $\rho_\lambda$  of Lemma 16, public generation failure, bounded sampling failure or incorrect output has probability at most

$$(N + 1)2^{-4d^2} + 8\rho_\lambda + 16N2^{-2\lambda}.$$

The terms charge uniform-coordinate and prime exhaustion, eight Gaussian couplings, and their primitive coefficient tails. Privacy is not conditioned on correctness or non-abort.

Key generation uses one ring product, the two encryptions four, and the two decryptions two: seven in total. Lemma 21 prices each at  $O(Nd^4)$ . Gaussian tables cost  $O(\sqrt{\lambda}d^7)$ , eight draws  $O(Nd^3)$ , and the one bounded uniform mask  $O(Nb_qd^2)$ . Four tensor codec calls cost  $O(N(256 + d))$ . Other arithmetic, ciphertext writes, forwarding and output cost  $O(Nb_q + Nd)$ . Kinoshita–Li [28, Theorem 1 and Section 2.3] permit the bit-cost multiplication bound  $O(Ld^2)$  to be substituted in clear composition, giving  $O(hLd^3)$ . This is charged nonlinear recipient work, not a homomorphic compilation of their algorithm. Summing proves the bound;  $\Pi_{\text{gen}}$  is polynomial in  $d$  and  $d^3/\sqrt{\lambda}$  is bounded. Transmission latency remains a separately scheduled charge. No Gaussian HE execution or finite crossover is asserted.

*A lower bound for the declared reference, not just two upper bounds.* For the balanced schedule,  $I = 2^{k+1} - 1 + \tau = \Omega(\sqrt{L}d)$ . Its specified initial ciphertext arrays write  $2INb_q$  bits. Hence even the complete warm batch has work  $\Omega(N\sqrt{L}d)$ . Lemma 20 also gives an  $\Omega(IN)$  expected and overwhelming-probability write bound for compressed encodings of these carriers that are correct and private over their full message domain. Fresh-state admission and Gaussian multi-query privacy cover arbitrary initial plaintext carriers. No entropy of the prepared image is assumed. Compared with the relay’s asymptotic  $O(Nd^4)$  upper bound, this gives a ratio growing at least as  $\Omega(\sqrt{L}/d^3)$  for that reference and encoding contract. It is not a measured speedup or a lower bound for every JetHE compiler.

Even changing  $k$  within the explicitly serialized reference does not remove a first-use volume floor. For  $x = 2^k$ ,  $g \geq 1$  and  $H = g(2^{d-k+1} + 3(d - k))$ ,

$$I + H \geq x + 2L/x \geq 2\sqrt{2L}.$$

Publishing the  $H$  two-polynomial hint rows and  $I$  inputs therefore requires  $\Omega(N\sqrt{L})$  bit writes. This statement is about the declared explicit format; it does not require retransmitting keys on warm batches or exclude compressed correlated evaluation material. The source-security comparison above uses only the original balanced modulus distribution, not a newly assumed family for other parameters.

*Prefix retuning over repeated fresh batches.* Consider the direct realization of Proposition 19 that materializes every full-ring product and applies each hint row separately. Its setup and complete fresh-batch ring-product counts are

$$R_{\text{setup}} = H + 2\tau + 2, \quad R_{\text{batch}} = 7M + 5\tau + 2H + 3.$$

The latter includes encryption, evaluation and decryption. Reusing setup for  $B$  fresh batches costs  $R_{\text{setup}} + BR_{\text{batch}}$  products: every batch still consumes all  $H$

hint rows. Since  $M = 2^k - 1$ ,

$$M + H \geq 2^k - 1 + 2gL/2^k \geq 2\sqrt{2gL} - 1,$$

this implementation writes  $\Omega(BN\sqrt{L})$  bits for every admitted prefix. The balanced fast-arithmetic upper bound is  $O(BN\sqrt{L}d^5)$ , including setup asymptotically. Thus optimizing only the prefix within these direct loops leaves a  $\tilde{\Theta}(BL^{3/2})$  best exponent for a polynomially bounded, admitted horizon. This does not lower-bound joint row algorithms, compressed arithmetic, input-dependent pruning or other compilers. The publication-only quantity  $H + BI$  omits recurring hint application and cannot establish a complete-work improvement by amortization alone.

Reuse also enlarges the honest Gaussian catalogue to  $G_B = H + 3\tau + 3 + 3BI$  in (25); the setup-only uniform count remains  $U = H + \tau + 1$ . The query hybrid must use a padded bound on  $2\tau + 2 + BI$ , with the corresponding simulator coupling. Changing the prefix requires its own admission and source-generator argument. Evaluating these formulas for 192 profiles and 1122 reuse horizons, including source-row projection to the original modulus interval, gives finite ring-count improvements for some profiles even after that projection; they are not timing measurements or a global optimality result.

The relay's nonlinear recipient computation lies outside linear recovery, so this adverse total-work result is consistent with the one-layer separation. It prevents that separation from establishing a broad composition advantage. Whether the quantified placement of nonlinear work is independently useful remains a substantive question.

*Recipient computation includes decryption.* For the same Gaussian modulus and coefficient-form arithmetic, let  $\mathcal{M}_q$  price a physical ring product,  $\mathcal{C}$  a tensor codec, and  $\mathcal{P}_K(L)$  clear composition of all  $h$  jobs. Native returns one two-component ciphertext; the relay returns two. Their recipient work is bounded respectively by

$$\begin{aligned} C_{\text{rec,native}} &\leq \mathcal{M}_q + \mathcal{C} + O(Nb_q), \\ C_{\text{rec,relay}} &\leq 2\mathcal{M}_q + 2\mathcal{C} + \mathcal{P}_K(L) + O(Nb_q). \end{aligned}$$

These bounds charge centering, parity, decoding and output movement; initial key/codec installation is also absorbed. By the same exact arithmetic and clear-composition bounds, both are  $O(Nd^4)$ . Writing the required explicit  $hL$  field coefficients costs at least  $hsL = N$  bit writes, with  $h = s = 16$ . Thus both recipient procedures have  $\tilde{\Theta}(L)$  work. This output lower bound rules out a polynomial-exponent advantage over the relay for this explicit-output task; it does not establish equal timings or exclude a polylogarithmic or constant-factor difference. Native avoids the relay's clear composition, one decryption and one codec call, and halves its two-component ciphertext payload at the common modulus. Those are precise operation and traffic savings, not a unique asymptotic recipient-work separation or an observed device constraint.

This compares protocols for one task. Standard HE compactness bounds decryption independently of the evaluated circuit; mere correctness also permits relaying a circuit and its encrypted inputs [23, Definition 2.1.2 and preceding discussion]. The present cost comparison does not replace that requirement, refute compact homomorphic evaluation or prove a lower bound for all compact schemes. Our linear-recovery model is an additional algorithmic restriction, not the definition of compactness. Indeed, for this fixed composition family with  $L = \Theta(\lambda)$ , the relay has a fixed polynomial-time recovery algorithm, so compactness alone does not automatically exclude it. This supplies no compact evaluation procedure for arbitrary circuits.

The specified measured terminal profile is different: it returns three components over two primes, with secret-square spectra prepared and charged during setup. Its recipient forms two spectral products, performs one inverse transform per limb, then CRT, parity and tensor decoding. It must not be assigned the Gaussian reference’s one-product decryption count. Nor do these two spectral products each require a fresh complete transform. The four-configuration recipient-time frontier in Appendix I.4 remains a separate recorded comparison; its hypothetical recipient weight is not an application fact.

## F.2 Joint Hasse Evaluation Across Fresh Batches

The limit in Section F.1 concerns separate full-ring row applications. Joint evaluation permits a different algorithm. We give its exact identity and complete work bound, then report finite implementation measurements, including their unfavorable preparation costs. The arithmetic uses Strassen’s established seven-product recursion [40]. Reductions from homomorphic linear algebra to ordinary matrix arithmetic, including relative-module decomposition, are prior art [4, Sections 1.1–1.2 and 4.4]. The result here is an application to unchanged Hasse hints with a charged composition ledger, rather than a new general matrix or key-switching primitive.

**F.2.1 An Exact Relative-Module Identity** Fix a common modulus  $q$  in the Gaussian construction and write

$$R_q = (\mathbb{Z}/q\mathbb{Z})[t, b]/(t^L + 1, \Phi_{257}(b)), \quad N = 256L.$$

At a tail stage of order  $r$ , put  $e = 2r$  and

$$S_e = (\mathbb{Z}/q\mathbb{Z})[Y, b]/(Y^{L/e} + 1, \Phi_{257}(b)), \quad Y = t^e.$$

Since  $e \mid L$ , the monic relations give a unique expansion  $c = \sum_{i < e} t^i c_i(Y, b)$ . Thus  $R_q$  is free of rank  $e$  over  $S_e$ . This is an arithmetic representation even for composite  $q$ ; it introduces no smaller-dimensional secret or sample distribution.

For  $B$  fresh inputs at this stage, decompose  $c_{1,z}$  in these coordinates and extract the original balanced radix digits  $d_{i,j,z} \in S_e$ , where  $j < g$  and  $z < B$ . Use the identical canonical representative and tie rule as the reference. Coordinate

selection commutes with coefficientwise digit extraction. Consequently  $d_j(c_{1,z}) = \sum_i t^i d_{i,j,z}$ , including for shared identity alignment. Let  $K_{i,j,v}$  denote component  $v \in \{0, 1\}$  of the unchanged hint row for  $H_r(t^i s)$ , and expand

$$K_{i,j,v} = \sum_{u < e} t^u A_{(v,u),(i,j)}.$$

Form  $A \in S_e^{(2e) \times (eg)}$  and  $D \in S_e^{(eg) \times B}$  with  $D_{(i,j),z} = d_{i,j,z}$ .

**Proposition 22 (Exact joint transport).** *If  $C = AD$ , the reference transport output in batch  $z$  is exactly*

$$\text{out}_{v,z} = \mathbf{1}_{v=0} H_r(c_{0,z}) + \sum_{u < e} t^u C_{(v,u),z} \quad \text{in } R_q.$$

*Replacing each stage's Hasse bank sum by this formula preserves every retained ciphertext on the same input, keys and random tapes.*

*Proof.* Substitution into  $\sum_{i,j} d_{i,j,z} K_{i,j,v}$  and commutation of the  $S_e$  coefficients with  $t^u$  give the displayed product. Multiplication in  $S_e$  includes the sign  $Y^{L/e} = -1$ . Adding  $H_r(c_{0,z})$  completes the reference formula. The shared digits reconstruct the original identity alignment. Prefix aggregation, raw products, product rekeys and stage order remain unchanged, so induction proves the last claim.

The entries of  $A$  are deterministic coordinates of published rows; they are not independent RLWE samples. Intermediate matrix sums are public arithmetic temporaries. Reducing them modulo  $q$  preserves the final phase exactly, so they incur no additional semantic noise term.

**F.2.2 Charged Matrix Arithmetic** For  $e = 2^s$ , full Strassen recursion uses  $7^s$  products and  $6(7^s - e^2)$  additions or subtractions over  $S_e$ . These follow from  $m_s = 7m_{s-1}$  and  $a_s = 7a_{s-1} + 18(e/2)^2$ , with  $m_0 = 1, a_0 = 0$ . There is no division, so zero divisors pose no problem. Partition  $A$  into two block rows and  $g$  block columns of width  $e$ , and zero-pad the final batch block of  $D$ . Accumulating each square product into an initially zero output block gives exact counts

$$\begin{aligned} M_e &= 2g \lceil B/e \rceil 7^s, \\ A_e &= 2g \lceil B/e \rceil [6(7^s - e^2) + e^2]. \end{aligned} \tag{30}$$

These counts include padding and output accumulation. For  $B < e$  the ceiling must be retained; a fully occupied estimate is inapplicable. Public dimensions may instead select the original direct algorithm.

An  $S_e$  scalar has  $n_e = N/e$  coefficients, not unit cost. Lemma 21 applies with negacyclic length  $L/e$  and the same fixed degree-256 extension factor. For  $w_e = 2b_q + \lceil \log_2 n_e \rceil + 1$ , its product bound is

$$P(n_e, b_q) = O(M_{\text{int}}(n_e w_e) + n_e w_e^2), \quad M_{\text{int}}(a) = O(a \log(a + 2)).$$

An addition costs  $O(n_e b_q)$ . If  $b_q = O(d^2)$ ,  $d = \log_2 L$ , then  $P(n_e, b_q) = O(n_e d^4)$  uniformly in  $e \leq L$ . Thus a stage costs  $O(gNe^{\omega-1} \lceil B/e \rceil d^4)$ , where  $\omega = \log_2 7$ . For  $B \geq e$  this is  $O(BgNe^{\omega-2} d^4)$ .

The public matrix contains  $2e^2 g$  elements of  $S_e$ , or  $2egN$  coefficients: precisely the volume of the original  $eg$  two-component full-ring rows. Copying or reordering it costs  $O(egNb_q)$  bit work; retaining both layouts incurs both storage costs. Forming  $D$  costs  $O(gBNb_q)$ , and forming outputs costs  $O(BNb_q)$ . Sequential recursive calls can use  $O(e^2)$  elements of  $S_e$  as matrix scratch, since  $S(e) \leq S(e/2) + Ce^2$ . Materializing one padded digit block costs  $O(geNb_q)$  bits, besides inputs, output and scalar-product scratch. No cache of all recursive public descendants is required by this asymptotic algorithm. These are storage bounds, not claims of low finite memory or free conversion.

**F.2.3 Complete Work and the Source Required by Retuning** Set  $x = 2^k$ ,  $\tau = d - k$ ,  $E = L/x$ ,  $M = x - 1$ ,  $I = 2M + 1 + \tau$  and  $H = g(2E + 3\tau)$ . The ranks of the tail stages are  $E, E/2, \dots, 2$ ; their sum is  $2E - 2$ , including the empty case  $\tau = 0$ . Replacing just their Hasse sums leaves

$$R_{\text{other}} = H + 2\tau + 2 \\ + B[7M + 5\tau + 3 + 2g(2 + 3\tau)]$$

full-ring products. The first line includes setup. The second includes all owner encryptions, raw products, rekeys, identity alignment and recipient decryption. As a check on the subtraction,

$$R_{\text{other}} + B \sum_e 2eg = R_{\text{setup}} + BR_{\text{batch}}$$

with the direct counts in Section F.1.

Put  $G_B = H + 3\tau + 3 + 3BI$  and  $U = H + \tau + 1$ . Using the public synthesis cost  $\Pi_{\text{gen}}$  and shared sampler-table cost  $T_{\text{tables}}$  of the Gaussian family, one complete bit-work bound is

$$\begin{aligned} & \Pi_{\text{gen}} + T_{\text{tables}} + O\left(R_{\text{other}} P(N, b_q) \right. \\ & \quad + \sum_e [M_e P(N/e, b_q) + A_e(N/e) b_q] \\ & \quad + B(2 + 3\tau) N b_q + B\tau g N b_q + g N b_q \sum_e e \\ & \quad + Bh(M - k) L d^2 + BhL(M + d) + B(I + 1) N (256 + d) \\ & \quad + G_B N d^3 + U N b_q d^2 \\ & \quad \left. + [H + \tau + 1 + B(M + \tau + g(2 + 3\tau) + I + 1)] N b_q\right). \end{aligned}$$

The lines charge ring and matrix work; shared decomposition, digit copying and public conversion; owner preparation and codecs; Gaussian and uniform sampling; and the remaining arrays, public maps and wires. The matrix operations

charge the removed Hasse-row reads and writes; they are not silently retained as unpriced per-row loops. Setup masks, payloads, secret squares, tables and every party's serialization are included. Network scheduling is separate from local computation.

**Proposition 23 (Attained joint-batch work bound).** *Use fixed radix,  $h = 16$ ,  $N = 256L$ ,  $L/2 < \lambda \leq L$  and a declared polynomial batch horizon. For each selected  $1 \leq k \leq d$ , use that profile's admitted Gaussian generator. If  $B \geq E$ , the complete work of the specified algorithm satisfies*

$$W_{\text{joint}}(B, k) = \tilde{O}(N[E + Bx + BE^\alpha]), \quad \alpha = \log_2 7 - 2.$$

Choosing the nearest admissible integer to  $\alpha d/(1 + \alpha)$  gives

$$\frac{W_{\text{joint}}(B, k)}{B} = \tilde{O}(L^{2-1/(\omega-1)}), \quad B \geq E = \Theta(L^{1/(\omega-1)}). \quad (31)$$

The work exponent is about 1.446705244 and the batch-threshold exponent about 0.553294756.

*Proof.* The existing synthesis and admission give  $b_q, g = O(d^2)$  for each profile, and  $T_{\text{tables}} = O(\sqrt{\lambda}d^7)$ . For  $B \geq E$ , equation (30) and  $\sum_e e^\alpha = O(E^\alpha)$  bound all matrix work by  $\tilde{O}(BNE^\alpha)$ . The setup and matrix conversion cost  $\tilde{O}(NE)$ ; the remaining ring, owner, sampler, codec and array terms are  $\tilde{O}(BNx)$ . The declared public-generation bound is absorbed as well. Since  $xE = L$ , the selected prefix has  $x = \Theta(L^{\alpha/(1+\alpha)})$  and  $E = \Theta(L^{1/(1+\alpha)})$ . Also  $E/B \leq 1$ , proving the bound.

Here  $B$  counts fresh batches, each containing  $h$  lanes, sharing a bank and available for joint evaluation. This is a throughput and total work result; it supplies no isolated-job latency bound. Keeping the old balanced prefix would retain its dominant owner-input cost and would not yield the improved complete exponent. Retuning the prefix therefore matters, as do its bank volume and buffering costs.

Exact output equivalence preserves the fixed profile's correctness, noise recurrence and privacy experiment by deterministic public postprocessing. Changing  $k$  and its admitted modulus interval is a different public generator. Its full-dimensional source remains a premise; no reduction to the original balanced source or certified 128-bit setting follows. In particular, with the existing simulator coupling errors, the source rows and query loss are

$$\begin{aligned} \bar{m} &= \begin{cases} \max(2, 2g), & \tau = 0, \\ \max(2, 3g, Eg + 1), & \tau \geq 1, \end{cases} \\ \bar{H} &= \text{nextpow}2(2\tau + 2 + BI), \\ \text{Adv}_{\text{scheme}} &\leq 2\bar{H} \text{Adv}_{\text{source}} + 2^{-\lambda} + 2\eta_{\text{hon}} + 4\bar{H}\eta_{\text{sim}}. \end{aligned}$$

The honest failure bound is  $(UN + 1)2^{-4d^2} + G_B\rho_\lambda + 2G_BN2^{-2\lambda}$ , using the sampler error  $\rho_\lambda$  from the Gaussian-family analysis. The polynomial horizon keeps

its sampling terms negligible. These failure exponents do not assign numerical security bits.

The result improves the best  $\tilde{\Theta}(BL^{3/2})$  exponent within the separately materialized direct row loops. It is not a lower bound or optimum for joint algorithms. Eligible BFV/BGV compilers can adopt exactly this arithmetic method. The broader recipient relay of Proposition 21 still has  $\tilde{O}(BL)$  complete work when its recovery interface is allowed. Thus neither the exponent nor batching establishes broad HE superiority for other reconstruction interfaces.

**F.2.4 Finite Cached-Matrix Measurements** The measured trit/CBD<sub>20</sub> backend is a separate realization. Its paid Hasse-8 bank admits a mixed spectral identity: for  $d = \sum_{i<8} t^i(u_i + t^8 v_i)$ , the bank sum is

$$\sum_{i<8} [u_i K_i + v_i (-t^8 K_i + t^i K_{\text{id}})].$$

This uses the original identity bank and digits. The matrix implementation forms these public virtual rows and performs one Strassen level on sixteen columns, one per batch, leaving the other stages and original arithmetic library unchanged. Precomputed residue and residue/quotient variants require 336 and 672 MiB respectively.

Sixteen distinct synthetic input bundles share the public banks. The experiment uses one CPU. Against the serial reference, the grouped baseline and each candidate agreed on all  $16 \cdot 22 = 352$  retained states, or 133,169,152 words per arm. Scalar reference computations, partial blocks and rejection tests also agreed. These inputs contain no freshly sampled secrets or encryption coins. After one untimed correctness/warm-up group per arm, nine groups ran in the predeclared order baseline, residue, quotient, quotient, residue, baseline, baseline, residue, quotient. Median evaluator times were 9.350624 seconds for the baseline, 8.982119 for residue and 8.594572 for quotient. Additional complete cache preparations took 0.370740 and 0.709187 seconds.

The predeclared selection thresholds were an evaluator speed ratio of at least 1.05 and a preparation-inclusive ratio of at least 1.02. Residue ratios were 1.041026 and 0.999761; quotient ratios were 1.087968 and 1.005037. Neither candidate met both thresholds: the quotient cache improved evaluator time but saved only about 0.5% after preparation. The measured process peak of 5505.97 MiB included test inputs, both caches and correctness traces; it is not isolated HE memory. These synthetic-input measurements do not establish a complete encrypted-workflow improvement.

A separate cache-construction experiment replaced  $\lfloor w2^{64}/p \rfloor$  by exact folds for  $p = 2^{60} - c$ ,  $0 \leq c < 2^{31}$ . Put  $b = 2^{60}$  and

$$\begin{aligned} t &= 16wc, & a &= \lfloor t/b \rfloor, & s &= (t \bmod b) + ac, \\ d &= \lfloor s/b \rfloor, & v &= (s \bmod b) + dc. \end{aligned}$$

For  $0 \leq w < p$ , the exact quotient is  $16w + a + d + \mathbf{1}_{v \geq p}$ . Indeed  $t = ap + s$ ,  $s = dp + v$  and  $0 \leq v < 2p$ ; for  $c > 0$ ,  $a < 16c$ ,  $s < b + 16c^2 < 65b$ ,  $d \leq 64$

and  $v < b + 64c < 2p$  because  $66c < b$ . The case  $c = 0$  is immediate. The native implementation uses 128-bit intermediates for  $t, s$ .

This constructor produced a byte-identical 672 MiB cache and identical matrix outputs. Its six complete-preparation samples were ordered division, folding, folding, division, division, folding, after one untimed construction per arm. Division/folding medians were 0.446372/0.422265 seconds, ratio 1.057090, below the predeclared 1.25 selection threshold. All allocation, zeroing, validation, powers, matrix formation and quotient writes were charged. These samples must not be combined with the other experiment's evaluator samples to produce a workflow ratio.

A sixteen-batch encrypted experiment would require the declared source-gap factor 1138, compared with factor 158 for two batches. No such encrypted experiment was run. The complete HE timing results therefore concern the fixed-multiplier implementation, not these synthetic matrix variants. The arithmetic measurements neither establish a substantial complete-workflow gain nor supply a numerical security level.

**F.2.5 A One-Layer Comparator May Also Batch Jointly** The work comparison must allow the competing compiler to exploit the same batch availability. Put  $J = hB$  and consider the ordered tuple  $(f_a(g_a) \bmod z^L)_{a < J}$  on the full Cartesian input domain. Each owner knows all of its  $J$  inputs before preparation. Allow arbitrary deterministic preparation across those inputs, shared input-independent public material, cross-batch affine mixing, mixed operands, one parallel layer of fixed binary-bilinear primitives, fixed binary-linear recovery and unilateral bypasses. Each operand carrier still has at most  $N = hsL$  plaintext bits. No blockwise preparation or separate evaluation is assumed. These are the exact clean-plaintext semantics of the existing one-layer model.

**Lemma 22 (Joint-batch coefficient and input widths).** *For  $K = \mathbb{F}_{2^s}$  and dyadic  $L \geq 8$ , let  $C_L = 5L^2/64 + (d+1)L/16$ . In this model the initial inner-owner width and the sum of all primitive operand widths are each at least  $hBC_L$  bits. Consequently mixed primitive calls and initial inner carriers number at least*

$$\left\lceil \frac{BC_L}{2sL} \right\rceil \quad \text{and} \quad \left\lceil \frac{BC_L}{sL} \right\rceil,$$

*respectively. Owner-separated primitives need only their inner operand widths on the left. The initial outer width is at least  $hBs(L-1)$  bits.*

*Proof.* Lemma 19 supplies  $C_L$  independent  $K$ -valued coefficient functions  $Q_t$  on one Boolean inner-input block, each vanishing at zero. On the full joint domain set  $Q_{a,t}(G) = Q_t(g_a)$ . In a relation  $\sum_{a,t} c_{a,t} Q_{a,t} = 0$ , set all  $g_b = 0$  except one chosen block  $a$ . Its single-block independence forces all  $c_{a,t} = 0$ . Repeating this for every block gives dimension  $hBC_L$ . This proves a direct sum of centered function spaces; no general tensor-rank additivity premise is used.

Write a primitive's mixed operands as  $A(F) + B(G)$  and  $C(F) + D(G)$ , where each map can depend across its owner's batches. With centered maps  $\Delta A(F) =$

$A(F) - A(0)$  and similarly for the others, its mixed four-input difference is

$$\gamma(\Delta A(F), \Delta D(G)) + \gamma(\Delta B(G), \Delta C(F)).$$

Fix  $F$  to have  $f_a = z^i$  in one job and zero elsewhere. The target's  $j$ -th difference is  $[z^j]g_a^i$ . After fixed binary-linear recovery, it lies in the  $K$ -span of the centered Boolean coordinate functions of all  $B, D$ . The number of generators is at most the summed operand widths, whereas the targets contain the  $hBC_L$  independent functions above. There is no additional factor  $s$  in this Boolean witness. Affine preprocessing places all those generators in the span of the initial inner coordinates, proving their width bound even with arbitrary joint preparation.

For the outer owner, take  $g_a = z$  in one job and all other inner inputs zero, and subtract the all-zero inner input. The recovered nonconstant coefficients are  $f_{a,j}$ . The span of Boolean initial coordinates is Frobenius-stable. The functions  $f_{a,j}^{2^u}$ ,  $j \geq 1$ ,  $0 \leq u < s$ , are independent over  $K$  on the full outer domain: they are distinct reduced coordinate monomials of degree below  $|K|$ . This gives  $hBs(L-1)$  dimensions. Dividing the width bounds by the maximum carrier widths completes the count statements.

For owner-separated initial carriers, their total number is therefore at least

$$\left\lceil \frac{BC_L}{sL} \right\rceil + \left\lceil \frac{B(L-1)}{L} \right\rceil.$$

Extra input-dependent public data belongs in these widths. Independence in the domain means independently admissible plaintext choices, not independent message draws or merely fresh encryption coins. On the publicly promised diagonal  $g_0 = \dots = g_{J-1}$ , the  $Q_{a,t}$  coincide, so their span has only  $C_L$  dimensions. Also, repeated uncentered constants span one dimension. These are genuine boundaries of the isolation argument.

**Corollary 11 (Complete work against a joint one-layer compiler).** *Set  $h = s = 16$ , let  $L/2 < \lambda \leq L$ , and fix a polynomially bounded batch horizon. Suppose the initial stream is correct and private over its full carrier message domain, with negligible errors as in Lemma 20, for an efficient prepared-input family. Every one-layer compiler above requires  $\Omega(BL^2)$  complete bit work except negligibly and in expectation. Explicit materialization of the prepared plaintext coordinates is an alternative sufficient contract giving a deterministic bound.*

*Under its own admitted Gaussian source and  $B \geq E$ , Proposition 23's retuned deeper compiler has complete work  $O(BL^{2-\beta} \log^7 L)$  for  $\beta = 1/(\log_2 7 - 1)$ . The ratio of the lower bound to this attained upper bound is  $\Omega(L^\beta / \log^7 L)$ .*

*Proof.* Apply the secure-length lemma once to the whole joint initial stream, with full message width  $V \geq hBC_L$  and  $t = \lceil V/2 \rceil$ . Combining decoding keys is needed only for this counting argument; all input-dependent descriptors are included in the encoded string. The probability of writing at most  $\lfloor V/2 \rfloor$  bits is at most  $\epsilon_B + \Delta_B + 2^{1-\lceil V/2 \rceil}$ . Thus the stream's writes alone cost  $\Omega(BL^2)$  except negligibly and in expectation. No independent stream per batch or entropy

bound on actual prepared messages is assumed. Fused preparation/encryption and joint compression remain covered when they retain the full-domain contract.

For an explicit bound, substitute  $b_q, g = O(d^2)$ ,  $\tau \leq d$  and  $P(N/e, b_q) = O((N/e)d^4)$  in Section F.2.3. Matrix work is  $O(BNE^\alpha d^6)$ . Remaining full-ring work is at most

$$O(N[Ed^6 + d^7 + Bxd^4 + Bd^7]).$$

The sampler, owner, codec, representation and array terms are bounded by  $O(N[E + Bx]d^7)$ . The  $O(\sqrt{\lambda}d^7)$  tables and the retained polynomial-in- $d$  public generation are absorbed for sufficiently large  $L$ . Hence the complete bound is  $O(N[E + Bx + BE^\alpha]d^7)$ . At the selected prefix and  $B \geq E$  this is  $O(BL^{2-\beta}d^7)$ . Dividing the actual lower bound by this attained upper bound proves the ratio. It does not compare two upper bounds or claim a measured crossover.

The lower bound allows joint batching by the comparison compiler. Its scope remains the exact binary one-layer model and the stated encoding contract. Privacy on a promised prepared image does not supply full-domain correctness. Nonlinear recipient recovery, deeper conventional compilers and bounded-error randomized functional programs outside this exact model are not excluded. The upper source is that of its retuned Gaussian generator, with the horizon-dependent loss and failure bound already stated; it is not a numerical security match or a reduction to the balanced-profile source.

At  $L = 256, B = 16$ , this weak bound gives eleven mixed calls and 37 initial carriers, supplying no favorable finite comparison. The matrix-cache measurements and the complete relay comparison are given in Appendices F.2 and F.1.

### F.3 A Sharper Joint Bound from Known Matrix Algorithms

The Strassen algorithm in Appendix F.2 is a concrete arithmetic choice, not a limit on the identity. We now apply established rational matrix algorithms to the same Hasse product and charge their integer arithmetic. This improves the theoretical upper bound without changing the one-layer lower-bound model or establishing a finite implementation gain.

**F.3.1 Exact Rational Kernels and Bit Costs** Let  $\omega_{\mathbb{Q}}$  be the matrix multiplication exponent over  $\mathbb{Q}$ , defined using exact tensor rank. For fixed  $\theta > \omega_{\mathbb{Q}}$ ,  $\theta > 2$ , fix integers  $a \geq 2, R$  and a rational rank- $R$  bilinear decomposition of the  $a$ -by- $a$  product with  $\log_a R < \theta$ . Such a fixed kernel follows from the definition of the exponent [3, Section 3.4]. Its size and coefficients depend on  $\theta$ , not on  $L, q, B$  or the security parameter. That paper works over an arbitrary field and gives  $\omega < 2.371339$ . We specialize to  $\mathbb{Q}$  and use  $\theta = 2.372$  with strict slack. The published theorem supplies a fixed rational kernel; its constants are independent of the cryptographic parameters.

**Lemma 23 (Integer transfer of a fixed rational kernel).** *Two  $m$ -by- $m$  integer matrices with entries of bit length at most  $K$  can be multiplied exactly in  $O(m^\theta M_{\text{int}}(K + \log(m + 2)))$  bit operations by this fixed kernel, where*

$M_{\text{int}}(u) = O(u \log(u + 2))$ . The kernel does not require division modulo any ciphertext modulus.

*Proof.* Write its scalar output as  $C_k = \sum_r w_{kr} (\sum_i u_{ri} A_i) (\sum_j v_{rj} B_j)$ . Choose a fixed positive common denominator  $D$ . The integer coefficients  $U = Du, V = Dv, W = Dw$  give  $D^3 C_k$ . Reconstruct and divide exactly by  $D^3$  in  $\mathbb{Z}$  at each recursive node. Induction proves that each child returns its integer block product, and the tensor identity makes every reconstruction numerator divisible by  $D^3$ . This works even when  $D$  is a nonunit modulo  $q$ .

Each descent applies a bounded-coefficient linear map separately to each input. After  $O(\log m)$  levels, leaf inputs have  $K + O(\log m)$  bits. Products have degree one in each input, and the fixed reconstruction maps increase coefficient heights by only a polynomial in  $m$ . Thus all intermediate integers have  $O(K + \log m)$  bits. Padding to the next power of  $a$  changes dimension by less than a fixed factor. Recursion uses  $O(m^\theta)$  scalar operations, including fixed-integer scalings and exact divisions. Their stated bit price follows from exact integer multiplication [27]; fixed-integer division is linear. The chosen kernel is part of the fixed algorithm description, so no per-instance search or synthesis is omitted.

**F.3.2 Packing the Whole Quotient-Ring Matrix Product** At a Hasse stage retain  $e \mid L$ ,  $n_e = N/e = 256\ell$ ,  $\ell = L/e$ , and  $S_e$  from Section F.2.1. Split the  $(2e)$ -by- $(eg)$  times  $(eg)$ -by- $B$  product into  $2g\lceil B/e \rceil$  square products. For one square product choose physical coefficients in  $[0, q)$  and put

$$w = 2b_q + \lceil \log_2(en_e) \rceil + 1, \quad b_q = \lceil \log_2 q \rceil.$$

Pack the coefficient of  $Y^i b^v$ ,  $i < \ell$ ,  $1 \leq v \leq 256$ , at position  $513i + v$  in base  $2^w$ . Each entry has  $O(n_e w)$  bits. In the entire square product, a summed convolution coefficient is at most  $en_e(q-1)^2 < 2^w$ . The factor  $e$  charges the matrix sum; the scalar-product carry bound alone would be insufficient. Since the largest  $b$  exponent is 512, no digit positions alias.

Apply Lemma 23 to the packed integer matrices. Although intermediate values can be signed, the final exact product has the ordinary nonnegative convolution digits. Unpack, fold  $Y^\ell = -1$  and  $b^{257} = 1$ , eliminate  $b^0$  using  $1 = -\sum_{v=1}^{256} b^v$ , and reduce modulo  $q$ . These operations return exactly the product over  $S_e$ . Padding with zeros does not increase the carry bound. No inverse modulo  $q$  is used.

For  $d = \log_2 L$ ,  $b_q = O(d^2)$ , the packed input length is  $K = O(n_e d^2)$  and  $\log(K + \log e) = O(d)$ . Integer matrix work is  $O(e^\theta n_e d^3)$ . Packing by direct digit writes, unpacking, quotient folds and modular long division cost  $O(e^2 n_e d^4)$ . Since  $\theta > 2$ , their sum is  $O(Ne^{\theta-1} d^4)$ . The whole stage therefore costs

$$O(gNe^{\theta-1} \lceil B/e \rceil d^4), \quad \text{or} \quad O(BgNe^{\theta-2} d^4) \quad (B \geq e).$$

Repeated packing, output accumulation and the last partial block are charged. The original public-matrix conversion and digit preparation remain in the ledger.

Sequential recursion has matrix-storage recurrence  $S(e) \leq S(e/a) + Ce^2$ ; including entry bits, it uses  $O(Ned^2)$  matrix scratch, besides integer-arithmetic scratch and the separately charged published rows, inputs and outputs. Keeping two representations incurs both buffers.

### F.3.3 Complete Work and the Retuned Source

**Proposition 24 (Rational-matrix joint work).** *Use the Gaussian-family assumptions and polynomial batch horizon of Proposition 23. Fix  $2 < \theta$  with  $\theta > \omega_{\mathbb{Q}}$ . For each chosen  $1 \leq k \leq d$ , use that profile's admitted generator. Put  $x = 2^k$ ,  $E = L/x$ . If  $B \geq E$ , the exact joint evaluator above has complete bit work*

$$W_{\theta}(B, k) = O(N[E + Bx + BE^{\theta-2}]d^7).$$

Choosing the nearest admissible integer to  $k = (\theta - 2)d/(\theta - 1)$  gives

$$W_{\theta}(B, k) = O(BL^{2-\beta} \log^7 L), \quad B \geq E = \Theta(L^{\beta}), \quad \beta = \frac{1}{\theta - 1}. \quad (32)$$

In particular,  $\theta = 593/250$  gives  $\beta = 250/343$ , prefix exponent 93/343, and work exponent 436/343  $\approx 1.271137$ .

*Proof.* Proposition 22 makes the retained ciphertexts identical for a fixed profile and random tape. Substitute the stage price above into the complete ledger of Section F.2.3. For  $\alpha = \theta - 2 > 0$ ,  $\sum_{e=E, E/2, \dots, 2} e^{\alpha} = O(E^{\alpha})$ . Since  $g = O(d^2)$ , all matrix work is  $O(BNE^{\alpha}d^6)$ . As in Corollary 11, all remaining terms are  $O(N[E + Bx]d^7)$ . This includes public parameter generation, sampler tables, every key and hint, both owners, all codecs, arrays, wires and recipient recovery. With  $xE = L$ , the chosen prefix gives  $x = \Theta(L^{1-\beta})$  and  $E = \Theta(L^{\beta})$ ;  $B \geq E$  absorbs setup and proves the displayed bound.

For each fixed profile the phases, noise and distributions are unchanged: matrix intermediates are deterministic public arithmetic, not fresh source samples. Retuning still changes the required source. Explicitly, with  $\tau = d - k$ ,  $I = 2x - 1 + \tau$ ,  $H = g(2E + 3\tau)$ , the source catalogue is

$$\bar{m} = \begin{cases} \max(2, 2g), & \tau = 0, \\ \max(2, 3g, Eg + 1), & \tau > 0, \end{cases} \quad \bar{H} = \text{nextpow2}(2\tau + 2 + BI).$$

The same reduction yields  $\text{Adv}_{\text{scheme}} \leq 2\bar{H} \text{Adv}_{\text{source}} + 2^{-\lambda} + 2\eta_{\text{hon}} + 4\bar{H}\eta_{\text{sim}}$ , evaluated at the complete new simulator time. For  $U = H + \tau + 1$  and  $G_B = H + 3\tau + 3 + 3BI$ , honest failure remains bounded by  $(UN + 1)2^{-4d^2} + G_B\rho_{\lambda} + 2G_BN2^{-2\lambda}$ . The polynomial horizon preserves negligibility. This does not reduce the retuned source to the balanced profile or the timed  $q_3/37$  law.

**Corollary 12 (Sharper scoped work ratio).** *Under the exact joint one-layer model, full Cartesian domain and full-domain private-encoding contract of Corollary 11, its  $\Omega(BL^2)$  lower bound divided by Proposition 24's attained upper bound is  $\Omega(L^{250/343}/\log^7 L)$  at the stated horizon and own source. Explicit materialization remains the alternative sufficient lower-bound contract.*

*Proof.* Keep the existing lower bound and substitute  $\theta = 593/250$  in equation (32).

The batching threshold rises from the Strassen specialization’s exponent 0.553295 to  $250/343 \approx 0.728863$ , with corresponding bank and buffering costs. The fixed rational kernel can have enormous constants. At  $L = 256$  the rounded prefix is  $k = 2$ ,  $E = 64$ ; the implementation instead uses  $k = 4$  and ordinary transforms. Thus the asymptotic improvement has no measured finite crossover in these experiments. Finite arithmetic checks use a rescaled Strassen kernel with denominator two to test exact division and packing, not the imported exponent or a fresh HE execution.

Known matrix algorithms supply this improvement to the existing compiler. Deeper conventional compilers can adopt it, and the unrestricted relay still costs  $\tilde{O}(BL)$  at this same retuned source (Proposition 25). Its attained upper bound is below this compiler’s necessary input-write work under the specified format (Corollary 13). No broader lower bound, new cryptographic primitive, numerical security qualification or a broader application guarantee follows. The finite matrix measurements are reported in Appendix F.2.

#### F.4 The Relay at the Retuned Gaussian Source

Proposition 21 fixes the balanced Gaussian generator. The joint compiler changes that generator when it retunes the prefix. We make the adverse comparison explicit at the new source and batch horizon, without a reduction between different prime laws.

**Proposition 25 (Same-source relay for a retuned profile).** *Fix a public prefix  $1 \leq k \leq d$  before sampling the modulus and a polynomial batch horizon  $B \geq 1$ . Use exactly that profile’s admitted ordinary-prime generator from Proposition 8. Charge computing the prefix as  $T_k$  and full public synthesis as  $\Pi_{\text{gen},k} = T_k + \text{poly}(d)$ . The existing encrypted-input relay solves  $B$  batches of the unrestricted task, under the same source assumption, in complete work*

$$\Pi_{\text{gen},k} + O((1+B)Nd^4 + \sqrt{\lambda}d^7 + BhLd^3).$$

*For the balanced and rational-matrix prefixes used here,  $T_k = \text{poly}(d)$ , so this is  $O(BNd^4)$  asymptotically. Its source has  $\bar{m}_R = 2$  and padded horizon  $\bar{H}_R = \text{nextpow}2(1+2B)$ , giving*

$$\text{Adv}_R \leq 2\bar{H}_R \text{Adv}_{\text{source},k} + 2^{-\lambda} + 2\eta_{\text{hon},R} + 4\bar{H}_R \eta_{\text{sim},R}. \quad (33)$$

*The source advantage includes the complete projection and simulator time; the coupling terms use the relay’s own full catalogues.*

*Proof.* Run the identical interval synthesis and bounded prime routine, retaining its modulus marginal and public-failure law. Publish one ordinary affine key and allow  $2B$  predeclared encryption queries, two unprepared operands per batch.

Forward, decrypt and compose as in Proposition 21. Every primitive Gaussian draw and uniform mask has the unchanged law. The honest algorithm does not generate a native bank.

The native source has at least two response rows. Retain the  $\lambda$  reserved pivots and two responses, discard surplus rows and project the public metadata to the one-vertex relay manifest. The real and null laws are exactly the source laws needed by the Gaussian theorem at this same modulus generator. A distinguisher for the projected source gives one for the original source at equal gap, with its row/metadata handling charged. This does not infer hardness for another prefix's prime distribution.

There is one row-bearing vertex and  $2B$  owner-query positions. Definition 2 therefore permits  $\bar{H}_R = \text{nextpow2}(1 + 2B)$ . For  $x = 2^k$ ,  $\tau = d - k$  and  $I = 2x - 1 + \tau \geq 3$ ,  $1 + 2B \leq 2\tau + 2 + BI$ , so this horizon fits within the native bound. Apply Theorem 10 and the projection above to obtain equation (33). The old one-batch factor eight is not reused for larger  $B$ ; for example  $B = 64$  gives factor 512. These factors are not security bits or a concrete-security match. No recipient privacy or decryption oracle is provided.

There are  $G_R = 2 + 6B$  Gaussian polynomial draws and one uniform mask. Each unprepared binary physical input satisfies the same fresh-phase bound  $F = (2\kappa\lambda + 1)\lambda$ , already admitted by the retuned generator. Honest failure is bounded by

$$(N + 1)2^{-4d^2} + G_R\rho_\lambda + 2G_RN2^{-2\lambda}.$$

This charges prime/uniform exhaustion, sampler coupling and coefficient tails. Privacy is not conditioned on correctness or nonabort.

The exact ring-product count is  $1 + 6B$ : one for setup, four per batch for encryption and two for decryption. Their bit price is  $O(Nd^4)$  by Lemma 21. The existing sampler ledger charges  $O(\sqrt{\lambda}d^7)$  for tables,  $O(G_RNd^3)$  for draws and  $O(Nb_qd^2)$  for the uniform mask. There are  $4B$  codec calls. Publication, inputs and forwarding serialize  $2 + 8B$  residue polynomials; all other arrays and output writes are charged as well. The existing clear-composition bound costs  $O(BhLd^3)$ . Summing gives the stated complete bound. For the prefixes used here, synthesis and the table term are asymptotically absorbed in  $O(BNd^4)$ . Physical network latency remains a separate scheduling cost.

**Corollary 13 (Input-work gap at the rational-matrix prefix).** *For Proposition 24's prefix with  $x = \Theta(L^{93/343})$  and  $B \geq E = \Theta(L^{250/343})$ , the specified joint compiler requires  $\Omega(BL^{436/343})$  input-write work under its explicit ciphertext format. The same bound holds except negligibly and in expectation under the full-domain correct and private initial-encoding contract of Lemma 20. Relative to the attained same-source relay bound, the work ratio is at least  $\Omega(L^{93/343}/\log^4 L)$ .*

*Proof.* Joint Hasse evaluation leaves all  $BI$  owner input ciphertexts intact. The explicit format writes  $2BINb_q$  bits. Alternatively, apply the secure-length lemma to the whole initial stream of full message width  $BIN$  for an efficient prepared-input family. Its writes cost  $\Omega(BIN)$  under the stated contract. Since  $I =$

$2x - 1 + \tau = \Theta(L^{93/343})$ , this proves the necessary native cost. Divide it by Proposition 25's attained  $O(BL \log^4 L)$  upper bound. Thus the ordering uses a lower bound and an upper bound, not two upper bounds.

The input-write statement concerns this prefix and the specified encoding contract. It does not lower-bound all JetHE algorithms, prepared-image-only encodings or other prefix choices. Nonlinear recipient recovery puts relay outside the one-layer comparator, so the scoped separation remains valid. No new HE execution, finite timing advantage, numerical security qualification or transfer to the timed  $q_3/37$  source follows.

## G Conventional controls and receiver algorithms

### G.1 One-prime control with the shared arithmetic backend

This control uses the same ring  $R_{256}$ , codec, ternary/CBD20 source laws, and arithmetic library as JetHE-102. It has no evaluation keys and one root key, at modulus  $p_0 = 1152921504002872321$ .

*Prepared functionality.* Write  $P_{i,j}(g) = [z^j]g^i$ . For each  $j \geq 1$ , pair the eligible outer indices  $i, k$  in increasing order and put  $x = P_{i,j}$ ,  $y = P_{k,j}$ . In characteristic two,

$$(f_i + y)(f_k + x) + f_i f_k + xy = f_i x + f_k y.$$

The joint product uses two sums of separately encrypted owner inputs. Each owner aggregates its unilateral corrections per output degree; the outer correction also includes  $f_0$ . A final unpaired term uses one separated product with zero padding on the unused inputs.

For fixed  $j$ , the eligible indices satisfy  $i \leq j$  and  $2^{v_2(i)} \mid j$ . Here  $v_2$  is the exponent of two in a positive integer. The series  $g^i$  has minimum degree  $i$  and all exponents divisible by  $2^{v_2(i)}$ . For an odd exponent after this Frobenius factor is removed, every higher degree has a nonzero formal coefficient, witnessed by a term with all but one factors of degree one. Thus their number is  $t_j = j - (\text{oddpart}(j) - 1)/2$ , where  $\text{oddpart}(j) = j/2^{v_2(j)}$ . At  $L = 256$ ,

$$\sum_{j=1}^{255} t_j = 21845, \quad \sum_{j=1}^{255} \lceil t_j/2 \rceil = 10986.$$

Across sixteen jobs this is 175776 paired scalar products.

Interpolate 128 scalar values at distinct public points of  $K$  into a degree-at-most-127 polynomial in each jet. The product degree is at most 254, so native truncation modulo  $z^{256}$  preserves the entire product. Terminal evaluation recovers the 128 pointwise products. Sixteen jets give 2048 effective scalar-product positions; 86 carriers accommodate the paired list. Each carrier uses four fresh input ciphertexts. Each owner's 4096 correction coefficients fit directly into one full native carrier. Hence the control uses  $4 \cdot 86 + 2 = 346$  inputs and returns 86 raw three-component products plus two two-component correction ciphertexts. Its recipient reconstruction consists of fixed field-linear evaluation and routing.

*Fixed-input ciphertext correctness.* The fresh-error calculation of Appendix B.3 with  $T = 16$  gives  $F_1 = \lceil 16\sqrt{80N + 10} \rceil = 36636$ . The probability that any fresh error exceeds this cap in a collection of at most 1024 fixed batches is at most

$$2N \cdot 346 \cdot 1024 \exp(-128) < 2^{-149}.$$

This is a union of marginal events and permits reuse of the root public-key error. On their complement, each mixed operand has integer signal cap 2 and error cap  $2F_1$ . Expanding its raw product and charging the canonical binary-representative carry gives

$$B_1 = 8\kappa(F_1^2 + F_1) + 2\kappa = 1404684555427328.$$

The argument is deterministic on the fresh-cap event and allows correlation between the product operands. Since  $p_0 - 2 - 4B_1 = 1147302765781163007 > 0$ , centered parity recovers every product and bypass. No relinearization or switching adds noise.

*Conditional source comparison.* Deleting to two primitive rows in  $D_{q_3,37}$  and reducing modulo  $p_0$  gives the control's source game: the ring map has uniform fibers and preserves the underlying integer ternary/CBD20 laws. The public-key and fresh-encryption hybrids give, for  $B$  batches,

$$\text{Adv}_{\text{control}}^{\text{gap}}(T) \leq 2(1 + 346B)\epsilon_{q_3,37}(T + T_{\text{sim,control}}) + 2\delta_{\text{control,B}}.$$

At two batches its factor is 1386, compared with 158 for JetHE. The simulator times and implementation deviations remain distinct. Both reductions are conditional and supply no numerical security match.

## G.2 An analytical conventional control for prepared composition

The executable HELib controls in Appendix I.2 do not exhaust conventional implementations of prepared composition. This subsection adapts the existing shrinking algorithm to the same owner-preparation permissions and a declared terminal interface. Its receiver is the degree  $N = 32768$  real subfield of conductor 65537, with 2048 full  $\mathbb{F}_{2^{16}}$  slots. The selected sufficient profile has *25,940,590,592 bytes of raw public rows*, an attained analytical upper bound for this construction. It supplies no conventional-HE lower bound, global optimum, receiver timing or security-matched ranking.

*Functionality and the recipient.* There are sixteen independent length-256 pairs  $(f, g)$  over the field defined by  $0x1100b$ , with  $g(0) = 0$ . Every job returns all coefficients of  $f(g(z)) \bmod z^{256}$ . The outer owner supplies two coefficient carriers and one first-odd carrier derived only from  $f$ . The inner owner prepares eight arrays derived only from  $g$ , occupying 1, 2, 3, 4, 4, 4, 4, 4 carriers. All 29 fresh encryptions use one root public key and independent coins; their plaintexts may be correlated.

Eight descending stages perform 26 field ciphertext products. Seven outer maps and seven encrypted return/bypass operations use independent later keys.

The final encrypted return is omitted: the evaluator emits four switched two-component product ciphertexts and two preceding coefficient ciphertexts. The authorized recipient decrypts all six and applies the public final return and bypass, obtaining 4096 field coefficients. It sees these intermediate plaintexts. The contract protects inputs from the evaluator; it makes no output-only privacy claim against the secret-key recipient. It imposes no low-modulus copy, second segment or encrypted continuation.

*Graph, source and admission.* There are 23 main and 21 helper secrets. Each helper lies strictly between its source and destination. The first product stage needs two nonconstant secret-payload families; each subsequent stage needs three, including both source identities. The resulting 23 ordinary families and 1323 semilinear-map families have respectively 77 and 3072 switch uses. This is a separate conventional realization; the universal single-layer Hasse comparison in Appendix A.2 is not a lower bound for it.

The source is the existing finite sequential tensor-Gaussian CDF256 law, with whole-vector cap-to-zero and an independent global sign. Its distance from the associated ideal Gaussian is below  $2^{-228}$  per vector. Public whole-mask signs give a simultaneous integral-map norm event. For new bank errors, repeated uses of the same primitive error are collected before their multiplier bounds are squared. The independent check reconstructs 22 ancestor-disjointness cases; within each linear pipeline, second-component digit inputs remain independent of its protected row errors. Propagated errors and rounding retain deterministic envelopes. Correctness concerns a fixed public program and owner data independent of protected bank errors, not arbitrary key-visible adaptive inputs.

The specified parameter grid has  $q$  widths 619 through 1131,  $P$  widths in the distinct set  $\{128, 196, \lfloor q_b/4 \rfloor, \lfloor q_b/2 \rfloor, q_b\}$ , ordinary digit counts  $\{1, 2, 4, 8, 16\}$  and map digit counts  $\{2, 4, 8, 16\}$ . Among 51220 tested policies, 35954 satisfy the sufficient recurrence. The lexicographic objective is raw public-row bytes, evaluated rows, then the widths and digit counts. The selected grid minimum has  $q_b = 780$ ,  $P_b = 390$  and two digits of each type. All 94 complete-state envelopes satisfy  $4B < 2^{779} - 1$ , including the six terminal ciphertexts. The 619-bit floor constrains this sufficient recurrence only.

The corrected full failure union is bounded by

$$\begin{aligned} \varepsilon_{\text{fail}} \leq & 2 \cdot 2^{-180} + 2824 \cdot 2^{-228} + 44(2^{-235} + 2^{-183} + 2^{-634}) \\ & + 29 \cdot 2^{-634} + 2693 \cdot 32768 \cdot 2^{-256} < 2^{-177} < 2^{-160}. \end{aligned} \quad (34)$$

These terms cover signed-map and scalar-injection events, finite-source coupling, receiving-secret coefficient/spectral/energy events, owner ephemeral energy, and uniform-row sampling abort. In particular, the 29 owner-energy events are included in the displayed union bound. Rational log bounds certify the displayed dyadic ceilings directly from the source-tail formulae.

*Material and complete conditional work.* Table 14 counts fixed-width coefficient arrays without object overhead. A root row uses modulus  $q$ ; evaluation rows use

**Table 14.** Prepared-composition analytical control. GB is decimal. Cache alternatives are optional; coefficient payloads are not RSS.

| Quantity                                 | Count or bytes                |
|------------------------------------------|-------------------------------|
| Root / evaluation rows; evaluated rows   | 1/2692; 6298                  |
| Raw public rows                          | 25,940,590,592                |
| Owner ciphertexts / bytes                | 29/186,253,312                |
| Terminal ciphertexts / bytes             | 6/38,535,168                  |
| Recovered field elements / bytes         | 4096/8192                     |
| Source vectors / scalar draws            | 2824/92,536,832               |
| Source random bits / base-pass weights   | 23,689,431,816/19,432,734,720 |
| Adjusted mask instances / sign bytes     | 58,380/7125                   |
| Compact / expanded mask-cache bytes      | 239,124,480/281,210,388,480   |
| Peak logical ciphertexts / payload bytes | 31/199,098,368                |
| Separate allocate-all linear workspace   | 5,809,176,576                 |
| All 44 capped-secret coefficient arrays  | 1,441,792                     |

$M = qP$ . Projection of a common  $M$  root challenge to  $q$  is consistent with this inventory. The ring and finite source differ from both measured implementations and the primary Gaussian family in Section 7; this admission supplies no concrete security assessment for the new exposed profile.

For clarity, write  $M_b$  for a complete receiver ring product at modulus  $b$ ,  $V_b$  for an upper bound on one vector addition/scaling/lift/reduction,  $A_b$  for an automorphism,  $D_b(g)$  for full balanced decomposition and  $R_{a,b}$  for exact vector scaling, rounding and reduction. Full field/period codecs are  $E, C$ ; field multiplication, squaring and addition are  $F, S, X$ . Let  $\Gamma(v; \Pi)$  charge all source arithmetic and random bits for  $v$  vectors, including every certified-CDF precision pass  $\Pi$ . The compact-mask policy has the following sufficient cost expression:

$$\begin{aligned}
T_{\text{setup}} &\leq \Gamma(2737; \Pi_s) + U_q(1) + U_M(2692) + M_q + 2692M_M \\
&\quad + 8M_{\text{small}} + 1323A_{\text{small}} + 10772V_M + T_{\text{masks}} + IO_s, \\
T_{\text{owner}} &\leq \Gamma(87; \Pi_o) + 28672S + 1982368(F + X) \\
&\quad + 29E + 58M_q + 116V_q + IO_o, \\
T_{\text{eval}} &\leq 103M_{q^2} + 129356M_M + 77D_q(2) + 3072D_M(2) \\
&\quad + 103R_{q^2,q} + 130R_{M,q} + 6144A_M + 213448V_M + IO_e, \\
T_{\text{recover}} &\leq 6(M_q + V_q + R_{q,2} + C) + 1030272(F + X) + IO_r.
\end{aligned}$$

Here  $U$  includes capped uniform-coordinate rejection.  $T_{\text{masks}}$  charges 2922112 field-entry writes, 58380 full encodes and binary automorphisms, and public-sign generation and application.  $IO$  includes the public plan, rows, masks and ciphertexts. The evaluation allowance includes compact-mask expansion and deliberately separate linear decompositions; no unimplemented hoisting saving is taken. The 103 tensor products use scaled-BFV semantics: one same-key tensor uses three products and 25 distinct-key tensors use four each. Modulo  $q^2$  suffices for the exact tensor rounding, since adding  $q^2$  changes  $\text{round}(2x/q)$  by  $2q$ . Ordinary mod- $q$  products would not implement this step.

These formulas account for every kernel, conditional on its specified implementation cost. The base-pass weight count alone does not bound the number or precision of CDF refinement passes. The following subsections supply a bounded table sampler and receiver implementations; the displayed symbolic ledger by itself supplies no numerical runtime bound. Memory must add actual ring, codec, digit and rounding scratch, CDF interval workspace, plan parsing and serialization buffers to the selected arrays. The expanded mask cache is optional; generating masks on demand instead moves the corresponding work into evaluation.

*Public verification of the contraction plan.* The plaintext verification checks 5934752 edges across 115 maps, all 4096 fixture outputs against the Horner oracle, and an adverse case with an omitted bypass. The complete union bound is (34); partial operation counts do not replace it. These checks establish correspondence of the public plan with composition. The source sampler, receiver execution and concrete source-security analysis are separate parts of the implementation.

**G.2.1 A bounded table sampler for the same ideal source** The preceding sequential-source cost is not a lower bound on source generation. Its real-period precision matrix is  $Q = I - 2J/p$ , with  $n = 32768$ ,  $p = 2n + 1 = 65537$ , and  $Q^{-1} = I + 2J$ . This permits a finite specialization of established Gaussian perturbation sampling [35]. It strengthens this conventional control, not JetHE’s exclusivity claim.

**Proposition 26 (Finite table alternative for the real-period source).**

*There is an implemented finite symmetric source  $\Gamma_{\text{tab}}$  on  $\mathbb{Z}^n$ , capped at coefficient magnitude 96 by a whole-vector cap-to-zero operation, whose distance from  $D_Q(x) \propto \exp(-x^T Q x / 18)$  is below  $2^{-228}$ . Its public precomputation uses 257 scalar CDF tables with 74678 weight positions. One vector then uses  $256(n+1)+1$  ideal random bits, at most sixteen latent-table comparisons and seven per coordinate, integer translations, a cap scan and a sign pass. It performs no online exponential evaluation or adaptive precision refinement. Serialized tables occupy 2383552 bytes.*

*Proof.* Put  $M = 256$ . In the auxiliary experiment,  $Y \in M^{-1}\mathbb{Z}$  and the integer coordinates  $X_i$  are independent given  $Y$ , with laws

$$\Pr[Y = y] \propto e^{-y^2/36}, \quad \Pr[X_i = x \mid Y] \propto e^{-(x-Y)^2/18}.$$

For  $s = \sum_i x_i$ ,

$$\frac{Y^2}{36} + \frac{\sum_i (x_i - Y)^2}{18} = \frac{x^T Q x}{18} + \frac{p(Y - 2s/p)^2}{36}. \quad (35)$$

The coordinate normalizer depends on  $Y$ . Poisson summation bounds its relative deviation from the continuous normalizer, uniformly in its center, by

$$2 \sum_{k \geq 1} e^{-18\pi^2 k^2} < \delta = 2^{-251}.$$

For the posterior latent index the variance parameter is

$$18M^2/p = 1179648/65537.$$

The same bound gives deviation  $\delta' = 2^{-500}$ . Relative to the ideal mass in (35), the unnormalized auxiliary multiplier lies between  $(1 - \delta')/(1 + \delta)^n$  and  $(1 + \delta')/(1 - \delta)^n$ . The prior latent normalizer is constant in  $x$ . Normalize the marginal and use  $\log((1 + d)/(1 - d)) \leq 4d$  and  $e^z - 1 \leq 2z$  in their indicated small ranges. The distance from  $D_Q$  is at most  $8(n\delta + \delta')$ . No independence of the normalizers is assumed.

Retain latent indices  $k = MY$  from  $-21723$  through  $21723$ , using  $21723 = \lceil 20\sqrt{18M^2} \rceil$ . Write  $k = Ma + j$  by Euclidean division,  $0 \leq j < M$ . Each coordinate equals  $a + Z$ , with a variance-nine integer Gaussian  $Z$  centered at  $j/M$ , retained on  $[-60, 60]$  for  $j = 0$  and  $[-60, 61]$  otherwise. There are 256 fractional centers. Every omitted scalar tail is at most  $4e^{-200}$ , giving joint truncation cost at most  $4(n + 1)e^{-200}$ .

For each retained weight  $e^{-x}$ ,  $0 \leq x < 214$ . Divide the argument by  $2^h$  until it is at most  $1/8$ , with  $h \leq 11$ . At fixed scale  $2^{512}$ , integer floor/ceiling arithmetic encloses each Taylor term. Alternating partial sums of degrees 129 and 128 bound the exponential from below and above. At most eleven outward-rounded squarings enclose the desired weight. For a prefix-sum interval  $[A_l, A_u]$  and normalizer interval  $[Z_l, Z_u]$ , the CDF is in  $[A_l/Z_u, A_u/Z_l]$ . At every boundary the implemented precomputation verifies that the floors of these endpoints times  $2^{256}$  differ by at most one. Store the lower floor and an implicit final threshold  $2^{256}$ . Thresholds are monotone and the scaled CDF error is less than two units. An  $L$ -outcome table therefore has distance at most  $2L/2^{256}$ .

All 74421 internal boundaries pass the integer enclosure check. Joint quantization costs at most  $2(43447 + 122n)/2^{256}$ . Map the whole vector to zero if any coefficient exceeds 96 in magnitude, then apply an independent random sign. The ideal coefficient variance proxy is 27, so the cap costs at most  $2ne^{-97^2/54} < 2^{-235}$ ; integral coordinates justify 97. Symmetry of  $D_Q$  ensures that the sign does not increase distance. Thus

$$\begin{aligned} \text{TV}(\Gamma_{\text{tab}}, D_Q) &\leq 8(n2^{-251} + 2^{-500}) + 4(n + 1)2^{-285} \\ &\quad + \frac{2(43447 + 122n)}{2^{256}} + 2^{-235} < 2^{-228}. \end{aligned} \tag{36}$$

The tail ceilings use rational inequalities with  $\pi > 3.14$  and  $\log 2 < 347/500$ ; the latter is certified by the degree-ten positive exponential Taylor polynomial. Exact rational evaluation of the right side gives base-two logarithm below  $-231.85$ .

Every draw uses one 256-bit word and binary CDF search. Support lengths 43447, 121 and 122 give the comparison counts. The file stores 257 pairs of 32-bit support descriptors, 74421 thresholds of 32 bytes and a 24-byte header. All precomputation is fixed finite integer work.

The table law differs from the sequential CDF law. Both are within  $2^{-228}$  of the same  $D_Q$ , so their per-vector distance is at most  $2^{-227}$ . One setup and

$K_{\text{batch}}$  batches use  $m = 2737 + 87K_{\text{batch}}$  source vectors. Coupling changes a transcript event probability by at most  $m2^{-227}$ , or a distinguishing gap by at most twice that. This charges transfer of an exact sequential-source premise; it asserts neither identical sources nor numerical hardness. The existing ideal-source correctness proof and its coupling allowance admit the same 780/390-bit profile. No modulus, map-count or public-row reduction follows from changing the sampler.

For one setup and batch, the 2824 vectors use 23690154760 ideal random bits. The prototype reads a whole sign byte, charging 2961271816 OS bytes in that inventory. Only table setup evaluates exponentials; the original 19432734720 per-coordinate base-pass weight count is not a necessary cost of this alternative. The table figure is serialized size, not Python object memory or process RSS.

The implementation crosschecks 41 exponential enclosures using separate 200-digit arithmetic and checks every CDF boundary. Eight vectors of dimension  $n$  cover 262144 sampled coefficients. Two public tapes test whole-vector capping and eight paired tapes test sign symmetry. These are source-prototype checks, not an encrypted receiver execution or empirical evidence for the distance bound. Diagnostic elapsed values have no affinity or clock controls and are not comparative timings. Constant-time lookup, OS randomness qualification and receiver arithmetic remain separate obligations.

**G.2.2 A bounded padded-CRT refinement of the control** The direct-map witness above does not exhaust structured implementations. We also compile the same shrinking recurrence using power-of-two additive subspaces, explicit sparse transform factors, and five fixed fusion widths. The result strengthens this conventional alternative. It is not a new FFT, a conventional lower bound, or a security-matched runtime result.

*Functionality and fixed layout.* For  $r = 2^j$ ,  $n = 256/r$  and  $m = n/2$ , write  $G_j(X) = XO_j(X^2) + X^2E_j(X^2)$ , where  $G_j(X) = \sum_{i < n} g_i^{2^j} X^i$  and  $g_0 = 0$ . The child recurrence is

$$R_{j,t}(X) = R_{j+1,t}(X^2) + G_j(X)R_{j+1,t+r}(X^2) \pmod{X^n}.$$

Multiply the  $m$ -coefficient odd child by  $O_j$  and its first  $m - 1$  coefficients by  $E_j$ . Evaluation domains have the next power-of-two sizes at least  $2m - 1$  and  $2m - 3$ , respectively. Constant products use one point; an absent  $E_j$  product is omitted. Exact interpolation, truncation and the even-child bypass recover the recurrence. The state is job-branch-coefficient ordered, and product blocks are kind-job-branch-point ordered. The eight product counts are 1, 3, 4, 4, 4, 4, 4, 4. Thus there are 28 inner-owner carriers, two outer coefficient carriers and one first-odd carrier: 31 encryptions in total. The final four product and two bypass carriers are decrypted and recovered linearly, with the same recipient-privacy boundary as the direct control.

Write the subspace polynomial as  $V = X^d + \sum_{e < d} v_e X^e$ . It is linearized, so all lower exponents are at most  $d/2$ . For  $f = f_{\text{low}} + X^d h$ , polynomial division has

**Table 15.** Finite parameter search for padded CRT. Each factored graph tests 51,300 policies. Rows minimize public-row bytes within their declared grids, not time. Ring products are at that row’s own  $M = qP$ . GB is decimal.

| Fusion     | $q_b/P_b$ | Digits | Public GB | $M_M$ calls |
|------------|-----------|--------|-----------|-------------|
| Unfactored | 780/390   | 2/2    | 25.941    | 129356      |
| 1          | 2646/2646 | 1/2    | 95.338    | 30532       |
| 2          | 1736/1736 | 1/2    | 57.838    | 31270       |
| 4          | 1598/799  | 2/2    | 35.874    | 31772       |
| 8          | 1247/623  | 2/2    | 25.498    | 33052       |
| 16         | 1066/533  | 2/2    | 20.587    | 33220       |

quotient  $q = h + Th$ , where  $(Th)_a = \sum_{e>a} v_e h_{d+a-e}$  and  $T^2 = 0$ . The remainder is  $r_a = (f_{\text{low}})_a + \sum_{e\leq a} v_e q_{a-e}$ . The two children are  $r + c_0 q$  and  $r + (c_0 + s)q$ , with public  $s \neq 0$ ; inversion starts with  $q = (\text{left} + \text{right})/s$ . These quotient/child maps give two elementary stages per recursion level. Remove exact identities and fuse  $w \in \{1, 2, 4, 8, 16\}$  consecutive stages *over the field before lifting*. Products of integer lifts are not identified with lifts of their field products.

Each encrypted layout, retained factor and return has a fresh main key. Both differently keyed return sources and all BSGS helper keys are charged. For a field-linear edge from slot  $s$  to  $t$ , the actual Galois exponent is  $\ell = s - t \pmod{32768}$  in the chart with representatives  $3^t$ . The equality  $3^t 3^\ell = \pm 3^s \pmod{65537}$  verifies twist zero; rotations modulo 2048 are not granted for free. The existing signed-mask, dependency-aware source and scaled-BFV bounds apply to this explicit graph.

*Admission and complete costs.* Each graph uses the preceding finite grid shifted to its own sufficient-recurrence floor. The five floors are 2645, 1735, 1233, 969 and 833 bits. There are 256,500 tested policies in total. Table 15 retains every outcome. The storage winner fuses each entire transform; finer encrypted FFT stages incur larger sufficient moduli. Its 70 independent secrets, 762 map families and 23 ordinary families give 1570 evaluation rows plus the root, occupying 20,587,085,824 bytes. All selected complete-state envelopes satisfy  $4B < 2^{q_b-1} - 1$ . For  $k$  secrets and  $a$  primitive rows, the full one-batch failure union is

$$2 \cdot 2^{-180} + (k+a+93)2^{-228} + k(2^{-235} + 2^{-183} + 2^{-634}) + 31 \cdot 2^{-634} + aN2^{-256} < 2^{-160}$$

for every selected profile. It includes the owner ephemeral-energy events. The table source of Proposition 26 is an alternative at the same ideal-source distance; the transfer between these two finite laws is  $(k+a+93K)2^{-227}$  over  $K$  batches.

For width 16, the evaluator’s conditional kernel charge is

$$\begin{aligned} T_{\text{eval}} \leq & 111M_{q^2} + 33220M_M + 83D_q(2) + 1866D_M(2) \\ & + 111R_{q^2,q} + 238R_{M,q} + 3732A_M + 62476V_M + 10e. \end{aligned}$$

Setup charges 1641 source vectors, the root and 1570 evaluation rows, eight small secret products and 762 small automorphisms, as well as source-table,

public-plan and mask construction. The compact mask cache has 6154 instances / 25,206,784 bytes; the expanded cache is optional. Owner work charges 93 source vectors, 31 encodes and 62 ring products. It may use width-one *public* factors with identical input carriers: 630464 field multiplications/additions, 8032 coefficient Frobenius calls, and array copying and initialization. Terminal recovery charges six complete decryptions/decodes, 303136 inverse-transform field multiplications/additions and 6128 return multiplications/additions. The complete cost accounting also includes uniform sampling, signs, all vector operations, transform-plan construction, and input/output processing and traffic, with workspace and arithmetic scratch space counted separately. Appendix G.2.3 executes receiver primitives and the complete compiled graph at these widths. Its diagnostic reference execution does not price an optimized complete warm workflow.

Independent arithmetic checks cover 87,381 monomial-matrix entries, nine inverse identities, all five complete 4096-coefficient fixtures against Horner evaluation, every physical edge, omitted-bypass failures and exact rational state bounds. The parameter search and these consistency checks are separate computations. The attained raw-row storage bound is 20.59 GB, compared with 25.94 GB for the unfactored control. Neither the smaller key inventory nor the lower number of more expensive ring products supplies a total-time ordering or source-hardness qualification.

**G.2.3 Executed Receiver Primitives** The control implements real-period ring products, full slot codec, exact rounding, ordinary switching, a single diagonal map and mixed-secret scaled BFV products. We report primitive checks and a complete 31-input composition execution.

*Exact ring arithmetic.* Put  $p = 65537$ ,  $n = (p - 1)/2$  and  $\rho_i = \zeta_p^i + \zeta_p^{-i}$  for  $1 \leq i \leq n$ . Since  $\sum_i \rho_i = -1$ , the scalar one has every coordinate  $-1$ . Embed a period vector  $a$  symmetrically in  $p$  cyclic coordinates:  $A_0 = 0$  and  $A_i = A_{p-i} = a_i$ . If  $D = A * B$  is cyclic convolution, reduction by  $\Phi_p = 1 + X + \cdots + X^{p-1}$  gives product coordinates  $D_i - D_0$ . For representatives in  $[0, m)$ , every ordinary convolution coefficient is at most  $p(m - 1)^2$ . A byte-aligned width  $w$  satisfying  $2^w > p(m - 1)^2$  therefore permits exact integer convolution without carries between coefficient blocks. Folding, subtracting  $D_0$  and reducing modulo  $m$  implements the product at arbitrary integer  $m$ . No transform roots or floating-point convolution are used. With  $l(t)$  the cost of a  $t$ -bit integer product and  $D(w, b)$  the cost of reducing  $w$  bits modulo a  $b$ -bit modulus, a sufficient kernel bound is

$$l(pw) + O(p(w + D(w + 1, b))), \quad w = O(b + \log p).$$

This counts packing and reduction; integer-multiplier scratch is additional to  $O(pw)$  bits. The implementation uses a bound GMP binary. This is standard arithmetic, not a new cryptographic contribution.

*Codec and phase checks.* The reference codec retains the degree-65536 priming CRT tree internally. Encoded subfield words have symmetric coefficients in the nonzero power basis; retaining the first half gives the period coordinates. Its larger internal degree and workspace remain charged. All 4096 input words are checked in their original slot order, all 65536 leaf coordinate basis images are checked, and both codec round trips and a complete 2048-slot product pass. Four product slots are also evaluated directly at the original field roots.

The fresh primitive trial uses  $q = 2^{1066} - 1$  and  $P = 2^{533} - 3$ , inside the selected bit intervals. They are odd and coprime to each other and to 65537; they are not asserted prime or independently secure. Ordinary switching uses  $q$ -digits and payloads  $PB^j f$ , whereas the single diagonal map decomposes the  $P$ -lifted component modulo  $qP$  and uses payloads  $B^j \sigma(s)$ . Both divide by  $P$  after accumulation. The independent-key product scales all four tensor components using  $\text{round}(2x/q) \bmod q$  with raw products modulo  $q^2$ , then switches both source identities and their secret product. Adding  $kq^2$  before rounding changes the result by  $2kq$ , justifying the  $q^2$  reduction. Reduction modulo  $q$  before scaling fails the independent integer adverse fixtures.

The bound execution records 23 fresh vectors from the preceding table source, 12 primitive public rows and two owner encryptions. Five decrypted outputs pass all 163840 binary-coordinate checks, including the ordinary switch, single diagonal and mixed-secret product. The actual encrypted product also matches all 2048 field slots. Omitting one identity family changes 16488 plaintext bits in the adverse fixture. No source cap occurred; these observations do not empirically validate a rare-event bound.

The elapsed time includes the independent reference computations and was measured without controlling competing activity. It therefore does not provide an isolated kernel time or a comparative total-time ratio.

*Complete compiled execution.* The streamed implementation evaluates all 1570 evaluation rows and the complete BSGS composition graph. All 45 nodes and 150 carriers pass their admitted phase checks, covering 4915200 binary coordinates. Six terminal ciphertexts recover all 4096 independently expected field coefficients. The final inventory contains 70 secrets, 1571 primitive rows including the root, 1734 finite vectors and 35128 instrumented ring products. Of the latter, 33220 are evaluator products modulo  $qP$ ; setup, validation, tensor and terminal products supply the remaining count.

Instrumented wall time is 20626.437 seconds; peak RSS is 1966892 KiB. The precompiled graph and prepared owner carriers, streamed keys and private phase checks differ from a complete warm-workflow experiment. No time ratio against native follows. The reported result describes this instrumented execution; its numerical summary does not constitute an independent decryption of the ciphertexts. The source-model concern in Appendix G.2.4 remains, and successful execution supplies no security qualification.

**G.2.4 Concrete Source Models and a Parameter Tradeoff** Correctness admission does not qualify the preceding control as a security-matched bench-

mark. Its actual first baby digit-zero row has phase  $s + e$  under an independent helper  $u$  at  $M = qP$ . For  $q = 2^{1066} - 1$  and  $P = 2^{533} - 3$ , this is a small perturbation at a 1599-bit modulus. The capped finite source gives  $\|u\|_\infty \leq 96$  and  $\|s + e\|_\infty \leq 192$ . Writing the body as  $c = Au + z \bmod M$  retains the actual structured ring multiplication matrix  $A$ ; its entries are not independent samples.

*Two explicit model interfaces.* For caps  $a, b$  and scale  $\gamma = b/a$ , the lattice

$$\{(\gamma x, Ax + my) : x, y \in \mathbb{Z}^n\}$$

has rank  $2n$  and determinant  $(\gamma m)^n$ . The target  $(0, c)$  has distance at most

$$R_\gamma = \sqrt{n(\gamma^2 a^2 + b^2)}$$

from its true-secret vector. If every actual Gram–Schmidt length exceeds  $2R_\gamma$ , nearest-plane recovers that vector. Substituting the heuristic GSA minimum  $\sqrt{\gamma m} \delta_\beta^{1-2n}$  gives a conditional recovery criterion. The pinned root-Hermite and work models are those of [30]; their numerical evaluation is not a lattice reduction or a hardness lower bound. For the stated root and helper rows, the first modeled blocks are 248 and 124, with classical work proxies approximately  $2^{107.816}$  and  $2^{71.872}$ , including their declared polynomial and repetition terms. Exact rational intervals check the threshold signs.

A source-metric dual test uses more of the actual law. Put  $G = I + 2J = Q^{-1}$  for the preceding real-period Gaussian. Its characteristic function satisfies  $\varphi(v) \geq \exp(-18\pi^2 v^T G v) / \Theta$ , where  $\Theta = \sum_{z \in \mathbb{Z}^n} \exp(-18\pi^2 z^T G z)$ . This is the zero-term Poisson bound, not an independent-coordinate Gaussian replacement. For a nonzero relation  $x + A^T y = 0 \bmod M$  chosen from  $A$  alone, the helper row gives

$$\mathbb{E} \cos(2\pi y^T c / M) \geq \frac{\exp[-18\pi^2(x^T G x + 2y^T G y) / M^2]}{\Theta^3}.$$

The two copies of  $y$  charge the independent source secret and error. With  $F = [I; \mathbf{1}^T; \mathbf{1}^T]$ , the integer embedding  $(x, y) \mapsto (Fx, Fy, Fy)$  has rank  $2n$ , ambient dimension  $3(n + 2)$  and covolume  $M^n 2^{n/2} 65537$ . Weighted squared length at most  $M^2 \log 2 / (18\pi^2)$  gives a Boolean distinguishing gap at least  $1/(4\Theta^3) - 3 \cdot 2^{-228} > 1/5$  for the implemented finite source. Indeed,  $18\pi^2 > 256 \log 2$  bounds the normalizer; the metric dominates the identity and the radius is below  $M$ , ensuring a nontrivial character even at composite  $M$ . The root version has two source factors, metric  $\text{diag}(G, G)$  and covolume  $q^n 65537$ . Finite cosine evaluation pays its approximation error separately. Finding the requisite short vector is the heuristic part. This dual test distinguishes a source/public-row distribution; it is not by itself an IND-CPA attack.

*Higher digits at smaller moduli.* A separate finite parameter search retains the same width-16 graph, source, signed maps, independent secrets and terminal interface. It tests  $q$  widths 833–896,  $P$  widths 8–96 in steps of four, and independent ordinary/map digit counts in  $\{2, 4, 8, 16, 32, 64\}$ : 52992 policies. Of

these, 18411 pass the sufficient correctness recurrence and 2538 also make all four preceding geometric criteria fail at block 317 at the upper modulus endpoints. Thus their first satisfying modeled blocks are at least 318. At rank 65536, the pinned overhead-inclusive classical exponential term then has logarithm  $0.292 \cdot 318 + 16.4 + \log_2(8 \cdot 65536) = 128.256$ . This is solely a filter in a named work model. The core-SVP term and quantum model are different quantities, and cheaper attacks outside these four tests have not been excluded.

Both minimum-public-bytes and minimum- $M$ -product selectors in this grid choose widths 840/16 and digit counts 16/16. All 150 selected state envelopes pass independent rational checks; the complete fixed-batch failure union, including the increased source count, remains below  $2^{-176}$ . Example moduli are  $q = 2^{840} - 1$  and  $P = 65531$ . The following are conditional ledger entries, not measured times:

|                           | Large modulus  | More digits    |
|---------------------------|----------------|----------------|
| $q/P$ widths              | 1066/533       | 840/16         |
| Primitive public rows     | 1 571          | 12 561         |
| Raw public bytes          | 20 587 085 824 | 88 082 022 400 |
| Evaluator products at $M$ | 33 220         | 87 792         |
| Setup source vectors      | 1 641          | 12 631         |
| Owner input bytes         | 272 236 544    | 213 319 680    |
| Terminal output bytes     | 52 690 944     | 41 287 680     |

For capped recovery, the root and helper blocks are 360 and 350, respectively. The corresponding dual blocks are 345 and 336. For the latter helper test, the classical and quantum work logarithms are 133.512 and 124.440, with classical core exponent 98.112. None is a number of scheme-security bits.

The source equations, exhaustive parameter search, selected bounds and complete costs establish a limited parameter tradeoff, not a new attack, optimized conventional compiler, security qualification or runtime ranking. The high-digit point has not been executed by the receiver trial, which uses the unchanged two-digit profile.

### G.3 Direct paired terminal receiver: admission and execution

This subsection gives an admitted encrypted realization of the paired conventional witness in Proposition 17. The realization uses one joint product layer and terminal linear recovery; it adds no exclusive JetHE capability. Its small public key must be read together with its large input and output traffic.

**G.3.1 A deterministic one-layer admission** Let  $p = 65537$ ,  $n = (p - 1)/2 = 32768$ , and  $R = \mathbb{Z}[\zeta_p + \zeta_p^{-1}]$ , in the basis  $\rho_i = \zeta_p^i + \zeta_p^{-i}$  for  $1 \leq i \leq n$ . Scalar one has all coordinates  $-1$ . The full CRT encoding identifies  $R/2R$  with 2048 copies of  $K = \mathbb{F}_{65536}$  using a fixed slot ordering. Plaintexts use the binary coordinate representative  $\mu \in \{0, 1\}^n$ .

The coefficient product constant is  $\kappa = 4n - 3 = 131069$ . Indeed, in any fixed output coordinate, the multiplication tensor has diagonal absolute sum  $2n - 1$ :

one diagonal entry is  $-1$ , the others are  $-2$ . Its off-diagonal absolute sum is  $2n - 2$ , from  $\rho_i \rho_j = \rho_{i+j} + \rho_{i-j}$ , with  $\rho_0 = 2$  and symmetric indices modulo  $p$ . Thus  $\|ab\|_\infty \leq \kappa \|a\|_\infty \|b\|_\infty$ .

Let  $\nu$  be exactly the finite source of Appendix G.2.1, including whole-vector capping at  $S = 96$  and the final independent sign. Source vectors are independent draws, while dependence among coordinates within each vector is retained. For odd  $q$ , choose uniform  $a \in R/qR$  and independent  $s, e \leftarrow \nu$ . Publish  $b = -as + 2e$  and  $a$ . Encryption uses independent  $u, e_0, e_1 \leftarrow \nu$  and returns

$$c_0 = bu + 2e_0 + \mu, \quad c_1 = au + 2e_1 \pmod{qR}.$$

This is an ordinary unscaled binary plaintext encryption interface.

**Proposition 27 (Direct terminal correctness).** *Put*

$$F = 2\kappa S^2 + S, \quad B_{\text{raw}} = 8\kappa(F^2 + F) + 2\kappa.$$

*If  $q > 2 + 4B_{\text{raw}}$ , every honest fresh ciphertext, sum of two fresh operands, and raw product of two such sums decrypts correctly by centering its phase coefficient-wise and taking parity. This holds for every source outcome and honest plaintext, without requiring input selection before setup. At the stated dimension and cap,  $q = 2^{89} - 1$  is admitted.*

*Proof.* The chosen integer phase congruence is  $c_0 + c_1 s \equiv \mu + 2E \pmod{qR}$ , with  $E = eu + e_0 + e_1 s$  and  $\|E\|_\infty \leq F$ . It is not an unreduced equality between canonical ciphertext representatives. For a sum of two inputs, keep the sum  $M$  of their binary lifts:  $\|M\|_\infty \leq 2$  and the summed error has cap  $2F$ .

Multiplying chosen phases gives

$$(M_1 + 2E_1)(M_2 + 2E_2) = M_1 M_2 + 2(M_1 E_2 + M_2 E_1 + 2E_1 E_2).$$

The parenthesized error has cap  $8\kappa(F + F^2)$ . The integer product  $M_1 M_2$  has cap  $4\kappa$ . If  $\mu_*$  is its binary representative, then  $(M_1 M_2 - \mu_*)/2$  is integral with cap  $2\kappa$ . This charges canonicalization once, yielding  $B_{\text{raw}}$ . The strict modulus inequality puts every coordinate of  $\mu_* + 2E_*$  inside the centered interval. It also dominates the fresh and mixed-operand bounds. Centering therefore recovers the chosen phase and parity gives the correct binary plaintext.

Here  $F = 2415863904$  and  $B_{\text{raw}} = 6119767220428748829808378$ . The exact admission checks

$$\begin{aligned} 2 + 4B_{\text{raw}} &= 24479068881714995319233514 \\ &< 618970019642690137449562111 = 2^{89} - 1. \end{aligned}$$

The prime exponent 89 and its 87-step Lucas–Lehmer chain are checked with exact integers, with final residue zero;  $q$  is also coprime to  $2p$ . Primality uses the usual Lucas–Lehmer criterion.

The raw product components are  $(a_0b_0, a_0b_1 + a_1b_0, a_1b_1)$ . The middle component is computed by subtracting the first and third from  $(a_0 + a_1)(b_0 + b_1)$ , using three ring products overall. No  $q^2$  arithmetic, scaled-product division, re-linearization or published  $s^2$  is needed. The recipient computes  $s^2$  locally once. This admission is for the direct terminal route, not deeper conventional shrinking or reusable FHE.

### G.3.2 Exact source optimization and conditional confidentiality

**Lemma 24 (Exact finite-CDF prefix sampler).** *For an integer CDF on uniform 256-bit inputs, partition its domain by the first 16 bits. Store a bin's output when its two endpoint outputs agree, and mark the bin ambiguous otherwise. Draw a uniform prefix; draw an independent 240-bit suffix only in an ambiguous bin and then use the original CDF comparison. This procedure has exactly the original scalar law. Applied to the latent and conditional coordinate tables, with the preserved cap and sign, it samples exactly  $\nu$ .*

*Proof.* Every 256-bit integer has a unique prefix/suffix decomposition. Within a resolved bin, monotonicity makes the CDF output independent of the suffix. Within an ambiguous bin, the fresh uniform suffix realizes the original conditional law. Thus each original output retains its integer probability mass, including aligned or repeated boundaries and zero-width outcomes. Equality of the latent law and of every conditional coordinate law gives equality of the joint raw vector law. Identical postprocessing by the cap and independent sign preserves that equality.

There are  $257 \cdot 65536 = 16842752$  bins, all checked together with their reconstructed original masses. The extra lookup uses 33685524 bytes, in addition to 2383552 bytes of finite CDF tables. The minimum and maximum random-byte budgets per vector are 65539 and 1048609. Table construction is a separate charged preprocessing operation; the optimization adds zero statistical distance. It makes no timing side-channel claim.

Assume ordinary decisional Ring-LWE for the exact  $R, q, \nu$ , with at most two independent masks and errors per secret; write its advantage bound as  $\varepsilon_2$ . One challenge row  $(A, As + e)$  simulates the public key through  $a = -2A$ ,  $b = 2(As + e)$ . Its null branch gives an independent uniform public key. With that public key distribution, two challenge rows  $(A_0, A_0u + e_0)$ ,  $(A_1, A_1u + e_1)$  simulate one encryption by setting  $b = 2A_0$ ,  $a = 2A_1$  and adding its plaintext to twice the first body. The null ciphertext is uniform independently of the message. Other encryptions can be generated locally from the public key with fresh independent coins, so the usual input-by-input hybrid applies.

For at most  $I$  submitted ciphertexts, each message endpoint is within  $(I+1)\varepsilon_2$  of a common input-independent endpoint; their absolute gap is at most  $2(I+1)\varepsilon_2$ . Deterministic evaluation and its public terminal transcript are postprocessing. The finite source law, including its internal correlations, belongs to this assumption; no numerical security level or worst-case reduction is asserted here. At

$q = 2^{89} - 1$ , bounded uniform sampling with 256 attempts per coordinate has exhaustion probability at most  $n2^{-89 \cdot 256} = 2^{-22769}$ . Coupling the two implemented message endpoints to their ideal-uniform versions adds at most twice this probability to the stated gap.

Private phase validation, cap/ambiguity counts, random-byte diagnostics and variable execution time are outside that protocol-public view. There is no decryption oracle, malicious-input guarantee, circuit privacy or output-only privacy against the recipient. In particular, the recipient recovers the returned paired statistics before routing them. The different deeper control’s helper-row model concern does not automatically transfer to this helper-free route; its absence also does not certify this route’s security.

**G.3.3 Complete fresh functional trial and charged costs** At the selected  $h = 16, L = 256, S_{\text{slots}} = 2048$  point, the direct route has  $C = 86$  product carriers,  $B = 2$  bypass carriers per owner and  $I = 4C + 2B = 348$  fresh inputs. Its source catalogue is  $2 + 3I = 1046$  finite vectors and one uniform root polynomial. All 86 raw products and four two-component bypasses are returned, with 266 output polynomials. At 12 bytes per coefficient the public-key, input and output payloads are respectively 786432, 273678336 and 104595456 bytes: 0.75, 261 and 99.75 MiB. These exclude framing; codecs, tables, owner preparation and all recipient work remain explicit costs.

| Instrumented ring-product phase   | Count |
|-----------------------------------|-------|
| Public key                        | 1     |
| Fresh input encryption            | 696   |
| Evaluator raw tensors             | 258   |
| Optional private input validation | 348   |
| Recipient secret square           | 1     |
| Terminal decryption               | 176   |
| Total                             | 1480  |

Both the 16-job, length-4 validation case and the full length-256 encrypted execution recovered the expected outputs. The full run checks 14352384 input/output binary coordinates and recovers all 4096 expected composition coefficients, byte-identical to the independent C++ Horner reference. The length-4 validation additionally uses independent Python Horner arithmetic and compares the three-product tensor with four direct products from the original ring kernel. Both runs check full-dimensional range arithmetic, small independent products and field/codec alignment. The full run has two additional diagnostic ring products beyond the displayed inventory. Omitting the outer or inner correction gives 4064 or 3545 incorrect nonconstant coefficients, respectively.

The full finite-source draws consume 68983124 OS-random bytes; the original fixed-draw budget for that many vectors is 1096845014. There are 14311 ambiguous scalar prefixes and no observed cap event. This is a byte-budget observation, not a rare-event probability estimate or a same-tape timing comparison.

The recorded instrumented wall time is 160.372 seconds. Named workflow phases sum to 127.060 seconds, including fresh setup, both owner preparations, codecs, source generation, encryption, counted serialization, evaluation and terminal recovery. Input encoding, encryption and evaluation contribute 47.626, 37.973 and 20.132 seconds respectively. Source timing is included in setup and encryption. Initial CDF, prefix-table and CRT-asset construction is excluded; their loading and retention is included. Ciphertext bytes go to counting hash sinks; network transport and framing are not implemented. Peak memory is 192168 KiB.

These are single-run Python/GMP functional diagnostics collected with competing process activity, rather than controlled comparative timings. They must not be divided into the native implementation's separately measured times to claim a speedup. Exact arithmetic checks confirm the error bounds, operation counts and recovered output. The execution demonstrates a conventional trade-off with a small public key and substantial traffic; it does not establish security-matched performance.

**G.3.4 A Control in the Native Ramified Algebra** A further control uses the native algebra  $(K[z]/z^{256})^{16}$ . It retains the measured independent trit/CBD20 source laws and the unchanged optimized native arithmetic backend. It has one ordinary two-prime public key and no evaluation keys. This isolates more of the representation and source differences than the preceding real-ring trial, while remaining a depth-one comparator.

**Proposition 28 (Native-algebra interpolation control).** *Interpolate 128 field values at distinct public  $K$  points into a polynomial of degree at most 127 in each native jet. Multiplication followed by terminal evaluation recovers 128 independent field products per jet. The paired contraction consequently uses 86 raw products, 346 input ciphertexts and 88 terminal ciphertexts at the stated point. The unchanged trit/CBD20 source is deterministically admitted at the first two native primes, with product modulus*

$$q = 1329227993889339670443893069050298881.$$

*Proof.* The product degree is at most 254, so reduction modulo  $z^{256}$  loses nothing. Public evaluation recovers the pointwise product. Sixteen jets provide 2048 effective product slots, and the paired list has 175776 entries. Its 86 batches need four operand ciphertexts each. Each owner's 4096 correction coefficients fit directly in one full native carrier; thus there are  $4 \cdot 86 + 2 = 346$  inputs and  $86 + 2 = 88$  outputs, of which 86 have three components and two have two components.

In the native integer basis  $\kappa = 130816$ . The fresh phase  $\mu + 2E$  has  $\|E\|_\infty \leq F = (2\kappa + 1)20 = 5232660$ . Adding owners gives signal bound 2 and error bound  $2F$ . Expanding a raw product and reducing its integer signal to the canonical binary lift gives error at most

$$B = 8\kappa(F^2 + F) + 2\kappa = 28654706788599782912.$$

The final term bounds the signal’s parity carry. The certified modulus above satisfies  $q > 2 + 4B$ ; bypasses satisfy the smaller fresh bound. The recipient evaluates the three-component phase using its local  $s^2$ , with no relinearization or scale-and-round operation.

Interpolation here is a one-product encoding, not an algebra embedding supporting arbitrary repeated field-slot multiplication. The recipient decrypts all products and corrections and performs fixed linear evaluation and routing. Its knowledge of prepared functions is within the stated evaluator-only privacy interface. Security remains conditional on the actual native source assumptions; no numerical security level follows from sharing the algebra and source laws.

A length-four validation case and the full 16-job length-256 trial both pass. The full trial checks 28442624 coefficients against the phase bounds and recovers all 4096 field words against the independent Horner oracle. It samples 1040 small source vectors, including 693 errors. Omitting an entire outer or inner correction gives 4080 or 3545 wrong output coefficients respectively. At two 64-bit RNS words per native coefficient, raw public-key, input and output payloads are 2, 692 and 262 MiB. The emitted serializer adds its framing; ciphertexts stream to counting/hash sinks, with no network transport or retained wire files.

Named complete-workflow phases sum to 15.696 seconds; instrumented wall time is 20.151 seconds, including 4.365 seconds of private diagnostics. Encryption takes 9.348 seconds, recipient decryption/native decoding 2.618, input encoding 1.740, serialization 0.790, inner-owner preparation 0.634 and evaluation 0.332. Setup, owner interpolation and recipient evaluation are included; kernel compilation is excluded. Peak RSS is 91888 KiB under a 2 GiB limit. This single-thread diagnostic shares the host with another computation. It supplies no controlled time ratio against the native measurement series or the real-ring trial.

This execution demonstrates the conventional small-key/large-traffic tradeoff under the stated correctness bounds. The controlled measurement series below supplies a separate complete-workflow comparison; deeper compilers remain eligible.

**G.3.5 A Common Ordinary Source Premise** The two native-algebra implementations can be compared under the same sufficient ordinary source premise, with different reduction losses. Let  $q_2 \mid q_3$  be the existing two- and three-prime products, and let  $\epsilon_{q_3,46}$  denote the exact trit-secret/CBD20-error source gap used in (15).

**Proposition 29 (Source alignment of the native-algebra control).** *For  $J$  batches with 346 fresh inputs each under the control’s single key, its evaluator-view gap satisfies*

$$\text{Adv}_{\text{control}}^{\text{gap}} \leq 2(1 + 346J)\epsilon_{q_3,46} + 2\delta_{\text{control}}(J),$$

*at adversary time plus the complete reduction overhead. The native compiler’s corresponding factor is  $2(9 + 35J)$ , as in (15). The source and implementation deviations  $\delta$  are retained separately for each implementation.*

*Proof.* Projection  $R/q_3R \rightarrow R/q_2R$  is a surjective ring homomorphism. Projecting two distinct primitive rows preserves uniform masks and null bodies exactly, together with the same integer trit secret and CBD20 errors. In particular, the common error across RNS channels is preserved; no derived row is treated as a new independent sample. Row deletion and projection therefore supply the ordinary two-row  $q_2$  game from the 46-row  $q_3$  game with no gap loss. Multiplication by the public unit 2 commutes with projection.

Replace the control's root public key by a uniform pair using one source hybrid. For each two-row encryption hybrid, receive the challenge masks  $A_0, A_1$  and set  $(b, a) = (2A_0, 2A_1)$ , inducing a uniform public key. Double the bodies and add the chosen plaintext to the first. This has exactly the required trit-ephemeral/CBD20-error law; the reduction does not choose masks to match a fixed key. Other inputs can be generated from the induced public key with independent local coins. The evaluator outputs are deterministic descendants of these inputs. The two message-world chains give  $2(1 + 346J)$  source terms, and the two implementation couplings give the stated deviation term.

At two batches the source-gap factors are 1386 for the control and 158 for the native compiler. This is a common sufficient assumption, not equal quantitative scheme security or evidence that the assumption is hard. Each reduction has its own simulation cost. Recipient disclosures and private diagnostic timings remain outside this view; correctness retains its separate input/depth restrictions.

The honest control source catalogue has two uniform prime-coordinate vectors,  $1 + 346J$  trit vectors and  $1 + 692J$  errors. Applying the existing eight-refill stopped-source coupling gives the bounded logical test budget  $B_{\text{control}}(J) = N(25 + 3460J)$  words. If  $\rho_2 = \max_{i < 2}(2^{64} \bmod p_i)/2^{64} < 2^{-30}$ , its ideal stop probability is at most

$$N(2\rho_2^8 + (1 + 346J)2^{-512}).$$

For  $J = 2$ , the word budget is 455147520 and this probability is strictly below  $2^{-222}$ ; the native catalogue is 264699904 words with its sufficient bound below  $2^{-207}$ . These are stopped logical source tests, not caps on the actual unchanged OS sampler, costs of the local privacy simulator, or attack-work estimates. The corresponding bounded word-source distinguishing gap and other implementation deviations still have to be charged.

Exact arithmetic checks verify these counts and inequalities, small projection fibers, unit-2 phase identities and the ideal CBD20 moments. These support the stated algebraic and probabilistic calculations, without supplying numerical hardness or matched-security timings. Separate encrypted validation executions check the phases of both implementations used in the following complete-workflow experiment.

**G.3.6 Controlled Complete-Workflow Evidence** This table uses the two-prime control. The one-prime correctness bounds and measurements are reported separately below; no samples are combined across them.

**Table 16.** Native composition and the two-prime control: median (minimum–maximum) over three fresh setups per arm. Times are seconds; RSS is MiB. All complete local batch work is charged. These are descriptive samples under the stated source assumptions, not a numerical security match.

| Metric       | Terminal JetHE            | One-layer control      |
|--------------|---------------------------|------------------------|
| Setup        | 3.196 (3.184–3.202)       | 0.100 (0.098–0.109)    |
| Cold batch   | 2.543 (2.533–2.550)       | 14.586 (14.488–14.626) |
| Warm batch   | 2.533 (2.497–2.670)       | 14.519 (14.504–14.559) |
| Setup + cold | 5.734 (5.729–5.745)       | 14.684 (14.597–14.726) |
| Peak RSS     | 593.957 (593.953–593.992) | 74.684 (74.551–74.684) |

The paired measurement series uses three fresh setups per arm in the fixed order native/control/control/native/native/control. Each setup processes cold and warm batches with fresh coins and recomputed owner preparations. All twelve batches recover the same full 4096-coefficient Horner oracle. Only public context, keys and public maps are reused. Both arms use the same field backend, serializer/counting sink and terminal decryption codec, with secret-square preparation charged during setup.

The host is an AMD Ryzen 9 9950X under WSL2. Sequential executions use CPU 0, one arithmetic thread and a 2 GiB address-space limit. Competing physical-host activity is uncontrolled. Correctness validation and timing use the same construction and arithmetic backend.

**Table 17.** Every setup in the predeclared native/control/control/native/native/control order for the two-prime comparison. Setup, cold, warm and first-use times are seconds; RSS is MiB. No observation is discarded.

| Index | Arm     | Setup | Cold   | Warm   | First use | RSS     |
|-------|---------|-------|--------|--------|-----------|---------|
| 0     | JetHE   | 3.184 | 2.550  | 2.670  | 5.734     | 593.992 |
| 1     | Control | 0.109 | 14.488 | 14.519 | 14.597    | 74.551  |
| 2     | Control | 0.100 | 14.626 | 14.504 | 14.726    | 74.684  |
| 3     | JetHE   | 3.196 | 2.533  | 2.533  | 5.729     | 593.953 |
| 4     | JetHE   | 3.202 | 2.543  | 2.497  | 5.745     | 593.957 |
| 5     | Control | 0.098 | 14.586 | 14.559 | 14.684    | 74.684  |

Setup wall time charges context/codec construction, public maps/tables, all secrets and keys, recipient preparation and public serialization. Each batch wall starts at original owner coefficients and ends at full recipient recovery. It includes owner preparation/interpolation, encoding, source generation and encryption, complete framing, evaluator arithmetic, decryption and decoding. Process startup/import, fixture/oracle construction, post-batch verification and physical network/disk transmission are excluded consistently. The control streams four

inputs and one raw product at a time; all bytes are charged. No private phase checks run inside measured batches.

Separate encrypted validation executions check 57 native and 434 control states, respectively 3735552 and 28442624 phase coefficients. These private phase checks are excluded from the timings. Every sample from the six independent executions is retained. Table 16 reports descriptive medians and ranges; these are not confidence intervals. The warm ratio is 5.730741 and the setup-plus-cold ratio is 2.560968. The control’s faster evaluator and smaller public material and RSS remain adverse dimensions of the native result.

The supplied observations retain every measured phase and permit recomputation of the summaries. The distinct source reductions above remain material: these measurements establish neither numerical hardness nor a general HE ranking.

**G.3.7 A One-Prime Control with Fixed-Batch Correctness** The two-prime control above uses an all-coins fresh-error cap. Applying the same fresh-encryption concentration used for the native terminal profile yields a smaller conventional admission without changing its algebra, independent source laws, preparation or recipient interface.

**Proposition 30 (One-prime terminal control).** *The 346-input, 86-product interpolation/polarization control is correct under ideal coins for a collection of at most 1024 fixed batches with failure probability below  $2^{-149}$  at  $q = p_0 = 1152921504002872321$ . Its fresh-error cap is  $F_1 = 36636$  and a sufficient raw-product cap is  $B_1 = 1404684555427328$ .*

*Proof.* Each fresh chosen phase is  $\mu + 2E$  modulo  $q$ , with  $E = e_{\text{pk}}u + e_0 + e_1s$ . Condition on the independent trit vectors  $s, u$  for that ciphertext. Lemma 6 and the CBD20 moment bound give coordinate variance proxy  $v = 80N + 10$ . At  $T = 16$ ,  $F_1 = \lceil T\sqrt{v} \rceil = 36636$ , with bad-event probability at most  $2N \exp(-128)$  for a fresh ciphertext. Reused public-key errors correlate encryptions; the uniform marginal bound and union over  $346 \cdot 1024$  encryptions require no independence between their events. A positive rational partial sum for  $\exp(128)$  certifies  $2N \cdot 346 \cdot 1024 \exp(-128) < 2^{-149}$ .

On the simultaneous fresh-cap event, each mixed operand has signal cap 2 and error cap  $2F_1$ . The existing deterministic product argument gives

$$B_1 = 8\kappa(F_1^2 + F_1) + 2\kappa.$$

The final term charges the canonical parity carry. This step imposes no independence on the raw operands and preserves their common root-key dependency. Bypasses have cap  $F_1$ . The exact margin  $p_0 - 2 - 4B_1 = 1147302765781163007 > 0$  proves centering at every output. No switching, scaling or relinearization adds an event. The unchanged degree-127 interpolation in length-256 jets preserves every product; the same terminal maps therefore recover every composition coefficient.

Exact arithmetic verifies the first prime’s complete factor/order certificate and the strict inequalities. The probability calculation uses 513 positive rational exponential terms, not an empirical rare-event measurement. Its original deterministic fresh cap fails the one-prime admission, and that adverse comparison is retained. The result applies to the stated fixed input collection; no adaptive correctness extension is claimed.

Projection of two primitive rows from  $q_3$  to  $p_0$  supplies the same sufficient source premise as Proposition 29. The two-batch source-gap factor remains 1386, with its own simulation and implementation deviations. One uniform prime-coordinate vector gives the logical stopped-word budget  $N(17 + 3460J)$ : 454623232 words at two batches, with ideal stop bound below  $2^{-223}$ . This is source accounting, not numerical hardness or a cap on the unchanged OS sampler.

One uint64 residue per coefficient gives raw public/input/output polynomial payloads of 1/346/131 MiB. The separately validated driver uses the unchanged arithmetic and source modules, a one-limb version of the same raw product, and a common final codec for both arms. A separate encrypted validation execution checks all 434 control states and 28442624 phase coefficients, recovering all 4096 Horner outputs. A separate native check checks all 57 states under the adapted shared decoding driver. Public zero-secret endpoint fixtures additionally distinguish centered parity from uncentered odd-modulus parity; they are not fresh encryption.

**G.3.8 Complete one-prime-control measurements** After these encrypted correctness checks, a separate six-setup experiment used the predeclared native/control/control/native/native/control order. Each setup performs a cold batch and a warm batch with fresh coins; all twelve batches recover all 4096 expected coefficients. No observation is discarded. Table 19 gives every sample, and Table 18 preserves its medians and observed ranges.

**Table 18.** Native composition and the one-prime control: median (minimum–maximum) over three fresh setups per arm. Times are seconds; RSS is MiB. All complete local batch work is charged. These are descriptive samples under the stated source assumptions, not a numerical security match.

| Metric       | Terminal JetHE            | One-prime control      |
|--------------|---------------------------|------------------------|
| Setup        | 3.210 (3.202–3.214)       | 0.088 (0.087–0.088)    |
| Cold batch   | 2.555 (2.546–2.624)       | 9.183 (9.176–9.218)    |
| Warm batch   | 2.521 (2.480–2.537)       | 9.203 (9.111–9.230)    |
| Setup + cold | 5.769 (5.748–5.834)       | 9.271 (9.263–9.306)    |
| Peak RSS     | 593.988 (593.941–594.137) | 62.508 (62.465–62.613) |

These are descriptive measurements with three setups per arm, not confidence intervals or an optimal-compiler comparison.

**Table 19.** Every setup in the predeclared native/control/control/native/native/control order for the one-prime comparison. Setup, cold, warm and first-use times are seconds; RSS is MiB. No observation is discarded.

| Index | Arm     | Setup | Cold  | Warm  | First use | RSS     |
|-------|---------|-------|-------|-------|-----------|---------|
| 0     | JetHE   | 3.214 | 2.555 | 2.537 | 5.769     | 593.941 |
| 1     | Control | 0.088 | 9.218 | 9.230 | 9.306     | 62.613  |
| 2     | Control | 0.087 | 9.176 | 9.203 | 9.263     | 62.508  |
| 3     | JetHE   | 3.210 | 2.624 | 2.521 | 5.834     | 593.988 |
| 4     | JetHE   | 3.202 | 2.546 | 2.480 | 5.748     | 594.137 |
| 5     | Control | 0.088 | 9.183 | 9.111 | 9.271     | 62.465  |

Arithmetic consistency checks cover 327680 phase coefficients without cryptographic coins. Separate encrypted validation executions check 3735552 native and 28442624 control phase coefficients. Measurements omit private phase diagnostics but retain complete output checks. The platform is an AMD Ryzen 9 9950X under WSL2 Linux 6.18.33.2, Python 3.14.4, with one arithmetic thread pinned to CPU 0. Competing physical-host activity is uncontrolled. Each execution has a 2 GiB address-space cap, 900-second CPU limit and 960-second wall limit.

The cost boundary is the same as the two-prime experiment: independent owner preparation, field encoding, encryption, complete serialization to a counting sink, evaluation, decryption and final recipient maps. Setup includes keys and terminal secret-square preparation. Physical transport is unmeasured. The native schedule, source laws and arithmetic backend are unchanged; the control uses one residue limb, and both arms use the same final field codec with explicit centered decoding.

Warm medians are 2.520930 and 9.202781 seconds, a control/native ratio of 3.650551. Setup plus cold medians are 5.769150 and 9.271179 seconds, a ratio of 1.607027. The control’s evaluator is faster: median 0.089218 versus 0.962098 seconds. Its public material is 1 versus 391 MiB and median peak RSS is 62.508 versus 593.988 MiB. Native encryption costs 1.324081 versus 5.071588 seconds; recipient decryption and the native codec cost 0.031509 versus 1.513215 seconds. Phase medians need not sum to the median complete batch. Public/input/output raw polynomial payloads are 391/103/3 MiB for native and 1/346/131 MiB for control.

The reduction factors remain different and neither profile has a numerical security certificate. The two-prime and one-prime measurements use separate setups and backend configurations, so their ratios do not isolate the effect of removing a prime. The fixed-backend comparison follows below.

**G.3.9 Direct comparison with JetHE-102** The optimized-backend comparison uses the same one-prime control algorithm and replaces its ring context with the fixed-multiplier backend used by JetHE. The adapter changes only the ring-context constructor. Arithmetic consistency checks verify agreement for the

one-prime transform, sampler, product and explicit-coin encryption; the same equivalence arguments apply at the first prime and in two/three-prime contexts. The native arm uses the unchanged 102-row implementation.

For the privacy comparison, delete to two distinct primitive rows in the JetHE-102  $q_3/37$  game and project modulo its first prime  $p_0$ . Since  $p_0 \mid q_3$ , the ring map is surjective with uniform fibers. Real rows retain the same integer trit secret and primitive  $\text{CBD}_{20}$  errors; independent uniform null rows project to independent uniform rows. Unit-2 normalization commutes with this map. Thus the two-row one-prime gap is at most the 37-row top-modulus gap at time including projection. This is the direction used in Proposition 29, now with the actual 37-row cap and first-prime modulus. It is not a numerical hardness assertion.

The JetHE-102 incoming catalogue (1, 8, 37, 12, 16, 9, 10, 9, 5) has maximum 37 and sum 107. Its existing acyclic simulation uses nine secrets and 35 fresh inputs per batch. The control uses one key and 346 fresh inputs. For  $J$  batches under one setup, their evaluator-view gap factors are  $2(9 + 35J)$  and  $2(1 + 346J)$ , plus their own source-deviation terms and complete simulation time. At  $J = 2$  the factors are 158 and 1386. For example, scheme gap at most  $1/4$  with zero source deviations would require common source gap at most  $1/5544$ , at adversary time plus the larger simulator cost. Neither zero deviation nor this source hardness is established. Each experimental setup defines a separate two-batch instance; the factor is not a joint-publication guarantee for all setups. Recipient privacy and adaptive key-dependent correctness are not added.

The existing stopped logical word-source bounds specialize to 213319680 native words and 454623232 control words for two batches. The control stop probability is below  $2^{-222}$ . These describe bounded logical source tests, not limits on the unchanged OS sampler or attack costs. Native and control fixed-input ideal correctness bounds remain below  $2^{-129}$  and  $2^{-149}$  for at most 1024 batches, respectively.

**Table 20.** Every setup in the predeclared JetHE/control order. Complete batch and setup times in seconds; all outputs pass.

| Index | Arm     | Setup    | First    | Warm Setup | + first  |
|-------|---------|----------|----------|------------|----------|
| 0     | JetHE   | 2.070366 | 1.764133 | 1.746354   | 3.834498 |
| 1     | Control | 0.089316 | 7.848439 | 7.724638   | 7.937754 |
| 2     | Control | 0.086764 | 7.695165 | 7.695535   | 7.781928 |
| 3     | JetHE   | 2.006243 | 1.772315 | 1.737026   | 3.778558 |
| 4     | JetHE   | 2.029513 | 1.768277 | 1.709757   | 3.797790 |
| 5     | Control | 0.086405 | 7.705324 | 7.729049   | 7.791730 |
| 6     | Control | 0.088149 | 7.772318 | 7.695951   | 7.860467 |
| 7     | JetHE   | 2.029574 | 1.780274 | 1.723741   | 3.809849 |

Separate encrypted validation executions check 57 native and 434 control phase states, 32178176 coefficients in total. All eight measured setups, in the

order shown, use fresh keys and two batches of fresh encryption coins. All 65536 measured output coefficients pass; there are no diagnostic phase checks in the measured runs. Both arms use one thread, CPU affinity zero, serial execution, a 2 GiB address-space cap, 900 CPU seconds and 960 wall seconds per execution. Competing physical-host activity is uncontrolled.

Setup charges context, codec and interpolation construction, keys and hints, the recipient key square and public serialization. Each complete batch charges both owners, codecs/interpolation, source sampling, encryption, serialization to a count sink, evaluation and recipient recovery. Lazy table construction is charged at its first timed use. First use is each setup plus its own first batch. Interpreter/import startup, fixture/oracle generation and physical transport are excluded. All observations and nonfatal launcher/observer diagnostics are retained. Four samples per arm give descriptive medians and ranges only.

Table 21 and Figure 1 use this experiment series alone. JetHE’s warm/first-use ratios are 4.456/2.057, while the control retains its faster evaluator, lower RSS and 1 MiB public material. Raw public/input/output bytes correspond to 297/103/3 MiB for JetHE and 1/346/131 MiB for the control. Serialized public, input and output sizes are 311428896, 108004000 and 3145864 bytes for JetHE, and 1048704, 362822688 and 137375408 bytes for the control. This compares implementations with a shared arithmetic backend; it is not an independent conventional implementation, a measurement of the Gaussian joint-batch exponent or an unrestricted performance ranking.

**Table 21.** 102-row JetHE and one-prime control using the same optimized arithmetic backend. Four fresh setups per arm, two batches each. Seconds except RSS; medians with observed ranges.

| Metric              | JetHE                      | One-prime control       |
|---------------------|----------------------------|-------------------------|
| Setup               | 2.030 [2.006, 2.070]       | 0.087 [0.086, 0.089]    |
| Complete warm       | 1.730 [1.710, 1.746]       | 7.710 [7.696, 7.729]    |
| Setup + first batch | 3.804 [3.779, 3.834]       | 7.826 [7.782, 7.938]    |
| Warm encryption     | 0.908 [0.900, 0.922]       | 3.690 [3.672, 3.697]    |
| Warm evaluator      | 0.597 [0.589, 0.605]       | 0.092 [0.092, 0.092]    |
| Peak RSS (MiB)      | 499.344 [498.645, 499.859] | 63.395 [63.121, 63.461] |

#### G.4 Recent SIMD comparators and exact-word terminal compilation

This subsection gives an algorithm-level eligibility analysis and an explicit terminal compiler. It supplies neither new encrypted measurements nor a security-qualified parameter choice. Ordinary approximate arithmetic is not the target functionality, but a discrete encoding built on CKKS remains an eligible exact-output competitor.

#### G.4.1 Polynomial evaluation: the hypotheses and the actual call site

Zhao et al. [44] give new multiplication bounds for non-power-of-two slot degree  $d$  with polynomial degree  $D \leq d$ , and for  $p > 2$ ,  $d < D \leq d \log p$ , respectively. These advertised regimes do not directly cover characteristic two,  $d = 16$ , and the degree-255 private outer polynomial of composition. Here  $p$  is the characteristic, not  $|K| = 65536$ . Moreover, the terminal product and circuit compiler evaluates its input polynomials at public points before encryption; its evaluator does not perform a high-degree univariate evaluation that could simply be replaced by this primitive.

The authors' implementation [43] makes the distinction concrete. Its norm-based evaluator receives plaintext integer-polynomial coefficients, forms  $T = x^4 + s_1x^3 + s_2x^2 + s_3x + s_4$ , and combines its 24 Frobenius conjugates. The norm stage uses exponents 1, 2, 3, 6, 12, with the first two hoisted together; power formation, public multipliers and a separate lifting polynomial are additional work. The digit-extraction call operates on unpacked integer slots. Interpreting that norm as coefficient-conjugate factor evaluation uses  $\sigma(x) = x$ , unlike an arbitrary binary-extension slot value. The implementation also offers encrypted-coefficient polynomial evaluation, but this interface does not establish that the norm-based bound applies to encrypted coefficients.

Composition permits owner-local preparation, including encrypted preparations of a private polynomial. Such an adaptation remains eligible, but making its private-dependent factors public changes the interface. A valid adaptation must price encrypted coefficients, maps, keys and preparation. The stated parameter regimes and the cited public implementation do not provide a directly substitutable bound for this private-composition task. An encrypted-coefficient adaptation requires its own complete cost analysis.

#### G.4.2 An exact-word compiler that needs no intermediate field reduction

The Gao–Zheng triangle encoding supports  $n_w$ -bit integer words with  $N_A/n_w$  word slots at ring dimension  $N_A$  under its stated encoding and correctness conditions [21,42]. Integer multiplication is not binary-extension-field multiplication. Nevertheless, terminal recovery permits the following classical carry-free Kronecker-substitution route [25]; Boolean emulation is not mandatory at every gate. The classical substitution is not our invention.

Let  $K = \mathbb{F}_2[Y]/(\psi)$ ,  $\deg \psi = d$ , and lift each field element to its unique binary polynomial  $\tilde{a} \in \mathbb{Z}[Y]_{<d}$ . Consider a circuit using addition and multiplication, and evaluate the same circuit on these integer-polynomial lifts. All coefficients are nonnegative. Suppose every lifted intermediate has degree at most  $D_Y$  and coefficients at most  $B_c$ . Choose

$$\omega = \max\{1, \lceil \log_2(B_c + 1) \rceil\}, \quad n_w \geq \omega(D_Y + 1), \quad \beta = 2^\omega > B_c,$$

with a word width admitted by the backend. Padding to a power of two is a sufficient choice for the triangle family, not a proof of noise admission.

**Proposition 31 (Carry-free recovery).** *Owners encode  $\tilde{a}(\beta)$  in  $\mathbb{Z}/2^{n_w}\mathbb{Z}$ , and the evaluator executes the integer circuit. Conditional on correct word evaluation*

and decryption, the recipient recovers the intended field output by taking the parity of each base- $\beta$  digit of the canonical unsigned output word and reducing the resulting binary polynomial modulo  $\psi$ .

*Proof.* Integer evaluation at  $Y = \beta$  commutes with additions and products. At each gate, the coefficient bound prevents digit carries, and the degree bound gives a value below  $\beta^{D_Y+1} \leq 2^{n_w}$ ; thus no word wrap occurs. The final digits are exactly the lifted coefficients. The map  $\mathbb{Z}[Y] \rightarrow \mathbb{F}_2[Y]/(\psi)$  is a ring homomorphism, so digit parity and polynomial reduction give the original field circuit's output.

For  $d = 16$ , use the field polynomial represented by `0x1100b` and put  $S = 1 + Y + \dots + Y^{15}$ . Nonnegative coefficientwise majorants for a product, the shallow circuit, and the deep pointwise circuit are  $S^2$ ,  $S^2 + S$  and  $(S^2 + S)^4$ . Their constituent intermediates are also bounded by these displayed maxima and degrees; in the deep case  $S^2 + S$  has constant coefficient two, so its square and fourth power dominate earlier nodes.

For composition, first prepare the field matrix  $A_{ji} = [z^j]g^i$  at its owner, as in Appendix E.2. Lift those field entries, not integer powers of a lifted  $g$ . Each output is the parity/reduction of the aggregate

$$\sum_{i=1}^{L-1} \tilde{A}_{ji} \tilde{f}_i + \mathbf{1}_{j=0} \tilde{f}_0.$$

Its majorant at  $L = 256$  is  $255S^2 + S$ . No individual product summands are delivered. Exact integer convolution gives the following sufficient widths. The slot column is conditional on an admitted triangle backend at  $N_A = 65536$ .

| Task           | $D_Y$ | $B_c$     | $\omega$ | $n_w$ | Word slots |
|----------------|-------|-----------|----------|-------|------------|
| composition    | 30    | 4081      | 12       | 512   | 128        |
| product        | 30    | 16        | 5        | 256   | 256        |
| affine product | 30    | 17        | 5        | 256   | 256        |
| balanced tree  | 120   | 160715046 | 28       | 4096  | 16         |

*Whole-workflow costs and a stronger composition packing.* For product and circuit, retain the terminal additive-CRT compiler from Appendix E.2. With  $P$  points,  $J$  jobs, and  $s_A = N_A/n_w$  word slots, put  $B_A = \lceil JP/s_A \rceil$ . Logical counts of word inputs, products, additions and outputs are  $(2, 1, 0, 1)B_A$ ,  $(3, 1, 1, 1)B_A$ , and  $(12, 7, 4, 1)B_A$ . At  $J = 16$ ,  $L = 256$ , this gives  $B_A = 32, 32, 2048$  for product, affine product and balanced tree. The same homomorphism argument permits deferring both jet truncation and field reduction until terminal decoding. These are attained logical counts under word correctness, not equally priced ciphertext operations.

One-output-per-word composition column contraction has  $B_A = \lceil JL/s_A \rceil$  and at most  $(2L-1)B_A$  inputs,  $(L-1)B_A$  word products and additions, and  $B_A$  aggregate outputs. This is not optimal packing. Since  $f_i$  is common to the output

rows of one job, pack  $r$  matrix entries as  $\sum_{j < r} \tilde{A}_{ji} Y^{(2d-1)j}$  while leaving  $\tilde{f}_i$  in the low-degree block. The products occupy disjoint degree blocks and need  $n_w \geq 372r$  at this profile. For  $n_w = 4096$ ,  $r = 11$ , sixteen jobs use  $\lceil 16 \lceil 256/11 \rceil / 16 \rceil = 24$  bundles, hence at most 6120 word products, rather than 8160 for the 512-bit one-output layout. Recover each row by separating its 31 radix digits and reducing that block modulo  $\psi$ , not by reducing the entire packed polynomial. Place  $f_0$  only in the block corresponding to global row zero.

The actual composition promise  $g(0) = 0$  makes this construction stronger still:  $A_{ji} = 0$  for  $i > j$ . In bundle  $k$ , whose last row is  $j_k = \min(11k + 10, 255)$ , only columns  $1 \leq i \leq j_k$  need products. The omission depends only on this public structural bound, not on whether a private matrix entry happens to vanish. With slots ordered by the sixteen jobs, all bundles can share the same 256 encrypted low-block  $f_i$  words. The resulting concrete ledger is

$$\sum_{k=0}^{23} j_k = 3268 \text{ word products}, \quad 3268 + 256 = 3524 \text{ input ciphertexts},$$

with 3245 word additions and 24 aggregate outputs, without encrypted rotations. Initialize each accumulator with its first product and add  $f_0$  only in the first bundle. Reuse requires the same admitted key/level and preservation of the shared input state; copies, alignment and correlated-noise admission remain charged. This is an explicit stronger upper bound, not an optimality claim. Different word widths change the per-product, modulus, key and encoding costs; the smaller count alone does not establish a faster implementation. Public zeros, diagonal regrouping and further packing remain permitted.

Let  $M_A, A_A$  price complete word operations at their actual levels, including scaling, rescaling and key policy. For example, the referenced full-integer multiplication routine executes a ciphertext product and rescaling, then a plaintext- $t$  product and another rescaling [20]; it is not a bare ring multiplication. The short path has a different input-encoding precondition: the binary digits of an integer lift do not establish that backend normalization invariant. A complete cost ledger adds public setup and keys, owner field preparation, the  $(c+1)J$  CRT transforms for the pointwise routes, word encoding/encryption, transfers and terminal decryption/recovery to the displayed word-operation counts. Bit placement costs  $O(n_w)$  per prepared word; digit/parity recovery and schoolbook field reduction cost  $O(n_w + dD_Y)$  bit operations per output word. For composition's packed rows this becomes  $O(n_w + rd(2d-1))$ , since the  $r$  field reductions are separate.

With  $I_A$  fresh two-component inputs and  $b_Q$ -bit coefficients, input traffic is  $2I_A N_A b_Q$  bits before metadata:  $I_A = cB_A$  for the pointwise product and circuit routes, but  $I_A = 3524$  for the strengthened composition example. Outputs use their actual component count and final modulus. Keeping  $w_A$  ciphertext registers costs  $O(w_A N_A b_Q)$  bits, plus all keys and backend scratch. Processing one bundle at a time bounds register count independently of the number of bundles; it does not remove the owner matrix or CRT workspace. The pointwise product and circuit cache uses  $O(cJPn_w)$  bits when all prepared words are

retained; composition separately retains its  $O(JL^2)$  field-element matrices and up to  $O(I_A s_A n_w)$  prepared-word bits, unless streamed. These dimensions, widths and costs are not matched-security parameters or measured memory.

The recipient obtains unreduced integer digits, which can reveal more than the field output. This route is eligible only under the stated absence of privacy against the recipient; it cannot silently implement a stronger output-privacy contract. Conversely, an encrypted coefficient map or an unbounded continuation outside the proved degree/majorant requires charged field recovery or a different compiler. The proposed 256/512/4096-bit leveled profiles still require concrete correctness and security parameters; the operation counts do not establish that this is the fastest ALU compilation.

**G.4.3 Conversion and reusable computation remain distinct** Wei et al. [42] complement the original batch-amortized conversion with a single-ciphertext route: overflow reset, multivalued bit extraction, carry-save contraction and carry lookahead. Their accounting uses three bootstrapping stages and  $O(\log \log K + \log n_w)$  additional levels, where  $K$  bounds reset overflow. Constant stage count is not constant bit complexity, state, LUT work or ciphertext modulus. Their specified layout uses two Boolean ciphertexts for the full payload of one arithmetic ciphertext and charges masks/movement. The original method can remain preferable for large batches; this improvement does not establish uniform dominance.

This improved conversion prevents excluding the ALU on the original single-ciphertext amortization limitation. Equally, our terminal integer route does not need that conversion merely to return its output. For genuine encrypted field reentry, Boolean AND/XOR circuits, digit extraction and polynomial reduction are eligible with their full depth, movement and refresh costs. An owner may start directly in Boolean form when useful. Neither a Boolean XOR in discrete CKKS nor a field-linear map is automatically a free ciphertext addition: the pinned XOR uses a ciphertext product. No published word-conversion timing is used as a full-jet measurement here.

## H Implementation profiles and native refinements

### H.1 Implementation profiles and alternative compilers

This subsection distinguishes measured implementations, analytic alternatives and comparison controls. Numerical parameters, implementation pins and security status belong to a row, not to the name of an HE scheme. None of the measured profiles below has an independent 128-bit qualification.

**H.1.1 Measured leveled profiles** The composition, product and circuit experiments use full extension-field coefficients, with polynomial 0x1100b, and terminal recovery. Native encoding uses the root/coset order and inverse confluent evaluation matrix of Appendix A.1. Conventional field-slot encoding uses its own

ordered evaluation points and interpolation matrix; the same field polynomial does not identify these layouts.

**Table 22.** HELib comparison profiles. The composition experiment has three fresh setups per arm; the product/circuit experiments have six native and 32 conventional configurations. Appendix J.1 describes the measurement protocol.

| Measured row               | Parameters and key structure                                                                                                                                       |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Native composition         | One sixteen-job carrier, $N = 65536$ , 35 inputs and one output; the exact independent-key graph and primes are in Table 9; this is the 203-row reference profile. |
| HELlib composition BSGS    | $m = 13107$ , 512 field slots, eight two-job bundles; 271 inputs/bundle and 16 switching matrices, including 15 direct rotations.                                  |
| HELlib composition column  | The same carrier and field layout, 511 inputs/bundle; one switching matrix and no encrypted slot rotations.                                                        |
| Native product and circuit | Six raw-terminal profiles in Appendix B.5, with $N = 256L$ ; balanced tree has chain $[3, 2, 1]$ and three independent secrets.                                    |
| HELlib terminal CRT        | 31 measured configurations, including the raw/lazy component policies; exact per-output prime sets and arities are retained.                                       |
| HELlib dense BSGS          | The separately measured length-256, sixteen-job product anchor; it is not the primary terminal-CRT compiler or a composition timing reuse.                         |

The HELlib generators are (4627, 12853), of orders (16, 32). Its ciphertext primes are

$$16212711268417537, \quad 17788947086180353,$$

and its special prime is 4025030748930049. Each of two gadget partitions contains one ciphertext prime. For the product and circuit CRT profiles, the default generators and their orders are respectively

| $m$   | generators | orders    |
|-------|------------|-----------|
| 4369  | (3, 317)   | (128, 2)  |
| 13107 | (5, 197)   | (128, 4). |

The records include both ciphertext and special moduli, all generators, partition indices and separately labeled library security estimates. Terminal moduli are specified separately, retaining legal small primes; the initial modulus is not a substitute. For each published switching matrix, the source-secret power, automorphism, destination secret and number of gadget columns are checked against the stated key policy.

*Different laws and admission rules.* Native uses the ternary/CBD20 tensor laws of Section 4. The composition sampler uses uncapped rejection. The core sam-

pler has eight-round and declared word-budget aborts. Their operational source interfaces therefore differ. Native deterministic arithmetic admission is not the conventional margin test: the latter combines public capacity checks, a ten-bit terminal margin and full output verification, not a matched quantified failure theorem.

The HELib 2.3.0 pin is commit `3e337a66a91a92d49de6a9505340826b0eb71081`. Its `skHwt=0` secrets and encryption ephemerals use the reduced, embedding-bounded small sampler. For these non-power-of-two conductors, it samples  $m$  signed/zero coordinates with nonzero probability  $\text{round}(2^{15}\varphi(m)/(2m))/2^{15}$ , reduces modulo  $\Phi_m$ , and applies an embedding rejection bound. The errors use radially clamped Box–Muller draws at scale  $3.2\sqrt{m}$ , integer rounding, cyclotomic reduction and embedding rejection. The bounded routines allow 1000 attempts before an exception. These are neither iid ternary physical coefficients nor exact discrete-Gaussian or CBD20 errors. These laws follow the key-generation and sampling routines in the pinned source tree.<sup>1</sup> Same-key switching differs from native’s acyclic graph. Neither its Q/QP screening number nor its successful outputs transfer the native conditional theorem to that transcript.

*Build and machine identity.* Composition and circuit experiments use a Ryzen 9950X, x86-64 WSL2 kernel `6.18.33.2-microsoft-standard-WSL2` and Python 3.14.4. The composition environment reports conda-forge GCC 16.2.0. Conventional adapters use Release `-03 -DNDEBUG`; HELib threading is disabled, with NTL 11.6.0 and GMP 6.3.0. An independent rebuild of the measured native shared library was not performed. Thread/affinity limits, exclusions and sample orders are those of Section 9. The accompanying artifact documentation identifies the recorded build configurations and executable versions.

### H.1.2 Eligible alternatives without matched profiles

**H.1.3 Public-polynomial specialization: an equal-count control** The retained Hermite-255 specification has one public dense degree-255 polynomial, sixteen distinct private-point fixtures and sixteen Hasse coefficients per point: 256 useful  $\mathbb{F}_{2^{16}}$  outputs. Its comparison field is fixed by codec factor zero, with explicit isomorphisms into the other fifteen factors. These explicit field maps are needed because the composition experiment uses a different polynomial convention.

The conventional owner repeats each point in sixteen field slots; public compilation supplies its sixteen derivative polynomials. The count compiler uses 15 baby-power products, three giant-power squares and 15 tree products, plus  $16 \cdot 15$  public-coefficient products. Independent plaintext checks compare Hasse coefficients, field maps, interpolation and encoding. These are compiler counts; no encrypted benchmark or security analysis for this public-polynomial profile is supplied.

<sup>1</sup> HELib source: <https://github.com/homenc/HELlib/tree/3e337a66a91a92d49de6a9505340826b0eb71081>.

**Table 23.** Eligibility is not measured competitiveness. Uninstantiated rows have no numerical modulus/noise/build/security certificate; absent entries are not zero costs. Primary applicability analysis is in Section 9 and Appendix G.4.

| Alternative                       | Functionality and status in this paper                                                                                                                                                          |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Other BFV/BGV backend             | Section 9.1 supplies an exact full-field OpenFHE adapter and fresh implementation timings. Other layouts and SEAL remain eligible; a shared concrete security level is not established.         |
| Generalized BFV                   | Principal plaintext modulus two admits this leveled quotient. Improved-GBFV reduced-slot bootstrap hypotheses do not allow direct parameter substitution; this does not exclude other adapters. |
| Recent SIMD polynomial evaluation | Reviewed regimes and public-coefficient code do not directly price private composition. An encrypted-coefficient adaptation remains eligible and requires a complete cost analysis.             |
| Exact-word SIMD                   | Appendix G.4 gives a terminal compiler and sufficient word widths, including shared-input triangular composition packing. Backend admission, parameters and matched timings are uninstantiated. |
| TFHE-rs Boolean/PBS               | A full exact-field circuit is a possible implementation, not a measured selected profile. A scalar PBS is not a full-jet result.                                                                |
| Ordinary approximate CKKS         | Changes the exact-output task. Discrete encodings and proved exact-recovery compositions remain eligible; they require their own full compiler and admission.                                   |

**H.1.4 Five-factor binary products** This OpenFHE 1.5.1 auxiliary experiment multiplies five binary length-16 unit fixtures per job and returns all sixteen low binary coefficients to their input owner. Native uses one job per ciphertext at dimension 8192. Conventional packing uses 76 evaluation points per job over  $\mathbb{F}_{65537}$ , then recipient interpolation, bounded integer lifting and parity. At dimension 16384 it holds 215 jobs, with 44 unused scalar slots. This is a different field and layout from the sixteen-lane composition construction. For the rows below, native uses  $5J$  inputs,  $4J$  products and  $J$  outputs; packing uses five inputs, four products and one output.

At 215 jobs, recorded median pipeline throughput is 15.093/2272.163 jobs/s for native/packed BGV and 19.031/1707.102 for BFV. The packed wins are genuine at this interface. Pipeline sums encoding, encryption, evaluation, decryption and checked decoding; setup, precomputation, serialization time and network are excluded. The run uses Windows 11/Ryzen 9950X, GCC 16.1.0 Release and one OpenMP thread. Its requested library security preset is not independent qualification of the acyclic JetHE key policy.

The complete experiment contains 92 cases, 644 retained trials and 158256 recorded output coefficients, with configurations, ranges, phase timings and re-

**Table 24.** Compiler-derived counts at equal dimension 4096 and capacity sixteen length-16 jets. These are semantic ciphertext counts, not encrypted measurements or a proof of optimality.

| Specified schedule             | Ct-ct Depth Ct-pt |   |     | Input/output |
|--------------------------------|-------------------|---|-----|--------------|
| Native radix-16 BSGS           | 33                | 8 | 240 | 1/1          |
| Conventional direct-Hasse BSGS | 33                | 8 | 240 | 1/1          |

**Table 25.** Sequential-product medians. Each configuration has one key setup, one excluded warmup and seven retained fresh-encryption trials on a fixed fixture. The native encoding in this experiment carries one binary jet per ciphertext; it does not use the sixteen-lane composition layout.

|     | Jobs BGV pipeline (ms) |        | BFV pipeline (ms) |         |
|-----|------------------------|--------|-------------------|---------|
|     | Native                 | Packed | Native            | Packed  |
| 1   | 54.602                 | 89.921 | 53.528            | 123.483 |
| 16  | 852.467                | 89.850 | 821.940           | 124.322 |
| 215 | 14245.037              | 94.624 | 11297.516         | 125.944 |

source measurements retained. A failed depth-three BFV configuration is included among the diagnostic results; the displayed five-factor measurements use common requested depth four. These are measurements of binary products and are separate from the composition experiments. The artifact documentation identifies the recorded data and procedures for recomputing their summaries.

## H.2 Stage-dependent native gadgets and complete measurements

This variant retains the actual basis, source laws, circuit, independent key graph and fixed-input contract of Appendix B.3. Only the radix changes by stage. Its banks are generated under fresh independent keys; joint publication of several variants under reused secrets is outside the stated proof.

**Proposition 32 (Fixed-chain stage admission and row minimum).** *At the prefix and tails 8, 4, 2, 1, respectively, use prime counts (3, 3, 3, 2, 2) and radix widths (45, 45, 60, 40, 60). Each stage uses one width for all its bank families. The corresponding digit counts are (4, 4, 3, 3, 2). Under the fixed-input ideal-coin contract of Theorem 8, all 22 evaluated states are admitted, with failure below  $2^{-129}$  over 1024 fixed batches. The profile uses 102 hint rows and 297 MiB of raw public material. Among widths  $2 \leq w \leq 60$  with this fixed chain, shared stage widths, the stated sufficient recurrence and no extra transformations, 102 rows and 284 row-prime pairs are minimal.*

*Proof.* At a stage with width  $w$ , write  $D_w = 2^{w-1}$  and  $g = \lceil \log_{2^w} q_a \rceil$ . Extract the first  $g-1$  digits in  $[-D_w, D_w)$  and keep the final signed quotient. For centered  $x$ ,  $|x| \leq (q_a - 1)/2$  and  $q_a < 2^{wg}$  imply that this last quotient lies in  $[-D_w, D_w]$ .

Its positive endpoint must be retained. Here  $D_w \leq 2^{59} < p_i$ , so the existing signed-word representation and final-quotient algorithm apply. The constructor scales row  $j$  by  $2^{wj}$  and the evaluator uses the same stage's decomposition.

Replace the two increments in (12) by

$$\Delta_a(b; w) = \left\lceil 15\sqrt{40ND_w^2gb} \right\rceil, \quad \Delta_{H,a}(w) = \left\lceil 15\sqrt{60ND_w^2g} \right\rceil.$$

The support counts and unused-bank independence are unchanged. In particular, Lemma 7 still removes the next helper secret from the chosen Hasse phase. Retain  $F = 34347$ , the prefix raw cap,  $P_T$  of (14), the drop before tail 2 and the final carry. Add the final shared identity-error caps without declaring independence. Exact evaluation gives the following tail bounds:

| $r$ | $B_r$                                |
|-----|--------------------------------------|
| 8   | 973616735988410940026120048          |
| 4   | 419522211267771624851839306616049674 |
| 2   | 168990270141831631071935287          |
| 1   | 72824530783593675310113562548789870  |

Every intervening state satisfies  $q_a > 2 + 4B$ . The final strict slack is

$$1037929870754964969203438818855139399.$$

The same 59 polynomial bad events per batch suffice because each increment sums over all banks and digits in that stage. The union bound is

$$1024 \cdot 59 \cdot 2N \exp(-225/2) < 2^{-129}.$$

For minimality, the width limit requires at least three digits at each three-prime stage and two at each two-prime stage. Three digits at the prefix or tail 8, or two digits at tail 2, require width 60. Exclude these three possibilities using optimistic continuations of the same recurrence. For the prefix, keep its actual raw cap and two-bank increment and set every later switching increment to zero. For tail 8 or tail 2, start with zero incoming error, retain its Hasse increment, and set all later switching increments to zero. Their respective final caps exceed  $10^{39}$ ,  $8 \cdot 10^{38}$  and  $4 \cdot 10^{39}$ , whereas admission requires  $B < (q_2 - 2)/4 < 4 \cdot 10^{35}$ . These are lower bounds on the sufficient recurrence: product, addition and prime-drop maps are nondecreasing and all omitted increments are nonnegative. They are not lower bounds on actual honest noise. Thus every admitted profile in this class has at least (4, 4, 3, 3, 2) digits, attained simultaneously by the stated widths. The bank-family counts (2, 12, 8, 6, 2) give

$$2 \cdot 4 + 12 \cdot 4 + 8 \cdot 3 + 6 \cdot 3 + 2 \cdot 2 = 102.$$

Weighting these rows by their creation prime counts gives 284 row-prime pairs, hence 284 MiB of two-polynomial hints. Five owner public keys add 13 MiB. This establishes both restricted minima.

The proof does not optimize other chains, unequal bank widths within a stage, larger-word arithmetic, tighter noise arguments or runtime. Exact integer evaluation of the recurrence covers all 22 states and identifies all three failed optimistic inequalities, including the unrounded values underlying the comparisons above.

*Public source and total costs.* There remain nine secrets, five public keys and 35 fresh encryptions per batch. Setup now uses 107 primitive errors and 116 small vectors. The incoming-row counts sum to 107; in topological order they are

$$(1, 8, 37, 12, 16, 9, 10, 9, 5).$$

By the same acyclic source projection and row deletion,  $q_3/37$  is a sufficient ordinary source premise; the  $q_3/46$  premise also suffices. The two-batch scheme gap factor remains  $2(9 + 70) = 158$ , at adversary time plus simulation cost. The stopped honest word-source test has budget  $65536[8(297+79)+247] = 213319680$  and ideal stop probability below  $2^{-207}$ . Neither statement assigns security bits to the source. Input and output raw payloads remain 103 and 3 MiB.

**H.2.1 Measured performance and alternative implementations** Both arms use the same lazy NTT arithmetic, based on redundant residue representatives as in Harvey [26]; this is an implementation technique, not a new cryptographic construction. The candidate backend additionally accepts widths 40 and 45. Public checks cover 655360 transform words, 262144 reconstructed integer coefficients, 655360 spectral words, 1048576 unchanged-width correspondence words and 26 deterministic zero-noise public payload rows. Separate encrypted validation executions check all 57 states and 3735552 phase coefficients per arm.

The eight-setup measurement series uses the predeclared order B, S, S, B, B, S, S, B, where B is the uniform-width lazy baseline. Each fresh setup runs a cold and a warm batch with fresh encryption coins, checking all 4096 outputs against the clear reference. All sixteen batches pass. Executions run serially on logical CPU 0, with one thread, 2 GiB address space, 180 seconds CPU and 210 seconds wall limits. Competing physical-host activity is uncontrolled. Measurement excludes private phase diagnostics and includes owner preparation, encoding, encryption, serialization, evaluation and recipient work. Setup is reported separately and in first use. Table 26 gives phase medians and Table 27 preserves every setup.

Additional implementation experiments quantify alternative choices. Lazy NTT alone changed warm medians from 2.520748 to 2.465445 seconds (ratio 1.022431, eight setups and sixteen batches). Architecture targeting and explicit transform scratch buffers were slower in public kernel comparisons; no encrypted timing experiment was performed for them. A separately admitted one-prime terminal uses 388 MiB public material, 102 MiB inputs and 1.5 MiB output; all 58 validation states and sixteen measurement batches pass. Its warm median is 2.480667 versus 2.471738 seconds for its lazy baseline, so it is retained as a bandwidth option. A uniform-width two-prime-start parameter search admits

**Table 26.** Uniform and stage-dependent gadget widths: median (minimum–maximum) across four fresh setups per arm. Times are seconds; RSS is MiB. Both arms use lazy NTT and the same complete local accounting.

| Metric          | Uniform width 44          | Stage widths              |
|-----------------|---------------------------|---------------------------|
| Setup           | 3.175 (3.165–3.235)       | 2.502 (2.473–2.573)       |
| Cold batch      | 2.532 (2.507–2.594)       | 2.342 (2.325–2.356)       |
| Warm batch      | 2.467 (2.453–2.491)       | 2.305 (2.268–2.324)       |
| Setup + cold    | 5.729 (5.674–5.785)       | 4.842 (4.803–4.926)       |
| Warm encryption | 1.289 (1.282–1.302)       | 1.304 (1.285–1.315)       |
| Warm evaluation | 0.953 (0.945–0.963)       | 0.776 (0.762–0.787)       |
| Peak RSS        | 593.977 (593.824–594.137) | 498.879 (498.703–499.133) |

**Table 27.** Every stage-dependent-gadget setup in its predeclared order. B is the uniform-width lazy baseline; S uses stage widths. Times are seconds; RSS is MiB. No sample is discarded.

| Index | Arm | Setup | Cold  | Warm  | First use | RSS     |
|-------|-----|-------|-------|-------|-----------|---------|
| 0     | B   | 3.235 | 2.549 | 2.491 | 5.785     | 593.824 |
| 1     | S   | 2.573 | 2.353 | 2.324 | 4.926     | 499.133 |
| 2     | S   | 2.516 | 2.325 | 2.268 | 4.841     | 498.715 |
| 3     | B   | 3.165 | 2.594 | 2.472 | 5.759     | 594.074 |
| 4     | B   | 3.167 | 2.507 | 2.453 | 5.674     | 594.137 |
| 5     | S   | 2.488 | 2.356 | 2.292 | 4.844     | 498.703 |
| 6     | S   | 2.473 | 2.331 | 2.318 | 4.803     | 499.043 |
| 7     | B   | 3.184 | 2.515 | 2.463 | 5.699     | 593.879 |

none of 225 tested profiles under the existing sufficient bounds; this is not a noise lower bound or impossibility result. These experiments have distinct samples. Their timing ratios are not multiplied together or into the native-algebra control comparison.

The accompanying artifact documentation identifies the build configuration, recorded observations and output checks. Recomputing those observations’ summary statistics does not perform fresh encryption or establish source hardness.

**H.2.2 Contiguous tensor-axis implementation** This implementation keeps the 102-row stage construction, its prime chain, sampling laws and admission from Appendix H.2. Only the order of the first-axis transform operations changes. The normal axis, source sampler and compiler options are unchanged.

**Proposition 33 (First-axis arithmetic equivalence).** *Let  $n$  be a power of two at most 256, and let  $x_{i,e}$  be canonical residues modulo one of the three measured primes, stored at index  $256i + e$ , where  $0 \leq i < n$  and  $0 \leq e < 256$ . Reordering the independent first-axis radix-two transforms to process all  $e$  within each butterfly returns exactly the same canonical forward and inverse transforms.*

*It uses  $256n$  temporary words and retains the same  $O(256n \log n + 256n)$  modular arithmetic bound per channel.*

*Proof.* Write  $\rho$  for the  $n$ -point bit-reversal permutation. The forward route initializes  $y_{j,e} = x_{\rho(j),e} \text{ twist}_{\rho(j)}$ ; the inverse initializes  $y_{j,e} = x_{\rho(j),e}$ . The span, block and twiddle indices then follow the original radix-two order, with  $e$  innermost. For each fixed  $e$ , every operation reads and writes the same two entries as its original column transform. Different  $e$  values have disjoint entries. Induction over butterfly stages therefore preserves every modular value. The inverse applies the original untwist, including inverse-length scaling, at the same final index  $i$ .

The integer representatives also remain valid. Each measured prime satisfies  $p < 2^{60}$ . Given  $0 \leq a, b < 4p$ , first reduce  $a$  into  $[0, 2p)$ . For a nontrivial twiddle  $0 \leq w < p$ , put  $Q = \lfloor 2^{64}w/p \rfloor$  and  $h = \lfloor bQ/2^{64} \rfloor$ . Since  $b < 4p < 2^{62}$ ,

$$0 \leq bw/p - bQ/2^{64} < b/2^{64} < 1/4.$$

Consequently  $h$  is either  $\lfloor bw/p \rfloor$  or one less, and  $b' = bw - hp$  lies in  $[0, 2p)$ . Unsigned 64-bit evaluation gives this same value even when its two products wrap, because their difference is nonnegative and less than  $2^{64}$ . For the unit-twiddle branch, reduce  $b$  directly into  $[0, 2p)$ . The two butterfly outputs  $a + b'$  and  $a + 2p - b'$  lie in  $[0, 4p)$ , below  $2^{62}$ . Final subtraction of  $2p$  and then  $p$  gives the unique canonical residue. Forward twists and inverse untwists use the unchanged canonical modular multiplier. The  $256n$ -word temporary and the original butterfly inventory give the stated space and work bounds, including  $n = 1$ .

At  $n = 256$  the temporary is 512 KiB. It is allocated and released inside each operation, and its cost is charged there. This reordering can improve memory access and allocation overhead; it does not prove a runtime reduction or change the asymptotic arithmetic count. It preserves the same lift, gadget, switching, dropping and fused-encryption outputs on identical inputs. The common integer trit/CBD<sub>20</sub> vectors across RNS channels and independent fresh coins across calls are unchanged. The 37-row source premise and factor-158 public-view reduction retain their original conditional scope.

*Encrypted correctness checks and complete measurements.* The arithmetic consistency checks compare all nine relative transform lengths under all three primes, with 784896 transform words and 63 independently evaluated sparse-polynomial values. It also checks the affected exported arithmetic, including 655360 fused-encryption words using deterministic public inputs. Six predeclared public kernel blocks yield a 1.238641 baseline/candidate ratio. These checks are separate from the fresh encrypted evidence and is not a complete-workflow timing result.

Both encrypted phase-validation executions check 57 states and 3735552 coefficients. The timing experiment uses eight fresh setups, each with a cold and a warm batch, in B,T,T,B,B,T,T,B order. Both arms retain nine secrets, five

**Table 28.** Column transforms and contiguous tensor lanes: median (minimum–maximum) over four fresh setups per arm. Both use the 102-row stage construction and identical complete accounting. Times are seconds; RSS is MiB.

| Metric          | Column transforms         | Contiguous lanes          |
|-----------------|---------------------------|---------------------------|
| Setup           | 2.576 (2.550–2.714)       | 2.330 (2.308–2.364)       |
| Cold batch      | 2.405 (2.389–2.435)       | 2.098 (2.092–2.138)       |
| Warm batch      | 2.354 (2.333–2.377)       | 2.057 (2.043–2.067)       |
| Setup + cold    | 4.982 (4.968–5.116)       | 4.426 (4.403–4.502)       |
| Warm encryption | 1.330 (1.318–1.349)       | 1.125 (1.115–1.128)       |
| Warm evaluation | 0.796 (0.791–0.802)       | 0.707 (0.704–0.711)       |
| Peak RSS        | 499.104 (498.699–499.355) | 498.762 (498.512–499.281) |

public keys, 102 hint rows and 107 setup error vectors. Each batch uses 35 fresh encryptions, 105 small vectors, 70 error vectors and 19 ciphertext products, and checks all 4096 output coefficients.

**Table 29.** Every native transform setup in predeclared order. B is the scalar-loop 102-row backend; T uses contiguous tensor lanes. Times are seconds; RSS is MiB. No sample is discarded.

| Index | Arm | Setup | Cold  | Warm  | First use | RSS     |
|-------|-----|-------|-------|-------|-----------|---------|
| 0     | B   | 2.561 | 2.408 | 2.377 | 4.968     | 498.699 |
| 1     | T   | 2.364 | 2.138 | 2.064 | 4.502     | 499.281 |
| 2     | T   | 2.308 | 2.094 | 2.043 | 4.403     | 498.512 |
| 3     | B   | 2.590 | 2.389 | 2.333 | 4.980     | 499.160 |
| 4     | B   | 2.550 | 2.435 | 2.362 | 4.985     | 499.355 |
| 5     | T   | 2.341 | 2.092 | 2.067 | 4.433     | 498.926 |
| 6     | T   | 2.318 | 2.101 | 2.050 | 4.419     | 498.598 |
| 7     | B   | 2.714 | 2.401 | 2.346 | 5.116     | 499.047 |

Executions run serially on CPU 0 of the same host, with one thread, 2 GiB address space, 180 CPU seconds and a 210-second wall limit. Competing physical-host activity is uncontrolled. Setup charges context/codecs, keys/hints, recipient preparation and public serialization. Each batch charges owner preparation, encoding, fresh encryption, counting-sink serialization, evaluation and recipient recovery. Process startup/import, public fixture/oracle preparation and checking, post-batch teardown and physical network transport are excluded in both arms. All samples are retained; first-use medians are formed from each setup-plus-cold sum.

The correctness checks and timing experiment use the same implementation. This experiment compares two native arithmetic backends; the conventional control is not measured here, and its times from another series are not combined with these observations.

**H.2.3 Extension-axis batching** The next implementation batches the independent normal-axis transforms. Let  $n = 256$  and let  $\ell$  be the relative first-axis length. For each row, the original transform computes  $QBDFPx$ :  $P$  is the multiplicative-index input permutation,  $F$  the forward  $n$ -point NTT,  $D$  the fixed diagonal convolution kernel,  $B$  the backward NTT and  $Q$  the output permutation. The kernel retains its original inverse- $n$  normalization. Forward and inverse normal transforms retain their respective original  $P, D, Q$ .

Store the workspace at index  $j\ell + i$ , where  $i$  selects a row. Initialize each entry with that row's  $Px$  at the forward bit-reversal index  $j$ . Run the original butterfly stages with  $i$  innermost. Different rows have disjoint writable entries, so induction over stages gives exactly  $FPx$  in each row. Canonicalize and multiply by the same kernel, obtaining  $DFPx$ . Swapping workspace rows  $j$  and its bit reversal once when  $j$  is smaller performs the backward transform's input permutation. The same batched butterflies then give  $BDFPx$ ; final canonicalization and  $Q$  return the original result. For  $\ell < 16$ , the implementation retains the original row transform.

Both butterfly passes use the integer invariant proved in Proposition 33: all intermediate values lie in  $[0, 4p)$  and final reductions return canonical residues. The kernel multiplier receives canonical values and is unchanged. There are still two 256-point NTTs and 256 kernel products per row. Asymptotic work is unchanged; the  $256\ell$ -word workspace is at most 512 KiB and is allocated inside the transform after the first-axis temporary has been released. Actual process memory is measured separately. No sampled vector, key dependency, source assumption or correctness admission changes.

*Fresh execution and accounting.* Arithmetic consistency checks cover every relative length and prime, independent sparse evaluations and all affected exported operations. Its six fixed-coin kernel blocks give a 1.260106 baseline/candidate ratio; this is separate from fresh secret encryption and complete timing. Both phase checks pass 57 states and 3735552 coefficients. Eight fresh setups then run in B,E,E,B,B,E,E,B order, each with a cold and warm batch, where B is the preceding tensor-axis implementation. Every 4096-symbol output passes. The same CPU, thread, limits and complete cost boundary described above apply; the tensor-axis baseline is measured afresh.

Both arms retain nine secrets, five public keys, 102 hint rows and 107 setup error vectors. Every batch uses 35 fresh encryptions and 19 products. Table 30 gives the phase and complete medians. The independent setups provide descriptive samples for this backend comparison; they do not establish population-level performance, numerical security or global optimality.

**H.2.4 Precomputed fixed transform multipliers** The current backend precomputes  $Q_w = \lfloor 2^{64}w/p \rfloor$  for each fixed dyadic twist, untwist and normal-axis convolution-kernel entry. For canonical  $0 \leq x, w < p < 2^{60}$ , put  $\hat{q} = \lfloor xQ_w/2^{64} \rfloor$ . Since  $0 \leq xw/p - xQ_w/2^{64} < 1$ , the approximate quotient is either  $\lfloor xw/p \rfloor$  or one smaller. Thus  $0 \leq r = xw - \hat{q}p < 2p$ , and a single conditional subtraction

**Table 30.** Batching the extension axis: median (minimum–maximum) over four fresh setups per arm. Both use the 102-row stage construction and identical complete accounting. Times are seconds; RSS is MiB.

| Metric          | Tensor baseline           | Both axes batched         |
|-----------------|---------------------------|---------------------------|
| Setup           | 2.326 (2.309–2.337)       | 2.125 (2.107–2.160)       |
| Cold batch      | 2.137 (2.102–2.159)       | 1.856 (1.845–1.870)       |
| Warm batch      | 2.048 (2.033–2.076)       | 1.805 (1.791–1.835)       |
| Setup + cold    | 4.456 (4.431–4.491)       | 3.978 (3.956–4.030)       |
| Warm encryption | 1.116 (1.109–1.125)       | 0.956 (0.947–0.981)       |
| Warm evaluation | 0.706 (0.698–0.725)       | 0.626 (0.623–0.632)       |
| Peak RSS        | 499.139 (498.789–499.387) | 499.066 (498.820–499.125) |

**Table 31.** Every extension-axis setup in predeclared order. B is the tensor-axis implementation; E also batches the extension axis. Times are seconds; RSS is MiB. No sample is discarded.

| Index | Arm | Setup | Cold  | Warm  | First use | RSS     |
|-------|-----|-------|-------|-------|-----------|---------|
| 0     | B   | 2.337 | 2.155 | 2.051 | 4.491     | 499.004 |
| 1     | E   | 2.160 | 1.870 | 1.811 | 4.030     | 499.016 |
| 2     | E   | 2.107 | 1.859 | 1.835 | 3.966     | 499.117 |
| 3     | B   | 2.309 | 2.159 | 2.033 | 4.468     | 499.273 |
| 4     | B   | 2.328 | 2.102 | 2.046 | 4.431     | 498.789 |
| 5     | E   | 2.138 | 1.852 | 1.799 | 3.990     | 499.125 |
| 6     | E   | 2.111 | 1.845 | 1.791 | 3.956     | 498.820 |
| 7     | B   | 2.323 | 2.120 | 2.076 | 4.443     | 499.387 |

returns  $xw \bmod p$ . Unsigned word wrap preserves this exact  $r < 2^{61}$ ; the product forming  $\hat{q}$  uses exact 128-bit arithmetic.

Each replaced call receives a canonical input. Kernel quotients are computed after their original inverse-length normalization, and twist quotients after their exact construction. Butterfly ordering, permutations, normalization and sampled coins are unchanged. Product-by-product equality therefore gives the original transforms and encrypted evaluation for the same inputs and coins. This reuses standard fixed-operand modular arithmetic; it changes no source or noise law and is not a new cryptographic lemma.

For each dyadic length  $\ell$ , two new arrays hold  $2\ell$  words per limb; the extension context adds  $2 \cdot 256$  words. Across all supported lengths  $1, 2, 4, \dots, 256$ , the three-limb quotient-array payload is at most  $3(2 \cdot 511 + 512) \cdot 8 = 36816$  bytes, plus container and allocator overhead. Construction remains inside context setup or the first timed use of a relative transform. Ciphertext/key arrays and asymptotic work are unchanged.

*Fresh execution and complete accounting.* The public correspondence check covers all relative lengths and primes, independent sparse polynomial evaluations and the affected exported operations. Its fixed-coin encryption-kernel ratio is

1.093793, separate from complete timings. Both fresh phase checks pass 57 states and 3735552 phase coefficients. Eight independent setups run in B, F, F, B, B, F, F, B order on CPU 0 with one thread, each with a cold and warm batch using fresh independent coins. B is the extension-axis baseline; F adds the fixed-multiplier quotients. Every 4096-symbol output passes. The setup/batch boundary and exclusions are identical to the preceding experiments, including charged precomputation and serialization.

**Table 32.** Every setup in the fixed-multiplier comparison in predeclared order. B is the extension-axis implementation; F precomputes fixed multiplier quotients. Times are seconds; RSS is MiB. No sample is discarded.

| Index | Arm | Setup | Cold  | Warm  | First use | RSS     |
|-------|-----|-------|-------|-------|-----------|---------|
| 0     | B   | 2.100 | 1.850 | 1.792 | 3.950     | 499.359 |
| 1     | F   | 2.039 | 1.795 | 1.767 | 3.834     | 498.848 |
| 2     | F   | 2.108 | 1.788 | 1.746 | 3.896     | 499.004 |
| 3     | B   | 2.159 | 1.855 | 1.868 | 4.014     | 499.344 |
| 4     | B   | 2.152 | 1.982 | 1.935 | 4.134     | 499.090 |
| 5     | F   | 2.159 | 1.822 | 1.777 | 3.980     | 498.941 |
| 6     | F   | 2.086 | 1.808 | 1.736 | 3.894     | 499.074 |
| 7     | B   | 2.174 | 1.870 | 1.805 | 4.044     | 499.535 |

Table 33 gives complete and phase medians. Every sample is retained. These measurements isolate the fixed-multiplier optimization within the native implementation. They do not provide a numerical-security match with conventional controls or establish an unrestricted performance ranking.

**Table 33.** Precomputation for fixed multipliers: median (minimum–maximum) over four fresh setups per arm. Both use the 102-row stage construction and identical complete accounting. Times are seconds; RSS is MiB.

| Metric          | Extension baseline        | Fixed multipliers         |
|-----------------|---------------------------|---------------------------|
| Setup           | 2.156 (2.100–2.174)       | 2.097 (2.039–2.159)       |
| Cold batch      | 1.863 (1.850–1.982)       | 1.802 (1.788–1.822)       |
| Warm batch      | 1.836 (1.792–1.935)       | 1.756 (1.736–1.777)       |
| Setup + cold    | 4.029 (3.950–4.134)       | 3.895 (3.834–3.980)       |
| Warm encryption | 0.969 (0.949–0.986)       | 0.920 (0.909–0.928)       |
| Warm evaluation | 0.636 (0.622–0.722)       | 0.612 (0.596–0.620)       |
| Peak RSS        | 499.352 (499.090–499.535) | 498.973 (498.848–499.074) |

### H.3 Effects of transport and retention choices

We quantify the effect of retaining intermediate ciphertexts, reducing Hasse payloads, choosing a prepared prefix, and selecting a BSGS split. Each experiment uses a fixed public composition fixture, with fresh setup and encryption coins. Repeated trials measure runtime variation for that workload rather than a distribution of plaintexts.

*Final-output retention.* The reference retains 24 intermediate ciphertext states for diagnostics. The revised evaluator performs the same ordered arithmetic and returns only the final ciphertext. A fresh setup supplies the same public bundle to both: their final two components agree on all 262144 uint64 words, and recovery agrees with all 4096 coefficient-Horner outputs. This paired same-process check establishes ciphertext correspondence, not an RSS gain. The symbolic operation-DAG check independently agrees on all 1189 nodes.

A separate experiment fixes T,F,F,T,T,F, with three fresh processes per arm and cold/warm batches. The traced and final-only warm medians are 4.792 and 4.814 seconds. Their median process peaks are 1092.488 and 1026.957 MiB, a 65.531 MiB reduction. All twelve batches pass and all samples remain, including the final-only 5.311-second warm observation. The result supports lower retained memory, with no established speed improvement. The 84.5 MiB raw diagnostic-payload estimate is not the measured RSS saving.

*Paid-identity transport.* Proposition 2 changes orders 8, 4 and 2; order 1 retains the reference bound. The paid-identity catalog has 149 rows in 33 families, versus 203 in 44. Its Hasse portion has 19 families instead of 30. Raw hint arrays fall from 754 to 542 MiB; the five public keys remain 17 MiB. Inputs and outputs remain 137 and 2 MiB of raw arrays. The ten-key acyclic graph is unchanged, with at most 46 incoming rows at any vertex instead of 81 (Appendix C.1).

The compact-spectrum implementation derives multipliers from the same already extracted digits. It charges an additional  $52N = 3407872$  modular products and additions, and a 112 KiB public power cache; it does not claim fewer NTTs or a lower sufficient error. Small-backend correspondence checks cover the multiplier and reused arithmetic APIs. A full-size encrypted validation run then checks all 35 input phases and 24 intermediate states, totaling 3866624 coordinates, and all 4096 terminal field coefficients. The checked states satisfy the complete admitted bounds. Diagnostic phase checks are outside timed batches.

The transport experiment uses the order R,P,P,R,R,P, where R uses the  $2r$  transport and P uses the paid-identity transport. Both arms use final-only evaluation and the same arithmetic library, including the public power-cache charge. Table 34 reports its three fresh setups per arm. The robust engineering change is lower public material and observed setup and process memory. Warm medians differ by about one percent; the experiment does not establish a steady-state speed improvement or significance.

*Finite native parameter search.* The unchanged reference recurrence was evaluated on 273 combinations of prefix  $k \in \{4, 5\}$ , radix widths 32/48/56 and

**Table 34.** Public-key transport comparison: three fresh setups per arm, both final-only and using the same arithmetic library. All twelve batches pass. Setup+cold uses declared setup phases plus cold workflow wall.

|                            | Reference 203 rows | Paid identity 149 rows |
|----------------------------|--------------------|------------------------|
| Raw public arrays, MiB     | 771                | 559                    |
| Serialized keys+hints, MiB | 771.003            | 559.002                |
| Warm wall median, s        | 4.817              | 4.762                  |
| Warm minimum–maximum, s    | 4.767–4.906        | 4.747–4.797            |
| Setup+cold wall, s         | 12.680             | 10.294                 |
| Warm evaluator median, s   | 1.959              | 1.942                  |
| Peak process RSS, MiB      | 1029.2             | 810.0                  |

nonincreasing limb chains of the prescribed length. Twenty-seven pass. A first-use raw-traffic objective selects  $k = 5$ , width 48, chain  $(4, 4, 3, 2)$ : 66 inputs, 34 products, depth four and 108 rows, with 650 MiB first-use traffic versus 910 MiB for the original  $k = 4$  reference. Repeat traffic instead rises from 139 to 263 MiB, and the existing modular-work proxy rises by 30.36%. Against that original reference, the first-use choice has lower cumulative raw traffic for at most three batches sharing setup. This comparison uses the original  $2r$  model, with the  $2r$  transport catalog. The changed prefix is an analytical endpoint and was not executed or called a speed gain.

*Conventional selection and joint comparison.* The existing diagonal BSGS formulas were checked over every divisor of 256. The full-field axis admits the executable subset 16/32/64. A declared work/material proxy selected this subset before timing, and its fixed order was 16,32,64,32,64,16,64,16,32. The public layout checks, three encrypted validation runs and all eighteen measured batches pass. Table 35 retains the slow observations: minimizing evaluator time alone would select 64 and worsen the complete workflow.

The comparison with native composition uses split 32, selected by complete warm median with ties broken by smaller split. Its N,C,C,N,N,C measurements use fresh processes and setups for every observation. The selection rule is fixed before this comparison. Both implementations check the same complete output vector against the public reference. These experimental correctness checks do not test isolation between deployment roles.

The transport identity reduces public material; its support analysis quantifies the additional noise and digit work. The retention experiment isolates memory effects, and the split experiment demonstrates why complete owner-to-recipient time is the relevant objective. The Gaussian asymptotic family is analyzed separately in Section 7.

#### H.4 Terminal-output and modulus tradeoffs

This is a separate four-configuration experiment on the same sixteen-job composition fixture. It uses separate observations from the OpenFHE and HELib

**Table 35.** Separate BSGS split-selection experiment. All sixteen jobs and all owner encryptions are charged. Three fresh setups per split; the complete warm median selects 32 for the native/HElib comparison.

|                           | $b = 16$      | $b = 32$      | $b = 64$      |
|---------------------------|---------------|---------------|---------------|
| Inputs per two-job bundle | 271           | 287           | 319           |
| Group rekeys / rotations  | 16 / 15       | 8 / 7         | 4 / 3         |
| Warm wall median, s       | 38.272        | 37.766        | 39.384        |
| Warm minimum–maximum, s   | 37.977–38.368 | 37.749–37.829 | 39.350–41.023 |
| Warm encryption median, s | 15.822        | 16.729        | 18.661        |
| Warm evaluator median, s  | 20.751        | 19.316        | 18.762        |
| Public keys, MiB          | 6.354         | 3.353         | 1.852         |
| Inputs per batch, MiB     | 542.529       | 574.561       | 638.623       |
| Peak process RSS, MiB     | 58.3          | 56.3          | 61.1          |

measurements. The paid-identity 149-row reference was executed with both the reference and optimized arithmetic backends. The two terminal variants use the optimized backend. Each configuration has three fresh setups, each with one cold and one warm batch; all 24 batches check all 4096 output symbols.

The fixed order is paid identity with the reference backend, paid identity with the optimized backend, optimized four-prime terminal, optimized three-prime terminal; the next round starts with optimized paid identity and the third with the four-prime terminal. Executions run serially on Ryzen 9950X WSL logical CPU 0, one thread, with 2 GiB address space, 840 CPU seconds and 900 wall seconds per execution. The implementation uses Python 3.14.4 and GCC 16.2.0. CPU frequency and competing host activity are not fixed. Every sample, including the slow control runs, is retained.

**Table 36.** All observed ranges in the twelve-setup terminal-delivery comparison. Each cell is median [minimum, maximum], in seconds. No slow sample is discarded.

| Configuration                    | Complete warm        | Setup plus first batch  |
|----------------------------------|----------------------|-------------------------|
| Paid identity, reference backend | 4.709 [4.673, 4.731] | 10.190 [10.181, 10.448] |
| Paid identity, optimized backend | 3.631 [3.445, 3.806] | 8.112 [8.044, 9.186]    |
| Four-prime terminal, optimized   | 3.493 [3.433, 4.592] | 8.033 [7.978, 10.450]   |
| Three-prime terminal, optimized  | 2.571 [2.542, 2.622] | 6.052 [5.893, 6.474]    |

Complete warm time includes owner preparation, encoding, fresh encryption, serialization to a counting sink, evaluation, decryption and linear recovery. Setup is the sum of context/codec, keys/hints and public-key serialization. Excluded costs are process startup/import, fixture and oracle construction, checking, post-batch object teardown, physical network and disk transfer. Setup plus first batch

is a declared-phase sum, not process-start latency. RSS includes setup and all roles. All configurations use the same accounting.

The three-prime warm median is 2.571 s, versus 4.709 s for the paid-identity reference, 3.631 s for paid identity with the optimized backend and 3.493 s for the optimized four-prime terminal. The resulting ratios are 1.832, 1.412 and 1.358. The predeclared twofold complete-workflow target was not met. The evaluator-only median, 0.984 s versus 1.930 s for the reference backend, does not replace that target. Three samples on one host give descriptive ratios, not a confidence bound or a population-speedup theorem.

**Table 37.** Material for the paid-identity and three-prime terminal profiles. One logical output has two and three components, respectively.

| Quantity                                | Paid identity | Three-prime terminal |
|-----------------------------------------|---------------|----------------------|
| Hint rows                               | 149           | 134                  |
| Raw keys and hints (MiB)                | 559           | 391                  |
| Serialized keys and hints (bytes)       | 586156560     | 409995552            |
| Raw inputs (MiB)                        | 137           | 103                  |
| Serialized inputs (bytes)               | 143655584     | 108004000            |
| Raw output (MiB)                        | 2             | 3                    |
| Serialized output (bytes)               | 2097280       | 3145864              |
| Owner public keys / independent secrets | 5 / 10        | 5 / 9                |

The terminal public spectral table adds 1 MiB. The paid-Hasse public power cache shrinks from 112 to 84 KiB with three channels. The optimized backend also stores precomputed NTT operand quotients. Their generation and memory are included in setup and RSS; they are not transmitted keys. The larger output must be charged when transport is modeled.

*Attribution of the computation saving.* The optimized backend uses exact reduction for the existing near- $2^{60}$  primes, precomputed NTT operand quotients, and fused encryption preserving the original coins and encryption equations. These are implementation improvements, not new cryptographic primitives. Applying them to the paid-identity reference provides the 3.631-second control. The four-prime terminal control isolates terminal switching elimination; comparison with it attributes the remaining change to the 44-bit gadget and three-prime schedule, rather than backend optimization alone.

At the reference two-prime terminal modulus and gadget length  $g = 3$ , switching elimination saves two decompositions,  $2g$  full lifts and  $(3g - 2)aN$  pointwise products after charging the recipient’s extra square and product, where  $a = 2$ . The transform counts give 12,461,056 saved modular multiplications in these classes. Additional output serialization and the one-time public table remain costs. This constant operation saving does not change the asymptotic composition class, and the measured total saving is not predicted by counting one operation class alone.

*Arithmetic and phase validation.* The backend checks compare 100256 modular products and 3309568 residue words, including encryption equality for identical fixture coins. The 44-bit gadget reconstructs 131072 coefficients covering 46 distinct boundary values. These are arithmetic checks, not encrypted benchmarks. Three separate encrypted validation executions check 11403264 phase coefficients in total; the three-prime execution covers 35 inputs and 22 intermediates, 3735552 coefficients. All timed batches check the full output.

Every admission entry is computed with exact integer arithmetic. Table 8 reports the lossless state bounds; all 22 strict inequalities hold. Numerical phase checks corroborate the recurrence on their sampled executions, while the tail calculation supplies the quantified fixed-batch failure bound. Library comparisons and implementation ablations retain their own measurement series; ratios from separate series are not multiplied to form a new observation.

## I Comparative measurements and resource tradeoffs

### I.1 Exact product and circuit measurements

The first two tables retain all 38 configurations, not only the main-text endpoints. Each row has three independently sampled key setups and two fresh-encryption batches per setup. Warm entries are median [minimum, maximum]; first use adds setup and keys to the first batch. Other resource entries are medians. Sizes are MiB and times seconds;  $b_Q$  is the actual initial ciphertext-modulus bit length, not the library bit request. Public keys include hints; input and output volumes are separated. Native has 60-bit primes and gadget width 48. HELib uses its own exact published-modulus and key policy. Neither side has a 128-bit qualification. Here  $P_{L,J}$  denotes product;  $S$  and  $D$  are affine-product/balanced-tree circuits at (256, 16). A conductor without a suffix uses raw three-part delivery; suffixes  $nn, ny, yn, yy$  are the deep policies of Appendix B.5. The BSGS row is the independent dense anchor, not the primary terminal-CRT endpoint. Combined-phase times, serialized sizes, individual kernel times and owner subphase times are retained for each observation; the artifact documentation identifies the full data.

*Complete phase and throughput accounting.* The following panels use the same six native/CRT endpoints as the main text. Each entry is native / CRT. Phase entries are medians across three warm batches; their sum need not equal the median complete local time. Residual plumbing time remains charged in the complete total. Evaluation includes all evaluator allocations, copies, alignment and switching, not only arithmetic kernels. Owner and recovery entries include the respective counting-sink serialization. Throughputs divide useful jobs or field coefficients by the warm local median; they are not steady-state or network measurements. Public setup and key generation/serialization are charged once per setup. Cache sizes count prepared-plaintext buffer payloads: native binary-coefficient bytearrays versus CRT uint16 evaluation arrays. They exclude con-

**Table 38.** All dense-product profiles and the BSGS anchor.

| Cell         | Compiler   | $b_Q$ | Warm [min,max]         | First use | Keys  | Inputs  | Outputs | RSS   |
|--------------|------------|-------|------------------------|-----------|-------|---------|---------|-------|
| $P_{16,1}$   | Native     | 60    | 0.0054 [0.0053,0.0054] | 0.0339    | 0.063 | 0.125   | 0.094   | 41.7  |
| $P_{16,1}$   | 4369       | 54    | 0.0144 [0.0132,0.0146] | 0.1060    | 0.063 | 0.125   | 0.094   | 41.7  |
| $P_{16,1}$   | 13107      | 54    | 0.0432 [0.0402,0.0472] | 0.3145    | 0.126 | 0.250   | 0.188   | 41.4  |
| $P_{16,1}$   | 21845      | 108   | 0.1193 [0.1176,0.1220] | 1.002     | 0.501 | 1.000   | 0.750   | 78.3  |
| $P_{16,1}$   | 65535      | 108   | 0.4067 [0.4003,0.4269] | 3.571     | 1.001 | 2.000   | 1.500   | 172.5 |
| $P_{16,16}$  | Native     | 60    | 0.0056 [0.0055,0.0076] | 0.0342    | 0.063 | 0.125   | 0.094   | 41.7  |
| $P_{16,16}$  | 4369       | 54    | 0.0295 [0.0286,0.0298] | 0.1236    | 0.063 | 0.251   | 0.188   | 41.7  |
| $P_{16,16}$  | 13107      | 54    | 0.0410 [0.0406,0.0427] | 0.3185    | 0.126 | 0.250   | 0.188   | 41.6  |
| $P_{16,16}$  | 21845      | 108   | 0.1234 [0.1187,0.1573] | 1.001     | 0.501 | 1.000   | 0.750   | 78.3  |
| $P_{16,16}$  | 65535      | 108   | 0.4114 [0.4023,0.5493] | 3.566     | 1.001 | 2.000   | 1.500   | 172.6 |
| $P_{256,1}$  | Native     | 120   | 0.1402 [0.1396,0.1414] | 0.2012    | 2.000 | 4.000   | 3.000   | 43.6  |
| $P_{256,1}$  | 4369       | 54    | 0.0302 [0.0299,0.0395] | 0.1275    | 0.063 | 0.251   | 0.188   | 41.4  |
| $P_{256,1}$  | 13107      | 54    | 0.0468 [0.0402,0.0508] | 0.3138    | 0.126 | 0.250   | 0.188   | 41.6  |
| $P_{256,1}$  | 21845      | 108   | 0.1190 [0.1169,0.1593] | 0.9953    | 0.501 | 1.000   | 0.750   | 78.4  |
| $P_{256,1}$  | 65535      | 108   | 0.4297 [0.4237,0.4422] | 3.573     | 1.001 | 2.000   | 1.500   | 172.4 |
| $P_{256,16}$ | Native     | 120   | 0.1439 [0.1438,0.1441] | 0.2053    | 2.000 | 4.000   | 3.000   | 43.7  |
| $P_{256,16}$ | 4369       | 54    | 0.4833 [0.4665,0.6455] | 0.5597    | 0.063 | 4.013   | 3.008   | 41.6  |
| $P_{256,16}$ | 13107      | 54    | 0.7011 [0.6834,0.9240] | 0.9600    | 0.126 | 4.007   | 3.004   | 41.6  |
| $P_{256,16}$ | 21845      | 108   | 0.9781 [0.9703,0.9897] | 1.877     | 0.501 | 8.004   | 6.002   | 78.4  |
| $P_{256,16}$ | 65535      | 108   | 1.677 [1.652,2.143]    | 4.868     | 1.001 | 8.002   | 6.001   | 172.4 |
| $P_{256,16}$ | BSGS 13107 | 108   | 36.743 [36.699,37.233] | 37.316    | 6.354 | 544.531 | 3.002   | 58.8  |

**Table 39.** All shallow/deep profiles. Nine-component delivery without evaluation keys remains visible alongside switching alternatives.

| Cell | Compiler | $b_Q$ | Warm [min,max]         | First use | Keys   | Inputs  | Outputs | RSS   |
|------|----------|-------|------------------------|-----------|--------|---------|---------|-------|
| $S$  | Native   | 120   | 0.1924 [0.1912,0.2522] | 0.2523    | 2.000  | 6.001   | 3.000   | 45.6  |
| $S$  | 4369     | 54    | 0.6152 [0.6120,0.7588] | 0.6849    | 0.063  | 6.020   | 3.008   | 41.7  |
| $S$  | 13107    | 54    | 0.8683 [0.8614,0.9607] | 1.132     | 0.126  | 6.010   | 3.004   | 41.6  |
| $S$  | 21845    | 108   | 1.163 [1.161,1.175]    | 2.049     | 0.501  | 12.006  | 6.002   | 78.3  |
| $S$  | 65535    | 108   | 1.971 [1.966,1.974]    | 5.130     | 1.001  | 12.003  | 6.001   | 172.4 |
| $D$  | Native   | 180   | 1.856 [1.836,1.860]    | 2.307     | 39.001 | 36.003  | 1.500   | 135.7 |
| $D$  | 13107/nn | 162   | 18.149 [18.053,18.187] | 18.569    | 0.376  | 288.217 | 108.052 | 58.2  |
| $D$  | 13107/ny | 108   | 16.696 [16.569,16.727] | 16.937    | 1.376  | 192.188 | 12.016  | 50.1  |
| $D$  | 13107/yy | 108   | 18.383 [18.335,18.401] | 18.603    | 0.626  | 192.188 | 12.016  | 47.7  |
| $D$  | 21845/nn | 162   | 19.287 [19.122,19.290] | 20.137    | 0.751  | 288.108 | 108.026 | 107.1 |
| $D$  | 21845/ny | 108   | 17.759 [17.600,17.770] | 18.680    | 2.751  | 192.094 | 12.008  | 91.2  |
| $D$  | 21845/yn | 162   | 21.610 [21.552,21.627] | 22.554    | 2.001  | 288.108 | 60.016  | 103.3 |
| $D$  | 21845/yy | 108   | 18.193 [18.145,18.284] | 19.254    | 1.251  | 192.094 | 12.008  | 86.8  |
| $D$  | 65535/nn | 162   | 28.693 [28.358,28.817] | 31.959    | 1.501  | 288.054 | 108.013 | 220.2 |
| $D$  | 65535/ny | 162   | 31.175 [31.046,31.413] | 34.704    | 9.002  | 288.054 | 24.005  | 229.9 |
| $D$  | 65535/yn | 162   | 32.776 [32.633,33.069] | 36.252    | 4.001  | 288.054 | 60.008  | 212.6 |
| $D$  | 65535/yy | 108   | 28.607 [28.533,28.876] | 31.823    | 2.501  | 192.047 | 12.004  | 178.8 |

tainer overhead, temporary buffers and fixed public tables; they are not peak RSS.

**Table 40.** Warm local phase medians and derived throughput for all six main-text endpoints.

| Cell         | Owner (s)       | Evaluation (s)    | Recovery (s)    | Jobs/s           |
|--------------|-----------------|-------------------|-----------------|------------------|
| $P_{16,1}$   | 0.0036 / 0.0050 | 0.000127 / 0.0026 | 0.0017 / 0.0065 | 185.76 / 69.48   |
| $P_{16,16}$  | 0.0036 / 0.0102 | 0.000125 / 0.0054 | 0.0018 / 0.0139 | 2882.37 / 542.24 |
| $P_{256,1}$  | 0.0942 / 0.0108 | 0.0034 / 0.0054   | 0.0419 / 0.0139 | 7.13 / 33.16     |
| $P_{256,16}$ | 0.0965 / 0.1695 | 0.0034 / 0.0878   | 0.0433 / 0.2229 | 111.20 / 33.10   |
| $S$          | 0.1442 / 0.2486 | 0.0036 / 0.1347   | 0.0439 / 0.2277 | 83.14 / 26.01    |
| $D$          | 0.7649 / 5.968  | 1.060 / 9.457     | 0.0287 / 1.226  | 8.62 / 0.96      |

**Table 41.** Separately charged preparation, prepared-plaintext buffer payloads and useful-output throughput.

| Cell         | Public setup (s) | Key setup (s)   | Cache (MiB)         | Field coefficients/s |
|--------------|------------------|-----------------|---------------------|----------------------|
| $P_{16,1}$   | 0.0274 / 0.0861  | 0.0011 / 0.0041 | 0.007812 / 0.000122 | 2972.18 / 1111.75    |
| $P_{16,16}$  | 0.0274 / 0.0881  | 0.0011 / 0.0039 | 0.007812 / 0.001953 | 46117.92 / 8675.85   |
| $P_{256,1}$  | 0.0272 / 0.0908  | 0.0287 / 0.0048 | 0.125000 / 0.001953 | 1825.45 / 8487.76    |
| $P_{256,16}$ | 0.0274 / 0.0875  | 0.0288 / 0.0040 | 0.125000 / 0.031250 | 28467.51 / 8474.87   |
| $S$          | 0.0274 / 0.0879  | 0.0289 / 0.0041 | 0.187500 / 0.046875 | 21283.72 / 6658.24   |
| $D$          | 0.0277 / 0.2744  | 0.3875 / 0.0238 | 0.750000 / 0.750000 | 2206.59 / 245.32     |

**Table 42.** Backend-specific nested timers, medians of three warm batches. These are non-exhaustive components of the complete evaluation times, not comparable primitive definitions or replacement totals. HELib multiplication also includes automatic modulus management; its separate addition timer has median 0.001413 seconds. Neither primary endpoint uses rotations.

| Deep-anchor nested timer                 | Native (s) | CRT 13107/ny (s) |
|------------------------------------------|------------|------------------|
| Product/add versus multiplication        | 0.029145   | 4.269850         |
| Independent rekey versus relinearization | 0.855693   | 3.554039         |
| Prime drop versus explicit modulus work  | 0.172648   | 1.605246         |

**I.1.1 Complete dense-core protocol** The six-cell experiment tests product for lengths 16 and 256, each with one or sixteen jobs. It tests arithmetic circuit at  $(L, J) = (256, 16)$  with two public circuits. The shallow control is  $ab + c$ ; the depth-three member is a balanced product of four factors  $a_i b_i + c_i$ , with twelve independently owned inputs. All field coefficients are permitted independently by the functionality. Six fixed synthetic fixtures exercise it; fresh keys and encryption coins do not make those independent plaintext-distribution samples. No arm bootstraps.

Terminal CRT evaluates the complete untruncated polynomial at 32, 512 or 2048 field points, as justified in Appendix E.2. Only the recipient interpolates and truncates; there are no encrypted rotations or interpolation in this primary route. This recipient may recover additional coefficients, which the declared interface permits. Native packs sixteen jets per ciphertext, even in the one-job cells. The fastest observed CRT carrier has 256 slots for product/shallow and 512 for balanced tree. Consequently, batched length 256 product uses one native product/output versus 32 CRT products/outputs, and balanced tree uses one native output versus 64 CRT outputs. These are unequal carrier capacities, not the normalized theorem’s experiment.

The finite admission in Appendix B.5 retains four conventional conductors and four deep relinearization policies, including five- and nine-component terminal delivery. It admits 32 conventional configurations; 31 rejected candidates comprise 23 public-noise stops and eight matching-output runs below the ten-bit margin, with no observed wrong-output rejection. These are sufficient admission rules, not parameter impossibility or a uniform failure theorem. Native uses one/two 60-bit primes for product, two for shallow, and chain  $[3, 2, 1]$  for deep. Its deep profile has three independent secrets, 14 hint rows, six rekeys and six prime drops.

The experiment includes 38 configurations: six native, 31 CRT and one dense-BSGS anchor. Three independent key setups per configuration each execute a first and a warm fresh-encryption batch. All 114 independent executions and 228 batches pass, totaling 581088 terminal-symbol comparisons. The three rounds interleave arms and reverse within-cell order in round two; no warm-up or slow sample is discarded. Executions run serially on the same Ryzen 9950X WSL host, CPU 0, with one thread, 2 GiB address space, 180 CPU seconds and 210 wall seconds per execution. Other host activity is uncontrolled.

Warm local time is complete batch wall time less separately timed validation. It charges every owner, full evaluator calls including copies/allocations, counting-sink serialization, raw decryption, decoding and unallocated plumbing. First use additionally charges public setup and all keys once. Process launch, library loading, fixture/oracle preparation and network transfer are excluded; neither column is network end-to-end or cold-machine latency. All roles share one process; native uses Python orchestration with compiled kernels, versus the C++ HELib adapter. The comparison does not assume equal optimization. The CRT endpoint minimizes the observed warm median in the admitted family, not

a global parameter search. Appendix I.1 retains all candidates, ranges, phase breakdowns, caches and throughputs.

## I.2 HELib composition measurements

The prepared composition and dense-core experiments show local-computation advantages for some workloads, with substantial resource tradeoffs. They do not establish a security-matched advantage, an application distribution, or a state-of-the-art FHE ranking. In particular, packed gains coexist with an underfilled native loss at length 256.

**I.2.1 Workload, implementations and parameter selection** Both owners supply sixteen length-256 series pairs over  $K = \mathbb{F}_2[X]/(X^{16} + X^{12} + X^3 + X + 1)$ . The inner series has zero constant coefficient. The fixed synthetic fixture contains  $g = 0, g = z$ , valuations two/four and twelve dense valuation-one lanes. Every batch returns all 4096 field coefficients of  $f(g(z)) \bmod z^{256}$ , checked against an independent coefficient-Horner specification. Owners may prepare from their own inputs; neither can use the other’s input. All routes may return terminal decoded coefficients. No encrypted-continuation requirement is imposed.

Native uses the exact selected graph from Table 9. The conventional controls use BGV in HELib 2.3.0, commit `3e337a66a91a92d49de6a9505340826b0eb71081`, with its stock key policy. Column contraction is the lower-key alternative to the stronger tested prepared-diagonal BSGS compiler, not the sole baseline. Appendix H.1 consolidates exact layouts, build/source pins, sampling interfaces and the uninstantiated-alternative roster.

The profile selection considered conductors 4369, 13107, 21845 and requested bit counts 20, 60 before timing. All six public layouts underwent full-field codec and automorphism checks. Eligible profiles were ranked by the declared tuple  $(\ell_q \log_2 n, (\ell_q + \ell_{sp}) \log_2 n, n, \text{requested bits})$ , then each arm selected its first encrypted validation run passing with at least ten reported capacity bits. This is a finite work proxy and margin policy, not parameter optimality. HELib’s security heuristic was only a preliminary filter. Six one-prime validation runs stopped at reported margins 5, 2, 0 below ten, *before decryption*: they establish neither correct nor incorrect outputs for those profiles. Both selected two-prime arms passed with capacity 56.

The resulting conventional carrier has  $m = 13107$ ,  $n = 8192$  and 512 full-field slots, so eight two-job bundles cover the same sixteen jobs. It uses two ciphertext primes and one special prime, with two gadget partitions. Charged evaluation removes special primes from outputs. Their retained modulus (108 bits) is

$$288407062877399519762840090050561.$$

The native top modulus is 240 bits; its output has two limbs. Equal useful outputs, not equal ring dimensions, nonlinear depths or security assumptions, are matched here.

**Table 43.** Executed compiler inventories, not interchangeable-cost API calls. Native additionally performs four Hasse transports, four alignments and two prime drops; each native result rekey uses both source-secret banks.

| All sixteen composition jobs       | Native | BSGS $b = 16$ | Column |
|------------------------------------|--------|---------------|--------|
| Physical bundles / jobs per bundle | 1 / 16 | 8 / 2         | 8 / 2  |
| Input ciphertexts                  | 35     | 2168          | 4088   |
| Ciphertext products                | 19     | 2040          | 2040   |
| Nonlinear depth                    | 5      | 1             | 1      |
| Group relinearizations             | 5      | 128           | 8      |
| Direct slot rotations              | 0      | 120           | 0      |
| Output ciphertexts                 | 1      | 8             | 8      |

**I.2.2 Measured latency, throughput and memory** The fixed order was N,C,B; C,B,N; B,N,C, on the same Ryzen 9950X Ubuntu WSL host, pinned to one common logical CPU, one process and one thread. Each arm has three fresh processes with independently generated keys. Each process runs a cold and a warm batch using fresh encryption coins, but the same public fixture. All eighteen batches pass, with no discarded or replaced timed sample. The 73728 symbol comparisons are repeated checks, not that many independent plaintext cases. Per-execution limits were 2 GiB address space, 900 seconds wall time and 840 seconds CPU.

**Table 44.** Reference 203-row compiler: medians of three samples per arm. Warm batch time includes owner preparation, codec, fresh encryption, evaluation, serialization and recipient recovery, but excludes one-time setup, verification and network. Throughputs are derived from the batch medians, not a saturation experiment.

| Seconds, unless indicated    | Native      | BSGS          | Column        |
|------------------------------|-------------|---------------|---------------|
| Warm batch median            | 4.674       | 36.739        | 48.601        |
| Warm minimum–maximum         | 4.662–4.792 | 36.545–36.965 | 48.497–49.772 |
| Setup+cold phase subtotal    | 12.193      | 36.993        | 49.000        |
| Key/hint setup               | 7.141       | 0.101         | 0.013         |
| Fresh encryption, warm       | 2.518       | 15.331        | 28.471        |
| Public evaluation, warm      | 1.920       | 19.945        | 17.489        |
| Decryption/decoding, warm    | 0.033       | 0.156         | 0.158         |
| Jobs/s from warm median      | 3.423       | 0.436         | 0.329         |
| Field coefficients/s         | 876.3       | 111.5         | 84.3          |
| Public keys+hints, MiB       | 771.003     | 6.354         | 0.626         |
| Inputs per batch, MiB        | 137.001     | 542.529       | 1022.998      |
| Outputs per batch, MiB       | 2.000       | 2.002         | 2.002         |
| Median peak process RSS, MiB | 1091.8      | 58.3          | 42.7          |

The warm ratio is  $36.739290643/4.674077982 \approx 7.86$ . BSGS is faster overall than column although its evaluator is slower: it saves owner encryptions at the cost of group relinearizations and rotations. Native wins this measured warm workflow, while BSGS has much smaller setup material and process memory. These conclusions concern the evaluated implementations and profile selection. They are not consequences of Theorem 11.

The setup+cold ratio is 3.03. This subtotal includes the recorded setup and cold batch phases, not literal process-start latency; imports and oracle preparation are excluded. Individual phase medians need not sum to the median total. Serialization uses non-storing counting sinks, not a measured network. Native intermediate diagnostic retention is charged to its measured memory. Table 47 separately accounts for named owner buffers from the implementation; the composition experiments do not measure total prepared cache size. These payload counts are not substitutes for the RSS row. Three samples support a descriptive range, not a statistical-significance claim or a trend fitted to depth/occupancy.

For illustration only, charge one public bundle and each input/output to a serial transfer model. The first native batch sends 376563120 more bytes than BSGS. Combining this with the 24.800013551-second setup+cold difference gives a modeled crossover near 121.47 Mbit/s: native wins that first-batch model above the threshold. At two batches sharing setup, its cumulative transfer is 48666416 bytes smaller. Latency, overlap and additional owner-key distribution are excluded. Neither calculation is network evidence or an authorization to extend the security reuse horizon.

**I.2.3 Paid-identity HELib comparison and terminal controls** The following experiment uses the 149-row paid-identity profile. Its setups and samples are independent of the 203-row experiment above.

In the following library comparisons, native uses the 149-row paid-identity profile of Proposition 2. Its ten independent secrets, ternary/CBD20 laws and chain remain distinct from the Gaussian theorem. Conventional BGV uses HELib 2.3.0, commit `3e337a66a91a92d49de6a9505340826b0eb71081`, with stock keys and errors. The finite conductor/modulus search selected  $m = 13107$ , dimension 8192, 512 full-field slots, two ciphertext primes and one special prime. The ciphertext modulus is 288407062877399519762840090050561 (108 bits); the native top modulus is 240 bits. Neither nominal bit counts nor HELib’s preliminary heuristic establish equal security. Appendix H.1 specifies the layouts and source laws; the 203-row experiment is specified above.

To compare BSGS layouts, we fixed  $b = 16, 32, 64$  before timing and ran three fresh setups per split. Every full-field correctness check and all eighteen batches passed with the declared output-capacity threshold of ten bits. Split 32 has the smallest complete warm median; split 64 reduces evaluator time but increases owner encryption and full-batch time. The selection uses the complete workflow, not the fastest individual phase.

**Table 45.** Executed inventories for all sixteen composition jobs. Calls in different carriers have different costs. Native also charges four Hasse maps, four bypass alignments and two prime drops.

|                                    | Native paid identity BSGS $b = 32$ |       |
|------------------------------------|------------------------------------|-------|
| Physical bundles / jobs per bundle | 1 / 16                             | 8 / 2 |
| Fresh input ciphertexts            | 35                                 | 2296  |
| Ciphertext products                | 19                                 | 2040  |
| Nonlinear depth                    | 5                                  | 1     |
| Group relinearizations             | 5                                  | 64    |
| Direct slot rotations              | 0                                  | 56    |
| Terminal ciphertexts               | 1                                  | 8     |

**I.2.4 HELib joint comparison and bounded native improvements** After selecting split 32, a separate experiment fixed the order N,C,C,N,N,C. Each of its six independent setups evaluated a cold and a warm batch with fresh encryption coins. Both arms ran sequentially on logical CPU 0 of the Ryzen 9950X Ubuntu WSL host, with one process/thread, 2 GiB address space, 840 seconds CPU and 900 seconds wall limits. Every output check passed; no sample was discarded or replaced. The preceding native and split-selection experiments supply no timing sample here.

**Table 46.** Native/HELlib composition comparison: medians of three independent setups per arm. Warm wall includes preparation, codec, fresh encryption, evaluation, serialization and recovery; excludes setup, verification and network. Byte rows are serialized objects. Source laws and key policies differ; no numerical security level is certified.

|                              | Native 149 rows | HELlib BSGS 32 |
|------------------------------|-----------------|----------------|
| Warm batch wall, s           | 4.781           | 37.976         |
| Warm minimum–maximum, s      | 4.709–5.285     | 37.947–38.034  |
| Setup+cold wall, s           | 10.295          | 38.265         |
| Setup+cold phase subtotal, s | 10.295          | 38.050         |
| Keys/hints setup, s          | 5.225           | 0.050          |
| Fresh encryption, warm s     | 2.588           | 16.795         |
| Public evaluation, warm s    | 1.948           | 19.446         |
| Decryption/codec, warm s     | 0.034           | 0.160          |
| Jobs/s from warm median      | 3.346           | 0.421          |
| Public keys+hints, MiB       | 559.002         | 3.353          |
| Inputs per batch, MiB        | 137.001         | 574.561        |
| Outputs per batch, MiB       | 2.000           | 2.002          |
| Peak process RSS, MiB        | 810.1           | 56.1           |

The ratio of complete warm medians is 7.94; this is a descriptive implementation result on one synthetic fixture. It is not a consequence of the restricted

multiplication lower bound. The conventional arm retains far smaller keys and process memory. Setup plus cold wall sums declared setup phases and the cold workflow wall; the phase subtotal instead sums the individually recorded phases. Neither measures literal process-start latency. Phase medians need not sum to the median total. Counting sinks charge serialization but measure no network, and RSS includes all roles, setup and both batches. Three samples do not establish significance.

Two separate native experiments isolate existing-mechanism changes. Returning only the final ciphertext, instead of retaining 24 diagnostic states, preserves the exact final ciphertext on the same public bundle and reduces median process RSS by 6.00%; its warm median does not improve. Next, both final-only arms use the same rebuilt arithmetic library: replacing 203 rows by 149 reduces raw public arrays by 27.50% and median RSS by 21.30%, while warm latency remains similar. The sufficient error bound and multiplier work increase; these costs are included in admission and measurement. Appendix H.3 gives their separate sample tables, the finite parameter search. These percentages are within-experiment comparisons and must not be multiplied into a combined speed or memory claim.

The measured contraction is not the only conventional algorithm allowed by this interface. Appendix G.2 adapts the existing shrinking recursion to owner preparation and terminal recovery: 29 inputs, 26 field products, six outputs and an admitted 25.94 GB public-row upper bound under its own finite Gaussian law. Its maps and exact envelopes pass public arithmetic checks. Its original sequential sampling cost remains conditional; Proposition 26 supplies an implemented finite table alternative within the same ideal-source distance allowance. A bounded padded-CRT refinement lowers the attained public-row bound to 20.59 GB with 31 inputs and 28 products (Appendix G.2.2). It retains larger sufficient moduli and explicit owner/recipient costs. The full 31-input graph now executes, with all 4915200 phase coordinates and 4096 terminal field coefficients checked. This diagnostic reference is not an optimized timing comparator. The paired depth-one control of Appendix E.4 also strengthens the conventional product count, at greater terminal traffic. These alternatives prevent treating dense contraction counts as a conventional lower bound. The larger key figure does not establish a security-matched native advantage.

### I.3 Exact extension-field computation with OpenFHE

The summed and terminal BGV adapters implement exact binary-extension-field composition through bounded integer lifting. Their arithmetic derivation is independent of the ciphertext modulus and switching implementation.

*Exact binary extension-field recovery.* Represent  $K = \mathbb{F}_2[Y]/(Y^{16} + Y^{12} + Y^3 + Y + 1)$  by binary coefficient polynomials of degree below 16. The inner owner computes  $P_{ij} = [z^j]g(z)^i$  in  $K$  and the outer owner holds  $f_i$ . Writing hats for

their binary integral representatives, form

$$S_j(Y) = \sum_{i=0}^{255} \hat{f}_i(Y) \hat{P}_{ij}(Y) \quad \text{in } \mathbb{Z}[Y].$$

Each coefficient is nonnegative and at most  $256 \cdot 16 = 4096$ , and  $\deg S_j \leq 30$ . Evaluate at the 32 powers of a primitive 32nd root in  $\mathbb{F}_{65537}$ , multiply and sum homomorphically, and apply the inverse transform after decryption. Since  $4096 < 65537$ , canonical integer lifting recovers every integer coefficient of  $S_j$ . Reducing coefficients modulo two and then by the defining degree-16 polynomial gives exactly  $[z^j]f(g(z))$ . This follows because the map  $\mathbb{Z}[Y] \rightarrow K$  is a ring homomorphism. The field arithmetic proof is separate from ciphertext correctness and source security.

The selected baby-step/giant-step contraction realizes these sums with eight packing bundles and baby width 16. Each bundle has 255 products, 16 group relinearizations and 15 rotations; the complete inventory is 2040 products, 128 relinearizations, 120 rotations, 2032 additions and eight output ciphertexts. Both owners' prepared inputs and all plaintext codec/recovery work are charged. Terminal recovery supplies no generic encrypted binary-field reentry procedure.

*Configurations and measurement units.* The finite selection contains twenty parameter/layout candidates. Two BFV configurations fail at the API sizing stage; these produce no ciphertext correctness or timing result. Final measurements use three independent setups per configuration, each followed by first and warm batches. The BGV programs use the library build and thread settings in Section 9.1. Its HYBRID switching material has a 120-bit auxiliary modulus; the summed configuration has a 92-bit ciphertext modulus. The JetHE-149 reference uses a 240-bit creation modulus, subsequent 180/120-bit levels, and one two-component output. The component count is distinguished from the logical ciphertext count throughout the tables. The paired JetHE-102/control series uses the three-prime schedule of Section 6.

Both output recovery and all serialized components are charged. The output vector has 4096 field symbols and is checked in full. The supplied measurement data include every first/warm observation and its configuration. Each encrypted run checks the complete output vector. Recomputing summary statistics from these observations is separate from executing the encrypted computation again.

#### I.4 Complete resource objectives for the measured implementations

The fixed recovery interface does not imply a bound on recipient work or output volume. This subsection gives two transparent cost models using the same four configurations in the OpenFHE comparison. Here “native” denotes the four-prime, 149-row JetHE profile. These models make the effect of resource weights explicit; they establish neither a common numerical security level nor a bound against unimplemented algorithms.

*Serialized key and ciphertext volume.* For arm  $a$ , write  $P_a$  for serialized public keys/hints and  $I_a, O_a$  for serialized input/output ciphertexts per batch. The object-volume model

$$V_a(K) = P_a + K(I_a + O_a)$$

counts each public object once. It excludes separately reported OpenFHE context metadata, replication, routing and transport framing. It is not a network-time measurement. Only two batches per setup were executed; the symbolic expression for other  $K$  requires a separately admitted reuse horizon and does not extend the cryptographic query caps.

Native has  $P = 586156560$  and  $I + O = 145752864$  bytes. The corresponding summed values are 42745683 and 1711313968, and terminal-products has 1573801 and 2411263140 bytes. Thus native pays more public material but less total modeled volume than either nonlinear OpenFHE arm for every integer  $K \geq 1$ : the respective equality points are about 0.3471 and 0.2580. Relay, with  $P = 131773$  and  $I + O = 395451$ , has less volume for every nonnegative  $K$ . These are finite arithmetic consequences of the recorded layouts, not communication lower bounds.

*The price of recipient work.* For warm sample  $i$  of arm  $a$ , let  $W_{ai}$  be its complete batch wall time and  $R_{ai}$  its decryption/recovery phase. Define the hypothetical resource objective

$$F_a(\alpha) = \text{median}_{i=1,2,3} [W_{ai} + (\alpha - 1)R_{ai}], \quad \alpha \geq 1.$$

Here  $\alpha$  varies the price of recipient work, with other recorded phases fixed. It is not a measured CPU slowdown or a prediction for another device. Weighting occurs within each sample before taking its median; independently weighted phase medians need not give this function.

Enumerating the exact rational intersections of the twelve sample lines gives the following lower envelope over all four recorded arms:

| $\alpha$ interval (rounded endpoints) | minimizing arm     |
|---------------------------------------|--------------------|
| $[1, 6.1344)$                         | recipient computes |
| $(6.1344, 435.0547)$                  | native             |
| $(435.0547, \infty)$                  | summed OpenFHE.    |

Adjacent arms tie at the exact endpoints. Terminal-products never minimizes this particular scalar objective, although its public material and process memory differ favorably from native. Including setup and cold recovery changes the analogous transitions to about 13.1163 and 228.3808. No value of  $\alpha$  is selected as an application fact or as evidence that JetHE must be preferred. Faster recipient algorithms, other encodings, hardware-specific prices and communication can change the frontier.

Keeping warm time, recipient time, keys, input/output bytes and peak RSS as separate median coordinates leaves native, summed OpenFHE and relay non-dominated among these four observations. Relay dominates terminal-products

in that broad-task cost vector. Removing relay under the restricted recovery interface changes the eligible set; it does not improve any measured cost. This distinguishes a specified compiler interface from an unrestricted performance claim.

The crossover pieces and byte identities are computed with exact rational arithmetic from the individual phase observations. They describe the specified cost model; changing network or client prices changes the resulting frontier.

## J Reproducibility and complete recorded samples

### J.1 Reproducibility and measurement interpretation

The manuscript source is self-contained. The accompanying measurement data record configuration, setup, batch state, complete time, resource counts, and output digests. The portable verifier recomputes exact integer/rational admission inequalities and checks the supplied data and manuscript dependencies. Mathematical proofs establish the scope of the statements; recorded executions establish the reported timings.

*Experimental units and correctness checks.* Each independent setup generates keys and is followed by two freshly encrypted batches on a fixed synthetic fixture. Complete output vectors are compared with coefficient-based reference algorithms. Where phase checks require secret keys, they run separately from timed evaluation. Measurements use the source and arithmetic configuration specified in each table. Changing keys without changing the fixture samples execution variation, rather than a distribution of applications or plaintexts. All observations of each reported configuration are included.

*Timing and resource accounting.* Complete local time includes owner preparation, input/output codecs, sampling, encryption, evaluation, serialization, decryption and required reconstruction. First-use time adds the setup for that batch. The supplement separately reports kernel-only and diagnostic experiments; their times are not substituted for complete encrypted workflows. Serialization into a counting sink measures processing and byte volume, with physical network transfer excluded. RSS covers each specified process, while array formulas count their named payloads. These measures answer different resource questions.

*Composition prepared-buffer accounting.* The composition experiments do not measure total prepared-cache size separately. The native implementation retains 35 distinct arrays of 4096 two-byte field symbols and, concurrently, 35 encoded arrays of 65536 bytes. The encoder stores each binary coefficient in one byte. The conventional implementation streams eight two-job bundles. Its BSGS branch retains all 256 powers for the current bundle, whereas its column branch retains one running power. Each field symbol is stored in an unsigned two-byte integer.

**Table 47.** Source-derived owner-buffer payloads, not measured total caches. Native covers sixteen jobs; each conventional row covers the two jobs in one streamed bundle, not eight simultaneously retained bundles.

| Composition | arm | Named retained payload (bytes)           | KiB  |
|-------------|-----|------------------------------------------|------|
| Native      |     | $35(4096 \cdot 2 + 65536) = 2580480$     | 2520 |
| BSGS        |     | $2 \cdot 256 \cdot 256 \cdot 2 = 262144$ | 256  |
| Column      |     | $2 \cdot 256 \cdot 2 = 1024$             | 1    |

Python containers, NTL slot/polynomial objects, allocator overhead, fixture and oracle storage, ciphertexts and transient codec/preparation scratch are additional. The table therefore supplies neither a total-cache measurement nor a total-memory ranking. These payload counts are derived from the retained arrays and their scalar types; process RSS additionally includes the objects listed above.

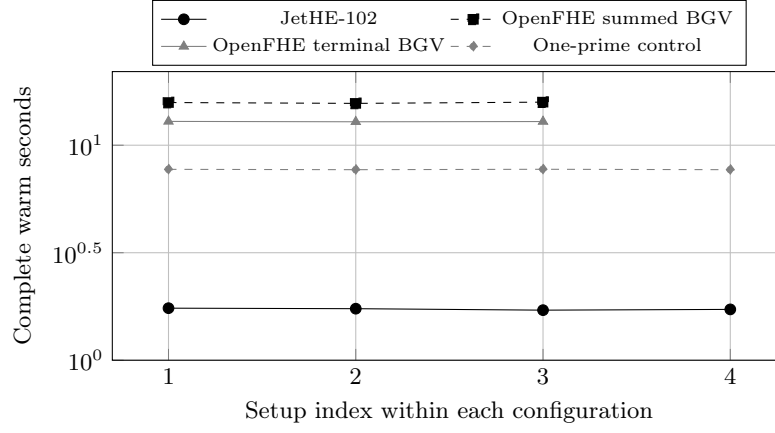
*Evidence for the analytical components.* Independent integer and rational calculations check the prime-drop and gadget recurrences, Gaussian sampler truncations, finite-law statistical distances, coefficient-space witnesses, and bilinear identities. Exact ciphertext comparisons test transform reordering and shared transport implementations. Public zero-noise matrix experiments are identified as such in Appendix F.2; they support arithmetic equivalence and cost measurements without being encrypted executions.

*Security and comparison parameters.* Every implementation profile retains its own ring, modulus, source laws, evaluation-key policy, and correctness admission. The Gaussian-family security theorem, finite ternary/CBD20 source transformations, and heuristic lattice-model outputs are separate results. Incomplete model runs are reported alongside completed predictions. Numerical security matching is not inferred from a library’s requested security setting, a decryption success rate, or a primitive-noise tail bound.

## J.2 Recorded samples and reproducibility

The table lists every setup of the five configurations. There are three setups each for the OpenFHE adapters and JetHE-149, and four each for JetHE-102 and the one-prime control. Every setup serves one first batch and one warm batch, both freshly encrypted. No measured batch was replaced or discarded. The public fixture has sixteen length-256 jobs and 4096 output field symbols. Every listed batch agrees with the complete reference output. Its canonical 8192-byte little-endian field-symbol vector has SHA-256 `d22a60188ba884b10626ae52a2902f003cc2535294053979c417c39be68fbda3`.

The accompanying data contain full-precision observations for every configuration in both measurement series. The reported statistics were recomputed from 20 recorded independent executions. The supplied verification procedures



**Fig. 1.** All warm observations for the selected configurations. Lines connect observations for visibility; indices are neither paired trials between measurement series nor a scaling variable.

recompute the statistics and the numerical correctness recurrence using exact integer and rational arithmetic; they do not perform fresh encrypted experiments. The artifact guide describes these procedures and identifies their inputs. Security and general correctness are established by the stated mathematical arguments.

**Table 48.** All recorded samples contributing to Tables 4 and 5, seconds. Setup identifiers are specific to each experiment series.

| Configuration           | Setup | Cold      | Warm      | First use |
|-------------------------|-------|-----------|-----------|-----------|
| JetHE-102               | 0     | 1.764133  | 1.746354  | 3.834498  |
| JetHE-102               | 3     | 1.772315  | 1.737026  | 3.778558  |
| JetHE-102               | 4     | 1.768277  | 1.709757  | 3.797790  |
| JetHE-102               | 7     | 1.780274  | 1.723741  | 3.809849  |
| OpenFHE summed BGV 1    |       | 15.852754 | 15.795136 | 16.012713 |
| OpenFHE summed BGV 6    |       | 15.703189 | 15.651759 | 15.858880 |
| OpenFHE summed BGV 8    |       | 15.871130 | 15.862125 | 16.025918 |
| OpenFHE terminal BGV 2  |       | 12.667837 | 12.903264 | 12.698492 |
| OpenFHE terminal BGV 5  |       | 12.643305 | 12.853722 | 12.675448 |
| OpenFHE terminal BGV 11 |       | 12.649719 | 12.884548 | 12.680919 |
| One-prime control       | 1     | 7.848439  | 7.724638  | 7.937754  |
| One-prime control       | 2     | 7.695165  | 7.695535  | 7.781928  |
| One-prime control       | 5     | 7.705324  | 7.729049  | 7.791730  |
| One-prime control       | 6     | 7.772318  | 7.695951  | 7.860467  |
| JetHE-149               | 0     | 4.878226  | 4.817546  | 10.391698 |
| JetHE-149               | 7     | 4.732058  | 4.690265  | 10.122360 |
| JetHE-149               | 9     | 4.801179  | 4.733799  | 10.239377 |